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MAX96752/MAX96754

GMSL2 Deserializer with Dual LVDS (OLDI) Output and Optional HDCP

General Description

The MAX96752/MAX96754 deserializers convert a single- or dual-link GMSL2 serial input to single or dual OLDI. They also send and receive side-channel and peripheral control data, enabling full-duplex, single wire, bidirectional transmission of video and data over 15 meter cables. Each GMSL2 link operates at a fixed rate of 3Gbps or 6Gbps in the forward direction and 1.5Gbps or 187.5Mbps in reverse. Data can be transmitted over low-cost 50Ω coax or 100Ω STP cables that meet the GMSL2 channel specification.

The OLDI output can be configured as single port (4 or 8 lanes) or dual port (2x4 lanes) for flexibility in driving displays with a variety of resolutions and pixel clock frequencies. Each port accommodates pixel clock rates of up to 150MHz, and in dual-port mode the MAX96752/MAX96754 support a combined pixel clock of up to 300MHz.

The MAX96754 adds HDCP v1.4 decryption, with on-chip secure key storage and a state machine for HDCP authentication and link monitoring.

The audio channel supports I²S stereo and up to eight channels in TDM mode. HD audio is supported at up to 32-bits/sample at a 192kHz sample rate for 7.1 channel in both forward and reverse directions.

The GMSL2 concurrent control channel operates at 9.6kbps to 1Mbps in UART mode and up to 1Mbps in I²C mode with clock-stretching. Using the control channel, a microcontroller (µC) can locally or remotely program serializer, deserializer, and peripheral device registers at any time, independent of video timing. Two additional pass-though I²C or UART channels and a pass-through 50MHz SPI channel are provided for peripheral control.

Operation is specified over the -40°C to +105°C automotive temperature range. The devices are AEC-Q100 qualified.

Applications

- Cluster and Heads-up Displays
- Central Information Displays
- Rear-Seat Infotainment Displays

Benefits and Features

- 1x4, 2x4, or 1x8 OLDI Output Lane Configurations
- Multiple GMSL2 Data Rates for System and Power Flexibility
  - 3Gbps or 6Gbps Forward Link Rates
  - 187Mbps or 1.5Gbps Reverse Link Rates
- Full-Duplex Capability Over a Single Wire
- Ideal for High-Definition Video Applications
  - Supports up to 300MHz PCLK
  - Supports High-Resolution Video Displays with 24-Bit Color Over a Single GMSL2 Link
  - Supports Video Replication and Dual-View Splitting for Driving Two Displays
- Uses Low-Cost 50Ω Coax or 100Ω STP Cables
- HDCP 1.4 Decryption (MAX96754 Only)
- Forward and Reverse I²S or 7.1 TDM Audio
- Optional Internal V_{DD} Regulator
- ASIL-Relevant Functional Safety Features
  - ASIL B Compliant
  - 16-Bit CRC Protection of Control Channel Data (I²C, UART, SPI, GPIO, Audio)
  - Retransmission of All Control Channel Data (I²C, UART, SPI, GPIO, Audio) Upon Error Detection
  - Optional 32-Bit CRC Protection of Video Line Data
  - Selectable Interrupts for Fault Detection
  - Video Watermark Detection
- Continuous Link Margin Monitoring and Optimization
  - Adaptive Equalization Allows 15m of Coax Cable with Multiple In-Line Connectors
  - Forward and Reverse Channel PRBS Generator and Checker for BER Testing of Serial I/O Links
  - Eye-Opening Monitor for Continuous Link Margin Diagnosis
- Concurrent Control Channel for Device Configuration and Communicating with Remote Peripherals
  - I²C, Pass-through I²C, UART, SPI and Tunneled or Register-Programmable GPIO
  - Settable Priority Levels
  - Eight Hardware Programmable Device Addresses
  - Up to 14 Programmable GPIO
  - Sleep Mode with Register State Retention
- Compact 8mm x 8mm TQFN Package with Exposed Pad

19-100422; Rev 0; 11/18

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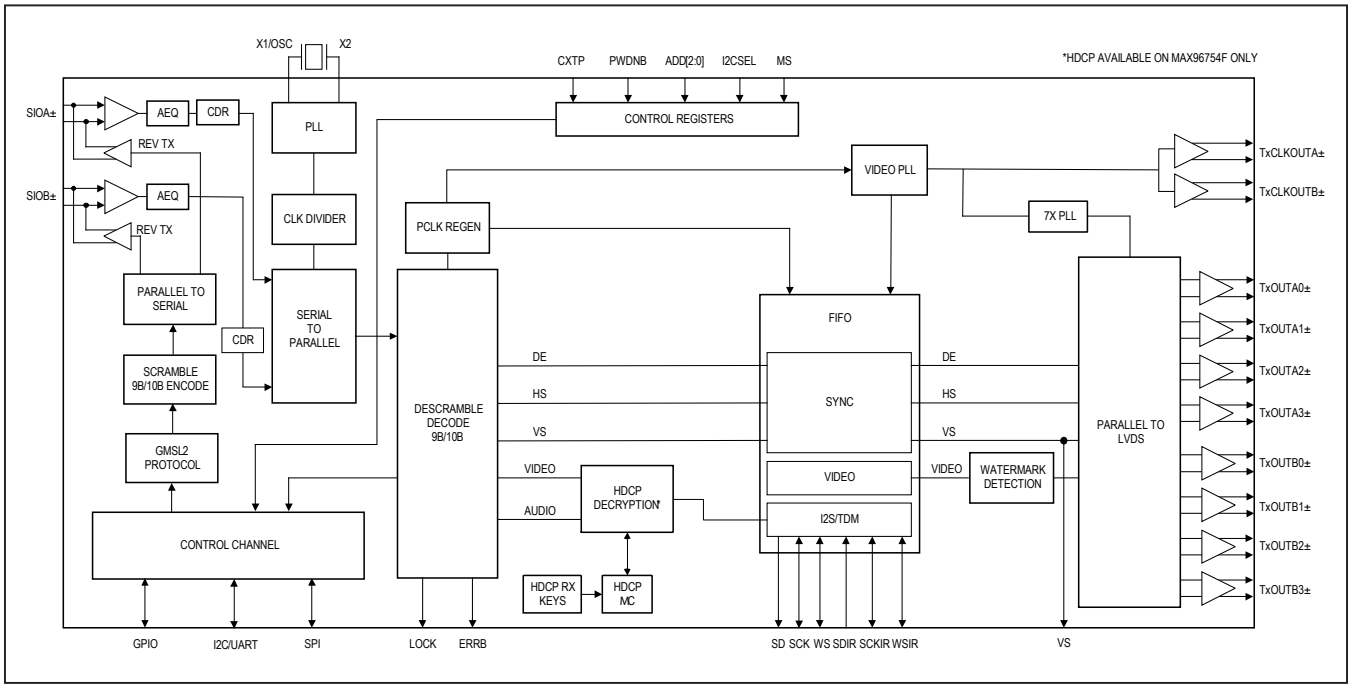
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Simplified Block Diagram



**HDCP decryption available only in MAX96754.*

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Absolute Maximum Ratings

(All voltages with respect to ground.)

V_{DDIO} .....-0.3V to +3.9V
V_{DD18} .....-0.3V to +2.0V
V_{DDD}, V_{DDA} .....-0.3V to +1.1V
V_{REG}, V_{REGA}.....-0.3V to +2.0V
SIO__ (Active State) (Note 1) .....(V_{DD18} - 1.1V) to V_{DD18}
SIO__ (Inactive State) (Note 1).....-0.3V to +1.1V
XRES, X2 .....-0.3V to (V_{DD18} + 0.3V)
All Other Pins (Note 2) .....-0.3V to (V_{DDIO} + 0.3V)

Continuous Power Dissipation (T_A = +70°C TQFN) (Single-layer board, derate 28.6mW/°C above +70°C) 1571mW
Continuous Power Dissipation (T_A = +70°C TQFN) (Multilayer board, derate 40mW/°C above +70°C) 2200mW
Storage Temperature Range ..... -40°C to +150°C
Soldering Temperature (reflow) .....+260°C

Note 1: Active state means the device is powered up and not in sleep or power-down modes. Inactive means the device is not powered up or powered up in sleep or power-down mode.
Note 2: Specified maximum voltage or 3.9V, whichever is lower.

Stresses beyond those listed under "Absolute Maximum Ratings" may cause permanent damage to the device. These are stress ratings only, and functional operation of the device at these or any other conditions beyond those indicated in the operational sections of the specifications is not implied. Exposure to absolute maximum rating conditions for extended periods may affect device reliability.

Package Information

56-Pin TQFN

Package Code	T5688+6
Outline Number	**21-0135**
Land Pattern Number	**90-100041**
Thermal Resistance, Single-Layer Board:	
Junction to Ambient (θ_{JA})	35°C/W
Junction to Case (θ_{JC})	1°C/W
Thermal Resistance, Four-Layer Board:	
Junction to Ambient (θ_{JA})	25°C/W
Junction to Case (θ_{JC})	1°C/W

For the latest package outline information and land patterns (footprints), go to **www.maximintegrated.com/packages**. Note that a "+", "#", or "-" in the package code indicates RoHS status only. Package drawings may show a different suffix character, but the drawing pertains to the package regardless of RoHS status.

Package thermal resistances were obtained using the method described in JEDEC specification JESD51-7, using a four-layer board. For detailed information on package thermal considerations, refer to **www.maximintegrated.com/thermal-tutorial**.

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Electrical Characteristics

(V_{DD18} = 1.7V to 1.9V, V_{DDD} = V_{DDA}= 0.95V to 1.05V or V_{REG} = V_{REGA} = 1.7V to 1.9V, V_{DDIO} = 1.7V to 3.6V, T_A = -40°C to +105°C, EP connected to PCB ground, typical values are at V_{DD18} = V_{DDIO} = 1.8V, V_{DDD} = V_{DDA} = 1.0V, T_A = 25°C, unless otherwise noted.) (Note 1)

PARAMETER	SYMBOL	CONDITIONS		MIN	TYP	MAX	UNITS
DC ELECTRICAL CHARACTERISTICS/GMSL2 DESERIALIZER REVERSE CHANNEL SERIAL OUTPUTS (SIO_P, SIO_N)—SEE Figure 1							
Output Voltage Swing (Single-ended)	V_O	R_L = 100Ω ±1%	Configured for 187.5Mbps	190	250	310	mV
			Configured for 1.5Gbps	280	370	460	
Output Voltage Swing (Differential)	V_{ODT}	R_L = 100Ω ±1%, peak-to-peak differential voltage	Configured for 187.5Mbps	380	500	620	mV
			Configured for 1.5Gbps	560	740	920	
Change in V_{OD} between Complementary Output States	ΔV_{OD}	R_L = 100Ω ±1%, $\lvert V_{OD(H)} - V_{OD(L)} \rvert$				25	mV
Differential Output Offset Voltage	V_{OS}	R_L = 100Ω ±1%, offset voltage in each output state	Configured for 187.5Mbps	V_{DD18} - 0.45	V_{DD18} - 0.3	V_{DD18} - 0.15	V
			Configured for 1.5Gbps (Note 6)	V_{DD18} - 0.55	V_{DD18} - 0.4	V_{DD18} - 0.25	
Change in V_{OS} between Complementary Output States	ΔV_{OS}	R_L = 100Ω ±1%, $\lvert (V_{OS(H)} - V_{OS(L)}) \rvert$				25	mV
Termination Resistance (Internal)	R_T	Any Pin to V_{DD18}		50	55	60	Ω
DC ELECTRICAL CHARACTERISTICS/LVDS OUTPUTS							
Differential Output Voltage	V_{OD}	Full swing enabled		250		450	mV
		Half swing enabled		125		225	
Change in VOD Between Complementary Output States	ΔV_{OD}					25	mV
Common Mode Voltage	V_{OS}			1.125		1.375	mV
Change in VOS Between Complementary Output States	ΔV_{OS}					25	mV
Output Short Circuit Current	I_{OS}	Tx_ pin shorted to 0V or V_{DD18}		-15		+15	mA
Magnitude of Differential Output Short Circuit Current	I_{OSD}	Differential output pair shorted together				15	mA
Output High Impedance Current	I_{OZ}	Output powered down, Tx_ pin shorted to 0V or V_{DD18}		-0.5		+0.5	µA
DC ELECTRICAL CHARACTERISTICS/I/O PINS							
High-Level Input Voltage	V_{IH}			0.7 x V_{DDIO}			V
Low-Level Input Voltage	V_{IL}					0.3 x V_{DDIO}	V

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Electrical Characteristics (continued)

(V_{DD18} = 1.7V to 1.9V, V_{DDD} = V_{DDA}= 0.95V to 1.05V or V_{REG} = V_{REGA} = 1.7V to 1.9V, V_{DDIO} = 1.7V to 3.6V, T_A = -40°C to +105°C, EP connected to PCB ground, typical values are at V_{DD18} = V_{DDIO} = 1.8V, V_{DDD} = V_{DDA} = 1.0V, T_A = 25°C, unless otherwise noted.) (Note 1)

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNITS
High-Level Output Voltage	V_{OH}	I_{OH} = -4mA	V_{DDIO} - 0.4			V
Low-Level Output Voltage	V_{OL}	I_{OL} = 4mA			0.4	V
Input Current	I_{IN}	All pullup/pulldown devices disabled. V_{IN} = 0V to V_{DDIO}			1	µA
Input Capacitance	C_{IN}			3		pF
Internal Pullup/Pulldown Resistance	R_{IN}	40kΩ enabled		40		kΩ
		1MΩ enabled		1		MΩ
DC ELECTRICAL CHARACTERISTICS/OPEN-DRAIN PINS						
High-Level Input Voltage	V_{IH}		0.7 x V_{DDIO}			V
Low-Level Input Voltage	V_{IL}				0.3 x V_{DDIO}	V
Low-Level Open-Drain Output Voltage	V_{OL}	I_{OL} = 4mA			0.4	V
Input Current	I_{IN}	All pullup/pulldown devices disabled, V_{IN} = 0V to V_{DDIO}			1	µA
Input Capacitance	C_{IN}			3		pF
Internal Pullup Resistance	R_{PU}	40kΩ enabled		40		kΩ
		1MΩ enabled		1		MΩ
DC ELECTRICAL CHARACTERISTICS/PWDNB INPUT						
High-Level Input Voltage	V_{IH}		0.7 x V_{DDIO}			V
Low-Level Input Voltage	V_{IL}				0.3 x V_{DDIO}	V
Input Current	I_{IN}	V_{IN} = 0V to V_{DDIO}			6	µA
Internal Pulldown Resistance	R_{PD}			1		MΩ
Input Capacitance	C_{IN}			3		pF
DC ELECTRICAL CHARACTERISTICS/PUSH-PULL OUTPUTS						
High-Level Output Voltage	V_{OH}	I_{OH} = -4mA	V_{DDIO} - 0.4			V
Low-Level Output Voltage	V_{OL}	I_{OL} = 4mA			0.4	V
DC ELECTRICAL CHARACTERISTICS/REFERENCE CLOCK INPUT REQUIREMENTS (CRYSTAL) (X1, X2)						
X1 Input Capacitance	C_{IN_X1}			3		pF
X2 Input Capacitance	C_{IN_X2}			1		pF
Internal X2 Limit Resistor	R_{LIM}			1.2		kΩ
Internal Feedback Resistor	R_{FB}			10		kΩ
Transconductance	g_m			28		mA/V

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Electrical Characteristics (continued)

(V_{DD18} = 1.7V to 1.9V, V_{DDD} = V_{DDA}= 0.95V to 1.05V or V_{REG} = V_{REGA} = 1.7V to 1.9V, V_{DDIO} = 1.7V to 3.6V, T_A = -40°C to +105°C, EP connected to PCB ground, typical values are at V_{DD18} = V_{DDIO} = 1.8V, V_{DDD} = V_{DDA} = 1.0V, T_A = 25°C, unless otherwise noted.) (Note 1)

PARAMETER	SYMBOL	CONDITIONS		MIN	TYP	MAX	UNITS
DC ELECTRICAL CHARACTERISTICS/REFERENCE CLOCK REQUIREMENTS (EXTERNAL INPUT ON X1, X2 UNCONNECTED)							
High-Level Input Voltage	V_{IH}			0.9			V
Low-Level Input Voltage	V_{IL}					0.4	V
Input Impedance	R_{IN}				10		kΩ
X1 Input Capacitance	C_{IN_X1}				5.5		pF
DC ELECTRICAL CHARACTERISTICS/POWER SUPPLY CURRENTS							
Supply Current (Note 4)	I_{DD}	Single 6Gbps link, 187.5Mbps reverse, 148.5MHz PCLK, dual OLDI output at 74.25MHz, color bar pattern	V_{DD18} = 1.9V		125	180	mA
			V_{DDD} = 1.05V		83	250	
			V_{DDA} = 1.05V		7	15	
			V_{REG} = 1.9V (Note 5)		75	250	
			V_{REGA} = 1.9V (Note 5)		8	20	
Maximum VDDIO Supply Current	I_{DDIO}	Per toggling GPIO, C_L = 20pF	V_{DDIO} at 1.9V		44		µA/MHz
			V_{DDIO} at 3.6V		81		
DC ELECTRICAL CHARACTERISTICS/POWER-DOWN CURRENT							
Maximum Power-Down Current	I_{DD}	V_{DDIO} at 3.6V	T_A = 25°C		4		µA
			T_A = 105°C		5		
		V_{DD18} at 1.9V	T_A = 25°C		8		
			T_A = 105°C		25		
		V_{DDD} at 1.05V	T_A = 25°C		7		
			T_A = 105°C		10		
		V_{DDA} at 1.05V	T_A = 25°C		4		
			T_A = 105°C		5		
		V_{REG} at 1.9V (Note 5)	T_A = 25°C		0		
			T_A = 105°C		1		
		V_{REGA} at 1.9V (Note 5)	T_A = 25°C		0.2		
			T_A = 105°C		1		

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Electrical Characteristics (continued)

(V_{DD18} = 1.7V to 1.9V, V_{DDD} = V_{DDA}= 0.95V to 1.05V or V_{REG} = V_{REGA} = 1.7V to 1.9V, V_{DDIO} = 1.7V to 3.6V, T_A = -40°C to +105°C, EP connected to PCB ground, typical values are at V_{DD18} = V_{DDIO} = 1.8V, V_{DDD} = V_{DDA} = 1.0V, T_A = 25°C, unless otherwise noted.) (Note 1)

PARAMETER	SYMBOL	CONDITIONS		MIN	TYP	MAX	UNITS
DC ELECTRICAL CHARACTERISTICS/SLEEP CURRENT							
Maximum Sleep Current	I_{DD}	V_{DDIO} at 1.9V	T_A = 25°C		4		µA
			T_A = 105°C		7		
		V_{DD18} at 1.9V	T_A = 25°C		9		
			T_A = 105°C		18		
		V_{DDD} at 1.05V	T_A = 25°C		5		
			T_A = 105°C		35		
		V_{DDA} at 1.05V	T_A = 25°C		4		
			T_A = 105°C		7		
		V_{REG} at 1.9V (Note 5)	T_A = 25°C		3.6		
			T_A = 105°C		4		
		V_{REGA} at 1.9V (Note 5)	T_A = 25°C		0.2		
			T_A = 105°C		1		
AC ELECTRICAL CHARACTERISTICS/FORWARD CHANNEL SWITCHING CHARACTERISTICS							
Lock Time	t_{LOCK}	From power-up, one-shot reset, or rising edge of PWDNB to rising edge of LOCK			25		ms
Maximum Video Initialization Time	$t_{VIDEOSTART}$	Time from GMSL2 video packet input at SIO± input to when pixels appear at OLDI outputs, assuming link and configuration are already established			0.2 + 45,000 x t_{PCLK}		ms
Maximum Video Latency	t_{VL}	Time from pixel within GMSL2 packet at SIO± to output at OLDI interface			400 x t_{PCLK}		s
PWDNB Hold Time	t_{HOLD_PWDNB}	The minimum duration PWDNB must be held low to reset the chip			1		µs
AC ELECTRICAL CHARACTERISTICS/DESERIALIZER REVERSE CHANNEL SERIAL OUTPUTS							
GMSL Reverse Channel Transmitter Rise/Fall Time	t_R, t_F	20% to 80%, V_O = 250mV, R_L = 100Ω, configured for 187.5Mbps			2300		ps
		20% to 80%, V_O = 370mV, R_L = 100Ω, configured for 1.5Gbps			400		
Total Serial Output p-p Jitter	t_{TSOJ2}	PRBS7, single-ended or differential output (Note 3)			0.15		UI
Deterministic Serial Output p-p Jitter	t_{DSOJ2}	PRBS7, single-ended or differential output (Note 3)			0.1		UI
AC ELECTRICAL CHARACTERISTICS/LVDS OUTPUTS							
LVDS Clock Frequency	$f_{TXCLKOUT}$	Each OLDI output (Note 2)		6.45		150	MHz
LVDS Output Rise Time	t_R	From 20% to 80%, C_L = 5pF, R_L = 100Ω (Note 2)			170	350	ps

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Electrical Characteristics (continued)

(V_{DD18} = 1.7V to 1.9V, V_{DDD} = V_{DDA}= 0.95V to 1.05V or V_{REG} = V_{REGA} = 1.7V to 1.9V, V_{DDIO} = 1.7V to 3.6V, T_A = -40°C to +105°C, EP connected to PCB ground, typical values are at V_{DD18} = V_{DDIO} = 1.8V, V_{DDD} = V_{DDA} = 1.0V, T_A = 25°C, unless otherwise noted.) (Note 1)

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNITS
LVDS Output Fall Time	t_F	From 20% to 80%, C_L = 5pF, R_L = 100Ω (Note 2)		170	350	ps
LVDS Output Pulse Position	t_{PPosN}	$f_{TXCLKOUT}$ = 150MHz (Note 2) (Figure 5)	N/7x t_{CLK} - 100	N/7x t_{CLK}	N/7x t_{CLK} + 100	ps
LVDS Output Jitter		$f_{TXCLKOUT}$ = 150MHz, BER = 10^{-12}		110		ps_{P-P}
LVDS Output Enable Time	t_{LVEN}	Measured from the last bit of UART packet that enables the LVDS to the point that LVDS output amplitude crosses 250mV (Note 2)			50	µs
LVDS Output Disable Time	t_{LVDS}	Measured from the last bit of UART packet that enables the LVDS to the point that LVDS output amplitude crosses 0V (Note 2)			50	µs
AC ELECTRICAL CHARACTERISTICS/I²C/UART PORT TIMING						
Output Fall Time	t_F	70% to 30%, C_L = 20pF to 100pF, 1kΩ pullup to V_{DDIO} (Note 2)	20 x V_{DDIO}/ 5.5V		150	ns
I²C/UART Wake Time		From power-up, or rising edge of PWDNB to local register access; for remote register access, I²C/UART wake time is the same as lock time (t_{LOCK})		1.1		ms
AC ELECTRICAL CHARACTERISTICS/I²C TIMING						
SCL Clock Frequency	f_{SCL}	Low f_{SCL} range: (I²C MST_BT = 010, I²C SLV_SH = 10)	9.6		100	kHz
		Mid f_{SCL} range: (I²C MST_BT = 101, I²C SLV_SH = 01)	100		400	
		High f_{SCL} range : (I²C MST_BT = 111, I²C SLV_SH = 00)	400		1000	
Start Condition Hold Time	$t_{HD:STA}$	f_{SCL} range, low	4			µs
		f_{SCL} range, mid	0.6			
		f_{SCL} range, high	0.26			
Low Period of SCL Clock	t_{LOW}	f_{SCL} range, low	4.7			µs
		f_{SCL} range, mid	1.3			
		f_{SCL} range, high	0.5			
High Period of SCL Clock	t_{HIGH}	f_{SCL} range, low	4			µs
		f_{SCL} range, mid	0.6			
		f_{SCL} range, high	0.26			
Repeated Start Condition Setup Time	$t_{SU:STA}$	f_{SCL} range, low	4.7			µs
		f_{SCL} range, mid	0.6			
		f_{SCL} range, high	0.26			

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Electrical Characteristics (continued)

(V_{DD18} = 1.7V to 1.9V, V_{DDD} = V_{DDA}= 0.95V to 1.05V or V_{REG} = V_{REGA} = 1.7V to 1.9V, V_{DDIO} = 1.7V to 3.6V, T_A = -40°C to +105°C, EP connected to PCB ground, typical values are at V_{DD18} = V_{DDIO} = 1.8V, V_{DDD} = V_{DDA} = 1.0V, T_A = 25°C, unless otherwise noted.) (Note 1)

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNITS
Data Hold Time	$t_{HD:DAT}$	f_{SCL} range, low	0			ns
		f_{SCL} range, mid	0			
		f_{SCL} range, high	0			
Data Setup Time	$t_{SU:DAT}$	f_{SCL} range, low	250			ns
		f_{SCL} range, mid	100			
		f_{SCL} range, high	50			
Setup Time for Stop Condition	$t_{SU:STO}$	f_{SCL} range, low	4			µs
		f_{SCL} range, mid	0.6			
		f_{SCL} range, High	0.26			
Bus Free Time	t_{BUF}	f_{SCL} range, low	4.7			µs
		f_{SCL} range, mid	1.3			
		f_{SCL} range, high	0.5			
Data Valid Time	$t_{VD:DAT}$	f_{SCL} range, low			3.45	µs
		f_{SCL} range, mid			0.9	
		f_{SCL} range, high			0.45	
Data Valid Acknowledge Time	$t_{VD:ACK}$	f_{SCL} range, low			3.45	µs
		f_{SCL} range, mid			0.9	
		f_{SCL} range, high			0.45	
Pulse Width of Spikes Suppressed	t_{SP}	f_{SCL} range, low			50	ns
		f_{SCL} range, mid			50	
		f_{SCL} range, high			50	
Capacitive Load on Each Bus Line	C_B	(Note 2)			100	pF
AC ELECTRICAL CHARACTERISTICS/SPI MASTER (Note 7)						
Operating Frequency	f_{MCK}	(Note 2)	0.588		50	MHz
SCLK Period	t_{MCK}			$1/f_{MCK}$		ns
SCLK Output Pulse-Width High/Low	t_{MCH}, t_{MCL}	C_L = 5pF (Note 2)	$t_{MCK}/2$ - 3	$t_{MCK}/2$		ns
MOSI Output Valid Time	t_{MOV}	After SCLK falling edge, C_L = 5pF (Note 2)	2.3		t_{MCK} - 2.3	ns
MISO Input Setup Time	t_{MIS}	Before programmed sampling edge (Note 2)	13.5			ns
MISO Input Hold Time	t_{MIH}	After programmed sampling edge (Note 2)	-2			ns
AC ELECTRICAL CHARACTERISTICS/SPI SLAVE (Note 7)						
Operating Frequency	f_{SCK}	(Note 2)			50	MHz
SCLK Period	t_{SCK}			$1/f_{SCK}$		ns
MISO Output Valid Time	t_{SOV}	After SCLK falling edge, C_L = 5pF (Note 2)	11.3		t_{SCK} - 1.5	ns

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Electrical Characteristics (continued)

(V_{DD18} = 1.7V to 1.9V, V_{DDD} = V_{DDA}= 0.95V to 1.05V or V_{REG} = V_{REGA} = 1.7V to 1.9V, V_{DDIO} = 1.7V to 3.6V, T_A = -40°C to +105°C, EP connected to PCB ground, typical values are at V_{DD18} = V_{DDIO} = 1.8V, V_{DDD} = V_{DDA} = 1.0V, T_A = 25°C, unless otherwise noted.) (Note 1)

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNITS
MOSI Input Setup Time	t_{SIS}	Before SCLK rising edge (Note 2)	5			ns
MOSI Input Hold Time	t_{SIH}	After SCLK rising edge (Note 2)	3			ns
AC ELECTRICAL CHARACTERISTICS/I²S/TDM MASTER TIMING (Note 2) (Note 7)						
WS Frequency	f_{WS}		8		192	kHz
Sample Word Length	n_{WS}		8		32	Bits
SCK Frequency	f_{SCK}	$f_{SCK} = f_{WS} \times n_{WS} \times (2 \text{ or } 8)$	0.512		49.152	MHz
SCK Clock High Time	t_{HC}	$V_{SCK} \geq V_{IH}$, $t_{SCK} = 1/f_{SCK}$, C_L = 5pF	0.35 x t_{SCK}			ns
SCK Clock Low Time	t_{LC}	$V_{SCK} \leq V_{IL}$, $t_{SCK} = 1/f_{SCK}$, C_L = 5pF	0.35 x t_{SCK}			ns
SD, WS Valid Time Before SCK	t_{MAVS}	$t_{SCK} = 1/f_{SCK}$, C_L = 5pF	0.2 x t_{SCK}	0.5 x t_{SCK}		ns
SD, WS Valid Time After SCK	t_{MAVH}	$t_{SCK} = 1/f_{SCK}$, C_L = 5pF	0.2 x t_{SCK}	0.5 x t_{SCK}		ns
AC ELECTRICAL CHARACTERISTICS/I²S/TDM SLAVE TIMING						
WS Frequency	f_{WS}	(Note 2)	8		192	kHz
Sample Word Length	n_{WS}	(Note 2)	8		32	Bits
SCK Frequency	f_{SCK}	$f_{SCK} = f_{WS} \times n_{WS} \times (2 \text{ or } 8)$ (Note 2)	0.512		49.152	MHz
SD, WS Setup Time	t_{SAS}	(Note 2)	4			ns
SD, WS Hold Time	t_{SAH}	(Note 2)	4			ns
AC ELECTRICAL CHARACTERISTICS/REFERENCE CLOCK INPUT REQUIREMENTS (CRYSTAL)(X1, X2) (Note 2)						
Frequency	f_{XTAL}			25		MHz
Frequency Stability + Frequency Tolerance	f_{TN}				±200	ppm
AC ELECTRICAL CHARACTERISTICS/REFERENCE CLOCK REQUIREMENTS (EXTERNAL CLOCK INPUT ON X1, X2 UNCONNECTED) (Note 2)						
Frequency	f_{REF}			25		MHz
Frequency Stability + Frequency Tolerance	f_{TN}				±200	ppm
Input Jitter		6Gbps/187.5Mbps, sinusoidal jitter < 1MHz (falling edge), upstream serializer using crystal reference			600	ps p-p
Input Duty Cycle	t_{DUTY}		40		60	%
Input Rise Time	T_R	20% to 80%			4	ns

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Electrical Characteristics (continued)

(V_{DD18} = 1.7V to 1.9V, V_{DDD} = V_{DDA}= 0.95V to 1.05V or V_{REG} = V_{REGA} = 1.7V to 1.9V, V_{DDIO} = 1.7V to 3.6V, T_A = -40°C to +105°C, EP connected to PCB ground, typical values are at V_{DD18} = V_{DDIO} = 1.8V, V_{DDD} = V_{DDA} = 1.0V, T_A = 25°C, unless otherwise noted.) (Note 1)

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNITS
ESD Protection						
SIO__	V_{ESD}	Human Body Model, R_D = 1.5kΩ, C_S = 100pF		±8		kV
		ISO10605, R_D = 330Ω, C_S = 150pF, Contact Discharge, Coax Configuration		±6		
		ISO10605, R_D = 330Ω, C_S = 150pF, Contact Discharge, STP Configuration		±4		
		ISO10605, R_D = 330Ω, C_S = 150pF, Air Discharge		±8		
		AEC-Q100-011 Rev-C1, Charged Device Model (CDM)		750		V
All Other Pins	V_{ESD}	Human Body Model, R_D = 1.5kΩ, C_S = 100pF		±4		kV
		AEC-Q100-011 Rev-C1, Charged Device Model (CDM)		750		V

Note 1: Limits are 100% tested at T_A = +105°C unless otherwise noted. Limits within the operating temperature range and relevant supply voltage range are guaranteed by design and characterization.

Note 2: Not production tested. Guaranteed by design.

Note 3: This parameter was measured applying the following transfer function to differential or single-ended serial outputs at 1.5Gbps.

$$H(s) = \frac{A_{DC}\omega_{P1}\omega_{P2}}{\omega_Z} \times \frac{s+\omega_Z}{(s+\omega_{P1})(s+\omega_{P2})}$$

where:

$$A_{DC} = 1,\ \omega_Z = 2\pi \cdot 350MHz,\ \omega_{P1} = 2\pi \cdot 750MHz,\ \omega_{P2} = 2\pi \cdot 2000MHz$$

Note 4: Color bar pattern. Maximum supply currents are measured at indicated supply voltages. Typical supply currents are measured at typical supply voltages.

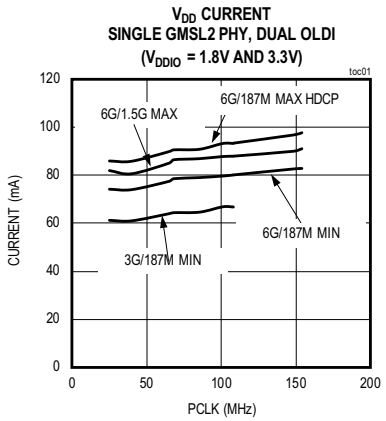
Note 5: If the the V_{DD} regulators regulators are used, use the V_{REG} and V_{REGA} currents instead of the V_{DDD} and V_{DDA} currents. If the regulators are not used, ignore the V_{REG} and V_{REGA} currents and use the V_{DDD} and V_{DDA} currents.

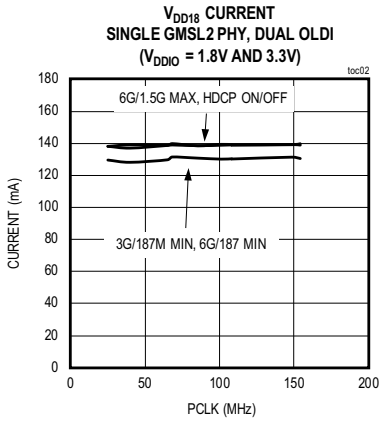
Note 6: Operation at 6Gbps forward and 1.5Gbps reverse bandwidth requires optimized configuration settings. Contact the factory for additional details.

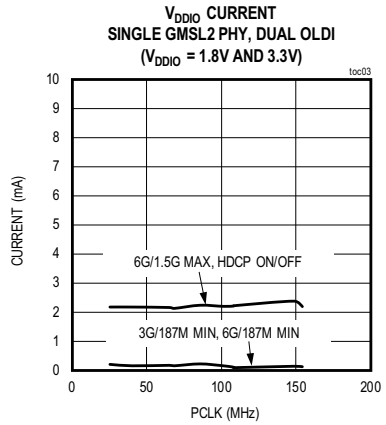
Note 7: Measured at 50MHz. See the *Speed Programming for SPI and I²S* section for pin programming recommendations.

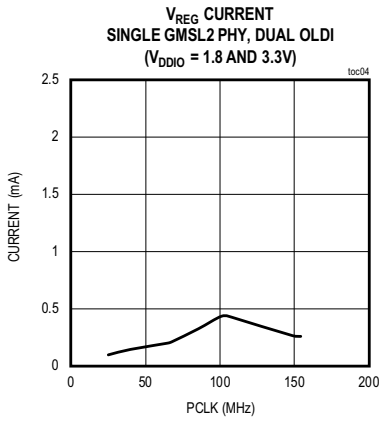
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Typical Operating Characteristics



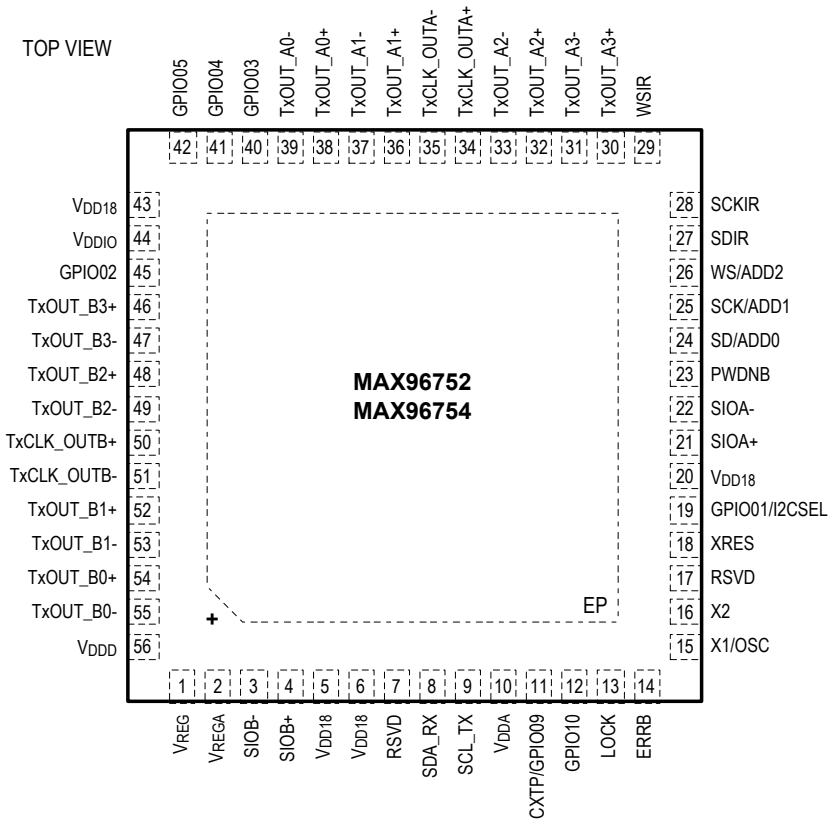






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Pin Configuration



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Pin Description

PIN	NAME	FUNCTION MODES SIGNAL NAME	DESCRIPTION
GMSL2 SERIAL LINK			
21	SIOA+	SIOA+	Noninverted Coax/Twisted-Pair Serial-Data Input 1
22	SIOA-	SIOA-	Inverted Twisted-Pair Serial-Data Input 1
4	SIOB+	SIOB+	Noninverted Coax/Twisted-Pair Serial-Data Input 2
3	SIOB-	SIOB-	Inverted Twisted-Pair Serial-Data Input 2
OLDI INTERFACE			
30	TxOUT_A3+	TxOUT_A3+	LVDS Data Output Port A
31	TxOUT_A3-	TxOUT_A3-	LVDS Data Output Port A
32	TxOUT_A2+	TxOUT_A2+	LVDS Data Output Port A
33	TxOUT_A2-	TxOUT_A2-	LVDS Data Output Port A
36	TxOUT_A1+	TxOUT_A1+	LVDS Data Output Port A
37	TxOUT_A1-	TxOUT_A1-	LVDS Data Output Port A
38	TxOUT_A0+	TxOUT_A0+	LVDS Data Output Port A
39	TxOUT_A0-	TxOUT_A0-	LVDS Data Output Port A
34	TxCLK_OUTA+	TxCLK_OUTA+	LVDS Clock Output Port A
35	TxCLK_OUTA-	TxCLK_OUTA-	LVDS Clock Output Port A
46	TxOUT_B3+	TxOUT_B3+	LVDS Data Output Port B
47	TxOUT_B3-	TxOUT_B3-	LVDS Data Output Port B
48	TxOUT_B2+	TxOUT_B2+	LVDS Data Output Port B
49	TxOUT_B2-	TxOUT_B2-	LVDS Data Output Port B
52	TxOUT_B1+	TxOUT_B1+	LVDS Data Output Port B
53	TxOUT_B1-	TxOUT_B1-	LVDS Data Output Port B
54	TxOUT_B0+	TxOUT_B0+	LVDS Data Output Port B
55	TxOUT_B0-	TxOUT_B0-	LVDS Data Output Port B
50	TxCLK_OUTB+	TxCLK_OUTB+	LVDS Clock Output Port B
51	TxCLK_OUTB-	TxCLK_OUTB-	LVDS Clock Output Port B
CONTROL AND GPIO (* denotes default state after power-up)			
19	GPIO01/I2CSEL	I2CSEL GPO01* VS	I2CSEL: Control-Channel Interface Select for Main I2C/UART with an Internal 1MΩ Pulldown to Ground. State is latched at power-up. Set I2CSEL high to select I2C interface. Set I2CSEL low to select UART interface. GPO01: Configurable General-Purpose Output. By default, configured to output the GPI01 value received from the serializer. VS: Vertical Sync Push-Pull Output

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Pin Description (continued)

PIN	NAME	FUNCTION MODES SIGNAL NAME	DESCRIPTION
45	GPIO02	GPIO02* WMD MS	GPIO02: Configurable General-Purpose Input or Output with and Internal 1MΩ Pulldown to Ground WMD: Watermark Detect Push-Pull Output. High when watermark is detected. MS: UART Mode Select with an Internal 1MΩ Pulldown to Ground. Set MS low to select base mode. Set MS high to select bypass mode. MS state can also be temporarily overwritten by a register write.
40	GPIO03	GPIO03* MOSI SDA1_RX1	GPIO03: Configurable General-Purpose Input or Output with an Internal 1MΩ Pulldown to Ground MOSI: SPI Master Out Slave In. When configured as the master, push-pull output that drives data to external slave. When configured as a slave, input with an internal 1MΩ pulldown to ground that receives data from an external master. SDA1_RX1: Pass-Through I^2C1 Serial-Data Input/Output or UART1 Receive. Internal 40kΩ pullup to V_{DDIO}. SDA1: I^2C serial data input/open-drain output. RX1: UART input.
41	GPIO04	GPIO04* MISO	GPIO04: Configurable General-Purpose Input or Output with an Internal 1MΩ Pulldown to Ground MISO: SPI Master In Slave Out. When configured as the master, input with an internal 1MΩ pulldown to ground that receives data from an external slave. When configured as a slave, push-pull output that drives data to an external master.
42	GPIO05	GPIO05* SCLK SCL1_TX1	GPIO05: Configurable General-Purpose Input or Output with an Internal 1MΩ Pulldown to Ground SCLK: SPI Clock. When configured as the master, push-pull clock output. When configured as a slave, clock input with an internal 1MΩ pulldown to ground. SCL1_TX1: Pass-Through I^2C1 Serial-Clock Input/Output or UART1 Transmit. Internal 40kΩ pullup to V_{DDIO}. SCL1: I^2C clock input/open-drain output. TX1: UART open-drain output.
27	SDIR	SDIR* GPIO06 SDA2_RX2	SDIR: I^2S/TDM Serial-Data Input with an Internal 1MΩ Pulldown to Ground. Supports reverse audio from deserializer to serializer . GPIO06: Configurable General-Purpose Input or Output with an Internal 1MΩ Pulldown to Ground SDA2_RX2: Pass-Through I^2C2 Serial-Data Input/Output or UART2 Receive. Internal 40kΩ Pullup to V_{DDIO}. SDA2: I^2C serial data input/open-drain output. RX2: UART input.
28	SCKIR	SCKIR* GPIO07	SCKIR: I^2S/TDM Serial-Clock Input with an Internal 1MΩ Pulldown to Ground. Supports reverse audio from the deserializer to the serializer. GPIO07: Configurable General-Purpose Input or Output with an Internal 1MΩ Pulldown to Ground

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Pin Description (continued)

PIN	NAME	FUNCTION MODES	DESCRIPTION
		SIGNAL NAME	
29	WSIR	WSIR* GPIO08 SCL2_TX2	WSIR: I²S/TDM Serial-Word Select Input with an Internal 1MΩ Pulldown to Ground. Supports reverse audio from the deserializer to the serializer. GPIO08: Configurable General-Purpose Input or Output with an Internal 1MΩ Pulldown to Ground. SCL2_TX2: Pass-Through I²C2 Serial-Clock Input/Output or UART2 Transmit. Internal 40kΩ pullup to V_{DDIO}. SCL2: I²C clock input/open-drain output. TX2: UART open-drain output.
11	GPIO09/CXTP	CXTP GPO09* BNE SS1	CXTP: Coax/Twisted-Pair Select Input with an Internal 1MΩ Pull-down to Ground. State is latched on power-up. Set CXTP high for coax cable drive. Set CXTP low for a twisted-pair cable. GPO09: Configurable General-Purpose Output BNE: Buffer Not Empty. When configured as slave, SPI BNE push-pull output. When BNE is high, indicates SPI data is available. SS1: SPI Slave Select 1. When configured as the master, slave 1 select push-pull output.
12	GPIO10	GPIO10* RO SS2	GPIO10: Configurable General-Purpose Input or Output with an Internal 1MΩ Pulldown to Ground RO: When configured as a slave, SPI mode-select input with an internal 1MΩ pulldown to ground. When RO is high, enables master read from MISO. When RO is low, enables the master write to MOSI. SS2: SPI Slave Select 2. When configured as the master, slave 2 select push-pull output.
24	SD/ADD0	ADD0 SD* GPO11	ADD0: Address Selection Input with an Internal 1MΩ Pulldown to Ground. Latched on power-up. SD: I²S/TDM Serial Data Push-Pull Output. Supports forward-channel audio from the serializer to the deserializer. GPO11: Configurable General-Purpose Output
25	SCK/ADD1	ADD1 SCK* GPO12	ADD1: Address Selection Input with an Internal 1MΩ Pulldown to Ground. Latched on power-up. SCK: I²S/TDM Serial Clock Push-Pull Output. Supports forward-channel audio from the serializer to the deserializer. GPO12: Configurable General-Purpose Output
26	WS/ADD2	ADD2 WS* GPO13	ADD2: Address Selection Input with an Internal 1MΩ Pulldown to Ground. Latched on power-up. WS: I²S/TDM Word Select Push-Pull Output. Supports forward-channel audio from the serializer to the deserializer. GPO13: Configurable General-Purpose Output
8	SDA_RX	SDA_RX* GPIO14	SDA_RX: I²C Serial-Data Input/Output or UART Receive. Internal 40kΩ pullup to V_{DDIO}. SDA: I²C serial data input/open-drain output. RX: UART input. GPIO14: Configurable General-Purpose Input or Open-Drain Output. Function available only if the MAX96752/MAX96754 is being programmed over the GMSL2 link and not locally. Internal 40kΩ pullup to V_{DDIO}.

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Pin Description (continued)

PIN	NAME	FUNCTION MODES	DESCRIPTION
		SIGNAL NAME	
9	SCL_TX	SCL_TX* GPIO15	SCL_TX: I²C Serial-Clock Input/Output or UART Transmit. Internal 40kΩ pullup to V_{DDIO}. SCL: I²C clock input/open-drain output. TX: UART open-drain output. GPIO15: Configurable General-Purpose Input or Open-Drain Output. Function available only if the MAX96752/MAX96754 is being programmed over the GMSL2 link and not locally. Internal 40kΩ pullup to V_{DDIO}.
13	LOCK	LOCK	Open-Drain Lock Indication Output with an Internal 40kΩ Pullup to V_{DDIO}
14	ERRB	ERRB	Open-Drain Error Indication Output with an Internal 40kΩ Pullup to V_{DDIO}. When ERRB is low, indicates that a data error or interrupt was detected. ERRB is high when PWDNB is low.
23	PWDNB	PWDNB	Active-Low, Power-Down Input with an Internal 1MΩ Pulldown to Ground. Set PWDNB low to enter power-down mode.
MISCELLANEOUS—SEE Table 2			
15	X1/OSC	X1/OSC	Crystal/Oscillator Input. Connect to either a 25MHz crystal or 25MHz external clock source.
16	X2	X2	Crystal Input. Connect to one terminal of a 25MHz crystal and connect a load capacitor from X1/OSC to ground (load capacitor value depends on crystal used).
18	XRES	XRES	Used to calibrate SIO output driver swings. Connect an external 402Ω ±1% resistor between XRES and ground.
7, 17	RSVD	RSVD	Reserved. Make no electrical connection to this pin.
POWER SUPPLIES—SEE Table 2			
1	V_{REG}	V_{REG}	Supply for Internal V_{DDD} Regulator. If using the regulator, connect a 1.8V ±5% supply. If not using the regulator, make no connection.
56	V_{DDD}	V_{DDD}	Core Digital 1V Supply. If using the V_{REG} regulator, do not apply any voltage to this pin. If not using the regulator, connect a 1.0V ±5% supply.
2	V_{REGA}	V_{REGA}	Supply for Internal V_{DDA} Regulator. If using the regulator, connect a 1.8V ±5% supply. If not using the regulator, make no connection.
10	V_{DDA}	V_{DDA}	Core Analog 1V Supply. If using the V_{REGA} regulator, do not apply any voltage to this pin. If not using the regulator, connect a 1.0V ±5% supply.
44	V_{DDIO}	V_{DDIO}	1.8V to 3.3V I/O Supply
5, 6, 20, 43	V_{DD18}	V_{DD18}	1.8V ±5% I/O Supply
—	EP	EP	Exposed Pad. EP is internally connected to device ground. EP must be connected to the PCB ground plane through an array of vias for proper thermal and electrical performance.

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Detailed Description

Table 1. Recommended Operating Conditions

			MIN	TYP	MAX	UNIT
Supply Range	V_{DDD18}		1.7	1.8	1.9	V
	V_{DDIO}	1.8	1.7	1.8	1.9	V
		3.3	3.0	3.3	3.6	V
	V_{DDD}, V_{DDA} (if driven externally)		0.95	1.0	1.05	V
	V_{REG}, V_{REGA} (if regulators used)		1.7	1.8	1.9	V
Maximum Supply Noise	V_{DDD18}			25		mV_{P-P}
	V_{DDIO}	1.8		50		mV_{P-P}
		3.3		100		mV_{P-P}
	V_{DDD}, V_{DDA}			25		mV_{P-P}
	V_{REG}, V_{REGA} (if regulators used)			50		mV_{P-P}
Operating Junction Temperature, T_J			-40		+125	ºC

Note: *Supply noise < 1MHz.*

External Component Requirements

See Figure 16 and Figure 17.

Table 2. External Component Requirements

COMPONENT	SYMBOL	CONDITION	VALUE	UNIT
XRES	R_{XRES}		402 ±1%. Use a single resistor.	Ω
Line Fault External Pulldown Resistor	R_{PD}		49.9 ±1%	kΩ
Line Fault Resistor	R_{EXT}		48.7 ±1%	kΩ
Link Isolation Capacitors	C_{LINK}	Place close to the SIO pins (21, 22, 4, 3) used in the application.	0.1	µF
Crystal		Place as close as possible to X1/OSC (pin 15) and X2 (pin 16).	25MHz ±200ppm	
Crystal Load Capacitors		Use crystal loading capacitor guidance from the crystal manufacturer. Select values which compensate for the X1 and X2 input and PCB node capacitances. Place the capacitors as close as possible X1/OSC (pin 15) and X2 (pin 16).		
V_{DDIO} Decoupling Capacitors		Place a 0.01µF capacitor as close as possible to V_{DDIO} (pin 44). Include a minimum of 10µF bulk decoupling on the PCB.		

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Table 2. External Component Requirements (continued)

COMPONENT	SYMBOL	CONDITION	VALUE	UNIT
V_{DD18} Decoupling Capacitors		Place a 0.01µF capacitor as close as possible to V_{DD18}. Place another 0.01µF capacitor as close as possible to V_{DD18} (pins 5 and 6) and run separate traces from each pin to the capacitor. Include a minimum of 10µF bulk decoupling on the PCB.		
V_{DDD}, V_{DDA} Decoupling Capacitors		Place 0.1µF capacitors as close as possible to V_{DDA} (pin 10) and V_{DDD} (pin 56). If driving with a 1V supply, include a minimum of 10µF bulk decoupling on the PCB. If using the internal regulators for V_{DDD} and V_{DDA}, also include 10µF bulk decoupling capacitors for V_{DDD} and V_{DDA}. See the configuration information in the *Power Supplies* section.		
V_{REG} , V_{REGA} Decoupling Capacitors		If using the internal regulators for V_{DDD} and V_{DDA}, place a 0.1uF capacitor as close as possible to V_{REG} (pin 1) and V_{REGA} (pin 2). If sharing the V_{DD18} supply for V_{REG} and V_{REGA}, no bulk decoupling capacitance is required. If not using the internal regulators,make no connection to these pins. See the configuration information in the *Power Supplies* section.		
Open-Drain Pullup Resistors		Application-specific. Quantity and values depend on multifunction GPIO pin configurations.		

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Functional Diagrams

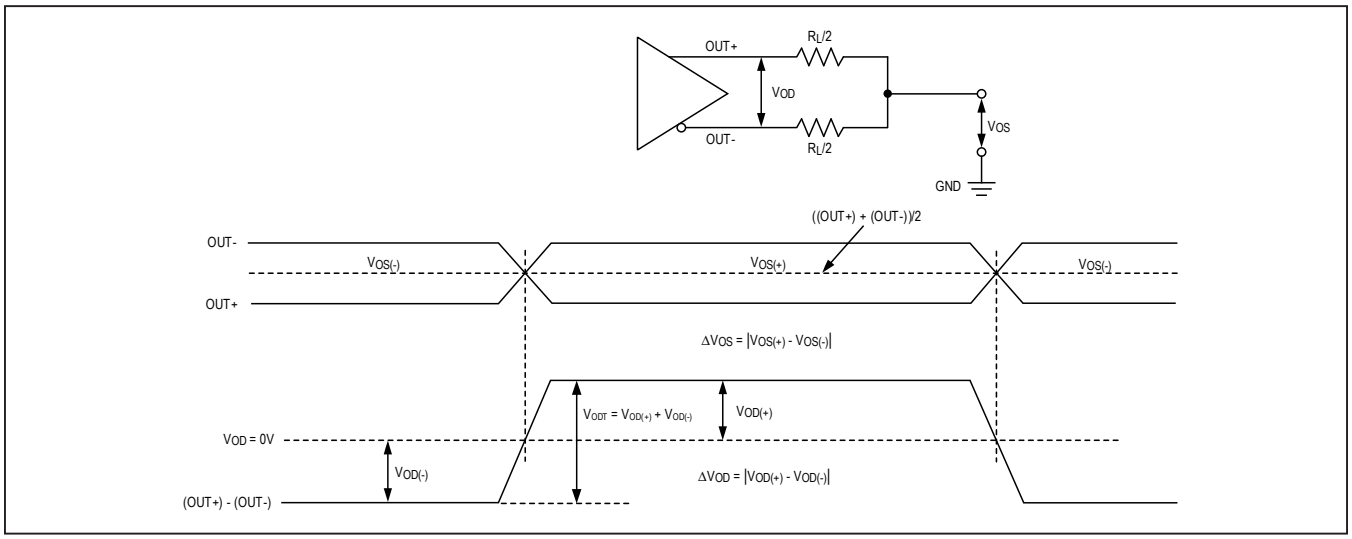


Figure 1. Serial Output Parameters

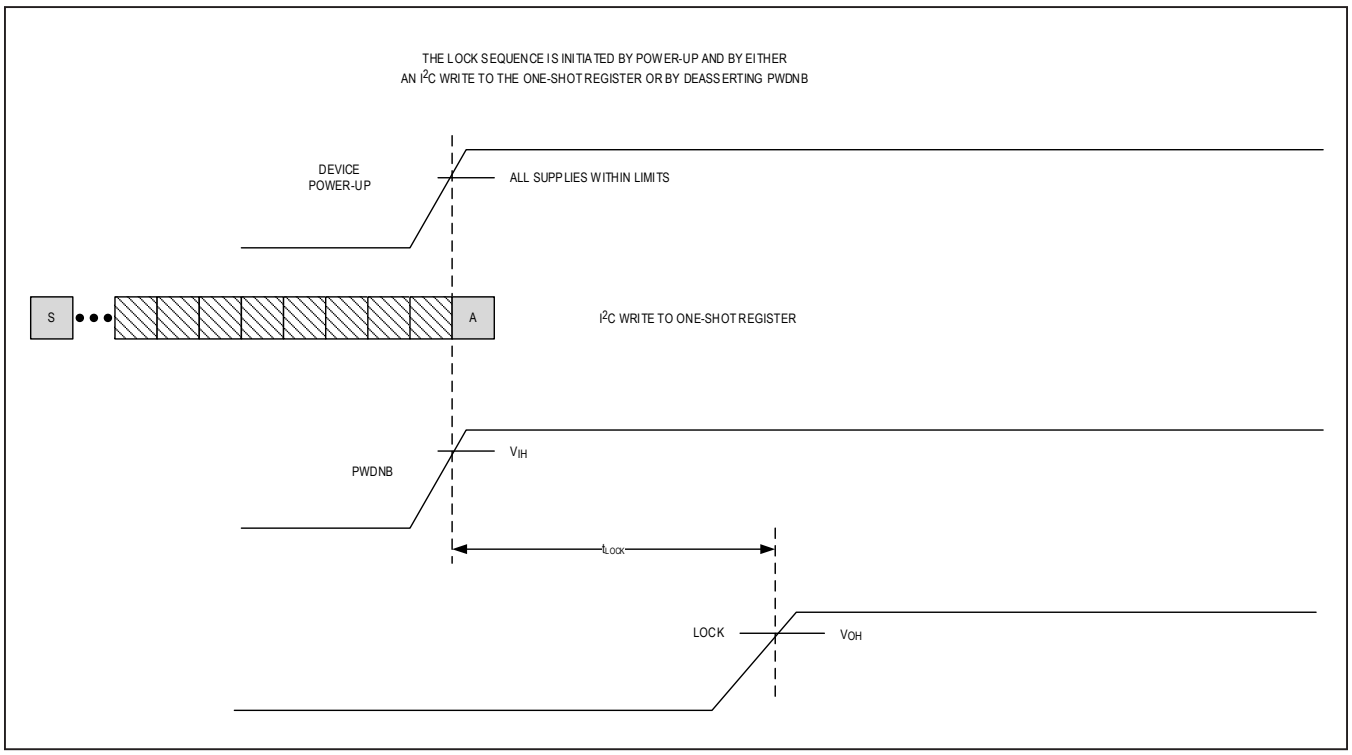


Figure 2. Lock Time (One-Shot Register Write or PWDNB Pin Transition)

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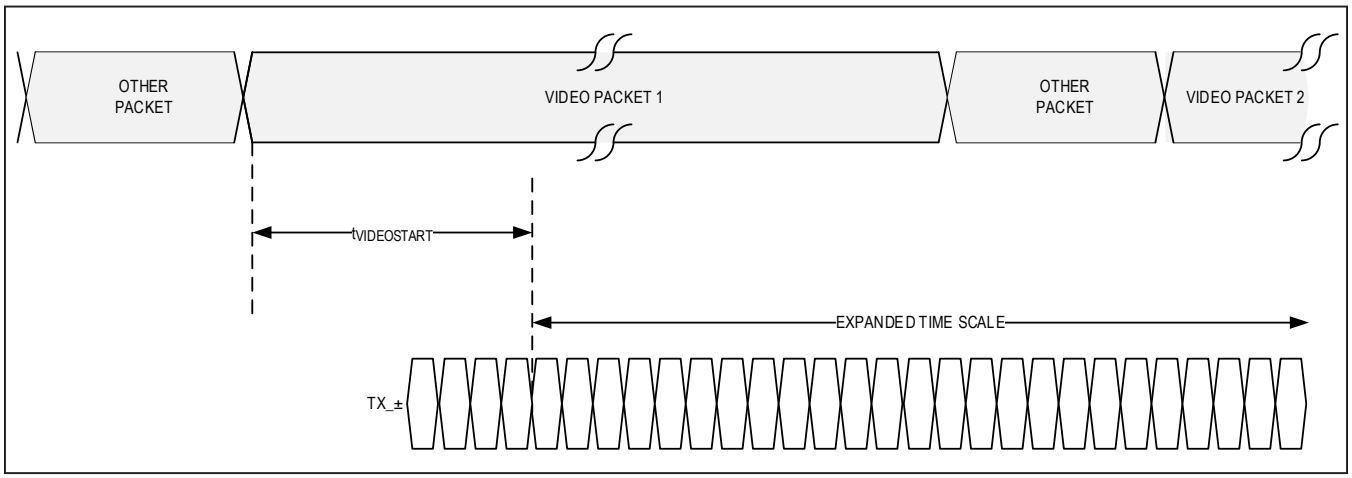


Figure 3. Video Initialization Time

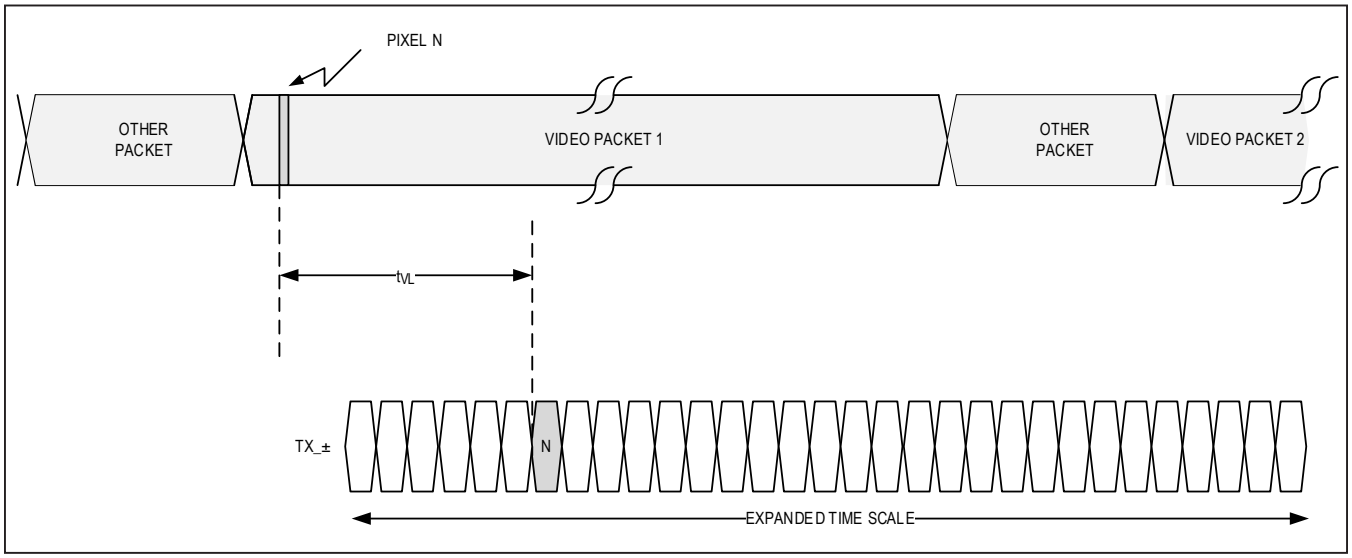


Figure 4. Video Latency

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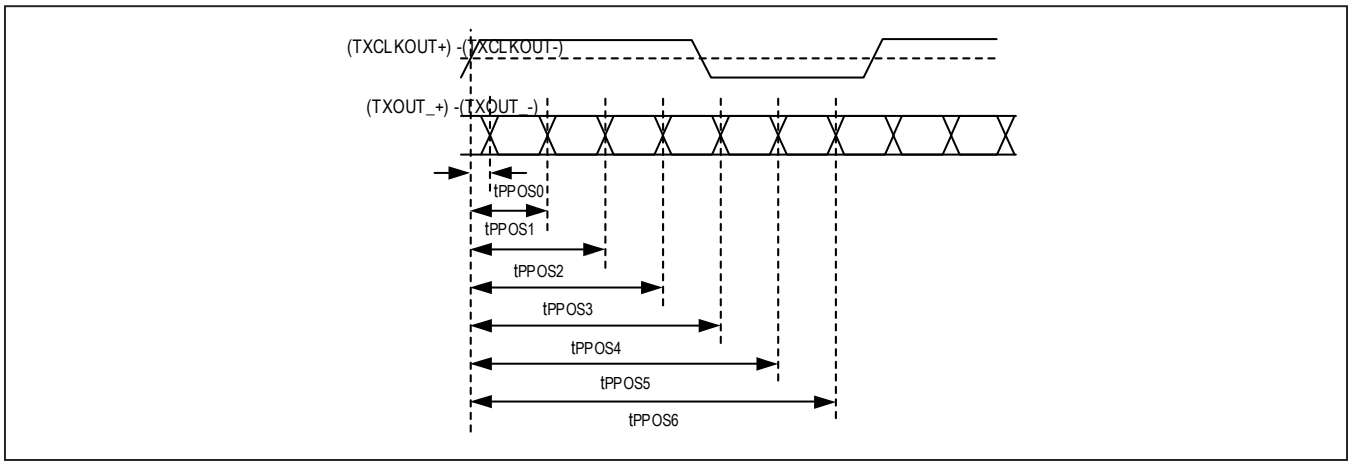


Figure 5. LVDS Output Pulse Positions

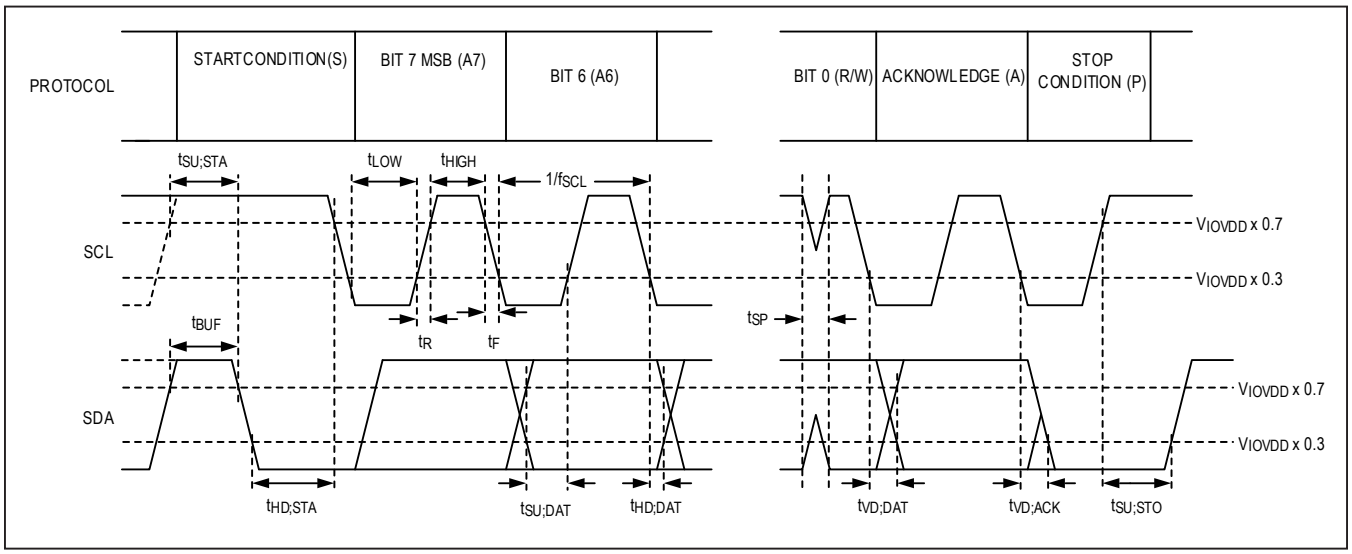


Figure 6. I²C Timing Parameters

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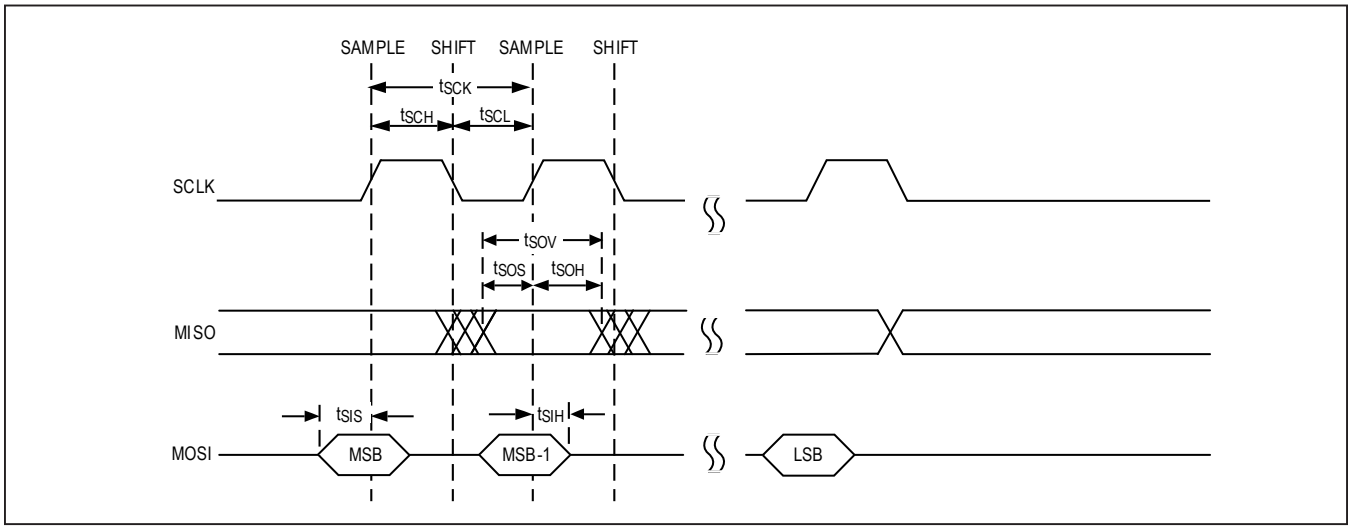


Figure 7. SPI Master Mode Timing Parameters

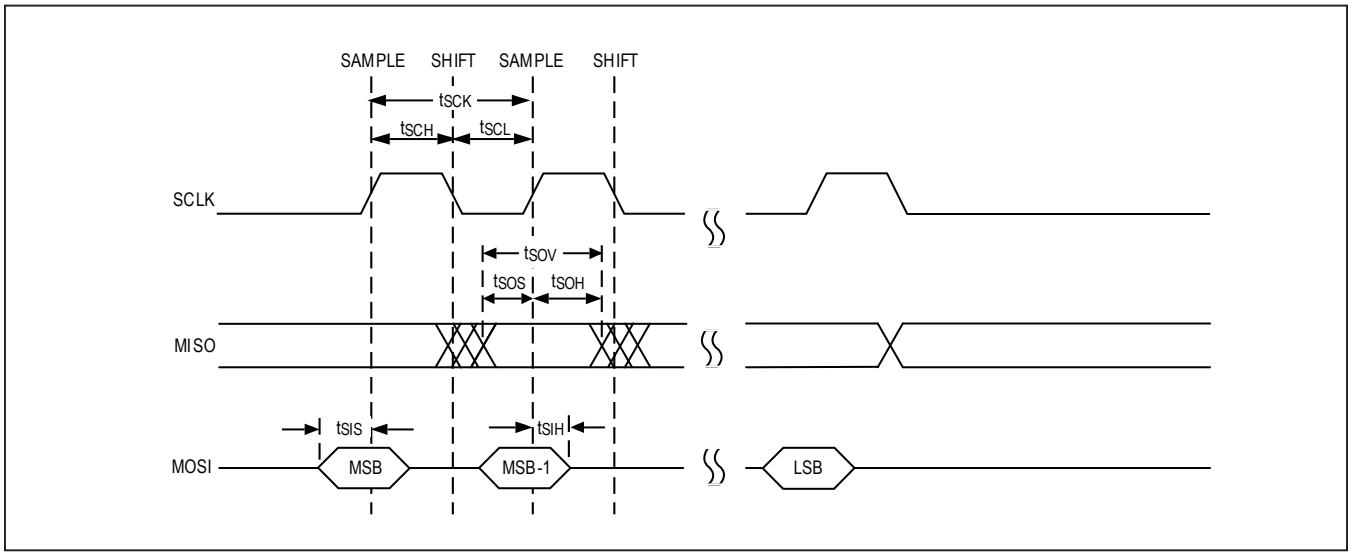


Figure 8. SPI Slave Mode Timing Parameters

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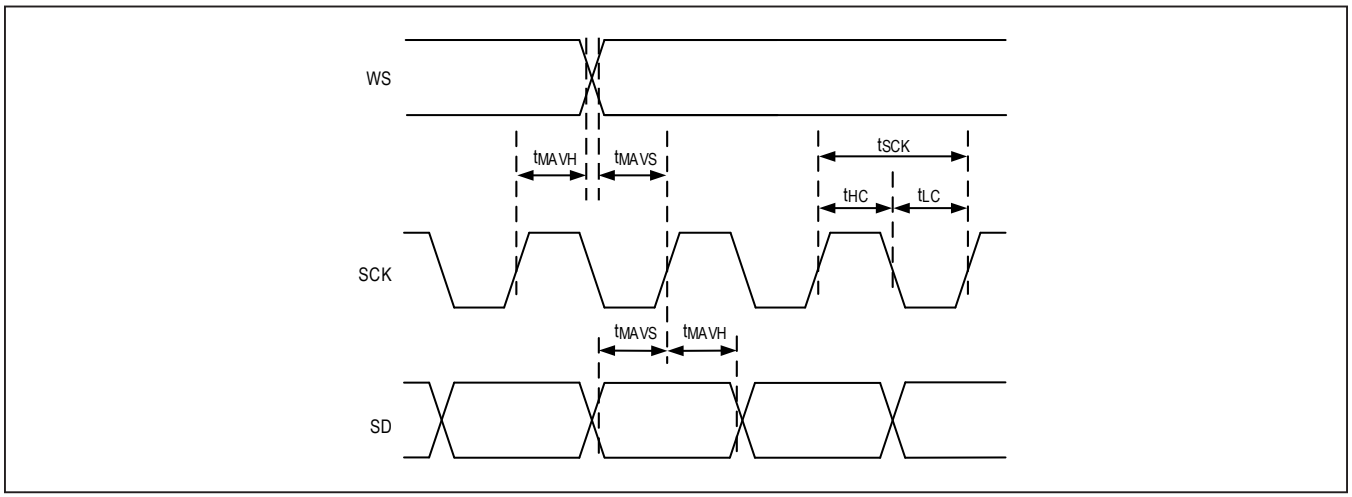


Figure 9. I²S Master Timing Diagram

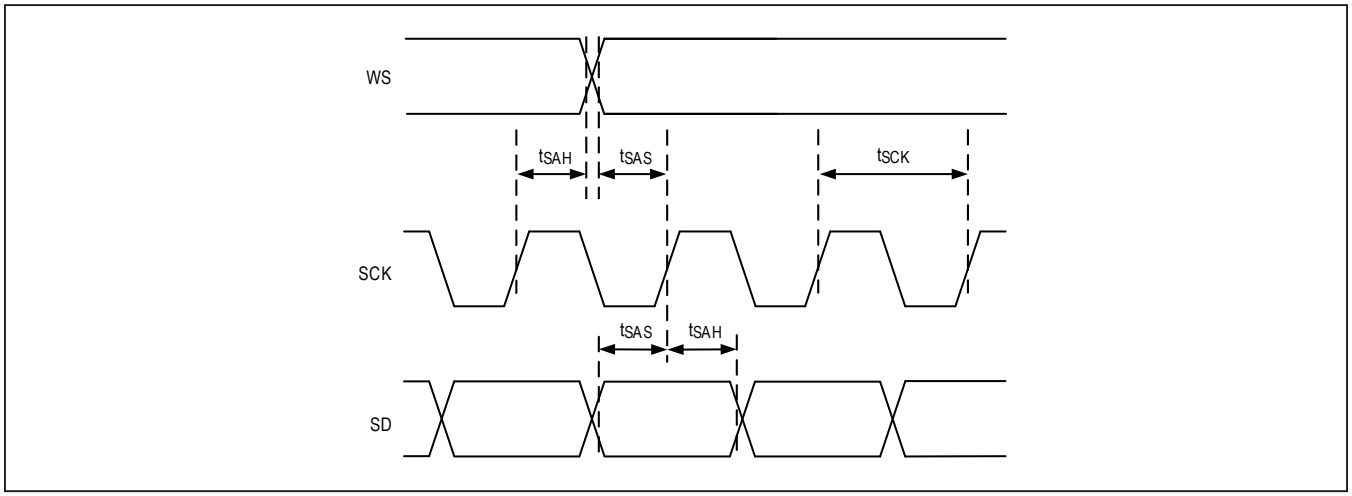


Figure 10. I²S Slave Timing Diagram

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Detailed Description

Introduction

Maxim's GMSL2 serializers and deserializers provide sophisticated link management for high-speed, low bit-error-rate, serial data transport. They support a comprehensive suite of display, camera, and communication interfaces over a single wire.

GMSL2 provides up to 6Gbps forward and 1.5Gbps reverse packetized data transmission over each fixed-speed link. Devices with two GMSL2 links provide a total capacity of 12Gbps and 3Gbps reverse in specific configurations.

The following sections provide a brief overview of the device functions and features. Contact the factory for additional information and details on the configuration of each function and feature.

Product Overview

The MAX96752/MAX96754 convert a single video stream received on 1 or 2 GMSL2 inputs to an LVDS (OLDI) output for driving display panels. Video data is mapped to OLDI output ports A, B, or both. The output can drive with OLDI or VESA formats.

For more information about the OLDI standard, refer to the Open LVDS Display Interface (OLDI) specification.

The MAX96754 adds HDCP 1.4 decryption. After HDCP is configured, the authentication and key exchange is fully automated without requiring additional external µC involvement.

OLDI Pixel Clock and Pixel Interleaving

The OLDI clock output is generated by the deserializer based on the tracked and measured clock frequency of its video input. When using a single output port, the maximum OLDI PCLK of 150MHz supports up to FHD (1920x1080, 24-bit/60Hz) or equivalent.

In dual OLDI mode, the interleaved even and odd pixels of the video stream are each routed to the A and B output ports, which are connected to a single display. This divides the video PCLK by a factor of 2 at the OLDI ports. FHD can thus be displayed with approximately 75MHz clocking on each port, keeping the PCLK below the FM radio band.

In dual OLDI mode, the MAX96752/MAX96754 support up to a 300MHz PCLK if both GMSL2 link inputs are used. This enables display of interleaved WQXGA (2560x1600, 24-bit/60Hz) or equivalent video. If only one GMSL link is used, the maximum PCLK is limited to approximately 217MHz by the 6Gbps link bandwidth. This enables display of 2560x1080, 24-bit/60Hz, or equivalent video.

Deserializer replication supports identical FHD, or equivalent, video to be sent to 2 displays.

Dual-view splitter mode supports 2 side-by-side FHD displays with different content split from a single video stream.

The MAX96752/MAX96754 provide flexible output configuration, a color lookup table for gamma correction, and programmable spread-spectrum clocking. Pixel mapping to the A and B ports can be swapped.

Deserializer Modes

The MAX96752/MAX96754 have a single video pipeline. Each device can support single video streams, video replication, and dual-view splitting to enable a single deserializer to support multiple system configurations as illustrated in Figure 11 through Figure 15. All use cases are shown with an HDMI source serializer, but these configurations are also possible with MIPI DSI and CSI sources as well.

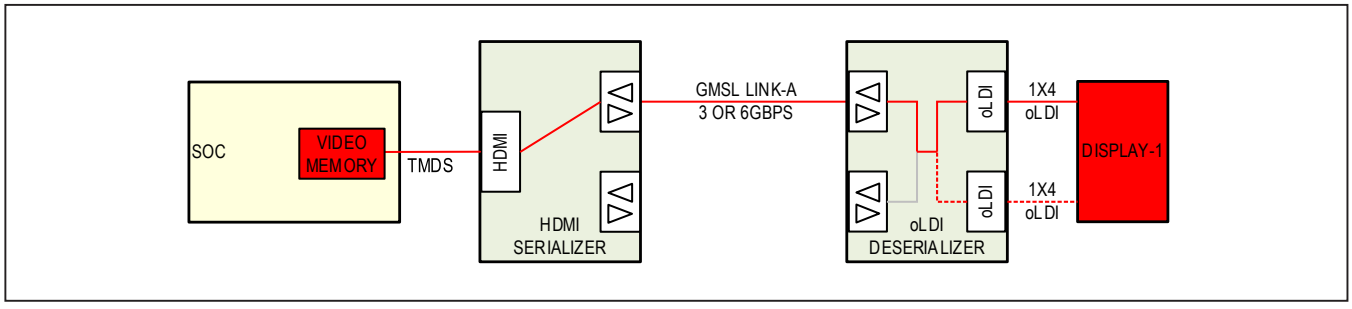


Figure 11. Single Video Stream, Single GMSL2 Link at Either 3Gbps or 6Gbps, MAX96752/MAX96754 Deserializer with Single or Dual OLDI Outputs

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Single Video Stream, Dual GMSL2 Links

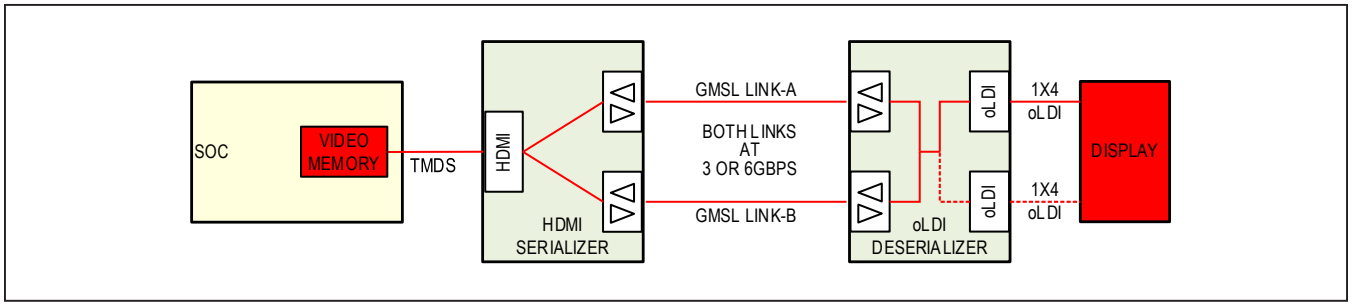


Figure 12. Single Video Stream, Dual GMSL2 Links with Both at Either 3Gbps or 6Gbps, MAX96752/MAX96754 Deserializer with Single or Dual OLDI Outputs

Single Video Stream with Serializer Replication

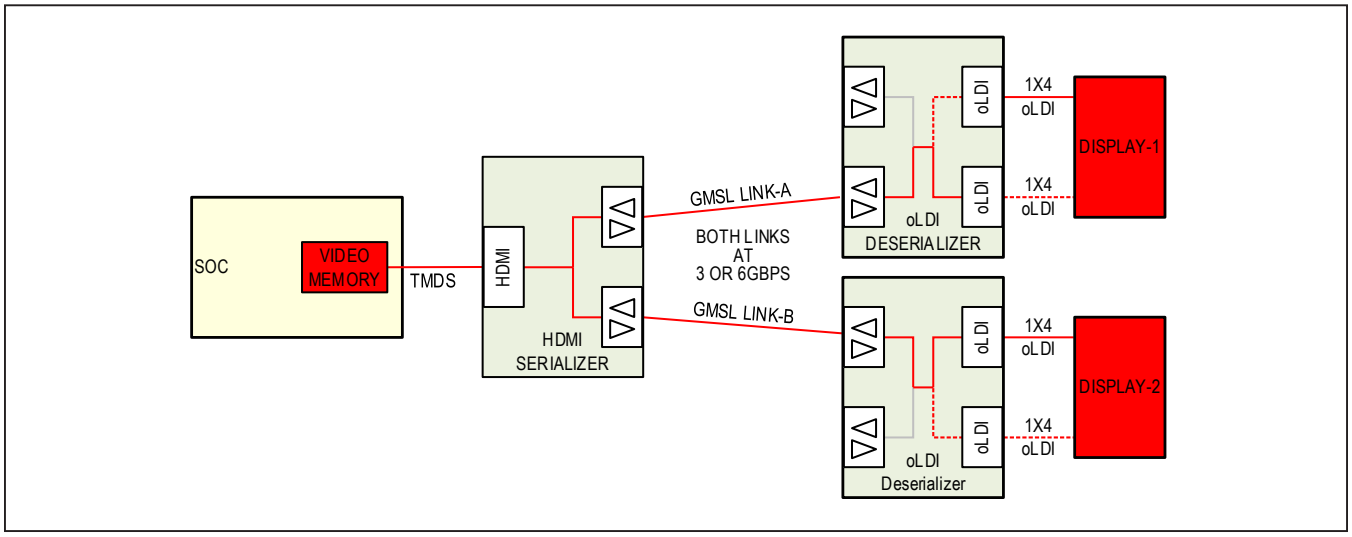


Figure 13. Serializer Replication, Dual GMSL2 Links, OLDI Deserializers with Single or Dual OLDI Outputs

Single Video Stream with OLDI Deserializer Replication

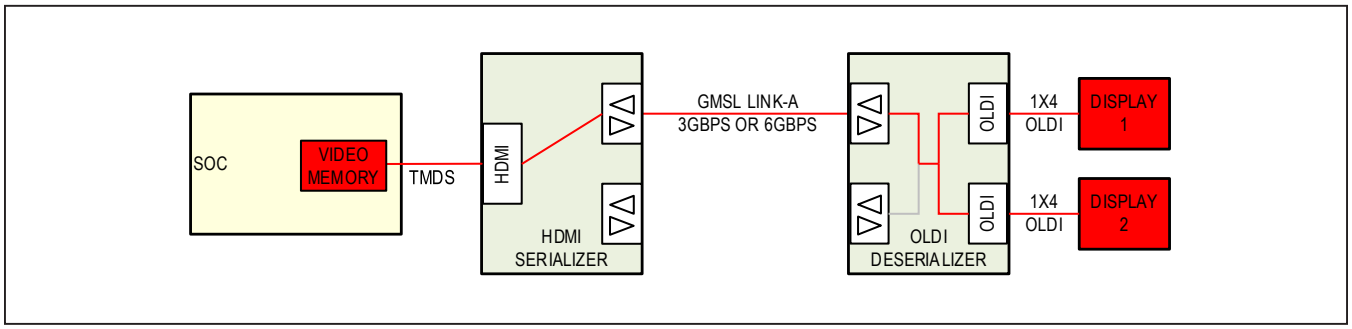


Figure 14. Single Stream OLDI Deserializer, Identical Video on Both Displays

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Serializer Dual-View Mode with OLDI Deserializer Splitter Mode

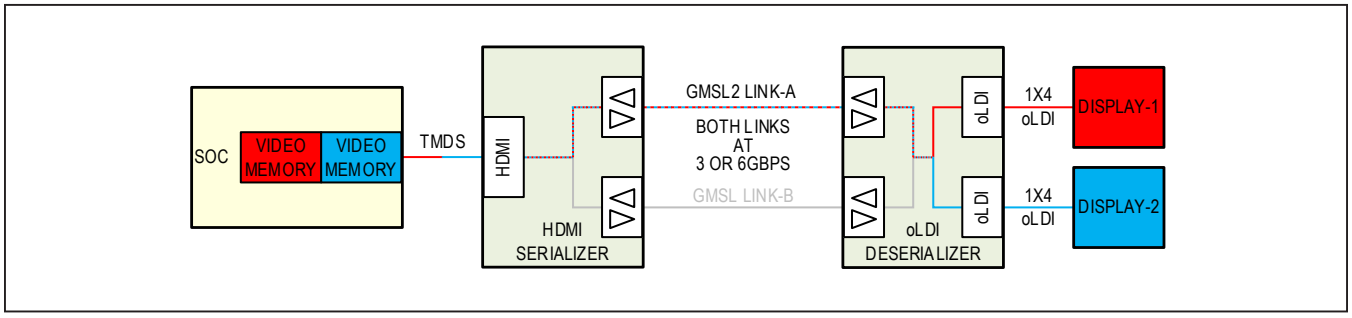


Figure 15. Serializer Dual-View Splitter Mode, Convert Dual-View to Pixel Interleave, Send Dual-Views Down Single GMSL2 Link, Dual OLDI Deserializer Splits Pixel Interleaved Dual-View, Each Video Stream (View) Must Have Identical Timing, Can Also Be Done in Dual GMSL2 Link Mode

Other Functions

GMSL2 serializers and deserializers have a main I²C/UART control channel interface that an ECU uses to access serializer and deserializer registers, as well as peripheral devices, from either end of the link. Each device also has two pass-through I²C/UART channels available for local or remote peripheral control. The pass-through I²C/UART channels do not have access to serializer and deserializer registers.

The MAX96752/MAX96754 support both forward (serializer to deserializer) and reverse (deserializer to serializer) audio channels. The audio channels support I²S stereo and up to 8 channels in TDM mode. Sample rates of 32kHz to 192kHz are supported with sample depth from 8 bits to 32 bits.

The MAX96752/MAX96754 additionally include an SPI master/slave interface, including two slave select pins, for peripheral control. The SPI interface enables a host SPI master on one side of the GMSL2 link to control a peripheral SPI slave on the opposite side. The host can be located at either end of the link or can swap ends by reprogramming the GMSL2 devices (a GMSL2 device can be configured as a SPI master or slave).

The MAX96752/MAX96754 provide up to 15 GPIOs, dependent on device feature utilization. GPIOs are typically used to tunnel low speed (< 100kbps) signals over the GMSL2 link, although rates in excess of 1MHz can also be supported. A GPIO tunnel can be set up in the forward or reverse direction.

The devices include a video crossbar and watermark generation or detection. The crossbar can be used to reorder the color and sync signals. It can also be used for HDMI TMDS lane remapping and phase inversion if desired. The watermark generation and detection is used for verifying that the video image is not frozen.

GMSL2 devices incorporate numerous link margin optimization and monitoring functions to ensure high link margin. Continuous (1Hz) adaptive equalization optimizes link margin to adapt to environmental changes and cable aging. An eye-opening monitor function for continuous link margin diagnosis with various threshold alarm levels is available for run-time alerts of link degradation. PRBS checking verifies correct link and video channel operation.

GMSL2 Protocol

GMSL2 is a fixed-rate transmission medium that is designed to carry multiple types of communication channels concurrently. The link bit rate is based on a constant-frequency link clock generated from the 25MHz crystal oscillator or external reference frequency. The link clock does not have any relation to the video pixel clock.

GMSL2 uses a packet-based protocol to seamlessly share the link bandwidth between communication channels in a flexible way. Bandwidth allocation is dynamic so that if a certain channel is not active, it does not consume any link bandwidth, and all the remaining active channels can share the full link bandwidth. Maximum packet size is limited to prevent a single channel from utilizing the link bandwidth for an extended time. In most of the cases available link bandwidth exceeds the bandwidth requirement. Idle packets are used to fill in the unused link bandwidth.

The same data protocol is used on forward and reverse channels, and for both video and control-channel data.

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GMSL2 Physical Layer

Maxim's GMSL2 family of serial links have transmitter and receiver capability enabled simultaneously, allowing full-duplex operation on a single wire. A single cable between the serializer and deserializer delivers data being transmitted from each end of the link. Forward transmission is data being sent from the serializer to the deserializer. Reverse transmission is data being sent from the deserializer to the serializer.

Forward rate options are fixed at 3Gbps or 6Gbps, with defaults of 6Gbps in coax mode and 3Gbps in STP mode. Reverse rate options are fixed at 187.5Mbps or 1.5Gbps, with the default of 187.5Mbps. To use a reverse data rate of 1.5Gbps, a forward data rate of 6Gbps must be selected. This mode requires optimized configuration settings. Contact the factory for additional details.

Both forward and reverse rates are doubled in dual-link configuration, in which a serializer and paired deserializer are connected using 2 links. Dual-link mode is not available in all GMSL2 device configurations.

Cabling Options

GMSL2 supports operation with either 50Ω coaxial or 100Ω shielded twisted pair (STP) cabling. Applications using 2 links can also use shielded twisted quad (STQ) cabling.

Cables must have sufficient return and insertion loss characteristics for best full-duplex link performance. The available link rates and GMSL2 dynamic link optimization enable support of a wide range of cabling options. Contact the factory for insertion and return loss guidelines.

Coax or STP operation is determined by the state of CXTP at power-up. Set CXTP high for coax cable drive (single-ended with default 6Gbps link rate). Set CXTP low for STP mode (differential with default 3Gbps link rate). See the *Latch-On Power-Up Pins* section.

In coax mode, use only the noninverted SIO pin. In STP configurations, both the noninverted and inverted SIO pins are enabled by default. Any unused SIO pins should be AC terminated with 50Ω to ground.

Following the GMSL2 design guidelines allows cable lengths of up to 15m for coax or up to 7.5m for STP commonly used in the automotive industry. The guidelines assume 2 in-line connectors in addition to the endpoint connectors, and 25mm of PCB trace at each end.

Line Fault

GMSL2 serializers include a novel line-fault detection circuit. It detects and reports open-circuit, short-to-battery, short-to-ground, and line-to-line short. The line fault monitor requires external resistors R_{EXT} and R_{PD} connected to the LMN pins as shown in Figure 16 and Figure 17.

The line fault monitor is enabled by default, and configuration options are available through registers. Its status can be read by register. If unmasked, a line fault condition asserts ERRB. Line fault detection cannot be used in conjunction with power over coax (POC).

The line fault monitor pins offer flexible connection and programming. On parts with multiple GMSL2 links, either pair (i.e., LMN1A/LMN0A) can be used with either link (i.e., SIOA+/SIOA-). In addition, the pins within each pair can be assigned to either polarity of the link.

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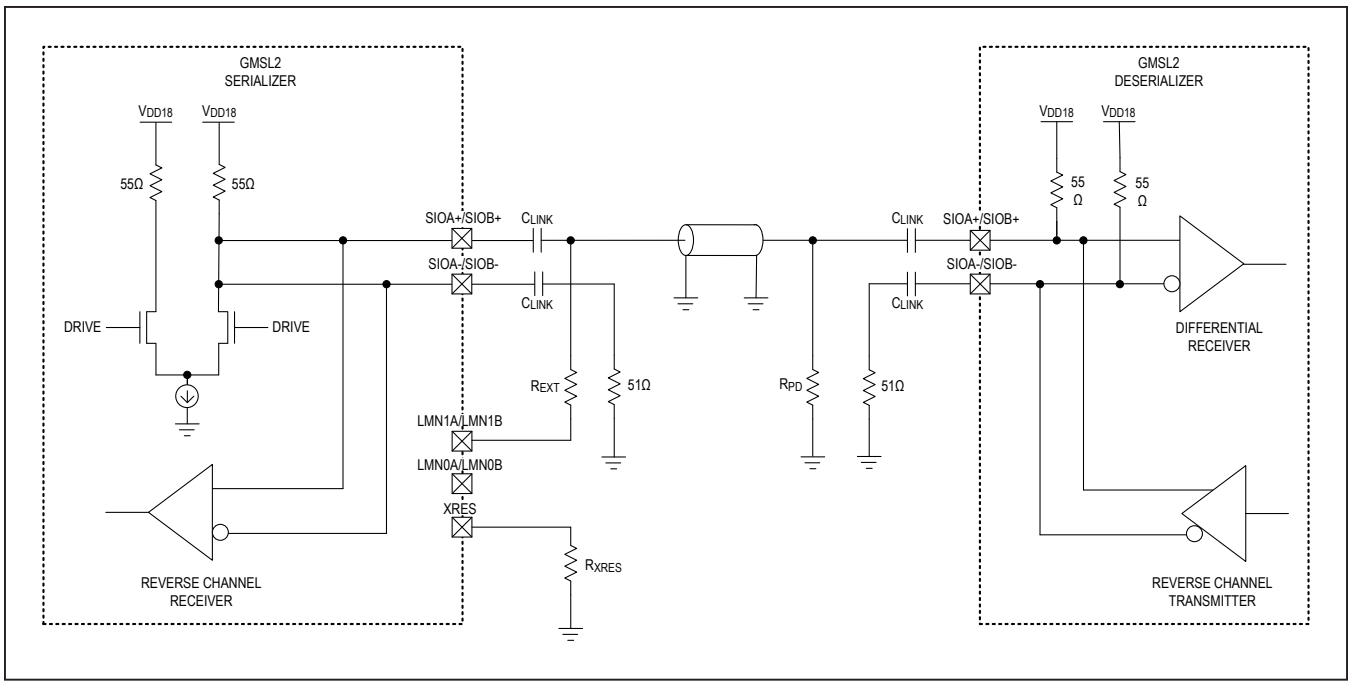


Figure 16. Typical GMSL2 Link Application Circuit for Coax Cable

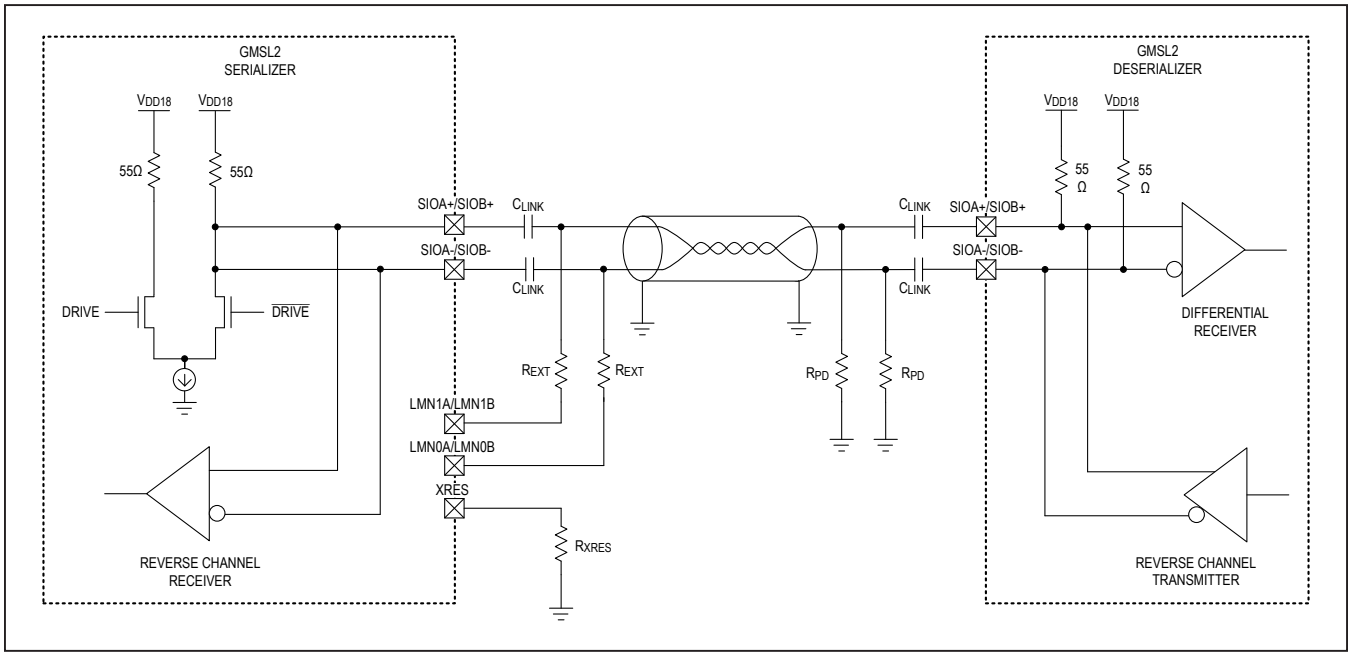


Figure 17. Typical GMSL2 Link Application Circuit for Twisted Pair

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GMSL2 Bandwidth Sharing

The GMSL forward bandwidth is shared between video, the I²C/UART control channel, pass-through I²C/UARTs, I²S/TDM audio, SPI, GPIOs, and RGMII/RMII ethernet, plus various protocol specific data exchanges (i.e., info frames, sync, and acknowledgements). The reverse channel bandwidth is also shared with all the above, with the exception of video packets.

The total link bandwidth used by all communication channels cannot exceed the fixed available link bandwidth.

Link bandwidth is shared flexibly between the various communication channels requesting the link for packet transmissions. This flexibility comes from packet-based transmission format and dynamic bandwidth allocation: if a certain channel is not active, it does not consume any link bandwidth, leaving the full link bandwidth available for all active communication channels to share. The packet-based protocol fulfills this sharing requirement. The maximum packet size is limited to prevent one single channel from monopolizing the link bandwidth and to ensure other channels are served.

The video and control channel packets can be assigned a priority level. There are four priority levels: low, normal, high, and urgent. The scheduler transmits the packet with the highest priority among the pending requests. Packets with maximum latency requirements can be assigned an increased priority.

GMSL2 Bandwidth Calculations

The GMSL2 forward link has a fixed link rate of 3Gbps or 6Gbps. The reverse link rate is also fixed at either 187.5Mbps or 1.5Gbps. The GMSL2 protocol overhead is roughly 14%. This leaves approximately 2.6Gbps or 5.2Gbps of data throughput in the forward direction and 162Mbps or 1.3Gbps in the reverse direction.

Users need to ensure that the worst-use cases do not exceed the available throughput of the forward and reverse links. Maxim's evaluation kit (EV kit) GUI includes a bandwidth (BW) calculator that can be used for initial bandwidth requirements estimates. Maxim also has other tools that are useful for calculating link bandwidth utilization. It is recommended that users consult the factory for high bandwidth use cases to ensure error free performance.

Table 3 provides estimates of the bandwidth utilization for each of the communication channels.

Table 3. Forward and Reverse Link Bandwidth Utilization

DATA	APPROXIMATE BANDWIDTH UTILIZATION
Display Video	H x V x fps x bpp x % blanking x 1.14 without VESA DSC compression
I²C	18 to 60 x I²C clock rate, depending on available link bandwidth
UART	6 x UART bit rate, 5.5 x when parity bit enabled
SPI	2.5 x SPI rate
GPIO	60 x GPIO transition rate without delay compensation 80 x GPIO transition rate with delay compensation enabled
I²S/TDM	BW = ws_freq x roundup ((bps x channelcnt + 27)/18) x 20

Definitions:

H = Horizontal resolution

V = Vertical resolution

fps = Frames per second

bpp = Bits per pixel

DSC = Display stream compression (only included in specific GMSL2 devices)

ws_freq = Word select frequency

bps = Bits per sample, per audio channel (must be the same for all channels)

channelcnt = Number of audio channels

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Power Supplies

The MAX96752/MAX96754 provide flexible power supply configurations. The V_{DDIO} supply for the GPIO pins can be 1.8V to 3.3V for flexibility in accommodating devices interfacing to the part. V_{DD18} is the primary analog supply and is fixed at 1.8V.

For best power efficiency, provide a 1.0V supply to both analog V_{DDA} and digital V_{DDD} pins. Make no connection to the V_{REGA} or V_{REG} pins.

Alternatively, the MAX96752/MAX96754 provides the capability of internally regulating 1.8V down to 1V for the V_{DDD} and V_{DDA} supplies. This eliminates the need to provide a separate 1.0V supply. Connect the 1.8V V_{DD18} supply to V_{REG} and V_{REGA}. Do not connect an external voltage to the V_{DDA} or V_{DDD} pins.

Internal logic senses the voltage at V_{REG} and appropriately enables or disables the two regulators. If 1.7V < V_{REG} < 1.9V, the regulators are enabled. Otherwise, they are disabled. The V_{REG} and V_{REGA} regulators cannot be used independently.

Proper power supply bypassing of all supplies is essential for high frequency circuit stability. See Table 2. See the Table 1 for power supply tolerances and noise requirements. Contact the factory for guidance on sharing supplies and optimizing supply decoupling.

Thermal Management

Power consumption of GMSL2 devices varies based on their use case. Care must be taken by the user to provide sufficient heat dissipation with proper board and cooling design techniques. The package exposed pad must be connected to the PCB ground plane by an array of vias. This approach simultaneously provides the lowest electrical and thermal impedance.

System thermal management must keep the operating junction temperature below 125°C to meet electrical specifications and avoid impacting device reliability.

For a given power dissipated by the device, the maximum allowed operating ambient temperature can be calculated using the following equation:

$$T_A = T_J - (\theta_{JA} \times \text{Power})$$

The maximum allowed operating case temperature can be calculated using the following equation:

$$T_C = T_J - (\theta_{JA} \times \text{Power})$$

Control Channel and Side Channels

A µC or other controller can send and receive control and side-channel data over the GMSL2 serial link simultaneously with high-speed video data. All GMSL2 devices support the following interfaces:

- Main I^2C/UART (internal access)
- Pass-through I^2C/UART
- SPI
- GPIO

Some GMSL2 devices also support the following interfaces:

- I^2S/TDM Audio
- RGMII/RMII

Data from all these interfaces tunnel through the GMSL2 link, but it is only through the main I^2C /UART interface that the GMSL2 device registers can be accessed and configured.

The side channel, with its various interfaces, is accessed via multifunction GPIO pins (MFPs on some devices) (with the exception of RGMII/RMII, which uses dedicated pins). Multi-function pins have a default function and can be programmed to an alternate function after power-up. Due to a practical limit in the number of pins available on a given device, not all interfaces can be simultaneously supported. See Table 6 and *Pin Description* section for default and alternate multi-function pin functions, as well as available combinations of interfaces.

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Main I2C/UART

The main I2C/UART is located on the SDA_RX and SCL_TX pins of each GMSL2 device. The I2C (SDA, SCL) or UART (Tx, Rx) interface is selected by the CFG0 or I2CSEL configuration pin state at power-up (see the *Latch-On Power-Up Pins* section). The selected interface provides master access to GMSL2 registers, as well as peripheral device registers, from either end of the link.

The master µC can be located on either end of the link (usually the serializer side for display applications and the deserializer side for camera applications). Dual master µC controllers are supported provided software arbitration (such as token passing) is used to prevent packet collisions. The control channel allows only one master µC to communicate at a time.

To configure peripheral devices over the link, the GMSL2 serializer and deserializer must use the same control channel interface (both I2C or both UART). Unlike GMSL1 devices, there is no I2C-to-UART conversion capability. I2C/UART outputs are open drain and require appropriately-sized external pullup resistors for proper operation.

Pass-Through I2C/UART

GMSL2 devices have two pass-through I2C/UART channels. These channels do not have access to registers in either the GMSL2 serializer or the deserializer; they simply tunnel the I2C or UART signal across the GMSL2 link. This can be useful for separating I2C channels so that multi-master conflicts do not occur. I2C/UART outputs are open drain and require appropriately-sized external pullup resistors for proper operation.

SPI

GMSL2 enables a host SPI master on one side of the GMSL2 link to control a peripheral SPI slave on the opposite side. Communication can be in either direction across the GMSL2 link.

The SPI clock range is 600kHz to 50MHz. Care must be taken to meet setup and hold time requirements when using at speeds higher than 20MHz.

I2S/TDM Audio

GMSL2 devices for display applications support I2S stereo and up to eight channels of audio in TDM mode. In GMSL2 mode most devices have two audio channels —one in the forward direction and one in the reverse direction. Sample rates of 8kHz to 192kHz and sample depths of 8- to 32-bits are supported. In GMSL2 mode the maximum SCK frequency is 49.152MHz. For system, flexibility, GMSL2 devices can act as either the slave or the master at either end of the link.

Table 4. Typical Control Channel Latencies

FUNCTION	FORWARD	REVERSE	NOTES
I2C	<10µs	<10µs	
UART	<10µs	<10µs	
Audio	<100µs	<100µs	For typical use cases
SPI	<10µs	<10µs	Round trip
Ethernet	<10µs	<10µs	Specific parts only

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General-Purpose Inputs and Outputs (GPIO)

GPIOs are typically used to tunnel low speed (< 100kbps) signals over the GMSL2 link. A GPIO tunnel can be set up in the forward (serializer to deserializer) or reverse (deserializer to serializer) direction. GPIO transmissions are transition based; a GPIO packet is created and transmitted on the GMSL2 link when a rising or falling edge transition is detected at a GPI pin. The transition is regenerated at a GPO on the other end of the link.

The multifunction pins GPIO can be programmed as GPI (input), GPO (push-pull output or open-drain output) or GPIO (bidirectional input/output). Each GPIO can also be programmed for 1MΩ or 40kΩ pullup or pulldown (or none). Although an internal pullup is provided, high-speed open-drain outputs require an appropriate value external pullup resistor to V_{DDIO}. Inputs cannot be left unconnected. Always ensure that every pin configured as an input has a pullup or pulldown programmed or is driven by another IC.

A GPI on one side of the serial link can be mapped to a single GPO or multiple GPOs on the other side of the link. Each GPI is assigned a pin ID with the destination GPO(s) on the other side of the link set to the same pin ID. By default, the ID mapping is GPIO0–GPIO0, GPIO1–GPIO1, GPIO2–GPIO2, etc. However, the GPIO mapping can be arbitrarily changed through register settings. GPI transitions can be transmitted in two modes: delay-compensated and non-delay-compensated. When delay compensation is enabled, the GPI-to-GPO delay across the link is a precise, fixed value. Latency increases, but jitter and skew decrease.The state of each GPIO can be read or written by register either locally or remotely over the GMSL2 link by a µC using the control channel I^2C/UART interface.

In non-delay-compensated mode, channel latency is not fixed. The GPI transition is sent as soon as possible based on priority and available link bandwidth. This variable delay is a result of multiple communication channels sharing the link. Non-delay-compensated mode should be used with signals tolerant to delay variation (i.e., µC interrupts).

Priority can be set for GPI pins by registers. If no priority is set, GPI transitions are transmitted in the order they occur. However, when priority is set transitions on GPI with higher priority are transmitted earlier. Typical GMSL2-only device delays for 6Gbps forward and 187Mbps reverse link rates are shown in Table 5..

Table 5. Typical GPIO Delays for Forward and Reverse Link Transmission

	DELAY COMPENSATION	DELAY
GPIO forwarding from serializer to deserializer (6Gbps forward channel)	0	720ns
	1	3.5µs
GPIO forwarding from deserializer to serializer (187Mbps reverse channel)	0	6µs
	1	15µs

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Multifunction Pin Assignments

Multiple functions are assigned to each multifunction pin. They cannot be used simultaneously. Some functions require only a single MFP, but most are implemented across a group of MFPs. For example, WMD is a single MFP, but SPI takes several MFPs. A user selects MFP functions to suit the use case by programming the appropriate registers.

The *Pin Description* indicates the function modes for each multifunction pin, listed in order of priority. Higher priority modes must be turned off when lower-priority modes are enabled, both through register writes. Table 6 indicates how latch-on power-up, I²C/UART, audio, SPI, and other functions are mapped to the multifunction pins. Pins are grouped by speed settings with different output rise and fall times to facilitate optimizing for different speed requirements of the different functions. Pins labeled A and B are pin speed groups. They can have their speeds changed as a group by register settings, i.e., the speed of the entire group B is changed in unison for all grouped pins, and for all functions on these pins with exception of I²C/UART. Each group can be programmed to one of four different speeds. See Table 7 for speed settings.

The main channel I²C/UART always uses I²C speed setting. Enabling the pass-through I²C 1 or pass-through I²C 2 sets these pins (2 each) to the I²C speed independent of the B pin speed group setting.

Table 6. GPIO Pin Function Map

PIN	LATCH ON POWER-UP	I²C/UART	AUDIO	SPI	OTHER FUNCTIONS	GPIO	SPEED GROUP	DEFAULT SPEED
GPIO01	I2CSEL				VS	GPIO01	A	11
GPIO02		MS			WMD	GPIO02	B	10
GPIO03		Pass-through 1 SDA/RX		MOSI		GPIO03	B/I²C	10
GPIO04				MISO		GPIO04	B	10
GPIO05		Pass-through 1 SCL/TX		SCLK		GPIO05	B/I²C	10
GPIO06		Pass-through 2 SDA/RX	Rev Path SD			GPIO06	B/I²C	10
GPIO07			Rev Path SCK			GPIO07	B	10
GPIO08		Pass-through 2 SCL/TX	Rev Path WS			GPIO08	B/I²C	10
GPIO09	CXTP			BNE, SS1		GPO09	B	10
GPIO10				RO, SS2		GPIO10	B	10
GPIO11	ADD0		Fwd Path SD			GPO11	B	10
GPIO12	ADD1		Fwd Path SCK			GPO12	B	10
GPIO13	ADD2		Fwd Path WS			GPO13	B	10
GPIO14		Main channel SDA/Rx				GPIO14	I²C	I²C
GPIO15		Main channel SCL/Tx				GPIO15	I²C	I²C

Table 7. Control- and Side-Channel Rise and Fall Times

PIN OR GPIO SPEED SETTING	RISE TIME		FALL TIME	
	V_{DDIO} = 1.8V	V_{DDIO} = 3.3V	V_{DDIO} = 1.8V	V_{DDIO} = 3.3V
ERRB, LOCK	1.0ns	0.6ns	0.8ns	0.5ns
00	1.0ns	0.6ns	0.8ns	0.5ns
01	2.1ns	1.1ns	2.0ns	1.1ns
10	4.0ns	2.3ns	4.3ns	2.3ns
11	9ns	5ns	10ns	5ns
I²C	***	***	40ns	30ns

**20 to 80%, 10pF load. Rise time is for push-pull output configuration.*
***80 to 20%, 10pF load.*
****Rise time for open-drain outputs depends on external pullup resistor value.*

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SPI and I²S Speed Programming

The SPI and I²S interfaces can be used over a wide range of frequencies. The MAX96752/MAX96754 provide flexible GPIO speed programming to maintain timing margins while minimizing radiated EMI. Table 8 and Table 9 provide guidance on recommended speed settings for various SPI and I²S operating frequencies and V_{DDIO} supply voltages.

See the *Multifunction Pin Assignments* section for GPIO pin speed programming information.

At lower frequencies, SPI data is typically latched on the opposite clock edge on which it is shifted. See Figure 7 and Figure 8. However, at higher frequencies the data must be latched on the same edge as the shift to meet setup and hold time requirements. Table 9 provides guidance on programming the latching clock edge.

Latch-On Power-Up Pins

The logic states at specific configuration input pins are latched on power-up. These states set initial register values and functional modes that cannot be easily programmed through I²C or UART after the device powers up.

Bias the pins to V_{DDIO} for a logic 1, or GND for a logic 0 using sufficiently high-value resistors (up to 10kΩ). Make sure that the voltage seen at the pin is at a valid logic level to ensure proper latching.

All latch-on power-up pins also double as general-purpose output-only pins after power-up. Before the pin states are latched, the device does not drive the pin. It is highly recommended to use the GPIO pins that function doubly as latch-on-power-up pins only as outputs, unless the user can guarantee that the proper pin logic state is present at the pin at power-up.

When the pin is used as an output, the pin should not be driven externally, and the power-up state is solely determined by the external resistor.

Here is the list and description of latch-on-power-up bits:

ADD2, ADD1, ADD0: These bits set the device address (DEV_ADDR) that is used as the slave address by I²C and UART. They also set the default value of the TX_SRC_ID registers of each low bandwidth channel (Audio, Info-Frame, I²C, UART, SPI, GPIO). Device addresses can be changed after power-up by writing to the DEV_ADDR register.

CXTP: This bit sets the device into coaxial or twisted-pair mode, with the value reflected in both register bits CXTP_A and CXTP_B. This bit can be changed after power-up, as one link can be set as coax and the other as twisted pair. CXTP = 0 sets differential mode operation at 3Gbps GMSL2 forward link rate, while CXTP = 1 sets single-ended coax operation at 6Gbps GMSL2 forward link rate. The default reverse channel rate is set to 187.5Mbps independent of CXTP setting.

I2CSEL: This bit selects either UART or I²C mode for internal register access and remote (over GMSL2 link) control channel communication. Set I2CSEL = 0 for UART or I2CSEL = 1 for I²C.

Table 8. Recommended I²S Pin Programming

FREQUENCY (MHz)	V_{DDIO} (V)	RECOMMENDED SPEED SETTING
< 25	1.7–2.24	MS
	2.25–3.6	LS
25–50	1.7–2.24	HS
	2.25–3.6	MS

Table 9. Recommended SPI Pin Programming

FREQUENCY (MHz)	V_{DDIO} (V)	LATCHING EDGE	RECOMMENDED SPEED SETTING
< 12.5	1.7– 2.24	Opposite from shift	MS
	2.25–3.6		LS
12.5–25	1.7–2.24	Opposite from shift	HS
	2.25–3.6		MS
25–50	1.7–2.24	Same as shift	HS
	2.25–3.6		MS

Table 10. Device Address

ADDRESS BITS ADD[2:0]	DEVICE ADDRESS
000	0x90
001	0x94
010	0x98
011	0xD0
100	0xD4
101	0xD8
110	0x50
111	0x54

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Clocking

GMSL Reference Clock

The GMSL2 devices require a reference clock source to generate the 6GHz line rate clock and associated internal clocks. Both the serializer and deserializer can be clocked with an external 25MHz crystal or an external clock source with a frequency accuracy of ±200ppm.

Spread-Spectrum Clocking

Maxim's GMSL2 links provide exceptional EMI performance. Optional spread-spectrum clocking (SSC) is available to mitigate electromagnetic interference emitted from devices and interconnections and provide additional margin.

SSC reduces peaks in the frequency spectrum by spreading the signal over a wider bandwidth. The spread has a 25kHz sawtooth modulation profile, programmable to deviate up to ±2500ppm from the center frequency.

Power-Up and Link Startup

GMSL2 devices are in power-down mode when PWDNB pin is low or when any of the power supplies are down. Register and configurations are set to default reset conditions.

The serializer and deserializer can power up in any order. After PWDNB is released and all power supplies are up, each device starts its power-up sequence and performs the following actions in sequence:

1) Latch-on-power-up pins register set. Set internal registers according to the selected configuration: I2CSEL, CXTP, ADD0, ADD1, ADD2, RCLKEN (if available).
2) Control channel (I^2C or UART) is functional on local side. Device registers are writable and readable.
3) Link is established based on settings as described:
   a. Single link auto selection mode (AUTO_LINK = 1 and LINK_CFG = 1 or 2): Automatically select which PHY to use to establish GMSL2 link by periodically trying to handshake using PHY A and PHY B.
   b. Single link manual selection mode (AUTO_LINK = 0 and LINK_CFG = 1 or 2): If LINK_CFG = 1, establish link using PHY A. If LINK_CFG = 2, establish link using PHY B.
   c. Dual link mode (LINK_CFG = 0): Establish link using both PHYs.
   d. Splitter (serializer)/aggregator (deserializer) mode (LINK_CFG = 3): Establish link using both PHYs (for specific applications only).
4) Each enabled PHY performs link calibration, equalizer adaptation and data channel locking. Both devices set their LOCK pin high.
5) Control channel is available from remote side.

This whole link up process, from the time that the last part's PWDNB input is brought high, takes approximately 20ms nominally and 100ms maximum for any channels that meet the GMSL2 channel specification.

After the devices are linked, they can be configured. This can be done locally or over the control channel, by a microcontroller on either the serializer side or the deserializer side.

Device Reset

There are 3 general reset options available through register writes:

1) RESET_ALL resets all blocks including all registers, digital and analog blocks. This is similar to driving the PWDNB pin low and then high.
2) Setting RESET_LINK resets all GMSL PHY related digital logic and all data pipeline. After this bit is set, all control registers are still accessible through local control channel. The link remains in RESET until RESET_LINK is cleared.
3) RESET_ONESHOT resets all GMSL PHY-related digital logic and all data pipeline and then automatically clears itself. This is similar to setting and clearing RESET_LINK.

Program registers which affect GMSL-2 link operation (i.e., TX_RATE, RX_RATE, CXTP_A/B, AUTO_LINK, LINK_CFG, GMSL2) first, followed by RESET_LINK or RESET_ONESHOT.

Link and Video Lock

Link Lock

Link lock happens automatically on power-up and is an indication that the cable is plugged in and the system is up and running. Lock is obtained with no interaction or commands from the µC to the GMSL parts. Both serializers and deserializers have an open-drain LOCK output pin and a related status register.

Link lock indicates that the PLLs for the GMSL2 link are locked to each other and the data receive paths are locked (forward channel in the serializer, reverse channel in the deserializer). Video and control channel functions (I^2C/UART, SPI, GPIO, audio, Ethernet) can be used immediately after link lock is asserted.

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In dual-link and splitter configurations, only one link locks automatically. The user must then program the serializer to the desired operating mode.

The GMSL2 link uses the crystal or external reference input as the reference clock for the GMSL2 link, so a valid video input (pixel clock) is not needed for the GMSL2 link to lock.

Video Lock

Video lock bit indicates that the deserializer is receiving valid video data. After the GMSL2 link has locked, the deserializer video output PLL starts its locking sequence. The deserializer normally starts outputting video data several milliseconds after it asserts line lock, provided that it is receiving video packets from the serializer. The video lock status is typically read from a register. However, the deserializer LOCK pin behavior can be changed by a register setting so that the LOCK pin is asserted only when the deserializer is outputting video.

Power Standby and Sleep Mode

A power manager block is present in all GSML2 products. Its primary function is to monitor supply voltages and control power-down (standby) and sleep modes.

There are two ways to go into low power mode while all power supplies are active: asserting the PWDNB pin or the invoking of the sleep state. Both states offer very low power supply currents.

Asserting the PWDNB pin (active-low) places the device in standby power mode and resets the digital registers and configurations to their default power-up condition. Any supply dropping below its internal threshold settings also places the device in power-down mode.

The sleep state offers the preservation of all critical register settings and configurations. The device can be put into the sleep state through an I^2C/UART command. The resume state restores the device back to the presleep condition without the need for additional register writes. Resume is invoked by an I^2C command or a low-frequency clock beacon transmitted from the master device over the GMSL2 link.

Error and Fault Condition Monitoring

Both the serializer and deserializer have an open-drain, multi-purpose error reporting and interrupt status output. The active-low ERRB pin is driven by the logical OR of a wide variety of error and event status indicators. The ability of each error condition to drive ERRB is maskable by register settings. Each error and event that can drive ERRB has a status flag within a sub-block of registers, so the reason for assertion of ERRB can be determined by reading the register status.

AEQ (Adaptive Equalization)

GMSL2 devices automatically adapt receiver characteristics to compensate for the insertion and return loss characteristics of the channel, which consists of the cables, connectors and PCBs. This optimizes performance on any channel that meets the GMSL2 channel spec.

The equalizer architecture makes GMSL2 links robust against noise, crosstalk and reflections. Initial adaptation is performed during link lock and then is invoked at a ~1Hz rate to track temperature and voltage variations. The adaptation process optimizes the equalizer coefficients to maximize the eye opening by using the built in eye-opening monitor.

Video Pipeline

Video Line CRC

A CRC32 polynomial is used to generate a 32-bit code at the end of each DE or HS pulse. This code is transferred to the receiver (deserializer) side using info frames. The CRC checker generates the same code on the receiver side and checks if the generated and received CRC codes are the same. If not, it asserts an error. The CRC check is done at every falling edge of DE on the receiver side even if the info frame is not received.

Rx Crossbar, Video PRBS Checker, and Watermark Checker

The device's VRX block contains functions including crossbar, watermark, and video PRBS checker.

Data received from the video FIFO is fed to a crossbar switch before it is passed on to the balance of the video pipeline. The crossbar is configured from the configuration registers and can be used to reorder the color and sync signal. It can also be used to invert the signals individually and force them to a constant value if desired.

The watermark block is used for checking that the video image is not frozen.

The video PRBS checker needs to be enabled after the video is locked.

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Eye-Opening Monitor

The eye-opening monitor (EOM) enables GMSL2 parts to monitor the link margin on an active link and generate an interrupt if it falls below an acceptable level. For example, if a cable is damaged the link can run error free, but have less link margin than desired. This allows the customer to proactively react to deteriorating cable performance before any link errors occur. GMSL2 devices can measure the horizontal or vertical eye opening of the equalizer's output. The measurement is activated automatically at a ~1Hz rate once a link is active. The EOM block compares the data sampled at the center of the eye with a sample offset in phase for the horizontal EOM or offset in voltage for the vertical EOM. The eye opening is then reported, and the EOM can trigger an interrupt or a reset if the opening falls below user-defined thresholds.

Watermarking

The watermarking block allows users to detect a frozen frame failure in a frame-based processing system between the generator and detector. This feature is specifically targeted to detect frozen frames caused by SOCs in safety-critical applications. It does not detect frozen frames that occur before the watermark generator or after the watermark detector. GMSL2 devices contain both a watermark generator and watermark detector. This allows both serializers and deserializers to insert a watermark or detect a watermark in a safety-relevant video stream. The watermark generator inserts a time-varying watermark that is highly redundant and robust to image processing and display stream compression. The watermark detector looks for this time-varying watermark, and if it fails to see all the generated watermarks, this indicates a frozen frame failure in a frame-based processing system between the generator and detector. Upon detection of this error condition, the watermark detector can generate an interrupt and/or blank the output video returning the display to a safe-state in less than 500ms.

Vertical Sync Output for Backlight PWM Synchronization

The deserializer can optionally synchronize the backlight PWM output with the vertical sync (VS) signal to ensure that video output is accurately dimmed and video data is properly aligned with the start of the video frame. Very predictable video performance is achieved by aligning the PWM output with the start of the video frame. Write 1 to the VS_OUT_EN register field to enable this function; write 0 to the same field to disable. Note that enabling VS output is a secondary function of a GPIO pin.

PRBS

Video PRBS

The video channel has a PRBS generator in the serializer and a PRBS checker in the deserializer for testing the video channel operation. The video PRBS generator can work with the PCLK recovered by the serializer from an HDMI, parallel, or MIPI input. An external PCLK can also be provided to a designated GPIO pin to run the video PRBS generator.

Note that all link bandwidth is not used by the video channel alone in GMSL2 mode, so it is possible to have a bit error on the link that does not cause a video PRBS error.

Audio PRBS

The audio channel includes a PRBS generator on the transmitter side and a checker at the receiver side for testing the audio channel operation. The audio PRBS generator can operate using the incoming I2S SCK and WS signals or using an internally generated reference clock. The detected audio channel bit errors are reported through a status register.

High-Bandwidth Digital Content Protection

Overview

High-bandwidth digital content protection (HDCP) is an encryption scheme to protect digital audio and digital video content in various protocols. The MAX96754 GMSL2 links support HDCP v1.4.

For power savings, the HDCP circuits are powered down by default. Once the user enables HDCP, the function powers up and performs authentication, after which link integrity checks are periodically performed by hardware through a dedicated control channel over the GMSL2 link. HDCP interrupts can be generated and status registers provide status of authentication progress indicating success and a ready state to the upstream video source.

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Each HDCP device has a factory-programmed unique device ID called a key selection vector (KSV) and associated private keys. They are licensed from DCP LLC.

Repeater Support

GMSL2 serializers and deserializers with HDCP capability can be configured as HDCP repeaters. An HDCP repeater receives and decrypts HDCP content and then encrypts and transmits on one or more downstream links. A maximum of 15 receivers are supported.

Auto HDCP Operation

GMSL2 has an auto HDCP (AHDCP) feature that fully automates the GMSL2 HDCP1.4 key exchange authentication without requiring external µC involvement. Authentication and link integrity checks are done by hardware through a dedicated control channel over the GMSL2 link.

GMSL2 AHDCP has an optional fast mode to perform link integrity check every frame. This reduces the video out of encryption synchronization duration down to as low as 2 frames.

Dual GMSL2 Links

There are two HDCP 1.4 engines in the serializer that work independently for the HDCP authentication and link integrity check. This supports configurations that have one GMSL2 serializer connected to two GMSL2 deserializers with HDCP operation.

OLDI Deserializer Video Output

Overview

The OpenLDI (OLDI) interface drives display panels from the deserializer. It receives video data from the video pipeline and maps it to OLDI output ports A, B, or both. The output can drive with OLDI or VESA formats. The OLDI clock output is generated by the deserializer based on the tracked and measured clock frequency of video input at the serializer. For more information about the OLDI standard, refer to the Open LVDS Display Interface (OpenLDI) specification.

Note: The OLDI_SPL_POL bit in register OLDI1 (0x1CE) must always be set to 0 for dual-OLDI applications.

Odd/Even Pixel Interleaving in Dual OLDI Mode

The OLDI output processor can split the output to the two OLDI ports at half the PCLK frequency by using the active-high or active-low versions of the DE or HS signals for even/odd pixel alignment or without any alignment at all.

Maximum OLDI Pixel Clock

TThe MAX96752/MAX96754 support a maximum pixel clock of 300MHz. The OLDI output ports each support 150MHz. Therefore, in single-output mode, a maximum of 150MHz is supported. In even/odd pixel interleaved or dual-view splitter modes, a single video stream of up to 300MHz PCLK frequency can be output from the device.

Flexible Output Programming

- Configurable as 1x4, 1x8, or 2x4 lanes
- 4 lanes (for RGB888) or optionally 3 lanes (for RGB666) of the OLDI ports can be enabled
- The OLDI output lanes can be inverted and/or swapped
- Vertical sync (VS) can be viewed on GPIO01. It includes an optional single-cycle glitch filter.

Color Lookup Table (LUT)

The LUT is intended mainly for gamma correction. It has 3 x 256, 8-bit registers that allow any color in the RGB domain to be mapped to any RGB color. This can be used for generating color filters or gamma correction maps by the user.

Spread Spectrum

The OLDI output port can be programmed to have 0.5% center spread spectrum clocking (SSC) with 20kHz to 40kHz triangular modulation frequency. It is possible to program the spread-spectrum percentage up to 4%. OLDI SSC mode programming is independent of the GMSL high-speed clock SSC mode.

Applications Information

Control Channel Programming

GMSL2 device registers can only be accessed and configured through the main I²C/UART interface.

Host-to-Peripheral Main I²C and Pass-Through I²C Communication

When communicating between a host and peripheral, main and pass-through I²C operation is the same. An I²C tunnel across the GMSL2 link connects the host's I²C master to the remote I²C slave. This logically connects separated I²C buses, enabling I²C transactions across the serial link to occur (with some delay) as if performed on the same physical I²C bus. The GMSL2 serializer and

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deserializer are intermediary devices; the host I2C master connects to a GMSL2 device I2C slave, and the peripheral I2C slave connects to a GMSL2 device I2C master. For example, when the host I2C master transacts on one side of the link (local-side), data is forwarded to the other side (remote-side) by the I2C slave of the local-side GMSL2 device. Data is then received by the I2C master of the remote-side GMSL2 device, which in turn generates the same I2C transaction with the peripheral slave I2C. The remote-side GMSL2 device sends back any I2C data expected by the local-side.

The I2C interface uses clock stretching (holding SCL low) to account for timing differences between master and slave and to allow time for data to be forwarded and received across the serial link. The host I2C master and peripheral I2C slave must support clock stretching by the GMSL2 device.

SDA and SCL lines operate as both an input and an open-drain output. Pullup resistors are required on SDA and SCL.

Each transmission consists of a START condition sent by a master, followed by the device's 7-bit slave address plus a R/W bit, register address bytes, one or more data bytes, and finally a STOP condition. Register addresses are 16-bits wide. Single or multiple data bytes can be written or read (by address autoincrements).

Device Address

Each device on the I2C UART control channel must have a unique address. The GMSL2 device address is set to one of several 7-bit addresses according to the voltage level of the ADD or CFG pins at power-up. See the *Latch-On Power-Up Pins* section. Note that device address can be changed after power-up by writing to the DEV_ADDR register.

Main I2C Host-to-GMSL2 Device Communication

The host I2C master has access to GMSL2 serializer and deserializer registers. The host can program GMSL2 device registers to change the pass-through I2C/UART interface from I2C to UART.

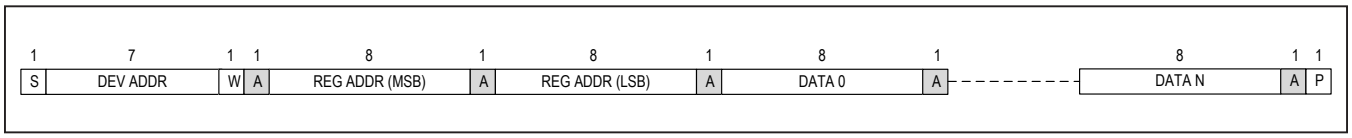


Figure 18. I2C Write Packet Format

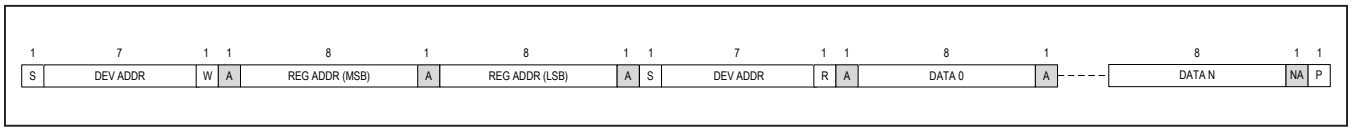


Figure 19. I2C Read Packet Format

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Main UART

When the main I2C/UART interface is configured as UART, there are two operating modes: base and bypass.

UART Base Mode

Base mode is the means of µC communication with the serializer and deserializer where registers in these devices can be accessed as well as registers in peripheral devices. Base mode is enabled by default at power-up. In base mode, the µC is the host and can access the registers of both the serializer and deserializer from either side of the link using the GMSL2 UART packet protocol. The µC can also program the peripherals on the remote side by sending the UART packets to the serializer or deserializer. The µC communicates with a UART peripheral in base mode (through INTTYPE register settings). The device addresses of the serializer and deserializer in this mode are programmable. In base mode, serializer, deserializer and peripheral registers can be written and read using the half-duplex GMSL2 UART protocol.

Figure 20 shows the UART protocol for writing and reading in base mode.

UART Bypass Mode

In bypass mode, the serializer/deserializer ignore UART commands from the µC, allowing the µC to communicate only with peripheral devices. The µC cannot access the serializer/deserializer's registers in this mode. The UART transitions are simply sent over the GMSL link. Ignoring UART transactions prevents inadvertent misprogramming of serializer and deserializer registers. The device addresses of the serializer and deserializer in this mode are not programmable.

Switching Between UART Base and Bypass Modes

There are two ways to switch between base mode and bypass mode: by programming the register or by using the MS pin.

Two modes can be set by programming: temporary or permanent. In temporary mode, bypass mode is active only as long as there is UART activity. When there is no UART activity for a selected timeout, both devices exit bypass mode and the bit is automatically cleared. In permanent mode, the devices stay in bypass mode until the next power-down.

When set by the MS pin, a high pin level puts the device into bypass mode, and a low level puts the device in base mode. MS is set on-the-fly and is not latched on power-up.

UART Frame Format

Regular UART frames with an even parity bit are used to carry one byte of data each. A frame consists of a low start bit followed by eight data bits, a parity bit, and is finished with a high stop bit. The parity bit is high if the number of ones in the 8 bits of data is odd, otherwise, the parity bit is low. There must be at least one high stop bit. If the next frame is in the same packet, there can be at most four high bits from the end of the stop bit to the beginning of the next start bit. Note that in the case of a parity bit error, the packet, starting from the frame with the error, is discarded. The start of each frame is always a high-to-low transition (i.e., the stop bit is high and the start bit is low). The phase of the internal UART bit clock is adjusted using the start bit of each frame. The framer calibrates the length of one UART bit in terms of the

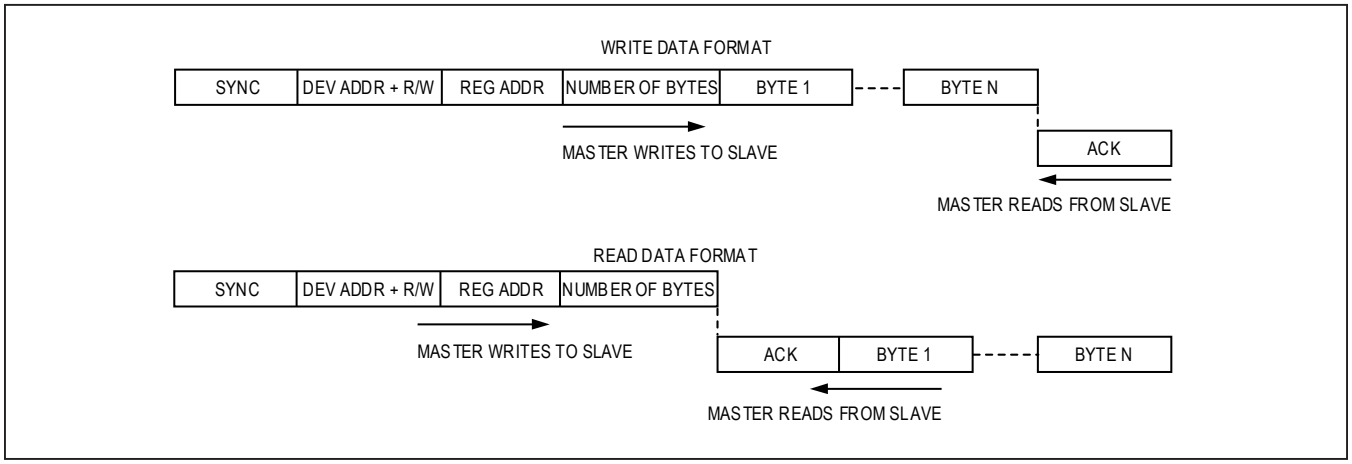


Figure 20. GMSL2 UART Protocol for Base Mode

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internal oscillator clock using the synchronization frame (i.e., the first frame of a UART packet transmission). In bypass mode, the parity bit is enabled by default, but the frames are not checked for parity errors. Either even or odd parity can be used. The parity bit is passed along with UART data transmissions; the recipient of the data must perform error checking. The parity bit can optionally be disabled in bypass mode before entering bypass mode. Note that the bit rate in bypass mode must be the same bit rate last used in base mode.

Synchronization Frame

The serializer/deserializer must calibrate internal bit length counters with the UART bit rate for proper recovery of UART frames. A sync frame (a regular UART frame with the value 0x79) is sent as the first frame of each data packet from the µC and is used to calibrate the bit length in terms of the device's internal 150MHz clock. Sync frames must be properly detected before the subsequent frames of the packet can be correctly received. When the line stays high for at least 32 bits, the packet boundary is reset and the framer begins waiting for the next sync frame.

Acknowledge Frame

When a packet is successfully received, the addressed device responds with an acknowledge frame to inform the µC that no errors were detected in the transmitted packet, and it was recognized as valid. This is sent after the last bit of a successfully recognized packet has been received. The acknowledge frame is a regular UART frame (value 0xC3). Data written to the serializer/deserializer registers do not take effect until after the acknowledge byte is sent.

Write Packet

Write packets consist of a five byte packet header followed by one or more data bytes. A packet is recognized as a write packet when the LSB of the device address frame is 0. The addressed device responds with an acknowledge frame if no errors were detected while receiving the write packet and the write packet is valid. Byte count indicates the number of data bytes to be written and it cannot be 0.

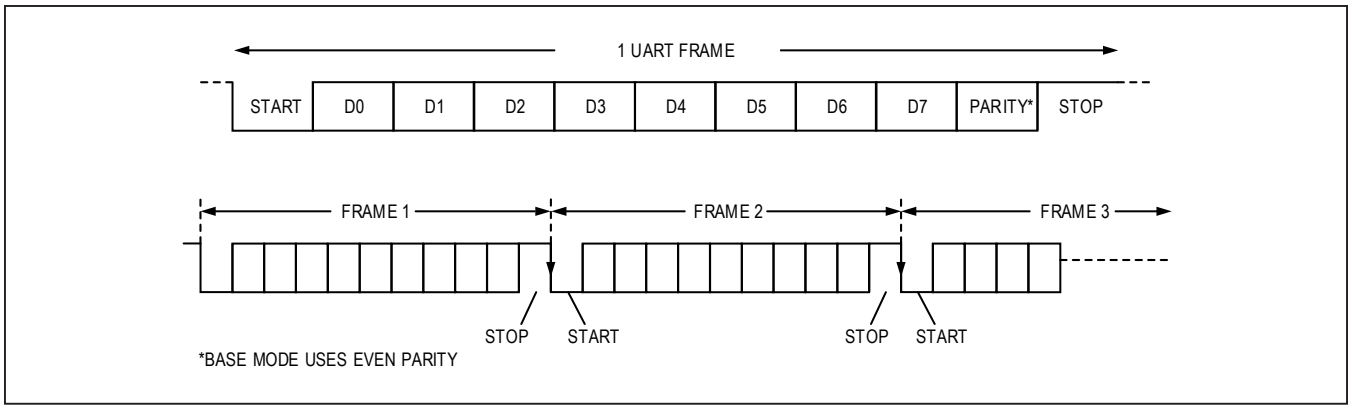


Figure 21. UART Data Format for Base Mode

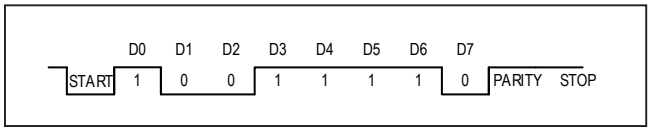


Figure 22. UART Synchronization Frame

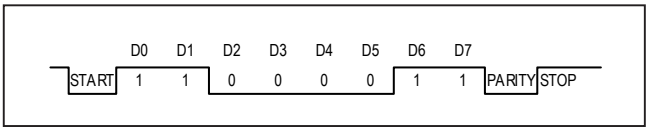


Figure 23. UART Acknowledge Frame

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Read Packet

Read packet consists of five bytes. The LSB of the device address frame is 1 for read packets. If no errors were detected while receiving the read packet and the packet is valid, the addressed device responds with an acknowledge frame followed by one or more data bytes. Byte count indicates the number of data bytes to be read and it cannot be 0.

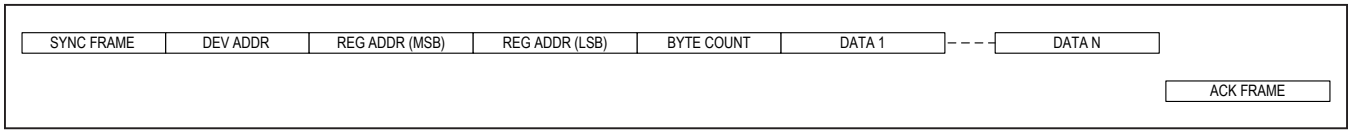


Figure 24. UART Write Packet Format

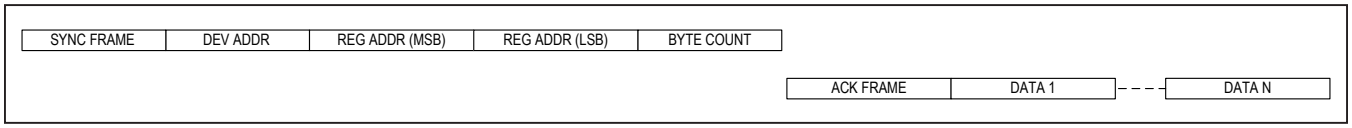


Figure 25. UART Read Packet Format

Ordering Information

PART	TEMP RANGE	PIN-PACKAGE	HDCP	TOP MARKING
MAX96752GTN/V+	-40°C to +105°C	56-Lead TQFN-EP*	No	MAXIM
MAX96752GTN/V+T	-40°C to +105°C	56-Lead TQFN-EP*	No	MAXIM
MAX96754GTN/V+*	-40°C to +105°C	56-Lead TQFN-EP**	Yes†	MAXIM
MAX96754GTN/V+T*	-40°C to +105°C	56-Lead TQFN-EP**	Yes†	MAXIM

/V Denotes an automotive qualified part.
+Denotes a lead(Pb)-free/RoHS-compliant package.
T = Tape and reel.
**Future product—contact factory for availability.*
***EP = Exposed pad.*
†HDCP parts require registration with Digital Content Protection, LLC.

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Revision History

REVISION NUMBER	REVISION DATE	DESCRIPTION	PAGES CHANGED
0	11/18	Initial release	—

For pricing, delivery, and ordering information, please visit Maxim Integrated's online storefront at https://www.maximintegrated.com/en/storefront/storefront.html.

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Reserved, Unused, and Read-Only Register Bits

Not all register bits in the register space are shown in the register table. Any bit not explicitly defined in the register table should be treated as reserved and should not be modified. When a write is required to a register with both defined and undefined register bits, first read the register's contents, then create a new register value by only changing the defined bits, and finally write the new byte to the register (read/replace/write).

In this document default values are provided for read-only register bits. Read-only bit states are changed at power-up according to the actual state of the device. To avoid overwriting these bits, treat read-only bits as undefined.

HDCP (MAX96754 Only)

The MAX96754 has HDCP while the MAX96752 does not. Register blocks (shown below), registers, and bitfields that refer to HDCP are reserved in the MAX96752 and should not be changed from the default state.

Table 1. HDCP Register Blocks (MAX96754 Only)

BLOCK NAME	REGISTER ADDRESS NAME
CFGL_AHDCP	0x0088–0x008F
HDCP_RX	0x0600–0x06FF

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Table 1. HDCP Register Blocks (MAX96754 Only). 1

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Register Map

MAX96752/MAX96754 GMSL2 oLDI Deserializers

ADDRESS	NAME	MSB							LSB
DEV									
0x00	REG0[7:0]	DEV_ADDR[6:0]							CFG_BLOCK
0x01	REG1[7:0]	LVDS_HALFSW	–	IIC_2_EN	IIC_1_EN	TX_RATE[1:0]		RX_RATE[1:0]	
0x02	REG2[7:0]	LOCK_CFG	VID_EN	DIS_LOCAL_CC	DIS_REM_CC	–	AUD_TX_EN	–	–
0x03	REG3[7:0]	–	–	–	I2CSEL	UART_2_EN	UART_1_EN	–	VIDEO_LOCK
0x0D	REG13[7:0]	DEV_ID[7:0]							
0x0E	REG14[7:0]	–	–	–	–	DEV_REV[3:0]			
0x0F	REG15[7:0]	–	–	SPEED_CPBL[1:0]		DV_CPBL	DUAL_CPBL	SPLTR_CPBL	HDCP_CPBL
0x38	IO_CHK0[7:0]	PIN_DRV_EN_0[7:0]							
0x39	IO_CHK1[7:0]	PIN_DRV_EN_1[7:0]							
0x3A	IO_CHK2[7:0]	PIN_DRV_SEL	–	–	–	–	–	PIN_DRV_EN_2[1:0]	
0x3B	IO_CHK3[7:0]	PIN_IN_0[7:0]							
0x3C	IO_CHK4[7:0]	PIN_IN_1[7:0]							
OVERLAP									
TCTRL									
0x09	PWR1[7:0]	OVERTEMP	–	–	–	–	–	–	–
0x0C	PWR4[7:0]	–	DIS_LOCAL_WAKE	WAKE_EN_B	WAKE_EN_A	–	–	–	–
0x10	CTRL0[7:0]	RESET_ALL	RESET_LINK	RESET_ONESHOT	AUTO_LINK	SLEEP	–	LINK_CFG[1:0]	
0x11	CTRL1[7:0]	–	–	–	CXTP_B	BACK_COMP_SPLTR	–	–	CXTP_A
0x13	CTRL3[7:0]	–	–	LINK_MODE[1:0]		LOCKED	ERROR	CMU_LOCKED	–
0x18	INTR0[7:0]	–	–	–	–	AUTO_ERR_RST_EN	DEC_ERR_THR[2:0]		
0x19	INTR1[7:0]	PKT_CNT_EXP[3:0]				AUTO_CNT_RST_EN	PKT_CNT_THR[2:0]		
0x1A	INTR2[7:0]	PHY_INT_OEN_B	PHY_INT_OEN_A	REM_ERR_OEN	–	–	IDLE_ERR_OEN	DEC_ERR_OEN_B	DEC_ERR_OEN_A

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ADDRESS	NAME	MSB							LSB
0x1B	INTR3[7:0]	PHY_INT_B	PHY_INT_A	REM_ERR_FLAG	–	–	IDLE_ERR_FLAG	DEC_ERR_FLAG_B	DEC_ERR_FLAG_A
0x1C	INTR4[7:0]	–	–	–	HDCP_INT_OEN	MAX_RT_OEN	RT_CNT_OEN	PKT_CNT_OEN	WM_ERR_OEN
0x1D	INTR5[7:0]	–	–	–	HDCP_INT_FLAG	MAX_RT_FLAG	RT_CNT_FLAG	PKT_CNT_FLAG	WM_ERR_FLAG
0x1E	INTR6[7:0]	VDDCMP_INT_OEN	PORZ_INT_OEN	VDDBAD_INT_OEN	VDD_OV_OEN	LCRC_ERR_OEN	VPRBS_ERR_OEN	APRBS_ERR_OEN	VID_PXL_CRC_ERR_OEN
0x1F	INTR7[7:0]	VDDCMP_INT_FLAG	PORZ_INT_FLAG	VDDBAD_INT_FLAG	VDD_OV_FLAG	LCRC_ERR_FLAG	VPRBS_ERR_FLAG	APRBS_ERR_FLAG	VID_PXL_CRC_ERR_FLAG
0x20	INTR8[7:0]	ERR_TX_EN	–	–	ERR_TX_ID[4:0]				
0x21	INTR9[7:0]	ERR_RX_EN	–	–	ERR_RX_ID[4:0]				
0x22	CNT0[7:0]	DEC_ERR_A[7:0]							
0x23	CNT1[7:0]	DEC_ERR_B[7:0]							
0x24	CNT2[7:0]	IDLE_ERR[7:0]							
0x25	CNT3[7:0]	PKT_CNT[7:0]							
0x26	CNT4[7:0]	VID_PXL_CRC_ERR[7:0]							
GMSL									
0x29	TX1[7:0]	LINK_PRBS_GEN	–	ERRG_EN_B	ERRG_EN_A	–	–	–	–
0x2A	TX2[7:0]	ERRG_CNT[1:0]		ERRG_RATE[1:0]		ERRG_BURST[2:0]			ERRG_PER
0x2B	TX3[7:0]	–	–	–	–	–	TIMEOUT[2:0]		
0x2C	RX0[7:0]	PKT_CNT_LBW[1:0]		–	–	PKT_CNT_SEL[3:0]			
0x30	GPIOA[7:0]	GPIO_RX_FAST_BIDIR_EN	GPIO_TX_CASC	GPIO_FWD_CDLY[5:0]					
0x31	GPIOB[7:0]	GPIO_TX_WNDW[1:0]		GPIO_REV_CDLY[5:0]					
CC									
0x40	I2C_0[7:0]	–	–	SLV_SH[1:0]		–	SLV_TO[2:0]		
0x41	I2C_1[7:0]	–	MST_BT[2:0]			–	MST_TO[2:0]		
0x42	I2C_2[7:0]	SRC_A[6:0]							–
0x43	I2C_3[7:0]	DST_A[6:0]							–
0x44	I2C_4[7:0]	SRC_B[6:0]							–
0x45	I2C_5[7:0]	DST_B[6:0]							–
0x46	I2C_6[7:0]	–	–	–	–	I2C_AUTO_CFG	I2C_SRC_CNT[2:0]		

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ADDRESS	NAME	MSB							LSB
0x47	I2C_7[7:0]	UART_RX_OVERFLOW	UART_TX_OVERFLOW	–	–	–	I2C_TIMED_OUT	REM_ACK_ACKED	REM_ACK_RECVED
0x48	UART_0[7:0]	ARB_TO_LEN[1:0]		REM_MS_EN	LOC_MS_EN	BYPASS_DIS_PAR	BYPASS_TO[1:0]		BYPASS_EN
0x49	UART_1[7:0]	BITLEN_LSB[7:0]							
0x4A	UART_2[7:0]	OUT_DELAY[1:0]		BITLEN_MSB[5:0]					
0x4C	I2C_PT_0[7:0]	–	–	SLV_SH_PT[1:0]		–	SLV_TO_PT[2:0]		
0x4D	I2C_PT_1[7:0]	MST_DBL_PT	MST_BT_PT[2:0]			–	MST_TO_PT[2:0]		
0x4E	I2C_PT_2[7:0]	XOVER_EN_2	I2C_TIMED_OUT_2	REM_ACK_ACKED_2	REM_ACK_RECVED_2	XOVER_EN_1	I2C_TIMED_OUT_1	REM_ACK_ACKED_1	REM_ACK_RECVED_1
0x4F	UART_PT_0[7:0]	BITLEN_MAN_CFG_2	DIS_PAR_2	UART_RX_OVERFLOW_2	UART_TX_OVERFLOW_2	BITLEN_MAN_CFG_1	DIS_PAR_1	UART_RX_OVERFLOW_1	UART_TX_OVERFLOW_1
CFGH VIDEO									
0x50	RX0[7:0]	RX_CRC_EN	–	–	–	–	–	STR_SEL[1:0]	
CFGL AUDIO									
0x58	TR0[7:0]	TX_CRC_EN	RX_CRC_EN	–	–	PRIO_VAL[1:0]		PRIO_CFG[1:0]	
0x59	TR1[7:0]	BW_MULT[1:0]		BW_VAL[5:0]					
0x5B	TR3[7:0]	–	–	TX_SPLT_MASK_B	TX_SPLT_MASK_A	–	TX_SRC_ID[2:0]		
0x5C	TR4[7:0]	RX_SRC_SEL[7:0]							
0x5D	ARQ0[7:0]	ARQ_AUTO_CFG	ACK_CNT	MATCH_SRC_ID	ACK_SRC_ID	ARQ_EN	–	–	–
0x5E	ARQ1[7:0]	–	MAX_RT[2:0]			–	–	MAX_RT_ERR_OEN	RT_CNT_OEN
0x5F	ARQ2[7:0]	MAX_RT_ERR	RT_CNT[6:0]						
CFGI INFOFR									
0x60	TR0[7:0]	TX_CRC_EN	RX_CRC_EN	–	–	PRIO_VAL[1:0]		PRIO_CFG[1:0]	
0x61	TR1[7:0]	BW_MULT[1:0]		BW_VAL[5:0]					
0x63	TR3[7:0]	–	–	TX_SPLT_MASK_B	TX_SPLT_MASK_A	–	TX_SRC_ID[2:0]		
0x64	TR4[7:0]	RX_SRC_SEL[7:0]							
CFGL SPI									
0x68	TR0[7:0]	TX_CRC_EN	RX_CRC_EN	–	–	PRIO_VAL[1:0]		PRIO_CFG[1:0]	
0x69	TR1[7:0]	BW_MULT[1:0]		BW_VAL[5:0]					

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ADDRESS	NAME	MSB							LSB
0x6B	TR3[7:0]	–	–	TX_SPLT_MASK_B	TX_SPLT_MASK_A	–	TX_SRC_ID[2:0]		
0x6C	TR4[7:0]	RX_SRC_SEL[7:0]							
0x6D	ARQ0[7:0]	ARQ_AUTO_CFG	ACK_CNT	MATCH_SRC_ID	ACK_SRC_ID	ARQ_EN	–	–	–
0x6E	ARQ1[7:0]	–	MAX_RT[2:0]			–	–	MAX_RT_ERR_OEN	RT_CNT_OEN
0x6F	ARQ2[7:0]	MAX_RT_ERR	RT_CNT[6:0]						
CFGC CC									
0x70	TR0[7:0]	TX_CRC_EN	RX_CRC_EN	–	–	PRIO_VAL[1:0]		PRIO_CFG[1:0]	
0x71	TR1[7:0]	BW_MULT[1:0]		BW_VAL[5:0]					
0x73	TR3[7:0]	–	–	TX_SPLT_MASK_B	TX_SPLT_MASK_A	–	TX_SRC_ID[2:0]		
0x74	TR4[7:0]	RX_SRC_SEL[7:0]							
0x75	ARQ0[7:0]	ARQ_AUTO_CFG	ACK_CNT	MATCH_SRC_ID	ACK_SRC_ID	ARQ_EN	–	–	–
0x76	ARQ1[7:0]	–	MAX_RT[2:0]			–	–	MAX_RT_ERR_OEN	RT_CNT_OEN
0x77	ARQ2[7:0]	MAX_RT_ERR	RT_CNT[6:0]						
CFGL GPIO									
0x78	TR0[7:0]	TX_CRC_EN	RX_CRC_EN	–	–	PRIO_VAL[1:0]		PRIO_CFG[1:0]	
0x79	TR1[7:0]	BW_MULT[1:0]		BW_VAL[5:0]					
0x7B	TR3[7:0]	–	–	TX_SPLT_MASK_B	TX_SPLT_MASK_A	–	TX_SRC_ID[2:0]		
0x7C	TR4[7:0]	RX_SRC_SEL[7:0]							
0x7D	ARQ0[7:0]	ARQ_AUTO_CFG	ACK_CNT	MATCH_SRC_ID	ACK_SRC_ID	ARQ_EN	–	–	–
0x7E	ARQ1[7:0]	–	MAX_RT[2:0]			–	–	MAX_RT_ERR_OEN	RT_CNT_OEN
0x7F	ARQ2[7:0]	MAX_RT_ERR	RT_CNT[6:0]						
CFGL AHDCP									
0x88	TR0[7:0]	TX_CRC_EN	RX_CRC_EN	–	–	PRIO_VAL[1:0]		PRIO_CFG[1:0]	
0x89	TR1[7:0]	BW_MULT[1:0]		BW_VAL[5:0]					
0x8B	TR3[7:0]	–	–	TX_SPLT_MASK	TX_SPLT_MASK	–	TX_SRC_ID[2:0]		

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ADDRESS	NAME	MSB							LSB
				_B	_A				
0x8C	TR4[7:0]	RX_SRC_SEL[7:0]							
0x8D	ARQ0[7:0]	ARQ_AUTO_CFG	ACK_CNT	MATCH_SRC_ID	ACK_SRC_ID	ARQ_EN	–	–	–
0x8E	ARQ1[7:0]	–	MAX_RT[2:0]			–	–	MAX_RT_ERR_OEN	RT_CNT_OEN
0x8F	ARQ2[7:0]	MAX_RT_ERR	RT_CNT[6:0]						
CFGL IIC_X									
0xA0	TR0[7:0]	TX_CRC_EN	RX_CRC_EN	–	–	PRIO_VAL[1:0]		PRIO_CFG[1:0]	
0xA1	TR1[7:0]	BW_MULT[1:0]		BW_VAL[5:0]					
0xA3	TR3[7:0]	–	–	TX_SPLT_MASK_B	TX_SPLT_MASK_A	–	TX_SRC_ID[2:0]		
0xA4	TR4[7:0]	RX_SRC_SEL[7:0]							
0xA5	ARQ0[7:0]	ARQ_AUTO_CFG	ACK_CNT	MATCH_SRC_ID	ACK_SRC_ID	ARQ_EN	–	–	–
0xA6	ARQ1[7:0]	–	MAX_RT[2:0]			–	–	MAX_RT_ERR_OEN	RT_CNT_OEN
0xA7	ARQ2[7:0]	MAX_RT_ERR	RT_CNT[6:0]						
CFGL IIC_Y									
0xA8	TR0[7:0]	TX_CRC_EN	RX_CRC_EN	–	–	PRIO_VAL[1:0]		PRIO_CFG[1:0]	
0xA9	TR1[7:0]	BW_MULT[1:0]		BW_VAL[5:0]					
0xAB	TR3[7:0]	–	–	TX_SPLT_MASK_B	TX_SPLT_MASK_A	–	TX_SRC_ID[2:0]		
0xAC	TR4[7:0]	RX_SRC_SEL[7:0]							
0xAD	ARQ0[7:0]	ARQ_AUTO_CFG	ACK_CNT	MATCH_SRC_ID	ACK_SRC_ID	ARQ_EN	–	–	–
0xAE	ARQ1[7:0]	–	MAX_RT[2:0]			–	–	MAX_RT_ERR_OEN	RT_CNT_OEN
0xAF	ARQ2[7:0]	MAX_RT_ERR	RT_CNT[6:0]						
VID_RX									
0x100	VIDEO_RX0[7:0]	LCRC_ERR	–	–	–	–	–	LINE_CRC_EN	–
0x103	VIDEO_RX3[7:0]	–	HD_TR_MODE	DLOCKED	VLOCKED	HLOCKED	DTRACKEN	VTRACKEN	HTRACKEN
0x104	VIDEO_RX4[7:0]	VID_OVERFLW_LIM_L[7:0]							
0x105	VIDEO_RX5[7:0]	VID_UNDERFLW_LIM_L[7:0]							

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ADDRESS	NAME	MSB							LSB
0x106	VIDEO_RX6[7:0]	VID_SAR_INC_H[2:0]			–	–	–	VID_UNDERFLW_LIM_H	VID_OVERFLW_LIM_H
0x108	VIDEO_RX8[7:0]	VID_BLK_LEN_ERR	VID_LOCK	VID_PKT_DET	VID_SEQ_ERR	–	–	–	–
0x10A	VIDEO_RX10[7:0]	–	MASK_VIDEO_DE	–	–	–	–	–	–
0x111	CLOCKGEN[7:0]	–	–	–	–	CLK_FREEZE_RNG[1:0]		CLK_FREEZE_CONS	CLK_FREEZE
AUD_TX AX									
0x120	AUDIO_TX0[7:0]	I2S_TDM_CFG[1:0]		I2S_CFG[1:0]		INV_SCK	INV_WS	FORCE_AUD	AUD_SINK_SRC
0x121	AUDIO_TX1[7:0]	AUD_PRIO[1:0]		AUD_STR_TX[1:0]		AUD_DRIFT_DET_EN	AUD_INF_PER	I2S_HDR_CFG[1:0]	
0x122	AUDIO_TX2[7:0]	AUD_HDR_L[7:0]							
0x123	AUDIO_TX3[7:0]	AUD_HDR_M[7:0]							
0x124	AUDIO_TX4[7:0]	AUD_HDR_H[7:0]							
0x125	AUDIO_TX5[7:0]	AUD_DRIFT_ERR	AUD_FIFO_WARN	AUD_OVERFLOW	ACLKDET	–	–	–	–
0x127	AUDIO_TX7[7:0]	PRBS_SEL	PRBSEN_AUD	–	–	–	–	–	–
0x128	AUDIO_TX8[7:0]	PRBS_WS_LEN	PRBS_WS_GEN	–	–	–	–	–	–
AUD_RX									
0x140	AUDIO_RX1[7:0]	AUD_RX_SINK_SRC	–	–	–	INV_SCK_RX	INV_WS_RX	–	AUD_EN_RX
0x143	AUDIO_RX4[7:0]	APRBS_ERR[7:0]							
0x146	AUDIO_RX7[7:0]	–	–	–	APRBS_CHK_EN	AUD_STRM[1:0]		–	–
0x148	AUDIO_RX9[7:0]	AUD_BLK_LEN_ERR	AUD_LOCK	AUD_PKT_DET	APRBS_VALID	–	–	–	–
SPI									
0x160	SPI_0[7:0]	SPI_LOC_ID[1:0]		SPI_CC_TRG_ID[1:0]		SPI_IGNR_ID	SPI_CC_EN	MST_SLVN	SPI_EN
0x161	SPI_1[7:0]	SPI_LOC_N[5:0]						SPI_BASE_PRIO[1:0]	
0x162	SPI_2[7:0]	REQ_HOLD_OFF[2:0]			FULL_SCK_SETUP	SPI_MOD3_F	SPI_MOD3	SPIM_SS2_ACT_H	SPIM_SS1_ACT_H

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ADDRESS	NAME	MSB							LSB
0x163	SPI_3[7:0]	SPIM_SS_DLY_CLKS[7:0]							
0x164	SPI_4[7:0]	SPIM_SCK_LO_CLKS[7:0]							
0x165	SPI_5[7:0]	SPIM_SCK_HI_CLKS[7:0]							
0x166	SPI_6[7:0]	–	–	BNE	SPIS_RWN	SS_IO_EN_2	SS_IO_EN_1	BNE_IO_EN	RWN_IO_EN
0x167	SPI_7[7:0]	SPI_RX_OVRFLW	SPI_TX_OVRFLW	–	SPIS_BYTE_CNT[4:0]				
0x168	SPI_8[7:0]	REQ_HOLD_OFF_TO[7:0]							
WM									
0x188	WM_0[7:0]	WM_LEN	WM_MODE[2:0]			WM_DET[1:0]		WME_EN	WM_EN
0x18A	WM_2[7:0]	–	–	–	–	HsyncPol	VsyncPol	WM_NPFILT[1:0]	
0x18B	WM_3[7:0]	–	WM_TH[6:0]						
0x18C	WM_4[7:0]	–	–	WM_YUV_IN[1:0]		WM_COLORADJ	–	WM_MASKMODE[1:0]	
0x18D	WM_5[7:0]	–	–	–	–	–	–	WM_DETOUT	WM_ERROR
0x18E	WM_6[7:0]	WM_TIMER[7:0]							
0x1AE	WM_WREN_0[7:0]	WM_WREN_L[7:0]							
0x1AF	WM_WREN_1[7:0]	WM_WREN_H[7:0]							
VRX									
0x1B0	CROSS_0[7:0]	–	CROSS0_I	CROSS0_F	CROSS0[4:0]				
0x1B1	CROSS_1[7:0]	–	CROSS1_I	CROSS1_F	CROSS1[4:0]				
0x1B2	CROSS_2[7:0]	–	CROSS2_I	CROSS2_F	CROSS2[4:0]				
0x1B3	CROSS_3[7:0]	–	CROSS3_I	CROSS3_F	CROSS3[4:0]				
0x1B4	CROSS_4[7:0]	–	CROSS4_I	CROSS4_F	CROSS4[4:0]				
0x1B5	CROSS_5[7:0]	–	CROSS5_I	CROSS5_F	CROSS5[4:0]				
0x1B6	CROSS_6[7:0]	–	CROSS6_I	CROSS6_F	CROSS6[4:0]				
0x1B7	CROSS_7[7:0]	–	CROSS7_I	CROSS7_F	CROSS7[4:0]				
0x1B8	CROSS_8[7:0]	–	CROSS8_I	CROSS8_F	CROSS8[4:0]				
0x1B9	CROSS_9[7:0]	–	CROSS9_I	CROSS9_F	CROSS9[4:0]				
0x1BA	CROSS_10[7:0]	–	CROSS10_I	CROSS10_F	CROSS10[4:0]				
0x1BB	CROSS_11[7:0]	–	CROSS1	CROSS1	CROSS11[4:0]				

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ADDRESS	NAME	MSB							LSB
			1_I	1_F					
0x1BC	CROSS_12[7:0]	–	CROSS12_I	CROSS12_F	CROSS12[4:0]				
0x1BD	CROSS_13[7:0]	–	CROSS13_I	CROSS13_F	CROSS13[4:0]				
0x1BE	CROSS_14[7:0]	–	CROSS14_I	CROSS14_F	CROSS14[4:0]				
0x1BF	CROSS_15[7:0]	–	CROSS15_I	CROSS15_F	CROSS15[4:0]				
0x1C0	CROSS_16[7:0]	–	CROSS16_I	CROSS16_F	CROSS16[4:0]				
0x1C1	CROSS_17[7:0]	–	CROSS17_I	CROSS17_F	CROSS17[4:0]				
0x1C2	CROSS_18[7:0]	–	CROSS18_I	CROSS18_F	CROSS18[4:0]				
0x1C3	CROSS_19[7:0]	–	CROSS19_I	CROSS19_F	CROSS19[4:0]				
0x1C4	CROSS_20[7:0]	–	CROSS20_I	CROSS20_F	CROSS20[4:0]				
0x1C5	CROSS_21[7:0]	–	CROSS21_I	CROSS21_F	CROSS21[4:0]				
0x1C6	CROSS_22[7:0]	–	CROSS22_I	CROSS22_F	CROSS22[4:0]				
0x1C7	CROSS_23[7:0]	–	CROSS23_I	CROSS23_F	CROSS23[4:0]				
0x1C8	CROSS_24[7:0]	–	CROSS24_I	CROSS24_F	CROSS24[4:0]				
0x1C9	CROSS_25[7:0]	–	CROSS25_I	CROSS25_F	CROSS25[4:0]				
0x1CA	CROSS_26[7:0]	–	CROSS26_I	CROSS26_F	CROSS26[4:0]				
0x1CB	CROSS_27[7:0]	–	CROSS27_I	CROSS27_F	CROSS27[4:0]				
0x1CC	PRBS_ERR[7:0]	VPRBS_ERR[7:0]							
0x1CD	OLDI0[7:0]	–	–	VPRBS_FAIL	VPRBS_CHK_EN	HS_OUT_EN	LUT_C_EN	LUT_B_EN	LUT_A_EN
0x1CE	OLDI1[7:0]	OLDI_OUTSEL	OLDI_FORMAT	OLDI_4TH_LANE	OLDI_SWAP_AB	OLDI_SPL_EN	OLDI_SPL_MODE[1:0]		OLDI_SPL_POL
0x1CF	OLDI2[7:0]	PD_LVDS_B	PD_LVDS_A	–	VS_OUT_EN	–	–	OLDI_DUP	SSEN
0x1D0	OLDI3[7:0]	LANE_INV_B0	LANE_SEL_B0[2:0]			LANE_INV_A0	LANE_SEL_A0[2:0]		
0x1D1	OLDI4[7:0]	LANE_INV_B1	LANE_SEL_B1[2:0]			LANE_INV_A1	LANE_SEL_A1[2:0]		
0x1D2	OLDI5[7:0]	LANE_INV_B2	LANE_SEL_B2[2:0]			LANE_INV_A2	LANE_SEL_A2[2:0]		
0x1D3	OLDI6[7:0]	LANE_IN	LANE_SEL_B3[2:0]			LANE_IN	LANE_SEL_A3[2:0]		

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ADDRESS	NAME	MSB							LSB
		V_B3				V_A3			
0x1D4	OLDI7[7:0]	LANE_INV_BCLK	LANE_SEL_BCLK[2:0]			LANE_INV_ACLK	LANE_SEL_ACLK[2:0]		
GPIO0 0									
0x200	GPIO_A[7:0]	RES_CFG	TX_PRIO	TX_COMP_EN	GPIO_OUT	GPIO_IN	GPIO_RX_EN	GPIO_TX_EN	GPIO_OUT_DIS
0x201	GPIO_B[7:0]	PULL_UPDN_SEL[1:0]		OUT_TYPE	GPIO_TX_ID[4:0]				
0x202	GPIO_C[7:0]	OVR_RES_CFG	–	–	GPIO_RX_ID[4:0]				
GPIO1 1									
0x203	GPIO_A[7:0]	RES_CFG	TX_PRIO	TX_COMP_EN	GPIO_OUT	GPIO_IN	GPIO_RX_EN	GPIO_TX_EN	GPIO_OUT_DIS
0x204	GPIO_B[7:0]	PULL_UPDN_SEL[1:0]		OUT_TYPE	GPIO_TX_ID[4:0]				
0x205	GPIO_C[7:0]	OVR_RES_CFG	–	–	GPIO_RX_ID[4:0]				
GPIO2 2									
0x206	GPIO_A[7:0]	RES_CFG	TX_PRIO	TX_COMP_EN	GPIO_OUT	GPIO_IN	GPIO_RX_EN	GPIO_TX_EN	GPIO_OUT_DIS
0x207	GPIO_B[7:0]	PULL_UPDN_SEL[1:0]		OUT_TYPE	GPIO_TX_ID[4:0]				
0x208	GPIO_C[7:0]	OVR_RES_CFG	–	–	GPIO_RX_ID[4:0]				
GPIO3 3									
0x209	GPIO_A[7:0]	RES_CFG	TX_PRIO	TX_COMP_EN	GPIO_OUT	GPIO_IN	GPIO_RX_EN	GPIO_TX_EN	GPIO_OUT_DIS
0x20A	GPIO_B[7:0]	PULL_UPDN_SEL[1:0]		OUT_TYPE	GPIO_TX_ID[4:0]				
0x20B	GPIO_C[7:0]	OVR_RES_CFG	–	–	GPIO_RX_ID[4:0]				
GPIO4 4									
0x20C	GPIO_A[7:0]	RES_CFG	TX_PRIO	TX_COMP_EN	GPIO_OUT	GPIO_IN	GPIO_RX_EN	GPIO_TX_EN	GPIO_OUT_DIS
0x20D	GPIO_B[7:0]	PULL_UPDN_SEL[1:0]		OUT_TYPE	GPIO_TX_ID[4:0]				
0x20E	GPIO_C[7:0]	OVR_RES_CFG	–	–	GPIO_RX_ID[4:0]				
GPIO5 5									
0x20F	GPIO_A[7:0]	RES_CFG	TX_PRIO	TX_COMP_EN	GPIO_OUT	GPIO_IN	GPIO_RX_EN	GPIO_TX_EN	GPIO_OUT_DIS
0x210	GPIO_B[7:0]	PULL_UPDN_SEL[1:0]		OUT_TYPE	GPIO_TX_ID[4:0]				
0x211	GPIO_C[7:0]	OVR_RES_CFG	–	–	GPIO_RX_ID[4:0]				
GPIO6 6									

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ADDRESS	NAME	MSB							LSB
0x212	GPIO_A[7:0]	RES_CFG	TX_PRIO	TX_COMP_EN	GPIO_OUT	GPIO_IN	GPIO_RX_EN	GPIO_TX_EN	GPIO_OUT_DIS
0x213	GPIO_B[7:0]	PULL_UPDN_SEL[1:0]		OUT_TYPE	GPIO_TX_ID[4:0]				
0x214	GPIO_C[7:0]	OVR_RES_CFG	–	–	GPIO_RX_ID[4:0]				
GPIO7 7									
0x215	GPIO_A[7:0]	RES_CFG	TX_PRIO	TX_COMP_EN	GPIO_OUT	GPIO_IN	GPIO_RX_EN	GPIO_TX_EN	GPIO_OUT_DIS
0x216	GPIO_B[7:0]	PULL_UPDN_SEL[1:0]		OUT_TYPE	GPIO_TX_ID[4:0]				
0x217	GPIO_C[7:0]	OVR_RES_CFG	–	–	GPIO_RX_ID[4:0]				
GPIO8 8									
0x218	GPIO_A[7:0]	RES_CFG	TX_PRIO	TX_COMP_EN	GPIO_OUT	GPIO_IN	GPIO_RX_EN	GPIO_TX_EN	GPIO_OUT_DIS
0x219	GPIO_B[7:0]	PULL_UPDN_SEL[1:0]		OUT_TYPE	GPIO_TX_ID[4:0]				
0x21A	GPIO_C[7:0]	OVR_RES_CFG	–	–	GPIO_RX_ID[4:0]				
GPIO9 9									
0x21B	GPIO_A[7:0]	RES_CFG	TX_PRIO	TX_COMP_EN	GPIO_OUT	GPIO_IN	GPIO_RX_EN	GPIO_TX_EN	GPIO_OUT_DIS
0x21C	GPIO_B[7:0]	PULL_UPDN_SEL[1:0]		OUT_TYPE	GPIO_TX_ID[4:0]				
0x21D	GPIO_C[7:0]	OVR_RES_CFG	–	–	GPIO_RX_ID[4:0]				
GPIO10 10									
0x21E	GPIO_A[7:0]	RES_CFG	TX_PRIO	TX_COMP_EN	GPIO_OUT	GPIO_IN	GPIO_RX_EN	GPIO_TX_EN	GPIO_OUT_DIS
0x21F	GPIO_B[7:0]	PULL_UPDN_SEL[1:0]		OUT_TYPE	GPIO_TX_ID[4:0]				
0x220	GPIO_C[7:0]	OVR_RES_CFG	–	–	GPIO_RX_ID[4:0]				
GPIO11 11									
0x221	GPIO_A[7:0]	RES_CFG	TX_PRIO	TX_COMP_EN	GPIO_OUT	GPIO_IN	GPIO_RX_EN	GPIO_TX_EN	GPIO_OUT_DIS
0x222	GPIO_B[7:0]	PULL_UPDN_SEL[1:0]		OUT_TYPE	GPIO_TX_ID[4:0]				
0x223	GPIO_C[7:0]	OVR_RES_CFG	–	–	GPIO_RX_ID[4:0]				
GPIO12 12									
0x224	GPIO_A[7:0]	RES_CFG	TX_PRIO	TX_COMP_EN	GPIO_OUT	GPIO_IN	GPIO_RX_EN	GPIO_TX_EN	GPIO_OUT_DIS
0x225	GPIO_B[7:0]	PULL_UPDN_SEL[1:0]		OUT_TYPE	GPIO_TX_ID[4:0]				

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ADDRESS	NAME	MSB							LSB
0x226	GPIO_C[7:0]	OVR_RES_CFG	–	–	GPIO_RX_ID[4:0]				
GPIO13 13									
0x227	GPIO_A[7:0]	RES_CFG	TX_PRIO	TX_COMP_EN	GPIO_OUT	GPIO_IN	GPIO_RX_EN	GPIO_TX_EN	GPIO_OUT_DIS
0x228	GPIO_B[7:0]	PULL_UPDN_SEL[1:0]		OUT_TYPE	GPIO_TX_ID[4:0]				
0x229	GPIO_C[7:0]	OVR_RES_CFG	–	–	GPIO_RX_ID[4:0]				
GPIO14 14									
0x22A	GPIO_A[7:0]	RES_CFG	TX_PRIO	TX_COMP_EN	GPIO_OUT	GPIO_IN	GPIO_RX_EN	GPIO_TX_EN	GPIO_OUT_DIS
0x22B	GPIO_B[7:0]	PULL_UPDN_SEL[1:0]		OUT_TYPE	GPIO_TX_ID[4:0]				
0x22C	GPIO_C[7:0]	OVR_RES_CFG	–	–	GPIO_RX_ID[4:0]				
GPIO15 15									
0x22D	GPIO_A[7:0]	RES_CFG	TX_PRIO	TX_COMP_EN	GPIO_OUT	GPIO_IN	GPIO_RX_EN	GPIO_TX_EN	GPIO_OUT_DIS
0x22E	GPIO_B[7:0]	PULL_UPDN_SEL[1:0]		OUT_TYPE	GPIO_TX_ID[4:0]				
0x22F	GPIO_C[7:0]	OVR_RES_CFG	–	–	GPIO_RX_ID[4:0]				
CMU									
0x244	CMU4[7:0]	–	–	–	–	GPIO_SPEED_B[1:0]		GPIO_SPEED_A[1:0]	
MISC									
0x24A	UART_PT_0[7:0]	BITLEN_PT_1_L[7:0]							
0x24B	UART_PT_1[7:0]	–	–	BITLEN_PT_1_H[5:0]					
0x24C	UART_PT_2[7:0]	BITLEN_PT_2_L[7:0]							
0x24D	UART_PT_3[7:0]	–	–	BITLEN_PT_2_H[5:0]					
0x252	I2C_PT_4[7:0]	SRC_A_1[6:0]							–
0x253	I2C_PT_5[7:0]	DST_A_1[6:0]							–
0x254	I2C_PT_6[7:0]	SRC_B_1[6:0]							–
0x255	I2C_PT_7[7:0]	DST_B_1[6:0]							–
0x256	I2C_PT_8[7:0]	SRC_A_2[6:0]							–
0x257	I2C_PT_9[7:0]	DST_A_2[6:0]							–
0x258	I2C_PT_10[7:0]	SRC_B_2[6:0]							–
0x259	I2C_PT_11[7:0]	DST_B_2[6:0]							–
RLMS A									
0x403	RLMS3[7:0]	AdaptEn	–	–	–	–	–	–	–
0x404	RLMS4[7:0]	EOM_CHK_AMOUNT[3:0]				EOM_CHK_THR[1:0]		EOM_PER_MODE	EOM_EN

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ADDRESS	NAME	MSB							LSB
0x405	RLMS5[7:0]	EOM_MAN_TRG_REQ	EOM_MIN_THR[6:0]						
0x406	RLMS6[7:0]	EOM_PV_MODE	EOM_RST_THR[6:0]						
0x407	RLMS7[7:0]	EOM_DONE	EOM[6:0]						
0x434	RLMS34[7:0]	EyeMonPerCntL[7:0]							
0x435	RLMS35[7:0]	EyeMonPerCntH[7:0]							
0x437	RLMS37[7:0]	–	–	–	EyeMonDone	EyeMonCntClr	EyeMonStart	EyeMonPh	EyeMonDPol
0x438	RLMS38[7:0]	EyeMonErrCntL[7:0]							
0x439	RLMS39[7:0]	EyeMonErrCntH[7:0]							
0x43A	RLMS3A[7:0]	EyeMonValCntL[7:0]							
0x43B	RLMS3B[7:0]	EyeMonValCntH[7:0]							
0x43D	RLMS3D[7:0]	ErrChPh[6:0]							ErrChPhTogEn
0x43E	RLMS3E[7:0]	ErrChPhSecTA	ErrChPhSec[6:0]						
0x43F	RLMS3F[7:0]	ErrChPhPriTA	ErrChPhPri[6:0]						
0x449	RLMS49[7:0]	–	–	–	–	–	ErrChPwrUp	–	–
0x458	RLMS58[7:0]	–	ErrChVTh1[6:0]						
0x459	RLMS59[7:0]	–	ErrChVTh0[6:0]						
0x464	RLMS64[7:0]	–	–	–	–	–	–	TxSSCMode[1:0]	
0x470	RLMS70[7:0]	–	TxSSCFrqCtrl[6:0]						
0x471	RLMS71[7:0]	–	TxSSCCenSprSt[5:0]						RSVD
0x472	RLMS72[7:0]	TxSSCPreSclL[7:0]							
0x473	RLMS73[7:0]	–	–	–	–	–	TxSSCPreSclH[2:0]		
0x474	RLMS74[7:0]	TxSSCPhL[7:0]							
0x475	RLMS75[7:0]	–	TxSSCPhH[6:0]						
0x476	RLMS76[7:0]	–	–	–	–	–	–	TxSSCPhQuad[1:0]	
0x495	RLMS95[7:0]	TxAmplManEn	–	TxAmplMan[5:0]					
0x4A4	RLMSA4[7:0]	AEQ_PER_MULT[1:0]	AEQ_PER[5:0]						
0x4AC	RLMSAC[7:0]	ErrChPhSecTAFR3G	ErrChPhSecFR3G[6:0]						
0x4AD	RLMSAD[7:0]	ErrChPhPriTAFR3G	ErrChPhPriFR3G[6:0]						
0x4AE	RLMSAE[7:0]	ErrChPhSecTAFR1G5	ErrChPhSecFR1G5[6:0]						

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ADDRESS	NAME	MSB							LSB
0x4AF	RLMSAF[7:0]	ErrChPhPriTAFR1G5	ErrChPhPriFR1G5[6:0]						
0x4B0	RLMSB0[7:0]	ErrChPhSecTASR1G5	ErrChPhSecSR1G5[6:0]						
0x4B1	RLMSB1[7:0]	ErrChPhPriTASR1G5	ErrChPhPriSR1G5[6:0]						
0x4B2	RLMSB2[7:0]	ErrChPhSecTASRG75	ErrChPhSecSRG75[6:0]						
0x4B3	RLMSB3[7:0]	ErrChPhPriTASRG75	ErrChPhPriSRG75[6:0]						
0x4B4	RLMSB4[7:0]	ErrChPhSecTASRG375	ErrChPhSecSRG375[6:0]						
0x4B5	RLMSB5[7:0]	ErrChPhPriTASRG375	ErrChPhPriSRG375[6:0]						
0x4B6	RLMSB6[7:0]	ErrChPhSecTASRG1875	ErrChPhSecSRG1875[6:0]						
0x4B7	RLMSB7[7:0]	ErrChPhPriTASRG1875	ErrChPhPriSRG1875[6:0]						
RLMS B									
0x503	RLMS3[7:0]	AdaptEn	–	–	–	–	–	–	–
0x504	RLMS4[7:0]	EOM_CHK_AMOUNT[3:0]				EOM_CHK_THR[1:0]		EOM_PER_MODE	EOM_EN
0x505	RLMS5[7:0]	EOM_MAN_TRG_REQ	EOM_MIN_THR[6:0]						
0x506	RLMS6[7:0]	EOM_PV_MODE	EOM_RST_THR[6:0]						
0x507	RLMS7[7:0]	EOM_DONE	EOM[6:0]						
0x534	RLMS34[7:0]	EyeMonPerCntL[7:0]							
0x535	RLMS35[7:0]	EyeMonPerCntH[7:0]							
0x537	RLMS37[7:0]	–	–	–	EyeMonDone	EyeMonCntClr	EyeMonStart	EyeMonPh	EyeMonDPol
0x538	RLMS38[7:0]	EyeMonErrCntL[7:0]							
0x539	RLMS39[7:0]	EyeMonErrCntH[7:0]							
0x53A	RLMS3A[7:0]	EyeMonValCntL[7:0]							
0x53B	RLMS3B[7:0]	EyeMonValCntH[7:0]							
0x53D	RLMS3D[7:0]	ErrChPh[6:0]							ErrChPh

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ADDRESS	NAME	MSB							LSB
									TogEn
0x53E	RLMS3E[7:0]	ErrChPhSecTA	ErrChPhSec[6:0]						
0x53F	RLMS3F[7:0]	ErrChPhPriTA	ErrChPhPri[6:0]						
0x549	RLMS49[7:0]	–	–	–	–	–	ErrChPwrUp	–	–
0x558	RLMS58[7:0]	–	ErrChVTh1[6:0]						
0x559	RLMS59[7:0]	–	ErrChVTh0[6:0]						
0x564	RLMS64[7:0]	–	–	–	–	–	–	TxSSCMode[1:0]	
0x570	RLMS70[7:0]	–	TxSSCFrqCtrl[6:0]						
0x571	RLMS71[7:0]	–	TxSSCCenSprSt[5:0]						RSVD
0x572	RLMS72[7:0]	TxSSCPreSclL[7:0]							
0x573	RLMS73[7:0]	–	–	–	–	–	TxSSCPreSclH[2:0]		
0x574	RLMS74[7:0]	TxSSCPhL[7:0]							
0x575	RLMS75[7:0]	–	TxSSCPhH[6:0]						
0x576	RLMS76[7:0]	–	–	–	–	–	–	TxSSCPhQuad[1:0]	
0x595	RLMS95[7:0]	TxAmplManEn	–	TxAmplMan[5:0]					
0x5A4	RLMSA4[7:0]	AEQ_PER_MULT[1:0]	AEQ_PER[5:0]						
0x5AC	RLMSAC[7:0]	ErrChPhSecTAFR3G	ErrChPhSecFR3G[6:0]						
0x5AD	RLMSAD[7:0]	ErrChPhPriTAFR3G	ErrChPhPriFR3G[6:0]						
0x5AE	RLMSAE[7:0]	ErrChPhSecTAFR1G5	ErrChPhSecFR1G5[6:0]						
0x5AF	RLMSAF[7:0]	ErrChPhPriTAFR1G5	ErrChPhPriFR1G5[6:0]						
0x5B0	RLMSB0[7:0]	ErrChPhSecTASR1G5	ErrChPhSecSR1G5[6:0]						
0x5B1	RLMSB1[7:0]	ErrChPhPriTASR1G5	ErrChPhPriSR1G5[6:0]						
0x5B2	RLMSB2[7:0]	ErrChPhSecTASRG75	ErrChPhSecSRG75[6:0]						
0x5B3	RLMSB3[7:0]	ErrChPhPriTASRG75	ErrChPhPriSRG75[6:0]						
0x5B4	RLMSB4[7:0]	ErrChPhSecTAS	ErrChPhSecSRG375[6:0]						

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ADDRESS	NAME	MSB							LSB
		RG375							
0x5B5	RLMSB5[7:0]	ErrChPhPriTASRG375	ErrChPhPriSRG375[6:0]						
0x5B6	RLMSB6[7:0]	ErrChPhSecTASRG1875	ErrChPhSecSRG1875[6:0]						
0x5B7	RLMSB7[7:0]	ErrChPhPriTASRG1875	ErrChPhPriSRG1875[6:0]						
HDCP_RX									
0x680	HDCP_0[7:0]	GMSL_BKSV_0[7:0]							
0x681	HDCP_1[7:0]	GMSL_BKSV_1[7:0]							
0x682	HDCP_2[7:0]	GMSL_BKSV_2[7:0]							
0x683	HDCP_3[7:0]	GMSL_BKSV_3[7:0]							
0x684	HDCP_4[7:0]	GMSL_BKSV_4[7:0]							
0x685	HDCP_5[7:0]	RIP_0[7:0]							
0x686	HDCP_6[7:0]	RIP_1[7:0]							
0x687	HDCP_7[7:0]	PJP[7:0]							
0x688	HDCP_8[7:0]	AN_0[7:0]							
0x689	HDCP_9[7:0]	AN_1[7:0]							
0x68A	HDCP_A[7:0]	AN_2[7:0]							
0x68B	HDCP_B[7:0]	AN_3[7:0]							
0x68C	HDCP_C[7:0]	AN_4[7:0]							
0x68D	HDCP_D[7:0]	AN_5[7:0]							
0x68E	HDCP_E[7:0]	AN_6[7:0]							
0x68F	HDCP_F[7:0]	AN_7[7:0]							
0x690	HDCP_10[7:0]	AKSV_0[7:0]							
0x691	HDCP_11[7:0]	AKSV_1[7:0]							
0x692	HDCP_12[7:0]	AKSV_2[7:0]							
0x693	HDCP_13[7:0]	AKSV_3[7:0]							
0x694	HDCP_14[7:0]	AKSV_4[7:0]							
0x695	HDCP_15[7:0]	–	–	–	–	–	–	AUTH_START	ENC_EN
0x696	HDCP_16[7:0]	–	–	–	–	–	–	HDCP_NEW_DEV_CONN	HDCP_READY
0x697	HDCP_17[7:0]	HDCP_BCAPS[6:0]							HDCP_REPEATER
0x699	HDCP_19[7:0]	–	–	–	–	–	–	–	DIS_AUDIO_ENC
0x69A	HDCP_1A[7:0]	PJFP_0[7:0]							

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ADDRESS	NAME	MSB							LSB
0x69B	HDCP_1B[7:0]	PJFP_1[7:0]							
0x69C	HDCP_1C[7:0]	PJFP_2[7:0]							
0x6A0	HDCP_20[7:0]	VH0_0[7:0]							
0x6A1	HDCP_21[7:0]	VH0_1[7:0]							
0x6A2	HDCP_22[7:0]	VH0_2[7:0]							
0x6A3	HDCP_23[7:0]	VH0_3[7:0]							
0x6A4	HDCP_24[7:0]	VH1_0[7:0]							
0x6A5	HDCP_25[7:0]	VH1_1[7:0]							
0x6A6	HDCP_26[7:0]	VH1_2[7:0]							
0x6A7	HDCP_27[7:0]	VH1_3[7:0]							
0x6A8	HDCP_28[7:0]	VH2_0[7:0]							
0x6A9	HDCP_29[7:0]	VH2_1[7:0]							
0x6AA	HDCP_2A[7:0]	VH2_2[7:0]							
0x6AB	HDCP_2B[7:0]	VH2_3[7:0]							
0x6AC	HDCP_2C[7:0]	VH3_0[7:0]							
0x6AD	HDCP_2D[7:0]	VH3_1[7:0]							
0x6AE	HDCP_2E[7:0]	VH3_2[7:0]							
0x6AF	HDCP_2F[7:0]	VH3_3[7:0]							
0x6B0	HDCP_30[7:0]	VH4_0[7:0]							
0x6B1	HDCP_31[7:0]	VH4_1[7:0]							
0x6B2	HDCP_32[7:0]	VH4_2[7:0]							
0x6B3	HDCP_33[7:0]	VH4_3[7:0]							
0x6B4	HDCP_34[7:0]	HDCP_MAX_DEV_EXCEEDED	HDCP_DEV_CNT[6:0]						
0x6B5	HDCP_35[7:0]	–	–	–	–	HDCP_MAX_CASCADE_EXCEEDED	HDCP_DEPTH[2:0]		
0x6B6	HDCP_36[7:0]	HDCP_GPMEM[7:0]							
0x6B7	HDCP_37[7:0]	HDCP_PD	AH_MODE	–	–	–	–	–	–
0x6B8	HDCP_38[7:0]	HDCP_INT	–	–	–	–	–	MASK_AUTH_ST	MASK_ENC_EN
0x6B9	HDCP_39[7:0]	LINK_LOCK	AH_VID_LOCK	AH_RESERVED[5:0]					
0x6BA	HDCP_3A[7:0]	KSV_LIST_0[7:0]							
0x6BB	HDCP_3B[7:0]	KSV_LIST_1[7:0]							
0x6BC	HDCP_3C[7:0]	KSV_LIST_2[7:0]							
0x6BD	HDCP_3D[7:0]	KSV_LIST_3[7:0]							
0x6BE	HDCP_3E[7:0]	KSV_LIST_4[7:0]							
0x6BF	HDCP_3F[7:0]	KSV_LIST_5[7:0]							

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ADDRESS	NAME	MSB							LSB
0x6C0	HDCP_40[7:0]	KSV_LIST_6[7:0]							
0x6C1	HDCP_41[7:0]	KSV_LIST_7[7:0]							
0x6C2	HDCP_42[7:0]	KSV_LIST_8[7:0]							
0x6C3	HDCP_43[7:0]	KSV_LIST_9[7:0]							
0x6C4	HDCP_44[7:0]	KSV_LIST_10[7:0]							
0x6C5	HDCP_45[7:0]	KSV_LIST_11[7:0]							
0x6C6	HDCP_46[7:0]	KSV_LIST_12[7:0]							
0x6C7	HDCP_47[7:0]	KSV_LIST_13[7:0]							
0x6C8	HDCP_48[7:0]	KSV_LIST_14[7:0]							
0x6C9	HDCP_49[7:0]	KSV_LIST_15[7:0]							
0x6CA	HDCP_4A[7:0]	KSV_LIST_16[7:0]							
0x6CB	HDCP_4B[7:0]	KSV_LIST_17[7:0]							
0x6CC	HDCP_4C[7:0]	KSV_LIST_18[7:0]							
0x6CD	HDCP_4D[7:0]	KSV_LIST_19[7:0]							
0x6CE	HDCP_4E[7:0]	KSV_LIST_20[7:0]							
0x6CF	HDCP_4F[7:0]	KSV_LIST_21[7:0]							
0x6D0	HDCP_50[7:0]	KSV_LIST_22[7:0]							
0x6D1	HDCP_51[7:0]	KSV_LIST_23[7:0]							
0x6D2	HDCP_52[7:0]	KSV_LIST_24[7:0]							
0x6D3	HDCP_53[7:0]	KSV_LIST_25[7:0]							
0x6D4	HDCP_54[7:0]	KSV_LIST_26[7:0]							
0x6D5	HDCP_55[7:0]	KSV_LIST_27[7:0]							
0x6D6	HDCP_56[7:0]	KSV_LIST_28[7:0]							
0x6D7	HDCP_57[7:0]	KSV_LIST_29[7:0]							
0x6D8	HDCP_58[7:0]	KSV_LIST_30[7:0]							
0x6D9	HDCP_59[7:0]	KSV_LIST_31[7:0]							
0x6DA	HDCP_5A[7:0]	KSV_LIST_32[7:0]							
0x6DB	HDCP_5B[7:0]	KSV_LIST_33[7:0]							
0x6DC	HDCP_5C[7:0]	KSV_LIST_34[7:0]							
0x6DD	HDCP_5D[7:0]	KSV_LIST_35[7:0]							
0x6DE	HDCP_5E[7:0]	KSV_LIST_36[7:0]							
0x6DF	HDCP_5F[7:0]	KSV_LIST_37[7:0]							
0x6E0	HDCP_60[7:0]	KSV_LIST_38[7:0]							
0x6E1	HDCP_61[7:0]	KSV_LIST_39[7:0]							
0x6E2	HDCP_62[7:0]	KSV_LIST_40[7:0]							
0x6E3	HDCP_63[7:0]	KSV_LIST_41[7:0]							
0x6E4	HDCP_64[7:0]	KSV_LIST_42[7:0]							
0x6E5	HDCP_65[7:0]	KSV_LIST_43[7:0]							
0x6E6	HDCP_66[7:0]	KSV_LIST_44[7:0]							
0x6E7	HDCP_67[7:0]	KSV_LIST_45[7:0]							
0x6E8	HDCP_68[7:0]	KSV_LIST_46[7:0]							

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ADDRESS	NAME	MSB							LSB
0x6E9	HDCP_69[7:0]	KSV_LIST_47[7:0]							
0x6EA	HDCP_6A[7:0]	KSV_LIST_48[7:0]							
0x6EB	HDCP_6B[7:0]	KSV_LIST_49[7:0]							
0x6EC	HDCP_6C[7:0]	KSV_LIST_50[7:0]							
0x6ED	HDCP_6D[7:0]	KSV_LIST_51[7:0]							
0x6EE	HDCP_6E[7:0]	KSV_LIST_52[7:0]							
0x6EF	HDCP_6F[7:0]	KSV_LIST_53[7:0]							
0x6F0	HDCP_70[7:0]	KSV_LIST_54[7:0]							
0x6F1	HDCP_71[7:0]	KSV_LIST_55[7:0]							
0x6F2	HDCP_72[7:0]	KSV_LIST_56[7:0]							
0x6F3	HDCP_73[7:0]	KSV_LIST_57[7:0]							
0x6F4	HDCP_74[7:0]	KSV_LIST_58[7:0]							
0x6F5	HDCP_75[7:0]	KSV_LIST_59[7:0]							
0x6F6	HDCP_76[7:0]	KSV_LIST_60[7:0]							
0x6F7	HDCP_77[7:0]	KSV_LIST_61[7:0]							
0x6F8	HDCP_78[7:0]	KSV_LIST_62[7:0]							
0x6F9	HDCP_79[7:0]	KSV_LIST_63[7:0]							
0x6FA	HDCP_7A[7:0]	KSV_LIST_64[7:0]							
0x6FB	HDCP_7B[7:0]	KSV_LIST_65[7:0]							
0x6FC	HDCP_7C[7:0]	KSV_LIST_66[7:0]							
0x6FD	HDCP_7D[7:0]	KSV_LIST_67[7:0]							
0x6FE	HDCP_7E[7:0]	KSV_LIST_68[7:0]							
0x6FF	HDCP_7F[7:0]	KSV_LIST_69[7:0]							

Register Details

REG0 (0x0)

BIT	7	6	5	4	3	2	1	0
Field	DEV_ADDR[6:0]							CFG_BLOCK
Reset	0b1001000							0b0
Access Type	Write, Read							Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
DEV_ADDR	7:1	Device Address - Default value is set by the ADD[2:0] pins as follows: ADD[2:0] Device Address 000 0b1001000 001 0b1001010 010 0b1001100 011 0b1101000 100 0b1101010 101 0b1101100 110 0b0101000	0b0000000: I²C write/read address is 0x00/0x01 0b0000001: I²C write/read address is 0x02/0x03 .: . .: . 0b1001000: I²C write/read address is 0x90/0x91 0b1001010: I²C write/read address is 0x94/0x95 0b1001100: I²Cwrite/read address is 0x98/0x99 0b1101000: I²C write/read address is 0xD0/0xD1 0b1101010: I²C write/read address is 0xD4/0xD5 0b1101100: I²C write/read address is 0xD8/0xD9

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BITFIELD	BITS	DESCRIPTION	DECODE
		111 0b0101010	0b0101000: I²C write/read address is 0x50/0x51 0b0101010: I²C write/read address is 0x54/0x55 .: . .: . 0b1111111: I²C write/read address is 0xFE/0xFF
CFG_BLOCK	0	Configuration block. When set, all registers become non-writable (read-only). This bit can be used to freeze the chip configuration.	0b0: Not Blocked 0b1: Blocked

REG1 (0x1)

BIT	7	6	5	4	3	2	1	0
Field	LVDS_HALFSW	–	IIC_2_EN	IIC_1_EN	TX_RATE[1:0]		RX_RATE[1:0]	
Reset	0x0	–	0x0	0x0	0x0		0x2	
Access Type	Write, Read	–	Write, Read	Write, Read	Write, Read		Write, Read	

BITFIELD	BITS	DESCRIPTION	DECODE
LVDS_HALFSW	7	LVDS driver amplitude setting	0x0: Full-swing amplitude (350mV typ.) 0x1: Half-swing amplitude (175mV typ.)
IIC_2_EN	5	Enables pass-through I²C Channel 2 (SDA2/RX2, SCL2/TX2)	0b0: Disable pass-through 0b1: Enable pass-through
IIC_1_EN	4	Enables pass-through I²C Channel 1 (SDA1/RX1, SCL1/TX1)	0b0: Disable pass-through 0b1: Enable pass-through
TX_RATE	3:2	Transmitter rate. When changed, becomes active after next link reset.	0b00: 187.5Mbps 0b01: 375Mbps 0b10: 750Mbps 0b11: 1.5Gbps
RX_RATE	1:0	Receiver rate. When changed, becomes active after next link reset. Default value is set by CXTP pin at power up: 6Gbps when CXTP = 1 (coaxial cable) and 3Gbps when CXTP = 0 (twisted-pair cable).	0b00: 1.5Gbps 0b01: 3Gbps 0b10: 6Gbps 0b11: Reserved

REG2 (0x2)

BIT	7	6	5	4	3	2	1	0
Field	LOCK_CFG	VID_EN	DIS_LOCAL_CC	DIS_REM_CC	–	AUD_TX_EN	–	–
Reset	0b0	0b1	0b0	0b0	–	0b1	–	–
Access Type	Write, Read	Write, Read	Write, Read	Write, Read	–	Write, Read	–	–

BITFIELD	BITS	DESCRIPTION	DECODE
LOCK_CFG	7	Configures LOCK pin behavior	0b0: GMSL2 link locked 0b1: GMSL2 link locked and MIPI output started
VID_EN	6	Video receive pipeline enable	0b0: Video disabled 0b1: Video enabled

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BITFIELD	BITS	DESCRIPTION	DECODE
DIS_LOCAL_CC	5	Disables control channel connection to RX/SDA and TX/SCL pins	0b0: RX/SDA and TX/XCL connected to control channel 0b1: RX/SDA and TX/SCL disconnected from control channel
DIS_REM_CC	4	Disables remote control channel link over GMSL2 connection	0b0: Remote control channel enabled 0b1: Remote control channel disabled
AUD_TX_EN	2	Audio transmit enable over reverse channel	0b0: Audio transmit channel X disabled 0b1: Audio transmit channel X enabled

REG3 (0x3)

BIT	7	6	5	4	3	2	1	0
Field	–	–	–	I2CSEL	UART_2_EN	UART_1_EN	–	VIDEO_LOCK
Reset	–	–	–	0b0	0x0	0x0	–	0b0
Access Type	–	–	–	Write, Read	Write, Read	Write, Read	–	Read Only

BITFIELD	BITS	DESCRIPTION	DECODE
I2CSEL	4	I^2C/UART selection. This bit is set according to the latched I2CSEL pin value at power-up.	0b0: UART 0b1: I^2C
UART_2_EN	3	Enables pass-through UART Channel 2 (SDA2/RX2, SCL2/TX2)	0b0: Disable pass-through 0b1: Enable pass-through
UART_1_EN	2	Enables pass-through UART Channel 1 (SDA1/RX1, SCL1/TX1)	0b0: Disable pass-through 0b1: Enable pass-through
VIDEO_LOCK	0	Indicates whether video channel is locked and outputting valid video data	0b0: Video channel not locked 0b1: Video channel locked

REG13 (0xD)

BIT	7	6	5	4	3	2	1	0
Field	DEV_ID[7:0]							
Reset	0x82							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION	DECODE
DEV_ID	7:0	Device identifier	0x82: MAX96752 0x84: MAX96754

REG14 (0xE)

BIT	7	6	5	4	3	2	1	0
Field	–	–	–	–	DEV_REV[3:0]			
Reset	–	–	–	–	0x01			
Access Type	–	–	–	–	Read Only			

BITFIELD	BITS	DESCRIPTION	DECODE
DEV_REV	3:0	Device revision	0xX: Device revision number

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REG15 (0xF)

BIT	7	6	5	4	3	2	1	0
Field	–	–	SPEED_CPBL[1:0]		DV_CPBL	DUAL_CPBL	SPLTR_CPBL	HDCP_CPBL
Reset	–	–	0x0		0b0	0b0	0b0	0b0
Access Type	–	–	Read Only		Read Only	Read Only	Read Only	Read Only

BITFIELD	BITS	DESCRIPTION	DECODE
SPEED_CPBL	5:4	Video resolution capability	0b00: No PCLK frequency limit 0b01: PCLK is limited to 240MHz 0b10: PCLK is limited to 160MHz 0b11: PLCK is limited to 104MHz
DV_CPBL	3	Dual-view video capability	0b0: Dual-view video splitting is not available 0b1: Dual-view video splitting is available
DUAL_CPBL	2	Dual-link capability	0b0: Dual-link mode is not available 0b1: Dual-link mode is available
SPLTR_CPBL	1	Splitter mode capability	0b0: Splitter mode is not available 0b1: Splitter mode is available
HDCP_CPBL	0	HDCP capability	0b0: HDCP not enabled on this device 0b1: HDCP enabled on this device

IO_CHK0 (0x38)

BIT	7	6	5	4	3	2	1	0
Field	PIN_DRV_EN_0[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
PIN_DRV_EN_0	7:0	Enables individual pin output buffers and drives the value selected by PIN_DRV_SEL to that pin so that the driven value can be read from the PIN_IN_0 register. This allows testing of IO buffer input and output paths. When the pin is driven using this register, it must not be externally driven by any other strong driver.	0bXXXXXXX0: Disable RG_MDC output buffer 0bXXXXXXX1: Enable RG_MDC output buffer 0bXXXXXX0X: Disable RG_MDIO output buffer 0bXXXXXX1X: Enable RG_MDIO output buffer 0bXXXXX0XX: Disable RG_RXCTL output buffer 0bXXXXX1XX: Enable RG_RXCTL output buffer 0bXXXX0XXX: Disable RG_RD3 output buffer 0bXXXX1XXX: Enable RG_RD3 output buffer 0bXXX0XXXX: Disable RG-RD2 output buffer 0bXXX1XXXX: Enable RG-RD2 output buffer 0bXX0XXXXX: Disable RG_RD output buffer 0bXX1XXXXX: Enable RG_RD output buffer 0bX0XXXXXX: Disable RG_RD0 output buffer 0bX1XXXXXX: Enable RG_RD0 output buffer 0b0XXXXXXX: Disable RG_RXC output buffer 0b1XXXXXXX: Enable RG_RXC output buffer

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IO_CHK1 (0x39)

BIT	7	6	5	4	3	2	1	0
Field	PIN_DRV_EN_1[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
PIN_DRV_EN_1	7:0	Enables individual pin output buffers and drives the value selected by PIN_DRV_SEL to that pin so that the driven value can be read from the PIN_IN_1 register. This allows testing of IO buffer input and output paths. When the pin is driven using this register, it must not be externally driven by any other strong driver.	0bXXXXXXX0: Disable RG_REFCLK output buffer 0bXXXXXXX1: Enable RG_REFCLK output buffer 0bXXXXXX0X: Disable RG_TXCTL output buffer 0bXXXXXX1X: Enable RG_TXCTL output buffer 0bXXXXX0XX: Disable RG_TD3 output buffer 0bXXXXX1XX: Enable RG_TD3 output buffer 0bXXXX0XXX: Disable RG_TD2 output buffer 0bXXXX1XXX: Enable RG_TD2 output buffer 0bXXX0XXXX: Disable RG-TD1 output buffer 0bXXX1XXXX: Enable RG-TD1 output buffer 0bXX0XXXXX: Disable RG_TD0 output buffer 0bXX1XXXXX: Enable RG_TD0 output buffer 0bX0XXXXXX: Disable RG_TXC output buffer 0bX1XXXXXX: Enable RG_TXC output buffer 0b0XXXXXXX: Disable GMSL2 output buffer 0b1XXXXXXX: Enable GMSL2 output buffer

IO_CHK2 (0x3A)

BIT	7	6	5	4	3	2	1	0
Field	PIN_DRV_SEL	–	–	–	–	–	PIN_DRV_EN_2[1:0]	
Reset	0x0	–	–	–	–	–	0x0	
Access Type	Write, Read	–	–	–	–	–	Write, Read	

BITFIELD	BITS	DESCRIPTION	DECODE
PIN_DRV_SEL	7	Selects polarity for driving dedicated IOs. When PIN_DRV_EN = 0, this register is used to enable IO buffer receivers for DEDIO pins.	0b0: Drive low level 0b1: Drive high level
PIN_DRV_EN_2	1:0	Enables individual pin output buffers and drives the value selected by PIN_DRV_SEL to that pin so that the driven value can be read from the PIN_IN_2 register. This allows testing of IO buffer input and output paths. When the pin is driven using this register, it must not be externally driven by any other strong driver.	0bX0: Disable LOCK output buffer 0bX1: Enable LOCK output buffer 0b0X: Disable ERRB output buffer 0b1X: Enable ERRB output buffer

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IO_CHK3 (0x3B)

BIT	7	6	5	4	3	2	1	0
Field	PIN_IN_0[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION	DECODE
PIN_IN_0	7:0	Pin level readback. Each bit contains the level at the pin, either driven externally or by the PIN_DRV_SEL value if the equivalent bit is enabled in the PIN_DRV_EN_0 register.	[0]: RG_MDC [1]: RG_MDIO [2]: RG_RXCTL [3]: RG_RD3 [4]: RG-RD2 [5]: RG_RD1 [6]: RG_RD0 [7]: RG_RXC

IO_CHK4 (0x3C)

BIT	7	6	5	4	3	2	1	0
Field	PIN_IN_1[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION	DECODE
PIN_IN_1	7:0	Pin level readback. Each bit contains the level at the pin, either driven externally or by the PIN_DRV_SEL value if the equivalent bit is enabled in the PIN_DRV_EN_1 register.	[0]: REFCLK [1]: TXCTL [2]: TD3 [3]: TD2 [4]: TD1 [5]: TD0 [6]: TXC [7]: GMSL2

PWR1 (0x9)

BIT	7	6	5	4	3	2	1	0
Field	OVERTEMP	–	–	–	–	–	–	–
Reset	0d0	–	–	–	–	–	–	–
Access Type	Read Only	–	–	–	–	–	–	–

BITFIELD	BITS	DESCRIPTION	DECODE
OVERTEMP	7	Temperature monitor overtemp indicator. Trip temperature T_{TRIP} set by SET_TEMP1P0[1:0] in CMU6	0x0: $T < T_{TRIP}$ 0x1: $T > T_{TRIP}$

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PWR4 (0xC)

BIT	7	6	5	4	3	2	1	0
Field	–	DIS_LOCAL_WAKE	WAKE_EN_B	WAKE_EN_A	–	–	–	–
Reset	–	0	0x0	0x1	–	–	–	–
Access Type	–	Write, Read	Write, Read	Write, Read	–	–	–	–

BITFIELD	BITS	DESCRIPTION	DECODE
DIS_LOCAL_WAKE	6	Disables wake up by local uC from SDA_RX pin	0b0: Local wake up disabled 0b1: Local wake up enabled
WAKE_EN_B	5	Enables wake up by remote chip connected to Link B	0b0: Link B remote wake up disabled 0b1: Link B remote wake up enabled
WAKE_EN_A	4	Enables wake up by remote chip connected to Link A	0b0: Link A remote wake up disabled 0b1: Link A remote wake up enabled

CTRL0 (0x10)

BIT	7	6	5	4	3	2	1	0
Field	RESET_ALL	RESET_LINK	RESET_ONESHOT	AUTO_LINK	SLEEP	–	LINK_CFG[1:0]	
Reset	0b0	0b0	0b0	0b1	0b0	–	0b01	
Access Type	Write, Read	Write, Read	Write Clears All, Read	Write, Read	Write, Read	–	Write, Read	

BITFIELD	BITS	DESCRIPTION	DECODE
RESET_ALL	7	Device reset. Writing 1 to this bit resets the device, returning all blocks and registers to their default values. This is equivalent to toggling the PWDNB pin. The bit is cleared when written.	0b0: No action 0b1: Activate chip reset
RESET_LINK	6	Resets the whole data path (keeping register settings). Write 1 to activate reset, write 0 to release reset.	0b0: Release link reset 0b1: Activate link reset
RESET_ONESHOT	5	Resets the whole data path (keeps register settings) one shot. Write 1 to activate reset. The bit self clears and automatically releases reset.	0b0: No action 0b1: Reset data path
AUTO_LINK	4	Automatically selects which link to enable (A or B). Dual-link and splitter modes are not automatic. For dual-link mode, set LINK_CFG = 00. For splitter mode, set LINK_CFG = 11.	0b0: Disable auto link configuration 0b1: Enable auto link configuration
SLEEP	3	Activates sleep mode	0b0: Sleep mode disabled 0b1: Sleep mode enabled
LINK_CFG	1:0	AUTO_LINK and this bitfield select the link configuration per the decode table	0b00: If AUTO_LINK = 0, dual-link is selected. If AUTO_LINK = 1, link mode is automatically selected. 0b01: If AUTO_LINK = 0, Link A is selected. If AUTO_LINK = 1, link mode is automatically selected. 0b10: If AUTO_LINK = 0, Link B is selected. If

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BITFIELD	BITS	DESCRIPTION	DECODE
			AUTO_LINK = 1, link mode is automatically selected. 0b11: Splitter mode

CTRL1 (0x11)

BIT	7	6	5	4	3	2	1	0
Field	–	–	–	CXTP_B	BACK_COMP_SPLTR	–	–	CXTP_A
Reset	–	–	–	0b0	0x0	–	–	0b0
Access Type	–	–	–	Write, Read	Write, Read	–	–	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
CXTP_B	4	Coaxial/Twisted-pair cable select for Link B. Bit is set according to the latched CXTP pin value at power up.	0b0: Shielded twisted-pair drive 0b1: Coaxial drive
BACK_COMP_SPLTR	3	Enables splitter mode compatibility	0b0: Compatibility disabled 0b1: Compatibility enabled
CXTP_A	0	Coaxial/Twisted-pair cable select for Link A. Bit is set according to the latched CXTP pin value at power up.	0b0: Shielded twisted-pair drive 0b1: Coaxial drive

CTRL3 (0x13)

BIT	7	6	5	4	3	2	1	0
Field	–	–	LINK_MODE[1:0]		LOCKED	ERROR	CMU_LOCKED	–
Reset	–	–	0x1		0	0	0	–
Access Type	–	–	Read Only		Read Only	Read Only	Read Only	–

BITFIELD	BITS	DESCRIPTION	DECODE
LINK_MODE	5:4	Active link mode	0b00: Dual-link 0b01: Link A 0b10: Link B 0b11: Splitter mode
LOCKED	3	GMSL-2 link locked (bi-directional)	0b0: GMSL2 link not locked 0b1: GMSL2 link locked
ERROR	2	Reflects error status (inverse of ERRB pin value)	0b0: ERRB not asserted (ERRB pin = 1) 0b1: ERRB asserted (ERRB pin = 0)
CMU_LOCKED	1	CMU (clock multiplier unit) lock status	0b0: CMU not locked 0b1: CMU locked

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INTR0 (0x18)

BIT	7	6	5	4	3	2	1	0
Field	–	–	–	–	AUTO_ERR_RST_EN	DEC_ERR_THR[2:0]		
Reset	–	–	–	–	0	0x0		
Access Type	–	–	–	–	Write, Read	Write, Read		

BITFIELD	BITS	DESCRIPTION	DECODE
AUTO_ERR_RST_EN	3	Automatically resets DEC_ERR_A, DEC_ERR_B, and IDLE_ERR registers after ERRB pin is asserted for 1µs	0b0: Auto-reset disabled 0b1: Auto-reset enabled
DEC_ERR_THR	2:0	Decoding and idle error reporting threshold. DEC_ERR_FLAG_A is asserted when DEC_ERR_A ≥ DEC_ERR_THR. DEC_ERR_FLAG_B is asserted when DEC_ERR_B ≥ DEC_ERR_THR. IDLE_ERR_FLAG is asserted when IDLE_ERR ≥ DEC_ERR_THR.	0b000: 1 error 0b001: 2 errors 0b010: 4 errors 0b011: 8 errors 0b100: 16 errors 0b101: 32 errors 0b110: 64 errors 0b111: 128 errors

INTR1 (0x19)

BIT	7	6	5	4	3	2	1	0
Field	PKT_CNT_EXP[3:0]				AUTO_CNT_RST_EN	PKT_CNT_THR[2:0]		
Reset	0x0				0	0x0		
Access Type	Write, Read				Write, Read	Write, Read		

BITFIELD	BITS	DESCRIPTION	DECODE
PKT_CNT_EXP	7:4	Packet count multiplier exponent. See the description of PKT_CNT register.	0bXXX: PKT_CNT exponent
AUTO_CNT_RST_EN	3	Automatically resets PKT_CNT register after ERRB pin is asserted for 1µs	0b0: Auto-reset disabled 0b1: Auto-reset enabled
PKT_CNT_THR	2:0	Packet count reporting threshold. (See PKT_CNT description.) PKT_CNT_FLAG is asserted when PKT_CNT ≥ PKT_CNT_THR.	0b000: 1 0b001: 2 0b010: 4 0b011: 8 0b100: 16 0b101: 32 0b110: 64 0b111: 128

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INTR2 (0x1A)

BIT	7	6	5	4	3	2	1	0
Field	PHY_INT_OEN_B	PHY_INT_OEN_A	REM_ERR_OEN	–	–	IDLE_ERR_OEN	DEC_ERR_OEN_B	DEC_ERR_OEN_A
Reset	0	0	0	–	–	0	1	1
Access Type	Write, Read	Write, Read	Write, Read	–	–	Write, Read	Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
PHY_INT_OEN_B	7	Enables reporting of PHY interrupt (PHY_INT_B) at ERRB pin	0b0: Reporting disabled 0b1: Reporting enabled
PHY_INT_OEN_A	6	Enables reporting of PHY interrupt (PHY_INT_A) at ERRB pin	0b0: Reporting disabled 0b1: Reporting enabled
REM_ERR_OEN	5	Enables reporting of remote error status (REM_ERR) at ERRB pin	0b0: Reporting disabled 0b1: Reporting enabled
IDLE_ERR_OEN	2	Enables reporting of idle word errors (IDLE_ERR_FLAG) at ERRB pin	0b0: Reporting disabled 0b1: Reporting enabled
DEC_ERR_OEN_B	1	Enables reporting of decoding errors (DEC_ERR_FLAG_B) at ERRB pin	0b0: Reporting disabled 0b1: Reporting enabled
DEC_ERR_OEN_A	0	Enables reporting of decoding errors (DEC_ERR_FLAG_A) at ERRB pin	0b0: Reporting disabled 0b1: Reporting enabled

INTR3 (0x1B)

BIT	7	6	5	4	3	2	1	0
Field	PHY_INT_B	PHY_INT_A	REM_ERR_FLAG	–	–	IDLE_ERR_FLAG	DEC_ERR_FLAG_B	DEC_ERR_FLAG_A
Reset	0	0	0	–	–	0	0	0
Access Type	Read Only	Read Only	Read Only	–	–	Read Only	Read Only	Read Only

BITFIELD	BITS	DESCRIPTION	DECODE
PHY_INT_B	7	PHY interrupt flag—Link B. See RLMS_B block (0x500) registers RLMS4C (0x54C), RLMS4D (0x54D), RLMS4E (0x54E) and RLMS4F (0x54F).	0b0: No Link B PHY interrupt 0b1: Link B PHY interrupt
PHY_INT_A	6	PHY interrupt flag—Link A. See RLMS_A block (0x400) registers RLMS4C (0x44C), RLMS4D (0x44D), RLMS4E (0x44E) and RLMS4F (0x44F).	0b0: No Link A PHY interrupt 0b1: Link A PHY interrupt
REM_ERR_FLAG	5	Received remote side error status (inverse of remote side ERRB pin level)	0b0: No remote side error 0b1: Remote side error
IDLE_ERR_FLAG	2	Idle word error flag. Asserted when IDLE_ERR ≥ DEC_ERR_THR.	0b0: Flag not asserted 0b1: Flag asserted
DEC_ERR_FLAG_B	1	Decoding error flag for Link B. Asserted when DEC_ERR_B ≥ DEC_ERR_THR.	0b0: Flag not asserted 0b1: Flag asserted
DEC_ERR_FLAG_A	0	Decoding error flag for Link. Asserted when DEC_ERR_A ≥ DEC_ERR_THR.	0b0: Flag not asserted 0b1: Flag asserted

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INTR4 (0x1C)

BIT	7	6	5	4	3	2	1	0
Field	–	–	–	HDCP_INT_OEN	MAX_RT_OEN	RT_CNT_OEN	PKT_CNT_OEN	WM_ERR_OEN
Reset	–	–	–	0	1	0	0	1
Access Type	–	–	–	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
HDCP_INT_OEN	4	Enables reporting of GMSL HDCP interrupt (HDCP_INT_FLAG) at ERRB pin	0b0: Reporting disabled 0b1: Reporting enabled
MAX_RT_OEN	3	Enables reporting of combined ARQ maximum re-transmission limit error flag (MAX_RT_FLAG) at ERRB pin	0b0: Reporting disabled 0b1: Reporting enabled
RT_CNT_OEN	2	Enables reporting of combined ARQ re-transmission event flag (RT_CNT_FLAG) at ERRB pin	0b0: Reporting disabled 0b1: Reporting enabled
PKT_CNT_OEN	1	Enables reporting of packet count flag (PKT_CNT_FLAG) at ERRB pin	0b0: Reporting disabled 0b1: Reporting enabled
WM_ERR_OEN	0	Enables reporting of watermark errors (WM_ERR_FLAG) at ERRB pin	0b0: Reporting disabled 0b1: Reporting enabled

INTR5 (0x1D)

BIT	7	6	5	4	3	2	1	0
Field	–	–	–	HDCP_INT_FLAG	MAX_RT_FLAG	RT_CNT_FLAG	PKT_CNT_FLAG	WM_ERR_FLAG
Reset	–	–	–	0	0	0	0	0
Access Type	–	–	–	Read Only	Read Only	Read Only	Read Only	Read Clears All

BITFIELD	BITS	DESCRIPTION	DECODE
HDCP_INT_FLAG	4	HDCP Interrupt Status. Asserted when there is a GMSL HDCP interrupt.	0b0: No HDCP interrupt 0b1: HDCP interrupt
MAX_RT_FLAG	3	Combined ARQ maximum re-transmission limit error flag. Asserted when any of the selected channels' ARQ re-transmission limit is reached. Selection is done by each channel's MAX_RT_ERR_OEN register bit.	0b0: Flag not asserted 0b1: Flag asserted
RT_CNT_FLAG	2	Combined ARQ re-transmission event flag. Asserted when any of the selected channels have done at least one ARQ re-transmission. Selection is done by each channel's RT_CNT_OEN register bit.	0b0: Flag not asserted 0b1: Flag asserted
PKT_CNT_FLAG	1	Packet count flag. Asserted when PKT_CNT ≥ PKT_CNT_THR.	0b0: Flag not asserted 0b1: Flag asserted
WM_ERR_FLAG	0	Watermark error flag. Asserted when a watermark error is detected.	0b0: Flag not asserted 0b1: Flag asserted

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INTR6 (0x1E)

BIT	7	6	5	4	3	2	1	0
Field	VDDCMP_INT_OEN	PORZ_INT_OEN	VDDBAD_INT_OEN	VDD_OV_OEN	LCRC_ERR_OEN	VPRBS_ERR_OEN	APRBS_ERR_OEN	VID_PXL_CRC_ERR_OEN
Reset	0b0	0b0	0b0	0x0	1	1	1	0x0
Access Type	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
VDDCMP_INT_OEN	7	Enables reporting of VDDCMP interrupt (VDDCMP_INT_FLAG) at ERRB pin	0b0: Reporting disabled 0b1: Reporting enabled
PORZ_INT_OEN	6	Enables reporting of PORZ interrupt (PORZ_INT_FLAG) at ERRB pin	0b0: Reporting disabled 0b1: Reporting enabled
VDDBAD_INT_OEN	5	Enables reporting of VDDBAD interrupt (VDDBAD_INT_FLAG) at ERRB pin	0b0: Reporting disabled 0b1: Reporting enabled
VDD_OV_OEN	4	Enables VDD overvoltage status on ERRB	0b0: Do not enable VDD overvoltage status 0b1: Enable VDD over voltage status
LCRC_ERR_OEN	3	Enables reporting of video line CRC errors (LCRC_ERR_FLAG) at ERRB pin	0b0: Reporting disabled 0b1: Reporting enabled
VPRBS_ERR_OEN	2	Enables reporting of video PRBS errors (VPRBS_ERR_FLAG) at ERRB pin	0b0: Reporting disabled 0b1: Reporting enabled
APRBS_ERR_OEN	1	Enables reporting of audio PRBS errors (APRBS_ERR_FLAG) at ERRB pin	0b0: Reporting disabled 0b1: Reporting enabled
VID_PXL_CRC_ERR_OEN	0	Enables reporting of video pixel CRC errors (VID_PXL_CRC_ERR_FLAG) at ERRB pin	0b0: Reporting disabled 0b1: Reporting enabled

INTR7 (0x1F)

BIT	7	6	5	4	3	2	1	0
Field	VDDCMP_INT_FLAG	PORZ_INT_FLAG	VDDBAD_INT_FLAG	VDD_OV_FLAG	LCRC_ERR_FLAG	VPRBS_ERR_FLAG	APRBS_ERR_FLAG	VID_PXL_CRC_ERR_FLAG
Reset	0	0	0	0x0	0	0	0	0x0
Access Type	Read Clears All	Read Clears All	Read Clears All	Read Clears All	Read Only	Read Only	Read Only	Read Only

BITFIELD	BITS	DESCRIPTION	DECODE
VDDCMP_INT_FLAG	7	VDDCMP interrupt flag	0b0: Flag not asserted 0b1: Flag asserted
PORZ_INT_FLAG	6	PORZ interrupt flag	0b0: Flag not asserted 0b1: Flag asserted
VDDBAD_INT_FLAG	5	VDD STATUS interrupt	0b0: Flag not asserted 0b1: Flag asserted
VDD_OV_FLAG	4	VDD overvoltage indication. This bit stays high when asserted and must be cleared by the user. It is set when VDD exceeds the overvoltage threshold. It is cleared when read. See OV_LEVEL register for over-voltage	0x0: VDD overvoltage not detected 0x1: VDD overvoltage detected

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BITFIELD	BITS	DESCRIPTION	DECODE
		threshold value.	
LCRC_ERR_FLAG	3	Video line CRC error flag. Asserted when a video line CRC error is detected.	0b0: Flag not asserted 0b1: Flag asserted
VPRBS_ERR_FLAG	2	Video PRBS error flag. Asserted when VPRBS_ERR > 0.	0b0: Flag not asserted 0b1: Flag asserted
APRBS_ERR_FLAG	1	Audio PRBS error flag. Asserted when APRBS_ERR > 0.	0b0: Flag not asserted 0b1: Flag asserted
VID_PXL_CRC_ERR_FLAG	0	Video pixel CRC error flag. Asserted when video pixel CRC error count > 0.	0b0: Flag not asserted 0b1: Flag asserted

INTR8 (0x20)

BIT	7	6	5	4	3	2	1	0
Field	ERR_TX_EN	–	–	ERR_TX_ID[4:0]				
Reset	0	–	–	0x1F				
Access Type	Write, Read	–	–	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
ERR_TX_EN	7	Transmit local error status (inverse of ERRB pin level) to remote side through GPIO channel	0b0: Transmit error status disabled 0b1: Transmit error status enabled
ERR_TX_ID	4:0	GPIO ID used for transmitting ERR_TX	0bXXXXX: Value of GPIO ID for transmitting ERR_TX

INTR9 (0x21)

BIT	7	6	5	4	3	2	1	0
Field	ERR_RX_EN	–	–	ERR_RX_ID[4:0]				
Reset	0	–	–	0x1F				
Access Type	Write, Read	–	–	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
ERR_RX_EN	7	Receive remote error status (inverse of ERRB pin level) through GPIO channel	0b0: Receive error status disabled 0b1: Receive error status enabled
ERR_RX_ID	4:0	GPIO ID used for receiving ERR_RX	0bXXXXX: Value of GPIO ID for receiving ERR_TX

CNT0 (0x22)

BIT	7	6	5	4	3	2	1	0
Field	DEC_ERR_A[7:0]							
Reset	0x00							
Access Type	Read Clears All							

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BITFIELD	BITS	DESCRIPTION	DECODE
DEC_ERR_A	7:0	Number of decoding (disparity) errors detected at Link A. Reset after reading or with the rising edge of LOCK.	0xXX: Number of Link A decoding errors detected

CNT1 (0x23)

BIT	7	6	5	4	3	2	1	0
Field	DEC_ERR_B[7:0]							
Reset	0x00							
Access Type	Read Clears All							

BITFIELD	BITS	DESCRIPTION	DECODE
DEC_ERR_B	7:0	Number of decoding (disparity) errors detected at Link B. Reset after reading or with the rising edge of LOCK.	0xXX: Number of Link B decoding errors detected

CNT2 (0x24)

BIT	7	6	5	4	3	2	1	0
Field	IDLE_ERR[7:0]							
Reset	0x00							
Access Type	Read Clears All							

BITFIELD	BITS	DESCRIPTION	DECODE
IDLE_ERR	7:0	Number of idle word errors detected. Reset after reading or with the rising edge of LOCK.	0xXX: Number of idle word errors detected

CNT3 (0x25)

BIT	7	6	5	4	3	2	1	0
Field	PKT_CNT[7:0]							
Reset	0x00							
Access Type	Read Clears All							

BITFIELD	BITS	DESCRIPTION	DECODE
PKT_CNT	7:0	Number of received packets of a selected type. Packet type is selected with PKT_CNT_SEL register. Reported packet count is a scaled value, such that actual packet count is ≥ PKT_CNT x ($2^{PKT_CNT_EXP}$) and < (PKT_CNT + 1) x ($2^{PKT_CNT_EXP}$). When maximum value is reported, packet count is greater or equal to the reported value.	0xXX: Scaled number of received packets

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CNT4 (0x26)

BIT	7	6	5	4	3	2	1	0
Field	VID_PXL_CRC_ERR[7:0]							
Reset	0x00							
Access Type	Read Clears All							

BITFIELD	BITS	DESCRIPTION	DECODE
VID_PXL_CRC_ERR	7:0	Number of video pixel CRC errors detected at video stream. Reset after each Read operation or with the rising edge of LOCK.	0bXXXXXXXX: Number of video pixel CRC errors detected

TX1 (0x29)

BIT	7	6	5	4	3	2	1	0
Field	LINK_PRBS_GEN	–	ERRG_EN_B	ERRG_EN_A	–	–	–	–
Reset	0	–	0	0	–	–	–	–
Access Type	Write, Read	–	Write, Read	Write, Read	–	–	–	–

BITFIELD	BITS	DESCRIPTION	DECODE
LINK_PRBS_GEN	7	Enables link PRBS-7 generator	0b0: Link PRBS generator disabled 0b1: Link PRBS generator enabled
ERRG_EN_B	5	Error generator enable for Link B	0b0: Link B error generator disabled 0b1: Link B error generator enabled
ERRG_EN_A	4	Error generator enable for Link A	0b0: Link A error generator disabled 0b1: Link A error generator enabled

TX2 (0x2A)

BIT	7	6	5	4	3	2	1	0
Field	ERRG_CNT[1:0]		ERRG_RATE[1:0]		ERRG_BURST[2:0]			ERRG_PER
Reset	0x0		0x2		0x0			0
Access Type	Write, Read		Write, Read		Write, Read			Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
ERRG_CNT	7:6	Number of errors to be generated	0b00: Continuous 0b01: 16 errors 0b10: 128 errors 0b11: 1024 errors
ERRG_RATE	5:4	Error generator average bit-error rate	0b00: 1 in 5120 bits 0b01: 1 in 81920 bits 0b10: 1 in 1310720 bits 0b11: 1 in 20971520 bits
ERRG_BURST	3:1	Selects the error generator burst-error length in bits. For example, if the bitfield value is 0b100, the burst-error length is 8 bits. Thus, each time	0b000: 1 bit 0b001: 2 bits 0b010: 3 bits 0b011: 4 bits

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BITFIELD	BITS	DESCRIPTION	DECODE
		an 8-bit error burst is generated, there is an error in the first and the last bit of the 8-bit window and the probability of an error in each of the remaining 6 bits is 50%.	0b100: 8 bits 0b101: 12 bits 0b110: 16 bits 0b111: 20 bits
ERRG_PER	0	Error generator error distribution selection	0b0: Pseudo random 0b1: Periodic

TX3 (0x2B)

BIT	7	6	5	4	3	2	1	0
Field	–	–	–	–	–	TIMEOUT[2:0]		
Reset	–	–	–	–	–	0b100		
Access Type	–	–	–	–	–	Write, Read		

BITFIELD	BITS	DESCRIPTION	DECODE
TIMEOUT	2:0	ARQ timeout duration multiplier. Multiplies a timeout base constant to set the ARQ timeout. The timeout base is set by the reverse channel link rate (RX_RATE) as follows: RX_RATE Timeout base 187.5 Mbps 8µs 350 Mbps 4µs 750 Mbps 3µs 1.5 Gbps 2µs	0b000: 0.5 x timeout base 0b001: 1.0 x timeout base 0b010: 1.5 x timeout base 0b011: 2.0 x timeout base 0b100: 2.5 x timeout base 0b101: 3.0 x timeout base 0b110: 3.5 x timeout base 0b111: 4.0 x timeout base

RX0 (0x2C)

BIT	7	6	5	4	3	2	1	0
Field	PKT_CNT_LBW[1:0]		–	–	PKT_CNT_SEL[3:0]			
Reset	0x0		–	–	0x0			
Access Type	Write, Read		–	–	Write, Read			

BITFIELD	BITS	DESCRIPTION	DECODE
PKT_CNT_LBW	7:6	Select the subtype of low bandwidth packets to count at PKT_CNT register	0b00: Count LBW data packets only 0b01: Count LBW acknowledge packets only 0b10: Count LBW data and acknowledge packets 0b11: Reserved
PKT_CNT_SEL	3:0	Select the type of received packets to count at PKT_CNT register	0x0: None 0x1: VIDEO 0x2: AUDIO 0x3: INFO Frame 0x4: SPI 0x5: I^2C 0x6: UART 0x7: GPIO 0x8: AHDCP 0x9: RGMII 0xA: Reserved

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BITFIELD	BITS	DESCRIPTION	DECODE
			0xB: Reserved 0xC: Reserved 0xD: Reserved 0xE: All 0xF: Unknown and packets with error

GPIOA (0x30)

BIT	7	6	5	4	3	2	1	0
Field	GPIO_RX_FAST_BIDIR_EN	GPIO_TX_CASC	GPIO_FWD_CDLY[5:0]					
Reset	0x0	0b1	0x01					
Access Type	Write, Read	Write, Read	Write, Read					

BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_RX_FAST_BIDIR_EN	7	GPIO fast direction switch for bi-directional IO	0b0: Fast direction switch disabled 0b1: Fast direction switch enabled
GPIO_TX_CASC	6	Allow multiple pin transitions to be transmitted in the same packet	0b0: Multiple pin transitions not allowed 0b1: Multiple pin transitions allowed
GPIO_FWD_CDLY	5:0	Compensation delay multiplier for the forward direction. This must be same value as GPIO_FWD_CDLY of the chip on the other side of the link. Total delay is the (value + 1) multiplied by 1.7µs. Default delay is 3.4µs.	0bXXXXXX: Forward compensation delay multiplier value

GPIOB (0x31)

BIT	7	6	5	4	3	2	1	0
Field	GPIO_TX_WNDW[1:0]		GPIO_REV_CDLY[5:0]					
Reset	0x2		0x08					
Access Type	Write, Read		Write, Read					

BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_TX_WNDW	7:6	Wait time after a GPIO transition to create a packet. This allows grouping transitions of different GPIO inputs in a single packet, increasing GPIO bandwidth usage efficiency.	0b00: Disabled 0b01: 200ns 0b10: 500ns 0b11: 1000ns
GPIO_REV_CDLY	5:0	Compensation delay multiplier for the reverse direction. This must be the same value as GPIO_REV_CDLY of the chip on the other side of the link. Total delay is the (value + 1) multiplied by 1.7µs. Default delay is 15.3µs.	0bXXXXXX: Reverse compensation delay multiplier value

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I2C_0 (0x40)

BIT	7	6	5	4	3	2	1	0
Field	–	–	SLV_SH[1:0]		–	SLV_TO[2:0]		
Reset	–	–	0b10		–	0b110		
Access Type	–	–	Write, Read		–	Write, Read		

BITFIELD	BITS	DESCRIPTION	DECODE
SLV_SH	5:4	I²C-to-I²C master bit rate setting. Configures the I²C bit rate used by the internal I²C master (in the device on the remote side from the external I²C master). Set this according to the I²C speed mode.	0b00: Set for I²C Fast-mode Plus speed 0b01: Set for I²C Fast-mode speed 0b10: Set for I²C standard-mode speed 0b11: Reserved
SLV_TO	2:0	I²C-to-I²C slave timeout setting. Internal GMSL2 I²C slave times out after the configured duration if it does not receive any response while waiting for a packet from remote device.	0b000: 16µs 0b001: 1ms 0b010: 2ms 0b011: 4ms 0b100: 8ms 0b101: 16ms 0b110: 32ms 0b111: Disabled

I2C_1 (0x41)

BIT	7	6	5	4	3	2	1	0
Field	–	MST_BT[2:0]			–	MST_TO[2:0]		
Reset	–	0x5			–	0b110		
Access Type	–	Write, Read			–	Write, Read		

BITFIELD	BITS	DESCRIPTION	DECODE
MST_BT	6:4	I²C-to-I²C master bit rate setting. Configures the I²C bit rate used by the internal I²C master (in the device on the remote side from the external I²C master). Set this according to the I²C speed mode.	0b000: 9.92Kbps - Set for I²C standard mode speed 0b001: 33.2Kbps - Set for I²C standard mode speed 0b010: 99.2Kbps - Set for I²C standard or Fast-mode speed 0b011: 123Kbps - Set for I²C Fast-mode speed 0b100: 203Kbps - Set for I²C Fast-mode speed 0b101: 397Kbps - Set for I²C Fast or Fast-mode Plus speed 0b110: 625Kbps - Set for I²C Fast-mode Plus speed 0b111: 980Kbps - Set for I²C Fast-mode Plus speed
MST_TO	2:0	I²C-to-I²C master timeout setting. Internal GMSL2 I²C master times out after the configured duration if it does not receive any response while waiting for a packet from remote device.	0b000: 16µs 0b001: 1ms 0b010: 2ms 0b011: 4ms 0b100: 8ms 0b101: 16ms 0b110: 32ms 0b111: Disabled

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I2C_2 (0x42)

BIT	7	6	5	4	3	2	1	0
Field	SRC_A[6:0]							–
Reset	0x0							–
Access Type	Write, Read							–

BITFIELD	BITS	DESCRIPTION	DECODE
SRC_A	7:1	I²C address translator source A. When I²C device address matches I2C_SRC_A, internal I²C master (on remote side) replaces the device address by I2C_DST_A.	0bXXXXXXX: Value of I²C SRC_A

I2C_3 (0x43)

BIT	7	6	5	4	3	2	1	0
Field	DST_A[6:0]							–
Reset	0x0							–
Access Type	Write, Read							–

BITFIELD	BITS	DESCRIPTION	DECODE
DST_A	7:1	I²C address translator destination A. See the description of I2C_SRC_A.	0bXXXXXXX: Value of I²C DST_A

I2C_4 (0x44)

BIT	7	6	5	4	3	2	1	0
Field	SRC_B[6:0]							–
Reset	0x0							–
Access Type	Write, Read							–

BITFIELD	BITS	DESCRIPTION	DECODE
SRC_B	7:1	I²C address translator source B. When I²C device address matches I2C_SRC_B, internal I²C master (on remote side) replaces the device address by I2C_DST_B.	0bXXXXXXX: Value of I²C SRC_B

I2C_5 (0x45)

BIT	7	6	5	4	3	2	1	0
Field	DST_B[6:0]							–
Reset	0x0							–
Access Type	Write, Read							–

BITFIELD	BITS	DESCRIPTION	DECODE
DST_B	7:1	I²C address translator destination B. See the description of I2C_SRC_B.	0bXXXXXXX: Value of I²C DST_B

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I2C_6 (0x46)

BIT	7	6	5	4	3	2	1	0
Field	–	–	–	–	I2C_AUTO_CFG	I2C_SRC_CNT[2:0]		
Reset	–	–	–	–	0x1	0b0		
Access Type	–	–	–	–	Write, Read	Write, Read		

BITFIELD	BITS	DESCRIPTION	DECODE
I2C_AUTO_CFG	3	When this bit = 1, I²C-to-I²C number of links is determined based on splitter mode automatically. In splitter mode, a response from two I²C channels is expected. Otherwise, a response from one I²C channel is expected.	0b0: Number of I²C-to-I²C links set by I2C_SRC_CNT[2:0] bits 0b1: Splitter mode automatically determines the number of I²C-to-I²C links
I2C_SRC_CNT	2:0	I²C-to-I²C number of links (valid when I2C_AUTO_SRC = 0). Set this to N - 1 when expecting I²C response from N remote I²C transmitters (usually the same as the number of remote devices connected to this device).	0b000: One deserializer connected 0b001: Two deserializers connected 0b010: Reserved 0b011: Reserved 0b100: Reserved 0b101: Reserved 0b110: Reserved 0b111: Reserved

I2C_7 (0x47)

BIT	7	6	5	4	3	2	1	0
Field	UART_RX_OVERFLOW	UART_TX_OVERFLOW	–	–	–	I2C_TIMED_OUT	REM_ACK_ACKED	REM_ACK_RECVED
Reset	0x0	0x0	–	–	–	0b0	0b0	0b0
Access Type	Read Clears All	Read Clears All	–	–	–	Read Only	Read Only	Read Only

BITFIELD	BITS	DESCRIPTION	DECODE
UART_RX_OVERFLOW	7	UART Rx FIFO overflow	0x0: No overflow occurred 0x1: Overflow occurred
UART_TX_OVERFLOW	6	UART Tx FIFO overflow	0x0: No overflow occurred 0x1: Overflow occurred
I2C_TIMED_OUT	2	Indicates whether internal I²C-to-I²C slave or master has timed out while receiving packet from remote device	0x0: Timeout has not occurred 0x1: Timeout has occurred
REM_ACK_ACKED	1	Inverse of the I²C acknowledge bit received from remote side	0x0: I²C acknowledge bit received as 1 0x1: I²C acknowledge bit received as 0
REM_ACK_RECVED	0	I²C acknowledge bit for any I²C byte is received from remote side for the previous I²C packet	0b0: I2C acknowledge bit not received 0b1: I²C acknowledge bit received

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UART_0 (0x48)

BIT	7	6	5	4	3	2	1	0
Field	ARB_TO_LEN[1:0]		REM_MS_EN	LOC_MS_EN	BYPASS_DIS_PAR	BYPASS_TO[1:0]		BYPASS_EN
Reset	0x1		0b0	0b0	0b0	0x1		0b0
Access Type	Write, Read		Write, Read	Write, Read	Write, Read	Write, Read		Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
ARB_TO_LEN	7:6	UART Rx source arbitration timeout duration. UART Rx processes packets from a single UART source at any time. When UART Rx does not receive any UART packets for this duration, it selects the next UART source according to the source ID of the next following received packet.	0b00: 1ms 0b01: 2ms 0b10: 8ms 0b11: 32ms
REM_MS_EN	5	Enables UART bypass mode control by remote GPIO pin. When set, remote chip's GPIO is used as MS pin (UART mode select). When MS is high, chip is in bypass mode. Otherwise, chip is in base mode.	0b0: UART bypass mode not controlled by remote MS pin 0b1: UART bypass mode controlled by remote MS pin
LOC_MS_EN	4	Enables UART bypass mode control by local GPIO pin. Set to use GPIO2 pin as MS pin (UART mode select). When MS is high, chip is in bypass mode. Otherwise,chip is in base mode.	0b0: UART bypass mode not controlled by local MS pin 0b1: UART bypass mode controlled by local MS pin
BYPASS_DIS_PAR	3	Selects whether to receive and send parity bit in bypass mode	0b0: Receive and send parity bit in bypass mode 0b1: Do not receive and send parity bit in bypass mode
BYPASS_TO	2:1	UART soft bypass timeout duration. When set to 11, BYPASS_EN is never cleared, so the device stays in bypass mode until next power-down.	0b00: 2ms 0b01: 8ms 0b10: 32ms 0b11: Disabled
BYPASS_EN	0	Enables UART soft bypass mode. Bypass mode remains active as long as there is UART activity. When there is no UART activity for the selected duration (configured by BYPASS_TO register), the device exits bypass mode and the bit is automatically cleared.	0b0: UART soft bypass mode disabled 0b1: UART soft bypass mode enabled

UART_1 (0x49)

BIT	7	6	5	4	3	2	1	0
Field	BITLEN_LSB[7:0]							
Reset	0x96							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION	DECODE
BITLEN_LSB	7:0	UART detected bit length in terms of internal 150MHz clock (LSB)	0xXX: UART detected bit length (LSB)

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UART_2 (0x4A)

BIT	7	6	5	4	3	2	1	0
Field	OUT_DELAY[1:0]		BITLEN_MSB[5:0]					
Reset	0x2		0x00					
Access Type	Write, Read		Read Only					

BITFIELD	BITS	DESCRIPTION	DECODE
OUT_DELAY	7:6	UART initial output delay. In base mode, the first received UART byte of a packet (sync or acknowledge frame) is delayed by the configured number of bit times in order to output the UART frames of the same packet back-to-back on remote side.	0b00: 0 bits 0b01: 4 bits 0b10: 8 bits 0b11: 1 bit
BITLEN_MSB	5:0	UART detected bit length in terms of internal 150MHz clock (6 MSbs)	0bXXXXXX: UART detected bit length (6 MSbs)

I2C_PT_0 (0x4C)

BIT	7	6	5	4	3	2	1	0
Field	–	–	SLV_SH_PT[1:0]		–	SLV_TO_PT[2:0]		
Reset	–	–	0x2		–	0x6		
Access Type	–	–	Write, Read		–	Write, Read		

BITFIELD	BITS	DESCRIPTION	DECODE
SLV_SH_PT	5:4	Pass-through I²C-to-I²C slave setup and hold time setting (setup, hold). Configures the interval between SDA and SCL transitions when driven by the internal I²C slave. Set according to the I²C speed mode shown in the decode table.	0b00: Fast-mode Plus 0b01: Fast-mode 0b10: Standard-mode 0b11: NA
SLV_TO_PT	2:0	Pass-through I²C-to-I²C slave timeout setting. Internal GMSL2 I²C slave times out after the configured duration if it does not receive any response while waiting for a packet from the remote device.	0bXXX: Slave timeout duration

I2C_PT_1 (0x4D)

BIT	7	6	5	4	3	2	1	0
Field	MST_DBL_PT	MST_BT_PT[2:0]			–	MST_TO_PT[2:0]		
Reset	0x0	0x5			–	0x6		
Access Type	Write, Read	Write, Read			–	Write, Read		

BITFIELD	BITS	DESCRIPTION	DECODE
MST_DBL_PT	7	Doubles the pass-through I²C-to-I²C master bit rate (MST_BT)	0x0: Do not double the pass-through I²C-to-I²C master bit rate 0x1: Double the pass-through I²C-to-I²C master bit

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BITFIELD	BITS	DESCRIPTION	DECODE
			rate
MST_BT_PT	6:4	Pass-through I²C-to-I²C master bit rate setting. Configures the I²C bit rate used by the internal I²C master (in the device on the remote side from the external I²C master). Set according to the I²C speed mode shown in the decode table. Fast-mode Plus = 101 to 111 Fast-mode = 010 to 100 Standard mode = 000 to 001	0b00x: Standard mode 0b01x: Fast-mode 0b100: Fast-mode 0b101: Fast-mode Plus 0b11x: Fast-mode Plus
MST_TO_PT	2:0	Pass-through I²C-to-I²C master timeout setting. Internal GMSL2 I²C master times out after the configured duration if it does not receive any response while waiting for a packet from the remote device.	

I2C_PT_2 (0x4E)

BIT	7	6	5	4	3	2	1	0
Field	XOVER_EN_2	I2C_TIMED_OUT_2	REM_ACK_ACKED_2	REM_ACK_RECVED_2	XOVER_EN_1	I2C_TIMED_OUT_1	REM_ACK_ACKED_1	REM_ACK_RECVED_1
Reset	0x0	0x0	0x0	0x0	0x0	0x0	0x0	0x0
Access Type	Write, Read	Read Only	Read Only	Read Only	Write, Read	Read Only	Read Only	Read Only

BITFIELD	BITS	DESCRIPTION	DECODE
XOVER_EN_2	7	Connect pass-through I²C/UART Channel 2 to main control channel on remote side	0x0: Do not connect pass-through I²C/UART Channel 2 to main control channel 0x1: Connect pass-through I²C/UART Channel 2 to main control channel
I2C_TIMED_OUT_2	6	Indicates that, in pass-through I²C Channel 2, internal I²C-to-I²C slave or master has timed out while receiving packet from remote device	0x0: Timeout has not occurred 0x1: Timeout has occurred
REM_ACK_ACKED_2	5	Indicates that, in pass-through I²C Channel 2, inverse of the I²C acknowledge bit was received from remote side	0x0: I²C acknowledge bit received as 1 0x1: Inverse I²C acknowledge bit received as 0
REM_ACK_RECVED_2	4	Indicates that, in pass-through I²C Channel 2, I²C acknowledge bit for any I²C byte was received from remote side for the previous I²C packet	0b0: I²C acknowledge bit not received 0b1: I²C acknowledge bit received
XOVER_EN_1	3	Connects pass-through I²C/UART Channel 1 to main control channel on remote side	0x0: Do not connect pass-through I²C/UART Channel 1 to main control channel 0x1: Connect pass-through I²C/UART Channel 1 to main control channel
I2C_TIMED_OUT_1	2	Indicates that, in pass-through I²C Channel 1, internal I²C-to-I²C slave or master has timed out while receiving packet from remote device	0x0: Timeout has not occurred 0x1: Timeout has occurred
REM_ACK_ACKED_1	1	Indicates that, in pass-through I²C channel 1, inverse of the I²C acknowledge bit was	0x0: I²C acknowledge bit received as 1 0x1: I²C acknowledge bit received as 0

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BITFIELD	BITS	DESCRIPTION	DECODE
		received from remote side	
REM_ACK_RECVED_1	0	Indicates that, in pass-through I²C channel 1, I²C acknowledge bit for any I²C byte was received from remote side for the previous I²C packet	0b0: I²C acknowledge bit not received 0b1: I²C acknowledge bit received

UART_PT_0 (0x4F)

BIT	7	6	5	4	3	2	1	0
Field	BITLEN_MAN_CFG_2	DIS_PAR_2	UART_RX_OVERFLOW_2	UART_TX_OVERFLOW_2	BITLEN_MAN_CFG_1	DIS_PAR_1	UART_RX_OVERFLOW_1	UART_TX_OVERFLOW_1
Reset	0x0	0x0	0x0	0x0	0x0	0x0	0x0	0x0
Access Type	Write, Read	Write, Read	Read Clears All	Read Clears All	Write, Read	Write, Read	Read Clears All	Read Clears All

BITFIELD	BITS	DESCRIPTION	DECODE
BITLEN_MAN_CFG_2	7	Enables the use of custom UART bit rate (BITLEN_PT_1 register) in pass-through UART Channel 2	0x0: Do not use custom UART bit rate in pass-through UART Channel 2 0x1: Use custom UART bit rate in pass-through UART Channel 2
DIS_PAR_2	6	Disables parity bit in pass-through UART Channel 2	0x0: Enable parity bit in pass-through UART Channel 2 0x1: Do not enable parity bit in pass-through UART Channel 2
UART_RX_OVERFLOW_2	5	Pass-through UART Rx FIFO overflow	0x0: No overflow occurred 0x1: Overflow occurred
UART_TX_OVERFLOW_2	4	Pass-through UART Tx FIFO overflow	0x0: No overflow occurred 0x1: Overflow occurred
BITLEN_MAN_CFG_1	3	Selects whether or not to use custom UART bit rate (BITLEN_PT_1 register) in pass-through UART Channel 1	0x0: Do not use custom UART bit rate in pass-through UART Channel 1 0x1: Use custom UART bit rate in pass-through UART Channel 1
DIS_PAR_1	2	Disables parity bit in pass-through UART Channel 1	0x0: Enable parity bit in pass-through UART Channel 1 0x1: Do not enable parity bit in pass-through UART Channel 1
UART_RX_OVERFLOW_1	1	Pass-through UART Rx FIFO overflow	0x0: No overflow occurred 0x1: Overflow occurred
UART_TX_OVERFLOW_1	0	Pass-through UART Tx FIFO overflow	0x0: No overflow occurred 0x1: Overflow occurred

RX0 (0x50)

BIT	7	6	5	4	3	2	1	0
Field	RX_CRC_EN	–	–	–	–	–	STR_SEL[1:0]	
Reset	0	–	–	–	–	–	0x0	
Access Type	Write, Read	–	–	–	–	–	Write, Read	

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BITFIELD	BITS	DESCRIPTION	DECODE
RX_CRC_EN	7	Receive packet CRC enable	0b0: No CRC on received packets 0b1: Received packets have CRC and checking is enabled
STR_SEL	1:0	Receive packets with selected stream ID	0bXX: Receive packets with this stream ID

TR0 (0x58, 0x60, 0x68, 0x78, 0x88, 0xA0, 0xA8)

BIT	7	6	5	4	3	2	1	0
Field	TX_CRC_EN	RX_CRC_EN	–	–	PRIO_VAL[1:0]		PRIO_CFG[1:0]	
Reset	1	1	–	–	0		0	
Access Type	Write, Read	Write, Read	–	–	Write, Read		Write, Read	

BITFIELD	BITS	DESCRIPTION	DECODE
TX_CRC_EN	7	When set, calculates and appends CRC to each packet transmitted from this port	0b0: Transmit CRC disabled 0b1: Transmit CRC enabled
RX_CRC_EN	6	When set, indicates that packets received at this port have appended CRC and that CRC checking should be performed at each packet	0b0: Receive CRC disabled 0b1: Receive CRC enabled
PRIO_VAL	3:2	Sets the priority for this channel's packet requests	0b00: Low priority 0b01: Normal priority 0b10: High priority 0b11: Urgent priority
PRIO_CFG	1:0	Adjusts the priority used for this channel	0b00: Current channel priority is used 0b01: Increase current channel priority 0b10: Decrease current channel priority 0b11: Reset to channel PRIO_VAL[1:0] setting

TR1 (0x59, 0x61, 0x69, 0x71, 0x79, 0x89, 0xA1, 0xA9)

BIT	7	6	5	4	3	2	1	0
Field	BW_MULT[1:0]		BW_VAL[5:0]					
Reset	0x2		0x30					
Access Type	Write, Read		Write, Read					

BITFIELD	BITS	DESCRIPTION	DECODE
BW_MULT	7:6	Channel bandwidth allocation multiplication factor	0b00: Multiply BW_VAL by 1 0b01: Multiply BW_VAL by 4 0b10: Multiply BW_VAL by 16 0b11: Multiply BW_VAL by 16
BW_VAL	5:0	Channel bandwidth allocation base. Fair bandwidth use ratio = BW_VAL x BW_MULT/10 as a percentage of total link bandwidth.	0bXXXXXX: Channel base bandwidth value

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TR3 (0x5B, 0x63, 0x6B, 0x73, 0x7B, 0x8B, 0xA3, 0xAB)

BIT	7	6	5	4	3	2	1	0
Field	–	–	TX_SPLT_MASK_B	TX_SPLT_MASK_A	–	TX_SRC_ID[2:0]		
Reset	–	–	0x1	0x1	–	0x0		
Access Type	–	–	Write, Read	Write, Read	–	Write, Read		

BITFIELD	BITS	DESCRIPTION	DECODE
TX_SPLT_MASK_B	5	Selects whether or not packets are transmited over GMSLB in splitter mode. When 0, packets from this port are not transmitted from port B. Otherwise, packets are transmitted from port B.	0b0: Packets are not transmitted over GMSLB in splitter mode 0b1: Packets are transmitted over GMSLB in splitter mode
TX_SPLT_MASK_A	4	Selects whether or not packets are transmited over GMSLB in splitter mode. When 0, packets from this port are not transmitted from port A. Otherwise, packets are transmitted from port A.	0b0: Packets are not transmitted over GMSLA in splitter mode 0b1: Packets are transmitted over GMSLA in splitter mode
TX_SRC_ID	2:0	Source identifier used in packets transmitted from this channel. Default value of the 2 MSbs are set by ADD2 and ADD1. Default value of the LSB is 0.	0bXXX: Source ID for packets from this channel

TR4 (0x5C, 0x64, 0x6C, 0x74, 0x7C, 0x8C, 0xA4, 0xAC)

BIT	7	6	5	4	3	2	1	0
Field	RX_SRC_SEL[7:0]							
Reset	0xFF							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
RX_SRC_SEL	7:0	Receive packets from selected sources. Each bit indicates whether packets with that source ID should be received or not. For example, when SRC_SEL = 00001001, then packets with source ID equal to 0 and 3 are received.	0x00: No packets received 0x01: Packets from source ID 1 received 0x02: Packets from source ID 2 received 0x03: Packets from source ID 1 and 2 received . . 0xFF: Packets from all source IDs received

ARQ0 (0x5D, 0x6D, 0x75, 0x7D, 0x8D, 0xA5, 0xAD)

BIT	7	6	5	4	3	2	1	0
Field	ARQ_AUTO_CFG	ACK_CNT	MATCH_SRC_ID	ACK_SRC_ID	ARQ_EN	–	–	–
Reset	0x1	0x0	0x0	0x1	0x1	–	–	–
Access Type	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read	–	–	–

BITFIELD	BITS	DESCRIPTION	DECODE
ARQ_AUTO_	7	Selects what values ARQ settings are based	0b0: ARQ settings are based on selections in

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BITFIELD	BITS	DESCRIPTION	DECODE
CFG		on. When this bit = 1, ARQ settings are automatically selected based on splitter mode	ACK_SRC_ID and MATCH_SRC_ID bitfields 0b1: ARQ settings are based on internal values
ACK_CNT	6	Selects whether to wait for one or two acknowledge packets. When ARQ_AUTO_CFG = 0, this bit selects whether to wait for one acknowledge packet or two acknowledge packets for each transmitted data packet	0b0: Wait for one acknowledge packet 0b1: Wait for two acknowledge packets
MATCH_SRC_ID	5	Acknowledge packet source ID checking method. The SRC_ID selected in this bitfield is used only when ARQ_AUTO_CFG = 0.	0b0: All received acknowledge packets are accepted 0b1: Acknowledge packet is accepted if SRC_ID of the received acknowledge packet matches TX_SRC_ID register
ACK_SRC_ID	4	Selects what to use as SRC_ID in transmitted acknowledge packets. The SRC_ID selected in this bitfield is used only when ARQ_AUTO_CFG = 0.	0b0: Use SRC_ID of the received data packet 0b1: Use TX_SRC_ID register
ARQ_EN	3	Enables ARQ	0b0: ARQ disabled 0b1: ARQ enabled

ARQ1 (0x5E, 0x6E, 0x76, 0x7E, 0x8E, 0xA6, 0xAE)

BIT	7	6	5	4	3	2	1	0
Field	–	MAX_RT[2:0]			–	–	MAX_RT_ERR_OEN	RT_CNT_OEN
Reset	–	0x7			–	–	1	0
Access Type	–	Write, Read			–	–	Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
MAX_RT	6:4	Maximum re-transmission limit. ARQ will stop re-transmiting after trying to re-transmit this many times for a single packet.	0bXXX: Maximum re-transmission limit
MAX_RT_ERR_OEN	1	Enables reporting of ARQ maximum re-transmission limit errors (MAX_RT_ERR) for this channel at ERRB pin	0b0: ARQ maximum re-transmission limit errors reporting at ERRB pin disabled 0b1: ARQ maximum re-transmission limit errors reporting at ERRB pin enabled
RT_CNT_OEN	0	Enables reporting of ARQ re-transmission event for this channel at ERRB pin. When enabled, ERRB is asserted when RT_CNT of this channel is greater than 0.	0b0: ARQ re-transmission count reporting at ERRB pin disabled 0b1: ARQ re-transmission count reporting at ERRB pin enabled

ARQ2 (0x5F, 0x6F, 0x77, 0x7F, 0x8F, 0xA7, 0xAF)

BIT	7	6	5	4	3	2	1	0
Field	MAX_RT_ERR	RT_CNT[6:0]						
Reset	0	0						
Access Type	Read Clears All	Read Clears All						

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BITFIELD	BITS	DESCRIPTION	DECODE
MAX_RT_ERR	7	Indicates whether or not the maximum re-transmission limit (MAX_RT) was reached for one packet in this channel	0b0: Maximum re-transmit limit not reached 0b1: Maximum re-transmit limit reached
RT_CNT	6:0	Total re-transmission count in this channel	0xXX: Count of re-transmissions for this channel

TR0 (0x70)

BIT	7	6	5	4	3	2	1	0
Field	TX_CRC_EN	RX_CRC_EN	–	–	PRIO_VAL[1:0]		PRIO_CFG[1:0]	
Reset	1	1	–	–	0		0	
Access Type	Write, Read	Write, Read	–	–	Write, Read		Write, Read	

BITFIELD	BITS	DESCRIPTION	DECODE
TX_CRC_EN	7	When set, calculates and appends CRC to each packet transmitted from this port	0b0: Transmit CRC disabled 0b1: Transmit CRC enabled
RX_CRC_EN	6	When set, indicates that packets received at this port have appended CRC and CRC checking should be performed at each packet.	0b0: Receive CRC disabled 0b1: Receive CRC enabled
PRIO_VAL	3:2	Sets the priority for this channel's packet requests	0b00: Low priority 0b01: Normal priority 0b10: High priority 0b11: Urgent priority
PRIO_CFG	1:0	Adjusts the priority used for this channel	0b00: Current channel priority is used 0b01: Increase current channel priority 0b10: Decrease current channel priority 0b11: Reset to channel PRIO_VAL[1:0] setting

VIDEO_RX0 (0x100)

BIT	7	6	5	4	3	2	1	0
Field	LCRC_ERR	–	–	–	–	–	LINE_CRC_EN	–
Reset	0x0	–	–	–	–	–	0b1	–
Access Type	Read Clears All	–	–	–	–	–	Write, Read	–

BITFIELD	BITS	DESCRIPTION	DECODE
LCRC_ERR	7	Video line CRC error flag	0b0: No video line CRC detected 0b1: Video line CRC detected
LINE_CRC_EN	1	Video line CRC enable	0b0: Video line CRC disabled 0b1: Video line CRC enabled

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VIDEO_RX3 (0x103)

BIT	7	6	5	4	3	2	1	0
Field	–	HD_TR_MODE	DLOCKED	VLOCKED	HLOCKED	DTRACKEN	VTRACKEN	HTRACKEN
Reset	–	0b1	0b0	0b0	0b0	0b0	0b0	0b0
Access Type	–	Write, Read	Read Only	Read Only	Read Only	Write, Read	Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
HD_TR_MODE	6	HS, DE Tracking enable full periodic pattern	0b0: Allow only partial periodic HS, DE 0b1: Allow partial periodic and full periodic HS, DE
DLOCKED	5	DE Tracking locked	0b0: DE tracking not locked 0b1: DE tracking locked
VLOCKED	4	VS Tracking locked	0b0: VS tracking not locked 0b1: VS tracking locked
HLOCKED	3	HS Tracking locked	0b0: HS tracking not locked 0b1: HS tracking locked
DTRACKEN	2	DE tracking enable (disabled if FSYNC = 1). The system observes DE pulses and can compensate for a limited number of missing pulses or suppress glitches when it locks on the pattern.	0b0: DE tracking disabled 0b1: DE tracking enabled
VTRACKEN	1	VS tracking enable (disabled if FSYNC = 1). The system observes VS pulses and can compensate for a limited number of missing pulses or suppress glitches when it locks on the pattern.	0b0: VS tracking disabled 0b1: VS tracking enabled
HTRACKEN	0	HS tracking enable (disabled if FSYNC = 1). The system observes HS pulses and can compensate for a limited number of missing pulses or suppress glitches when it locks on the pattern.	0b0: HS tracking disabled 0b1: HSYNC tracking enabled

VIDEO_RX4 (0x104)

BIT	7	6	5	4	3	2	1	0
Field	VID_OVERFLW_LIM_L[7:0]							
Reset	0x0C							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
VID_OVERFLW_LIM_L	7:0	Lower byte of 9-bit video FIFO overflow threshold during lock	0xXX: Video FIFO overflow threshold during lock (LSB)

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VIDEO_RX5 (0x105)

BIT	7	6	5	4	3	2	1	0
Field	VID_UNDERFLW_LIM_L[7:0]							
Reset	0xD0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
VID_UNDERFLW_LIM_L	7:0	Lower byte of 9-bit video FIFO underflow threshold during lock	0xXX: Video FIFO underflow threshold during lock (LSB)

VIDEO_RX6 (0x106)

BIT	7	6	5	4	3	2	1	0
Field	VID_SAR_INC_H[2:0]			–	–	–	VID_UNDERFLW_LIM_H	VID_OVERFLW_LIM_H
Reset	0x0			–	–	–	0b1	0b0
Access Type	Write, Read			–	–	–	Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
VID_SAR_INC_H	7:5	Upper bits of 10-bit video clock regeneration SAR start point	0bXXX: Video clock regeneration SAR start point (3 MSbs)
VID_UNDERFLW_LIM_H	1	Upper bit of 9-bit video FIFO underflow threshold during lock	0bX: Video FIFO underflow threshold during lock (MSb)
VID_OVERFLW_LIM_H	0	Upper bit of 9-bit video FIFO overflow threshold during lock	0bX: Video FIFO overflow threshold during lock (MSb)

VIDEO_RX8 (0x108)

BIT	7	6	5	4	3	2	1	0
Field	VID_BLK_LEN_ERR	VID_LOCK	VID_PKT_DET	VID_SEQ_ERR	–	–	–	–
Reset	0b0	0b0	0b0	0b0	–	–	–	–
Access Type	Read Clears All	Read Only	Read Only	Read Clears All	–	–	–	–

BITFIELD	BITS	DESCRIPTION	DECODE
VID_BLK_LEN_ERR	7	Video RX block length error detection	0b0: No error detected 0b1: Video RX block length error detected
VID_LOCK	6	Video pipeline locked	0b0: Video pipeline not locked 0b1: Video pipeline locked
VID_PKT_DET	5	Sufficient video Rx packet throughput detected	0b0: Not enough throughput 0b1: Sufficient throughput detected
VID_SEQ_ERR	4	Video Rx sequence error detection	0b0: No error detected 0b1: Error detected

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VIDEO_RX10 (0x10A)

BIT	7	6	5	4	3	2	1	0
Field	–	MASK_VIDEO_DE	–	–	–	–	–	–
Reset	–	0x0	–	–	–	–	–	–
Access Type	–	Write, Read	–	–	–	–	–	–

BITFIELD	BITS	DESCRIPTION	DECODE
MASK_VIDEO_DE	6	Mask video with DE	0x0: Do not mask video with DE 0x1: Mask video with DE

CLOCKGEN (0x111)

BIT	7	6	5	4	3	2	1	0
Field	–	–	–	–	CLK_FREEZE_RNG[1:0]		CLK_FREEZE_CONS	CLK_FREEZE
Reset	–	–	–	–	0x0		0b0	0b0
Access Type	–	–	–	–	Write, Read		Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
CLK_FREEZE_RNG	3:2	Selects the frequency when outputting a preset clock frequency on Tx clock outputs	0b00: 50MHz 0b01: 100MHz 0b10: 200MHz 0b11: 300MHz
CLK_FREEZE_CONS	1	Selects Tx clock output when video not locked and enabled by CLK_FREEZE	0b0: Output last clock frequency 0b1: Output frequency selected by CLK_FREEZE_RNG
CLK_FREEZE	0	Selects whether or not to always output video clock on Tx clock outputs even when receiver is not locked	0b0: Tx clock outputs disabled when video not locked 0b1: Always output video clock

AUDIO_TX0 (0x120)

BIT	7	6	5	4	3	2	1	0
Field	I2S_TDM_CFG[1:0]		I2S_CFG[1:0]		INV_SCK	INV_WS	FORCE_AUD	AUD_SINK_SRC
Reset	0x2		0x3		0b0	0b0	0b0	0b0
Access Type	Write, Read		Write, Read		Write, Read	Write, Read	Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
I2S_TDM_CFG	7:6	Audio format configuration (HDMI audio only)	0b00: Force stereo 0b01: Force TDM 0b10: Auto select stereo or TDM 0b11: Reserved
I2S_CFG	5:4	Audio channel configuration (HDMI audio only). Must be set the same on both I^2S Tx adapters.	0b00: Use HDMI as audio source 0b01: Reserved 0b10: Reserved 0b11: Use local I^2S input pins as audio source
INV_SCK	3	Inverts SCK (when using local I^2S interface)	0b0: Do not invert SCK

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BITFIELD	BITS	DESCRIPTION	DECODE
			0b1: Invert SCK
INV_WS	2	Inverts WS (when using local I2S interface)	0b0: Do not invert WS 0b1: Invert WS
FORCE_AUD	1	Forces audio data to zero	0b0: Normal audio data 0b1: Audio data forced to zero
AUD_SINK_SRC	0	Sink-sourced mode enable for audio TX. For use on the deserializer side when forward audio clock is the master.	0b0: Transmit audio in normal mode 0b1: Transmit audio in sink-sourced mode

AUDIO_TX1 (0x121)

BIT	7	6	5	4	3	2	1	0
Field	AUD_PRIO[1:0]		AUD_STR_TX[1:0]		AUD_DRIFT_DET_EN	AUD_INF_PER	I2S_HDR_CFG[1:0]	
Reset	0x1		0x0		0b1	0b1	0x2	
Access Type	Write, Read		Write, Read		Write, Read	Write, Read	Write, Read	

BITFIELD	BITS	DESCRIPTION	DECODE
AUD_PRIO	7:6	Scheduler audio priority	0b00: Priority 0 (low) 0b01: Priority 1 0b10: Priority 2 0b11: Priority 3
AUD_STR_TX	5:4	Audio stream ID	0b00: Use stream ID 0 0b01: Use stream ID 1 0b10: Use stream ID 2 0b11: Use stream ID 3
AUD_DRIFT_DET_EN	3	Audio clock drift detection enable. Resets sub system on SCK frequency drift and reports it.	0b0: Audio clock drift detection disabled 0b1: Audio clock drift detection enabled
AUD_INF_PER	2	Audio periodic infromation frame transmission enable	0b0: Audio periodic information frame transmission disabled 0b1: Audio periodic information frame transmission enabled
I2S_HDR_CFG	1:0	Audio header configuration (only applicable for TDM audio) (HDMI audio only)	0b00: Use audio info received on HDMI link 0b01: Use header programmed to register 0b10: No header 0b11: Reserved

AUDIO_TX2 (0x122)

BIT	7	6	5	4	3	2	1	0
Field	AUD_HDR_L[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
AUD_HDR_L	7:0	FIrst byte of audio header data (HDMI audio only)	0xX: First byte of audio header data

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AUDIO_TX3 (0x123)

BIT	7	6	5	4	3	2	1	0
Field	AUD_HDR_M[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
AUD_HDR_M	7:0	Second byte of audio header data (HDMI audio only)	0xX: Second byte of audio header data

AUDIO_TX4 (0x124)

BIT	7	6	5	4	3	2	1	0
Field	AUD_HDR_H[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
AUD_HDR_H	7:0	Third byte of audio header data (HDMI audio only)	0xX: Third byte of audio header data

AUDIO_TX5 (0x125)

BIT	7	6	5	4	3	2	1	0
Field	AUD_DRIFT_ERR	AUD_FIFO_WARN	AUD_OVERFLOW	ACLKDET	–	–	–	–
Reset	0b0	0b0	0b0	0b0	–	–	–	–
Access Type	Read Clears All	Read Clears All	Read Clears All	Read Only	–	–	–	–

BITFIELD	BITS	DESCRIPTION	DECODE
AUD_DRIFT_ERR	7	Indicates whether audio clock frequency drift was detected	0b0: Audio clock frequency drift not detected 0b1: Audio clock frequency drift detected
AUD_FIFO_WARN	6	Indicates whether audio FIFO is more than half full	0b0: Audio FIFO is less than or equal to half full 0b1: Audio FIFO is more than half full
AUD_OVERFLOW	5	Indicates whether audio buffer overflow was detected	0b0: Audio buffer overflow not detected 0b1: Audio buffer overflow detected
ACLKDET	4	Audio clock detected	0b0: Audio clock not detected 0b1: Audio clock detected

AUDIO_TX7 (0x127)

BIT	7	6	5	4	3	2	1	0
Field	PRBS_SEL	PRBSEN_AUD	–	–	–	–	–	–
Reset	0b0	0b0	–	–	–	–	–	–
Access Type	Write, Read	Write, Read	–	–	–	–	–	–

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BITFIELD	BITS	DESCRIPTION	DECODE
PRBS_SEL	7	Audio PRBS clock selection (program to 0 in sink-sourced mode)	0b0: SCK selected 0b1: Internal oscillator clock selected (150MHz)
PRBSEN_AUD	6	Audio PRBS enable	0b0: Audio PRBS disabled 0b1: Audio PRBS enabled

AUDIO_TX8 (0x128)

BIT	7	6	5	4	3	2	1	0
Field	PRBS_WS_LEN	PRBS_WS_GEN	–	–	–	–	–	–
Reset	0b0	0b1	–	–	–	–	–	–
Access Type	Write, Read	Write, Read	–	–	–	–	–	–

BITFIELD	BITS	DESCRIPTION	DECODE
PRBS_WS_LEN	7	Selects the length of the generated WS pulse	0b0: 256 data bits per WS cycle 0b1: 32 data bits per WS cycle
PRBS_WS_GEN	6	Audio PRBS WS generation enable	0b0: Audio PRBS WS generation disabled 0b1: Audio PRBS WS generation enabled

AUDIO_RX1 (0x140)

BIT	7	6	5	4	3	2	1	0
Field	AUD_RX_SINK_SRC	–	–	–	INV_SCK_RX	INV_WS_RX	–	AUD_EN_RX
Reset	0b0	–	–	–	0b0	0b0	–	0b1
Access Type	Write, Read	–	–	–	Write, Read	Write, Read	–	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
AUD_RX_SINK_SRC	7	Sink-sourced mode enable for audio Rx (for use on the serializer side)	0b0: Received audio in normal mode 0b1: Received audio in sink-sourced mode
INV_SCK_RX	3	Inverts SCK at I^2S output	0b0: Do not invert SCK 0b1: Invert SCK
INV_WS_RX	2	Inverts WS at I^2S output	0b0: Do not invert WS 0b1: Invert WS
AUD_EN_RX	0	Audio receiver adapter enable	0b0: Audio receiver disabled 0b1: Audio receiver enabled

AUDIO_RX4 (0x143)

BIT	7	6	5	4	3	2	1	0
Field	APRBS_ERR[7:0]							
Reset	0x00							
Access Type	Read Clears All							

BITFIELD	BITS	DESCRIPTION	DECODE
APRBS_ERR	7:0	Audio PRBS error counter, clears on read	0xXX: Number of audio PRBS errors

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AUDIO_RX7 (0x146)

BIT	7	6	5	4	3	2	1	0
Field	–	–	–	APRBS_CHK_EN	AUD_STRM[1:0]		–	–
Reset	–	–	–	0b0	0x0		–	–
Access Type	–	–	–	Write, Read	Write, Read		–	–

BITFIELD	BITS	DESCRIPTION	DECODE
APRBS_CHK_EN	4	Enables audio PRBS checker	0b0: Audio PRBS checker disabled 0b1: Audio PRBS checker enabled
AUD_STRM	3:2	Selects audio stream for reception	0b00: Receive stream ID 0 0b01: Receive stream ID 1 0b10: Receive stream ID 2 0b11: Receive stream ID 3

AUDIO_RX9 (0x148)

BIT	7	6	5	4	3	2	1	0
Field	AUD_BLK_LEN_ERR	AUD_LOCK	AUD_PKT_DET	APRBS_VALID	–	–	–	–
Reset	0b0	0b0	0b0	0b0	–	–	–	–
Access Type	Read Clears All	Read Only	Read Only	Read Only	–	–	–	–

BITFIELD	BITS	DESCRIPTION	DECODE
AUD_BLK_LEN_ERR	7	Indicates whether or not audio Rx block length error was detected	0b0: Received audio block length error not detected 0b1: Received audio block length error detected
AUD_LOCK	6	Indicates whether or not audio pipeline frequency is locked	0b0: Audio pipeline frequency is not locked 0b1: Audio pipeline frequency is locked
AUD_PKT_DET	5	Indicates whether or not sufficient audio Rx packet throughput is detected	0b0: Valid audio stream not detected 0b1: Valid audio stream detected
APRBS_VALID	4	Indicates whether or not audio PRBS is running	0b0: Audio PRBS is not running 0b1: Audio PRBS is running

SPI_0 (0x160)

BIT	7	6	5	4	3	2	1	0
Field	SPI_LOC_ID[1:0]		SPI_CC_TRG_ID[1:0]		SPI_IGNR_ID	SPI_CC_EN	MST_SLVN	SPI_EN
Reset	0x0		0x0		0x1	0x0	0x0	0x0
Access Type	Write, Read		Write, Read		Write, Read	Write, Read	Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
SPI_LOC_ID	7:6	Programs to local ID if filtering packets based on header ID	0b00: ID = 0 0b01: ID = 1 0b10: ID = 2 0b11: ID = 3
SPI_CC_TR	5:4	ID for GMSL2 header in SPI control channel	0b00: ID = 0

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BITFIELD	BITS	DESCRIPTION	DECODE
G_ID		bridge mode	0b01: ID = 1 0b10: ID = 2 0b11: ID = 3
SPI_IGNR_ID	3	Selects if SPI should use or ignore header ID to determine packet acceptance	0b0: Accept only packets with proper ID 0b1: Ignore ID and accept all packets
SPI_CC_EN	2	Enables control channel SPI bridge function	0b0: SPI bridge disabled 0b1: SPI bridge enabled
MST_SLVN	1	Selects if SPI is master or slave	0b0: SPI slave 0b1: SPI master
SPI_EN	0	Enables SPI channel	0b0: SPI channel disabled 0b1: SPI channel enabled

SPI_1 (0x161)

BIT	7	6	5	4	3	2	1	0
Field	SPI_LOC_N[5:0]						SPI_BASE_PRIO[1:0]	
Reset	0x07						0x1	
Access Type	Write, Read						Write, Read	

BITFIELD	BITS	DESCRIPTION	DECODE
SPI_LOC_N	7:2	Sets the packet size ((2N + 1) bytes) for GMSL2 SPI packets. If this is programmed to a value greater than 7, ARQ of the SPI channel must be disabled.	0b000000: Packet size is 1 byte 0b000001: Packet size is 3 bytes . . 0b111111: Packet size is 127 bytes
SPI_BASE_PRIO	1:0	Starting GMSL2 request priority. Advances by 1 (space permitting) if Tx buffer is over half full.	0b00: Priority 0 (low) 0b01: Priority 1 0b10: Priority 2 0b11: Priority 3

SPI_2 (0x162)

BIT	7	6	5	4	3	2	1	0
Field	REQ_HOLD_OFF[2:0]			FULL_SCK_SETUP	SPI_MOD3_F	SPI_MOD3	SPIM_SS2_ACT_H	SPIM_SS1_ACT_H
Reset	0x0			0x0	0x0	0x0	0x1	0x1
Access Type	Write, Read			Write, Read	Write, Read	Write, Read	Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
REQ_HOLD_OFF	7:5	Hold off GMSL2 request until this number of extra bytes is received on SPI port	0b000: Request after every byte received 0b001: Request after 2 bytes received 0b010: Request after 3 bytes received . . 0b111: Request after 8 bytes received
FULL_SCK_SETUP	4	Sample MISO after half or full SCK period	0b0: MISO sampled after half SCK period 0b1: MISO sampled after full SCK period

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BITFIELD	BITS	DESCRIPTION	DECODE
SPI_MOD3_F	3	Allows the suppression of an extra SCK prior to SS deassertion when SPI mode 3 is selected	0b0: Extra SCK present prior to SS deassertion when in SPI mode 3 0b1: Extra SCK suppressed prior to SS deassertion when in SPI mode 3
SPI_MOD3	2	Selects SPI mode 0 or 3	0b0: SPI mode 0 0b1: SPI mode 3
SPIM_SS2_ACT_H	1	Sets the polarity for SS2 when the SPI is a master.	0b0: Slave Select 2 is active low 0b1: Slave Select 2 is active high
SPIM_SS1_ACT_H	0	Sets the polarity for SS1 when the SPI is a master.	0b0: Slave select 1 is active low 0b1: Slave select 1 is active high

SPI_3 (0x163)

BIT	7	6	5	4	3	2	1	0
Field	SPIM_SS_DLY_CLKS[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
SPIM_SS_DLY_CLKS	7:0	Number of 300MHz clocks to delay between each the following: – Assertion of SS and start of SCK pulses – End of SCK pulses and de-assertion of SS – De-assertion of SS and re-assertion of SS (if necessary)	0xXX: Number of clock delays

SPI_4 (0x164)

BIT	7	6	5	4	3	2	1	0
Field	SPIM_SCK_LO_CLKS[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
SPIM_SCK_LO_CLKS	7:0	Number of 300MHz clocks for SCK low time	0xXX: Number of clocks for SCK low time

SPI_5 (0x165)

BIT	7	6	5	4	3	2	1	0
Field	SPIM_SCK_HI_CLKS[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
SPIM_SCK_HI_CLKS	7:0	Number of 300MHz clocks for SCK high time	0xXX: Number of clocks for SCK high time

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SPI_6 (0x166)

BIT	7	6	5	4	3	2	1	0
Field	–	–	BNE	SPIS_RWN	SS_IO_EN_2	SS_IO_EN_1	BNE_IO_EN	RWN_IO_EN
Reset	–	–	0x0	0x0	0x0	0x0	0x0	0x0
Access Type	–	–	Read Only	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
BNE	5	Alternate GPU Status register to use for BNE status if GPIO is not available	0b0: Buffer empty 0b1: Buffer not empty
SPIS_RWN	4	Alternate GPU Control register to use for Rd/WrN control if GPIO is not available.	0b0: Write 0b1: Read
SS_IO_EN_2	3	Enables GPIO for use as Slave Select 2 output.	0b0: GPIO not used for SPI SS2 function 0b1: GPIO used for SPI SS2 function
SS_IO_EN_1	2	Enables GPIO for use as Slave Select 1 output	0b0: GPIO not used for SPI SS1 function 0b1: GPIO used for SPI SS1 function
BNE_IO_EN	1	Enables GPIO for use as BNE output for SPI data available status	0b0: GPIO not used for SPI BNE function 0b1: GPIO used for SPI BNE function
RWN_IO_EN	0	Enables GPIO for use as RO input for control of SPI data movement	0b0: GPIO not used for SPI RO function 0b1: GPIO used for SPI RO function

SPI_7 (0x167)

BIT	7	6	5	4	3	2	1	0
Field	SPI_RX_OVRFLW	SPI_TX_OVRFLW	–	SPIS_BYTE_CNT[4:0]				
Reset	0b0	0b0	–	0x0				
Access Type	Read Clears All	Read Clears All	–	Read Only				

BITFIELD	BITS	DESCRIPTION	DECODE
SPI_RX_OVRFLW	7	SPI Rx buffer overflow flag	0b0: No SPI Rx buffer overflow 0b1: SPI Rx buffer overflow
SPI_TX_OVRFLW	6	SPI Tx buffer overflow flag	0b0: No SPI Tx buffer overflow 0b1: SPI Tx buffer overflow
SPIS_BYTE_CNT	4:0	Number of SPI data bytes available for reading from Rx Buffer.	0bXXXXX: Number of bytes available

SPI_8 (0x168)

BIT	7	6	5	4	3	2	1	0
Field	REQ_HOLD_OFF_TO[7:0]							
Reset	0x00							
Access Type	Write, Read							

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BITFIELD	BITS	DESCRIPTION	DECODE
REQ_HOLD_OFF_TO	7:0	Timeout delay (in 100nS increments) for GMSL2 Req Hold-Off (0 is disable).	0xXX: Number of 100nS delay increments for GMSL2 request hold-off

WM_0 (0x188)

BIT	7	6	5	4	3	2	1	0
Field	WM_LEN	WM_MODE[2:0]			WM_DET[1:0]		WME_EN	WM_EN
Reset	0b0	0x0			0x0		0x0	0b0
Access Type	Write, Read	Write, Read			Write, Read		Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
WM_LEN	7	Watermark length	0b0: 32-bit 0b1: 63-bit
WM_MODE	6:4	Watermark mode	0b000: Insert WM-/WM+ every frame, temporal dither enabled 0b001: Reserved 0b010: Insert WM0/WM1 every frame, temporal dither enabled, polarity disabled (Error generation mode) 0b011: Insert WM0/WM1 every frame, do not reset WM, temporal dither enabled 0b100: Insert WM0/WM1 every frame, temporal dither disabled 0b101: Reserved 0b110: Reserved 0b111: Reserved
WM_DET	3:2	Watermark detection/generation	0b00: Insert watermark in video stream 0b01: Detect watermark and remove from outgoing video stream 0b10: Reserved 0b11: Reserved
WME_EN	1	Enables WM error to be driven to GPIO pin	0x0: Disabled 0x1: Enabled
WM_EN	0	Watermark enable	0b0: Watermarking disabled 0b1: Watermarking enabled

WM_2 (0x18A)

BIT	7	6	5	4	3	2	1	0
Field	–	–	–	–	HsyncPol	VsyncPol	WM_NPFILT[1:0]	
Reset	–	–	–	–	0x0	0x0	0x0	
Access Type	–	–	–	–	Write, Read	Write, Read	Write, Read	

BITFIELD	BITS	DESCRIPTION	DECODE
HsyncPol	3	HS polarity (only effective for watermark block)	0b0: Non-inverting 0x1: Inverting
VsyncPol	2	VS polarity (only effective for watermark block)	0b0: Non-inverting 0x1: Inverting
WM_NPFILT	1:0	Phase accumulator terminal count	0bXX: Phase accumulator terminal count

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WM_3 (0x18B)

BIT	7	6	5	4	3	2	1	0
Field	–	WM_TH[6:0]						
Reset	–	0x14						
Access Type	–	Write, Read						

BITFIELD	BITS	DESCRIPTION	DECODE
WM_TH	6:0	Matched filter threshold	0bXXXXXXX: Matched filter threshold

WM_4 (0x18C)

BIT	7	6	5	4	3	2	1	0
Field	–	–	WM_YUV_IN[1:0]		WM_COLORADJ	–	WM_MASKMODE[1:0]	
Reset	–	–	0x1		0b0	–	0x0	
Access Type	–	–	Write, Read		Write, Read	–	Write, Read	

BITFIELD	BITS	DESCRIPTION	DECODE
WM_YUV_IN	5:4	Input color format	0b0: RGB 8-bit color data 0b1: YUV444 8-bits each
WM_COLORADJ	3	Color adjust	0b0: Masked off LSBs of U. Use if temporal dithering is enabled. 0b1: Set U[Ko:0] = 1/2 WM gain. Use this mode if temporal dithering is disabled
WM_MASKMODE	1:0	Sets watermark mask for the device	0b00: Mask if WM is detected 0b01: Mask if WM is detected, blank if error is detected 0b10: Reserved b011: Reserved

WM_5 (0x18D)

BIT	7	6	5	4	3	2	1	0
Field	–	–	–	–	–	–	WM_DETOUT	WM_ERROR
Reset	–	–	–	–	–	–	0b0	0b0
Access Type	–	–	–	–	–	–	Read Only	Read Only

BITFIELD	BITS	DESCRIPTION	DECODE
WM_DETOUT	1	Live frame-based detection output	0b0: Watermark not detected 0b1: Watermark detected
WM_ERROR	0	Live active-high watermark error	0b0: No watermark error 0b1: Watermark error active, bit automatically clears when error clears

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WM_6 (0x18E)

BIT	7	6	5	4	3	2	1	0
Field	WM_TIMER[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
WM_TIMER	7:0	Time in 2ms steps that the frozen-frame condition must be observed before an error is generated. 0 = no filter.	0xXX: Number of milliseconds

WM_WREN_0 (0x1AE)

BIT	7	6	5	4	3	2	1	0
Field	WM_WREN_L[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
WM_WREN_L	7:0	Works in conjunction with WM_WREN_H (HEX 1AF) to enable writing to watermark registers. When 0xBA is written to WM_WREN_L and 0xDC is written to WM_WREN_H, watermark registers are write-enabled. Otherwise, watermark registers are read-only.	0xBA: Enables writing to watermark registers (only if WM_WREN_H contains the value 0xDC) All other values: Watermark registers are read -only

WM_WREN_1 (0x1AF)

BIT	7	6	5	4	3	2	1	0
Field	WM_WREN_H[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
WM_WREN_H	7:0	Works in conjunction with WM_WREN_L (HEX 1AE) to enable writing to watermark registers. When 0xBA is written to WM_WREN_L and 0xDC is written to WM_WREN_H, watermark registers are write-enabled. Otherwise, watermark registers are read-only.	0xDC: Enables writing to watermark registers (only if WM_WREN_L contains the value 0xBA) All other values: Watermark registers are read-only

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CROSS_0 (0x1B0)

BIT	7	6	5	4	3	2	1	0
Field	–	CROSS0_I	CROSS0_F	CROSS0[4:0]				
Reset	–	0x0	0x0	0x00				
Access Type	–	Write, Read	Write, Read	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
CROSS0_I	6	Inverts CrossX	0b0: Do not invert bit 0b1: Invert bit
CROSS0_F	5	Forces CrossX to 0 before inversion	0b0: Do not force bit to zero 0b1: Force bit to zero
CROSS0	4:0	Maps incoming bit position set by this field to the outgoing bit position 0	0bXXXXX: Incoming bit position

CROSS_1 (0x1B1)

BIT	7	6	5	4	3	2	1	0
Field	–	CROSS1_I	CROSS1_F	CROSS1[4:0]				
Reset	–	0x0	0x0	0x01				
Access Type	–	Write, Read	Write, Read	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
CROSS1_I	6	Inverts CrossX	0b0: Do not invert bit 0b1: Invert bit
CROSS1_F	5	Forces CrossX to 0 before inversion	0b0: Do not force bit to zero 0b1: Force bit to zero
CROSS1	4:0	Maps incoming bit position set by this field to the outgoing bit position 1	0bXXXXX: Incoming bit position

CROSS_2 (0x1B2)

BIT	7	6	5	4	3	2	1	0
Field	–	CROSS2_I	CROSS2_F	CROSS2[4:0]				
Reset	–	0x0	0x0	0x02				
Access Type	–	Write, Read	Write, Read	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
CROSS2_I	6	Inverts CrossX	0b0: Do not invert bit 0b1: Invert bit
CROSS2_F	5	Forces CrossX to 0 before inversion	0b0: Do not force bit to zero 0b1: Force bit to zero
CROSS2	4:0	Maps incoming bit position set by this field to the outgoing bit position 2	0bXXXXX: Incoming bit position

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CROSS_3 (0x1B3)

BIT	7	6	5	4	3	2	1	0
Field	–	CROSS3_I	CROSS3_F	CROSS3[4:0]				
Reset	–	0x0	0x0	0x03				
Access Type	–	Write, Read	Write, Read	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
CROSS3_I	6	Inverts CrossX	0b0: Do not invert bit 0b1: Invert bit
CROSS3_F	5	Forces CrossX to 0 before inversion	0b0: Do not force bit to zero 0b1: Force bit to zero
CROSS3	4:0	Maps incoming bit position set by this field to the outgoing bit position 3	0bXXXXX: Incoming bit position

CROSS_4 (0x1B4)

BIT	7	6	5	4	3	2	1	0
Field	–	CROSS4_I	CROSS4_F	CROSS4[4:0]				
Reset	–	0x0	0x0	0x04				
Access Type	–	Write, Read	Write, Read	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
CROSS4_I	6	Inverts CrossX	0b0: Do not invert bit 0b1: Invert bit
CROSS4_F	5	Forces CrossX to 0 before inversion	0b0: Do not force bit to zero 0b1: Force bit to zero
CROSS4	4:0	Maps incoming bit position set by this field to the outgoing bit position 4	0bXXXXX: Incoming bit position

CROSS_5 (0x1B5)

BIT	7	6	5	4	3	2	1	0
Field	–	CROSS5_I	CROSS5_F	CROSS5[4:0]				
Reset	–	0x0	0x0	0x05				
Access Type	–	Write, Read	Write, Read	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
CROSS5_I	6	Inverts CrossX	0b0: Do not invert bit 0b1: Invert bit
CROSS5_F	5	Forces CrossX to 0 before inversion	0b0: Do not force bit to zero 0b1: Force bit to zero
CROSS5	4:0	Maps incoming bit position set by this field to the outgoing bit position 5	0bXXXXX: Incoming bit position

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CROSS_6 (0x1B6)

BIT	7	6	5	4	3	2	1	0
Field	–	CROSS6_I	CROSS6_F	CROSS6[4:0]				
Reset	–	0x0	0x0	0x06				
Access Type	–	Write, Read	Write, Read	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
CROSS6_I	6	Inverts CrossX	0b0: Do not invert bit 0b1: Invert bit
CROSS6_F	5	Forces CrossX to 0 before inversion	0b0: Do not force bit to zero 0b1: Force bit to zero
CROSS6	4:0	Maps incoming bit position set by this field to the outgoing bit position 6	0bXXXXX: Incoming bit position

CROSS_7 (0x1B7)

BIT	7	6	5	4	3	2	1	0
Field	–	CROSS7_I	CROSS7_F	CROSS7[4:0]				
Reset	–	0x0	0x0	0x07				
Access Type	–	Write, Read	Write, Read	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
CROSS7_I	6	Inverts CrossX	0b0: Do not invert bit 0b1: Invert bit
CROSS7_F	5	Forces CrossX to 0 before inversion	0b0: Do not force bit to zero 0b1: Force bit to zero
CROSS7	4:0	Maps incoming bit position set by this field to the outgoing bit position 7	0bXXXXX: Incoming bit position

CROSS_8 (0x1B8)

BIT	7	6	5	4	3	2	1	0
Field	–	CROSS8_I	CROSS8_F	CROSS8[4:0]				
Reset	–	0x0	0x0	0x08				
Access Type	–	Write, Read	Write, Read	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
CROSS8_I	6	Inverts CrossX	0b0: Do not invert bit 0b1: Invert bit
CROSS8_F	5	Forces CrossX to 0 before inversion	0b0: Do not force bit to zero 0b1: Force bit to zero
CROSS8	4:0	Maps incoming bit position set by this field to the outgoing bit position 8	0bXXXXX: Incoming bit position

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CROSS_9 (0x1B9)

BIT	7	6	5	4	3	2	1	0
Field	–	CROSS9_I	CROSS9_F	CROSS9[4:0]				
Reset	–	0x0	0x0	0x09				
Access Type	–	Write, Read	Write, Read	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
CROSS9_I	6	Inverts CrossX	0b0: Do not invert bit 0b1: Invert bit
CROSS9_F	5	Forces CrossX to 0 before inversion	0b0: Do not force bit to zero 0b1: Force bit to zero
CROSS9	4:0	Maps incoming bit position set by this field to the outgoing bit position 9	0bXXXXX: Incoming bit position

CROSS_10 (0x1BA)

BIT	7	6	5	4	3	2	1	0
Field	–	CROSS10_I	CROSS10_F	CROSS10[4:0]				
Reset	–	0x0	0x0	0x0A				
Access Type	–	Write, Read	Write, Read	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
CROSS10_I	6	Inverts CrossX	0b0: Do not invert bit 0b1: Invert bit
CROSS10_F	5	Forces CrossX to 0 before inversion	0b0: Do not force bit to zero 0b1: Force bit to zero
CROSS10	4:0	Maps incoming bit position set by this field to the outgoing bit position 10	0bXXXXX: Incoming bit position

CROSS_11 (0x1BB)

BIT	7	6	5	4	3	2	1	0
Field	–	CROSS11_I	CROSS11_F	CROSS11[4:0]				
Reset	–	0x0	0x0	0x0B				
Access Type	–	Write, Read	Write, Read	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
CROSS11_I	6	Inverts CrossX	0b0: Do not invert bit 0b1: Invert bit
CROSS11_F	5	Forces CrossX to 0 before inversion	0b0: Do not force bit to zero 0b1: Force bit to zero
CROSS11	4:0	Maps incoming bit position set by this field to the outgoing bit position 11	0bXXXXX: Incoming bit position

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CROSS_12 (0x1BC)

BIT	7	6	5	4	3	2	1	0
Field	–	CROSS12_I	CROSS12_F	CROSS12[4:0]				
Reset	–	0x0	0x0	0x0C				
Access Type	–	Write, Read	Write, Read	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
CROSS12_I	6	Inverts CrossX	0b0: Do not invert bit 0b1: Invert bit
CROSS12_F	5	Forces CrossX to 0 before inversion	0b0: Do not force bit to zero 0b1: Force bit to zero
CROSS12	4:0	Maps incoming bit position set by this field to the outgoing bit position 12	0bXXXXX: Incoming bit position

CROSS_13 (0x1BD)

BIT	7	6	5	4	3	2	1	0
Field	–	CROSS13_I	CROSS13_F	CROSS13[4:0]				
Reset	–	0x0	0x0	0x0D				
Access Type	–	Write, Read	Write, Read	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
CROSS13_I	6	Inverts CrossX	0b0: Do not invert bit 0b1: Invert bit
CROSS13_F	5	Forces CrossX to 0 before inversion	0b0: Do not force bit to zero 0b1: Force bit to zero
CROSS13	4:0	Maps incoming bit position set by this field to the outgoing bit position 13	0bXXXXX: Incoming bit position

CROSS_14 (0x1BE)

BIT	7	6	5	4	3	2	1	0
Field	–	CROSS14_I	CROSS14_F	CROSS14[4:0]				
Reset	–	0x0	0x0	0x0E				
Access Type	–	Write, Read	Write, Read	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
CROSS14_I	6	Inverts CrossX	0b0: Do not invert bit 0b1: Invert bit
CROSS14_F	5	Forces CrossX to 0 before inversion	0b0: Do not force bit to zero 0b1: Force bit to zero
CROSS14	4:0	Maps incoming bit position set by this field to the outgoing bit position 14	0bXXXXX: Incoming bit position

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CROSS_15 (0x1BF)

BIT	7	6	5	4	3	2	1	0
Field	–	CROSS15_I	CROSS15_F	CROSS15[4:0]				
Reset	–	0x0	0x0	0x0F				
Access Type	–	Write, Read	Write, Read	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
CROSS15_I	6	Inverts CrossX	0b0: Do not invert bit 0b1: Invert bit
CROSS15_F	5	Forces CrossX to 0 before inversion	0b0: Do not force bit to zero 0b1: Force bit to zero
CROSS15	4:0	Maps incoming bit position set by this field to the outgoing bit position 15	0bXXXXX: Incoming bit position

CROSS_16 (0x1C0)

BIT	7	6	5	4	3	2	1	0
Field	–	CROSS16_I	CROSS16_F	CROSS16[4:0]				
Reset	–	0x0	0x0	0x10				
Access Type	–	Write, Read	Write, Read	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
CROSS16_I	6	Inverts CrossX	0b0: Do not invert bit 0b1: Invert bit
CROSS16_F	5	Forces CrossX to 0 before inversion	0b0: Do not force bit to zero 0b1: Force bit to zero
CROSS16	4:0	Maps incoming bit position set by this field to the outgoing bit position 16	0bXXXXX: Incoming bit position

CROSS_17 (0x1C1)

BIT	7	6	5	4	3	2	1	0
Field	–	CROSS17_I	CROSS17_F	CROSS17[4:0]				
Reset	–	0x0	0x0	0x11				
Access Type	–	Write, Read	Write, Read	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
CROSS17_I	6	Inverts CrossX	0b0: Do not invert bit 0b1: Invert bit
CROSS17_F	5	Forces CrossX to 0 before inversion	0b0: Do not force bit to zero 0b1: Force bit to zero
CROSS17	4:0	Maps incoming bit position set by this field to the outgoing bit position 17	0bXXXXX: Incoming bit position

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CROSS_18 (0x1C2)

BIT	7	6	5	4	3	2	1	0
Field	–	CROSS18_I	CROSS18_F	CROSS18[4:0]				
Reset	–	0x0	0x0	0x12				
Access Type	–	Write, Read	Write, Read	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
CROSS18_I	6	Inverts CrossX	0b0: Do not invert bit 0b1: Invert bit
CROSS18_F	5	Forces CrossX to 0 before inversion	0b0: Do not force bit to zero 0b1: Force bit to zero
CROSS18	4:0	Maps incoming bit position set by this field to the outgoing bit position 18	0bXXXXX: Incoming bit position

CROSS_19 (0x1C3)

BIT	7	6	5	4	3	2	1	0
Field	–	CROSS19_I	CROSS19_F	CROSS19[4:0]				
Reset	–	0x0	0x0	0x13				
Access Type	–	Write, Read	Write, Read	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
CROSS19_I	6	Inverts CrossX	0b0: Do not invert bit 0b1: Invert bit
CROSS19_F	5	Forces CrossX to 0 before inversion	0b0: Do not force bit to zero 0b1: Force bit to zero
CROSS19	4:0	Maps incoming bit position set by this field to the outgoing bit position 19	0bXXXXX: Incoming bit position

CROSS_20 (0x1C4)

BIT	7	6	5	4	3	2	1	0
Field	–	CROSS20_I	CROSS20_F	CROSS20[4:0]				
Reset	–	0x0	0x0	0x14				
Access Type	–	Write, Read	Write, Read	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
CROSS20_I	6	Inverts CrossX	0b0: Do not invert bit 0b1: Invert bit
CROSS20_F	5	Forces CrossX to 0 before inversion	0b0: Do not force bit to zero 0b1: Force bit to zero
CROSS20	4:0	Maps incoming bit position set by this field to the outgoing bit position 20	0bXXXXX: Incoming bit position

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CROSS_21 (0x1C5)

BIT	7	6	5	4	3	2	1	0
Field	–	CROSS21_I	CROSS21_F	CROSS21[4:0]				
Reset	–	0x0	0x0	0x15				
Access Type	–	Write, Read	Write, Read	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
CROSS21_I	6	Inverts CrossX	0b0: Do not invert bit 0b1: Invert bit
CROSS21_F	5	Forces CrossX to 0 before inversion	0b0: Do not force bit to zero 0b1: Force bit to zero
CROSS21	4:0	Maps incoming bit position set by this field to the outgoing bit position 21	0bXXXXX: Incoming bit position

CROSS_22 (0x1C6)

BIT	7	6	5	4	3	2	1	0
Field	–	CROSS22_I	CROSS22_F	CROSS22[4:0]				
Reset	–	0x0	0x0	0x16				
Access Type	–	Write, Read	Write, Read	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
CROSS22_I	6	Inverts CrossX	0b0: Do not invert bit 0b1: Invert bit
CROSS22_F	5	Forces CrossX to 0 before inversion	0b0: Do not force bit to zero 0b1: Force bit to zero
CROSS22	4:0	Maps incoming bit position set by this field to the outgoing bit position 22	0bXXXXX: Incoming bit position

CROSS_23 (0x1C7)

BIT	7	6	5	4	3	2	1	0
Field	–	CROSS23_I	CROSS23_F	CROSS23[4:0]				
Reset	–	0x0	0x0	0x17				
Access Type	–	Write, Read	Write, Read	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
CROSS23_I	6	Inverts CrossX	0b0: Do not invert bit 0b1: Invert bit
CROSS23_F	5	Forces CrossX to 0 before inversion	0b0: Do not force bit to zero 0b1: Force bit to zero
CROSS23	4:0	Maps incoming bit position set by this field to the outgoing bit position 23	0bXXXXX: Incoming bit position

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CROSS_24 (0x1C8)

BIT	7	6	5	4	3	2	1	0
Field	–	CROSS24_I	CROSS24_F	CROSS24[4:0]				
Reset	–	0x0	0x0	0x18				
Access Type	–	Write, Read	Write, Read	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
CROSS24_I	6	Inverts CrossX	0b0: Do not invert bit 0b1: Invert bit
CROSS24_F	5	Forces CrossX to 0 before inversion	0b0: Do not force bit to zero 0b1: Force bit to zero
CROSS24	4:0	Maps incoming bit position set by this field to the outgoing bit position 24	0bXXXXX: Incoming bit position

CROSS_25 (0x1C9)

BIT	7	6	5	4	3	2	1	0
Field	–	CROSS25_I	CROSS25_F	CROSS25[4:0]				
Reset	–	0x0	0x0	0x19				
Access Type	–	Write, Read	Write, Read	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
CROSS25_I	6	Inverts CrossX	0b0: Do not invert bit 0b1: Invert bit
CROSS25_F	5	Forces CrossX to 0 before inversion	0b0: Do not force bit to zero 0b1: Force bit to zero
CROSS25	4:0	Maps incoming bit position set by this field to the outgoing bit position 25	0bXXXXX: Incoming bit position

CROSS_26 (0x1CA)

BIT	7	6	5	4	3	2	1	0
Field	–	CROSS26_I	CROSS26_F	CROSS26[4:0]				
Reset	–	0x0	0x0	0x1A				
Access Type	–	Write, Read	Write, Read	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
CROSS26_I	6	Inverts CrossX	0b0: Do not invert bit 0b1: Invert bit
CROSS26_F	5	Forces CrossX to 0 before inversion	0b0: Do not force bit to zero 0b1: Force bit to zero
CROSS26	4:0	Maps incoming bit position set by this field to the outgoing bit position 26	0bXXXXX: Incoming bit position

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CROSS_27 (0x1CB)

BIT	7	6	5	4	3	2	1	0
Field	–	CROSS27_I	CROSS27_F	CROSS27[4:0]				
Reset	–	0x0	0x0	0x1B				
Access Type	–	Write, Read	Write, Read	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
CROSS27_I	6	Inverts CrossX	0b0: Do not invert bit 0b1: Invert bit
CROSS27_F	5	Forces CrossX to 0 before inversion	0b0: Do not force bit to zero 0b1: Force bit to zero
CROSS27	4:0	Maps incoming bit position set by this field to the outgoing bit position 27	0bXXXXX: Incoming bit position

PRBS_ERR (0x1CC)

BIT	7	6	5	4	3	2	1	0
Field	VPRBS_ERR[7:0]							
Reset	0x00							
Access Type	Read Clears All							

BITFIELD	BITS	DESCRIPTION	DECODE
VPRBS_ERR	7:0	Video PRBS error counter, clears on read	0xXX: Number of video PRBS errors since last read

OLDI0 (0x1CD)

BIT	7	6	5	4	3	2	1	0
Field	–	–	VPRBS_FAIL	VPRBS_CHK_EN	HS_OUT_EN	LUT_C_EN	LUT_B_EN	LUT_A_EN
Reset	–	–	0b0	0b0	0b0	0x0	0x0	0x0
Access Type	–	–	Read Only	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
VPRBS_FAIL	5	Video PRBS check pass/fail	0b0: Video PRBS check passed 0b1: Video PRBS check failed
VPRBS_CHK_EN	4	Enables video PRBS checker	0b0: Video PRBS checker disabled 0b1: Video PRBS checker enabled
HS_OUT_EN	3	Output HSYNC from GPIO	0b0: HSYNC not output 0b1: Output HSYNC from GPIO
LUT_C_EN	2	Enables color C lookup table (LUT) [23:16]	0b0: Color C LUT disabled 0b1: Color C LUT enabled
LUT_B_EN	1	Enables color B lookup table (LUT) [15:8]	0b0: Color B LUT disabled 0b1: Color B LUT enabled
LUT_A_EN	0	Enables color A lookup table (LUT) [7:0]	0b0: Color A LUT disabled 0b1: Color A LUT enabled

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OLDI1 (0x1CE)

BIT	7	6	5	4	3	2	1	0
Field	OLDI_OUTSEL	OLDI_FORMAT	OLDI_4TH_LANE	OLDI_SWAP_AB	OLDI_SPL_EN	OLDI_SPL_MODE[1:0]		OLDI_SPL_POL
Reset	0x0	0x0	0x0	0x0	0x0	0x3		0x1
Access Type	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read		Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
OLDI_OUTSEL	7	Selects which OLDI port to use for data output	0b0: Port A 0b1: Port B
OLDI_FORMAT	6	Output format of LVDS port	0b0: OLDI 0b1: VESA
OLDI_4TH_LANE	5	OLDI lane count	0b0: 4 lanes 0b1: 3 lanes
OLDI_SWAP_AB	4	Swaps OLDI ports A and B	0b0: Ports A and B not swapped 0b1: Ports A and B swapped
OLDI_SPL_EN	3	OLDI splitter enable	0b0: Splitter disabled 0b1: Splitter enabled
OLDI_SPL_MODE	2:1	OLDI splitter mode	0b00: Random 0b01: Split with HS 0b10: Split with VS 0b11: Split with DE
OLDI_SPL_POL	0	Set to 0 for dual OLDI applications	0b0: Falling Edge 0b1: Rising Edge

OLDI2 (0x1CF)

BIT	7	6	5	4	3	2	1	0
Field	PD_LVDS_B	PD_LVDS_A	–	VS_OUT_EN	–	–	OLDI_DUP	SSEN
Reset	0b0	0b0	–	0b0	–	–	0b0	0b0
Access Type	Write, Read	Write, Read	–	Write, Read	–	–	Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
PD_LVDS_B	7	Selects power-up/power-down status for LVDS output driver B	0b0: LVDS ouptut driver B powered up 0b1: LVDS output driver B powered down
PD_LVDS_A	6	Selects power-up/power-down status for LVDS output driver A	0b0: LVDS ouput driver A powered up 0b1: LVDS output driver A powered down
VS_OUT_EN	4	Selects whether or not to output VSYNC from GPIO	0b0: VSYNC not output 0b1: Output VSYNC from GPIO
OLDI_DUP	1	Selects whether or not to duplicate OLDI at both ports	0b0: OLDI not duplicated 0b1: OLDI duplicated at both ports
SSEN	0	Enables spread spectrum OLDI output clock. Spread percentage is adjusted from OLDIPLL registers. Default percentage is 0.5%.	0b0: Spread spectrum disabled 0b1: Spread spectrum enabled

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OLDI3 (0x1D0)

BIT	7	6	5	4	3	2	1	0
Field	LANE_INV_B0	LANE_SEL_B0[2:0]			LANE_INV_A0	LANE_SEL_A0[2:0]		
Reset	0b0	0x0			0b0	0x0		
Access Type	Write, Read	Write, Read			Write, Read	Write, Read		

BITFIELD	BITS	DESCRIPTION	DECODE
LANE_INV_B0	7	Lane B0 polarity inversion	0b0: Lane polarity not inverted 0b1: Lane polarity inverted
LANE_SEL_B0	6:4	Selects OLDI Port B Lane 0 output pins	0b000: Output from B0 0b001: Output from B1 0b010: Output from B2 0b011: Output from B3 0b100: Output from BCLK 0b101: Reserved 0b110: Reserved 0b111: Reserved
LANE_INV_A0	3	Lane A0 polarity inversion	0b0: Lane polarity not inverted 0b1: Lane polarity inverted
LANE_SEL_A0	2:0	Selects OLDI Port A Lane 0 output pins	0b000: Output from A0 0b001: Output from A1 0b010: Output from A2 0b011: Output from A3 0b100: Output from ACLK 0b101: Reserved 0b110: Reserved 0b111: Reserved

OLDI4 (0x1D1)

BIT	7	6	5	4	3	2	1	0
Field	LANE_INV_B1	LANE_SEL_B1[2:0]			LANE_INV_A1	LANE_SEL_A1[2:0]		
Reset	0b0	0x1			0b0	0x1		
Access Type	Write, Read	Write, Read			Write, Read	Write, Read		

BITFIELD	BITS	DESCRIPTION	DECODE
LANE_INV_B1	7	Lane B1 polarity inversion	0b0: Lane polarity not inverted 0b1: Lane polarity inverted
LANE_SEL_B1	6:4	Selects OLDI Port B Lane 1 output pins	0b000: Output from B0 0b001: Output from B1 0b010: Output from B2 0b011: Output from B3 0b100: Output from BCLK 0b101: Reserved 0b110: Reserved 0b111: Reserved
LANE_INV_A1	3	Lane A1 polarity inversion	0b0: Lane polarity not inverted 0b1: Lane polarity inverted

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BITFIELD	BITS	DESCRIPTION	DECODE
LANE_SEL_A1	2:0	Selects OLDI Port A Lane 1 output pins	0b000: Output from A0 0b001: Output from A1 0b010: Output from A2 0b011: Output from A3 0b100: Output from ACLK 0b101: Reserved 0b110: Reserved 0b111: Reserved

OLDI5 (0x1D2)

BIT	7	6	5	4	3	2	1	0
Field	LANE_INV_B2	LANE_SEL_B2[2:0]			LANE_INV_A2	LANE_SEL_A2[2:0]		
Reset	0b0	0x2			0b0	0x2		
Access Type	Write, Read	Write, Read			Write, Read	Write, Read		

BITFIELD	BITS	DESCRIPTION	DECODE
LANE_INV_B2	7	Lane B2 polarity inversion	0b0: Lane polarity not inverted 0b1: Lane polarity inverted
LANE_SEL_B2	6:4	Selects OLDI Port B Lane 2 output pins	0b000: Output from B0 0b001: Output from B1 0b010: Output from B2 0b011: Output from B3 0b100: Output from BCLK 0b101: Reserved 0b110: Reserved 0b111: Reserved
LANE_INV_A2	3	Lane A2 polarity inversion	0b0: Lane polarity not inverted 0b1: Lane polarity inverted
LANE_SEL_A2	2:0	Selects OLDI Port A Lane 2 output pins	0b000: Output from A0 0b001: Output from A1 0b010: Output from A2 0b011: Output from A3 0b100: Output from ACLK 0b101: Reserved 0b110: Reserved 0b111: Reserved

OLDI6 (0x1D3)

BIT	7	6	5	4	3	2	1	0
Field	LANE_INV_B3	LANE_SEL_B3[2:0]			LANE_INV_A3	LANE_SEL_A3[2:0]		
Reset	0b0	0x3			0b0	0x3		
Access Type	Write, Read	Write, Read			Write, Read	Write, Read		

BITFIELD	BITS	DESCRIPTION	DECODE
LANE_INV_B3	7	Lane B3 polarity inversion	0b0: Lane polarity not inverted 0b1: Lane polarity inverted

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BITFIELD	BITS	DESCRIPTION	DECODE
LANE_SEL_B3	6:4	Selects OLDI Port B Lane 3 output pins	0b000: Output from B0 0b001: Output from B1 0b010: Output from B2 0b011: Output from B3 0b100: Output from BCLK 0b101: Reserved 0b110: Reserved 0b111: Reserved
LANE_INV_A3	3	Lane A3 polarity inversion	0b0: Lane polarity not inverted 0b1: Lane polarity inverted
LANE_SEL_A3	2:0	Selects OLDI Port A Lane 3 output pins	0b000: Output from A0 0b001: Output from A1 0b010: Output from A2 0b011: Output from A3 0b100: Output from ACLK 0b101: Reserved 0b110: Reserved 0b111: Reserved

OLDI7 (0x1D4)

BIT	7	6	5	4	3	2	1	0
Field	LANE_INV_BCLK	LANE_SEL_BCLK[2:0]			LANE_INV_ACLK	LANE_SEL_ACLK[2:0]		
Reset	0b0	0x4			0b0	0x4		
Access Type	Write, Read	Write, Read			Write, Read	Write, Read		

BITFIELD	BITS	DESCRIPTION	DECODE
LANE_INV_BCLK	7	Lane BCLK polarity inversion	0b0: Lane polarity not inverted 0b1: Lane polarity inverted
LANE_SEL_BCLK	6:4	Selects OLDI Port B Clock Lane output pins	0b000: Output from B0 0b001: Output from B1 0b010: Output from B2 0b011: Output from B3 0b100: Output from BCLK 0b101: Reserved 0b110: Reserved 0b111: Reserved
LANE_INV_ACLK	3	Lane ACLK polarity inversion	0b0: Lane polarity not inverted 0b1: Lane polarity inverted
LANE_SEL_ACLK	2:0	Selects OLDI Port A Clock Lane output pins	0b000: Output from A0 0b001: Output from A1 0b010: Output from A2 0b011: Output from A3 0b100: Output from ACLK 0b101: Reserved 0b110: Reserved 0b111: Reserved

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GPIO_A (0x200)

BIT	7	6	5	4	3	2	1	0
Field	RES_CFG	TX_PRIO	TX_COMP_EN	GPIO_OUT	GPIO_IN	GPIO_RX_EN	GPIO_TX_EN	GPIO_OUT_DIS
Reset	0b1	0x0	0x0	0b0	0b0	0b0	0b1	0b1
Access Type	Write, Read	Write, Read	Write, Read	Write, Read	Read Only	Write, Read	Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
RES_CFG	7	Resistor pullup/pulldown strength	0b0: 40kΩ 0b1: 1MΩ
TX_PRIO	6	Priority for GPIO scheduling	0b0: Low priority 0b1: High priority
TX_COMP_EN	5	Jitter minimization compensation enable	0b0: Jitter compensation disabled 0b1: Jitter compensation enabled
GPIO_OUT	4	GPIO pin output drive value when GPIO_RX_EN = 0	0b0: This GPIO pin output is driven to 0 0b1: This GPIO pin output is driven to 1
GPIO_IN	3	GPIO pin input level	0b0: This GPIO pin value is 0 0b1: This GPIO pin value is 1
GPIO_RX_EN	2	GPIO Out source control	0b0: This GPIO source disabled for GMSL2 reception 0b1: This GPIO source enabled for GMSL2 reception
GPIO_TX_EN	1	GPIO TX source control	0b0: This GPIO source disabled for GMSL2 transmission 0b1: This GPIO source enabled for GMSL2 transmission
GPIO_OUT_DIS	0	Disable GPIO output driver	0b0: Output driver enabled 0b1: Output driver disabled

GPIO_B (0x201)

BIT	7	6	5	4	3	2	1	0
Field	PULL_UPDN_SEL[1:0]		OUT_TYPE	GPIO_TX_ID[4:0]				
Reset	0x2		0b1	0x00				
Access Type	Write, Read		Write, Read	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
PULL_UPDN_SEL	7:6	Buffer pullup/pulldown configuration	0b00: None 0b01: Pullup 0b10: Pulldown 0b11: Reserved
OUT_TYPE	5	Driver type selection	0b0: Open-drain 0b1: Push-pull
GPIO_TX_ID	4:0	GPIO ID for pin while transmitting	0bXXXXX: GPIO transmit ID

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GPIO_C (0x202)

BIT	7	6	5	4	3	2	1	0
Field	OVR_RES_CFG	–	–	GPIO_RX_ID[4:0]				
Reset	0x0	–	–	0x00				
Access Type	Write, Read	–	–	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
OVR_RES_CFG	7	Overrides non-GPIO port function IO setting. When set, RES_CFGr and PULL_UPDN_SEL are effective when pin is configured as non-GPIO. When cleared, non-GPIO pin function determines IO type.	0b0: Non-GPIO function determines IO type when alternate function is selected 0b1: RES_CFG and PULL_UPDN_SEL determine IO type for non-GPIO configuration
GPIO_RX_ID	4:0	GPIO ID for pin while receiving	0bXXXXX: This GPIO receive ID

GPIO_A (0x203)

BIT	7	6	5	4	3	2	1	0
Field	RES_CFG	TX_PRIO	TX_COMP_EN	GPIO_OUT	GPIO_IN	GPIO_RX_EN	GPIO_TX_EN	GPIO_OUT_DIS
Reset	0b1	0x0	0x0	0b0	0b0	0b1	0b0	0b0
Access Type	Write, Read	Write, Read	Write, Read	Write, Read	Read Only	Write, Read	Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
RES_CFG	7	Resistor pullup/pulldown strength	0b0: 40kΩ 0b1: 1MΩ
TX_PRIO	6	Priority for GPIO scheduling	0b0: Low priority 0b1: High priority
TX_COMP_EN	5	Jitter minimization compensation enable	0b0: Jitter compensation disabled 0b1: Jitter compensation enabled
GPIO_OUT	4	GPIO pin output drive value when GPIO_RX_EN = 0	0b0: This GPIO pin output is driven to 0 0b1: This GPIO pin output is driven to 1
GPIO_IN	3	GPIO pin input level	0b0: This GPIO pin value is 0 0b1: This GPIO pin value is 1
GPIO_RX_EN	2	GPIO Out source control	0b0: This GPIO source disabled for GMSL2 reception 0b1: This GPIO source enabled for GMSL2 reception
GPIO_TX_EN	1	GPIO TX source control	0b0: This GPIO source disabled for GMSL2 transmission 0b1: This GPIO source enabled for GMSL2 transmission
GPIO_OUT_DIS	0	Disables GPIO output driver	0b0: Output driver enabled 0b1: Output driver disabled

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GPIO_B (0x204)

BIT	7	6	5	4	3	2	1	0
Field	PULL_UPDN_SEL[1:0]		OUT_TYPE	GPIO_TX_ID[4:0]				
Reset	0x2		0b1	0x01				
Access Type	Write, Read		Write, Read	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
PULL_UPDN_SEL	7:6	Buffer pull up/down configuration	0b00: None 0b01: Pullup 0b10: Pulldown 0b11: Reserved
OUT_TYPE	5	Driver type selection	0b0: Open-drain 0b1: Push-pull
GPIO_TX_ID	4:0	GPIO ID for pin while transmitting	0bXXXXX: GPIO transmit ID

GPIO_C (0x205)

BIT	7	6	5	4	3	2	1	0
Field	OVR_RES_CFG	–	–	GPIO_RX_ID[4:0]				
Reset	0x0	–	–	0x01				
Access Type	Write, Read	–	–	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
OVR_RES_CFG	7	Overrides non-GPIO port function IO setting. When set, RES_CFGr and PULL_UPDN_SEL are effective when pin is configured as non-GPIO. When cleared, non-GPIO pin function determines IO type.	0b0: Non-GPIO function determines IO type when alternate function is selected 0b1: RES_CFG and PULL_UPDN_SEL determine IO type for non-GPIO configuration
GPIO_RX_ID	4:0	GPIO ID for pin while receiving	0bXXXXX: GPIO receive ID

GPIO_A (0x206)

BIT	7	6	5	4	3	2	1	0
Field	RES_CFG	TX_PRIO	TX_COMP_EN	GPIO_OUT	GPIO_IN	GPIO_RX_EN	GPIO_TX_EN	GPIO_OUT_DIS
Reset	0b1	0x0	0x0	0b0	0b0	0b0	0b0	0b1
Access Type	Write, Read	Write, Read	Write, Read	Write, Read	Read Only	Write, Read	Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
RES_CFG	7	Resistor pullup/pulldown strength	0b0: 40kΩ 0b1: 1MΩ
TX_PRIO	6	Priority for GPIO scheduling	0b0: Low priority 0b1: High priority
TX_COMP_EN	5	Jitter minimization compensation enable	0b0: Jitter compensation disabled 0b1: Jitter compensation enabled
GPIO_OUT	4	GPIO pin output drive value when GPIO_RX_EN = 0	0b0: This GPIO pin output is driven to 0 0b1: This GPIO pin output is driven to 1

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BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_IN	3	GPIO pin input level	0b0: This GPIO pin value is 0 0b1: This GPIO pin value is 1
GPIO_RX_EN	2	GPIO Out source control	0b0: This GPIO source disabled for GMSL2 reception 0b1: This GPIO source enabled for GMSL2 reception
GPIO_TX_EN	1	GPIO TX source control	0b0: This GPIO source disabled for GMSL2 transmission 0b1: This GPIO source enabled for GMSL2 transmission
GPIO_OUT_DIS	0	Disables GPIO output driver	0b0: Output driver enabled 0b1: Output driver disabled

GPIO_B (0x207)

BIT	7	6	5	4	3	2	1	0
Field	PULL_UPDN_SEL[1:0]		OUT_TYPE	GPIO_TX_ID[4:0]				
Reset	0x2		0b1	0x02				
Access Type	Write, Read		Write, Read	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
PULL_UPDN_SEL	7:6	Buffer pullup/pulldown configuration	0b00: None 0b01: Pullup 0b10: Pulldown 0b11: Reserved
OUT_TYPE	5	Driver type selection	0b0: Open-drain 0b1: Push-pull
GPIO_TX_ID	4:0	GPIO ID for pin while transmitting	0bXXXXX: GPIO transmit ID

GPIO_C (0x208)

BIT	7	6	5	4	3	2	1	0
Field	OVR_RES_CFG	–	–	GPIO_RX_ID[4:0]				
Reset	0x0	–	–	0x02				
Access Type	Write, Read	–	–	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
OVR_RES_CFG	7	Overrides non-GPIO port function IO setting. When set, RES_CFGr and PULL_UPDN_SEL are effective when pin is configured as non-GPIO. When cleared, non-GPIO pin function determines IO type.	0b0: Non-GPIO function determines IO type when alternate function is selected 0b1: RES_CFG and PULL_UPDN_SEL determine IO type for non-GPIO configuration
GPIO_RX_ID	4:0	GPIO ID for pin while receiving	0bXXXXX: GPIO receive ID

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GPIO_A (0x209)

BIT	7	6	5	4	3	2	1	0
Field	RES_CFG	TX_PRIO	TX_COMP_EN	GPIO_OUT	GPIO_IN	GPIO_RX_EN	GPIO_TX_EN	GPIO_OUT_DIS
Reset	0b1	0x0	0x0	0b0	0b0	0b0	0b0	0b1
Access Type	Write, Read	Write, Read	Write, Read	Write, Read	Read Only	Write, Read	Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
RES_CFG	7	Resistor pullup/pulldown strength	0b0: 40kΩ 0b1: 1MΩ
TX_PRIO	6	Priority for GPIO scheduling	0b0: Low priority 0b1: High priority
TX_COMP_EN	5	Jitter minimization compensation enable	0b0: Jitter compensation disabled 0b1: Jitter compensation enabled
GPIO_OUT	4	GPIO pin output drive value when GPIO_RX_EN = 0	0b0: This GPIO pin output is driven to 0 0b1: This GPIO pin output is driven to 1
GPIO_IN	3	GPIO pin input level	0b0: This GPIO pin value is 0 0b1: This GPIO pin value is 1
GPIO_RX_EN	2	GPIO Out source control	0b0: This GPIO source disabled for GMSL2 reception 0b1: This GPIO source enabled for GMSL2 reception
GPIO_TX_EN	1	GPIO TX source control	0b0: This GPIO source disabled for GMSL2 transmission 0b1: This GPIO source enabled for GMSL2 transmission
GPIO_OUT_DIS	0	Disables GPIO output driver	0b0: Output driver enabled 0b1: Output driver disabled

GPIO_B (0x20A)

BIT	7	6	5	4	3	2	1	0
Field	PULL_UPDN_SEL[1:0]		OUT_TYPE	GPIO_TX_ID[4:0]				
Reset	0x2		0b1	0x03				
Access Type	Write, Read		Write, Read	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
PULL_UPDN_SEL	7:6	Buffer pullup/pulldown configuration	0b00: None 0b01: Pullup 0b10: Pulldown 0b11: Reserved
OUT_TYPE	5	Driver type selection	0b0: Open-drain 0b1: Push-pull
GPIO_TX_ID	4:0	GPIO ID for pin while transmitting	0bXXXXX: GPIO transmit ID

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GPIO_C (0x20B)

BIT	7	6	5	4	3	2	1	0
Field	OVR_RES_CFG	–	–	GPIO_RX_ID[4:0]				
Reset	0x0	–	–	0x03				
Access Type	Write, Read	–	–	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
OVR_RES_CFG	7	Overrides non-GPIO port function IO setting. When set, RES_CFGr and PULL_UPDN_SEL are effective when pin is configured as non-GPIO. When cleared, non-GPIO pin function determines IO type.	0b0: Non-GPIO function determines IO type when alternate function is selected 0b1: RES_CFG and PULL_UPDN_SEL determine IO type for non-GPIO configuration
GPIO_RX_ID	4:0	GPIO ID for pin while receiving	0bXXXXX: GPIO receive ID

GPIO_A (0x20C)

BIT	7	6	5	4	3	2	1	0
Field	RES_CFG	TX_PRIO	TX_COMP_EN	GPIO_OUT	GPIO_IN	GPIO_RX_EN	GPIO_TX_EN	GPIO_OUT_DIS
Reset	0b1	0x0	0x0	0b0	0b0	0b0	0b0	0b1
Access Type	Write, Read	Write, Read	Write, Read	Write, Read	Read Only	Write, Read	Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
RES_CFG	7	Resistor pullup/pulldown strength	0b0: 40kΩ 0b1: 1MΩ
TX_PRIO	6	Priority for GPIO scheduling	0b0: Low priority 0b1: High priority
TX_COMP_EN	5	Jitter minimization compensation enable	0b0: Jitter compensation disabled 0b1: Jitter compensation enabled
GPIO_OUT	4	GPIO pin output drive value when GPIO_RX_EN = 0	0b0: This GPIO pin output is driven to 0 0b1: This GPIO pin output is driven to 1
GPIO_IN	3	GPIO pin input level	0b0: This GPIO pin value is 0 0b1: This GPIO pin value is 1
GPIO_RX_EN	2	GPIO Out source control	0b0: This GPIO source disabled for GMSL2 reception 0b1: This GPIO source enabled for GMSL2 reception
GPIO_TX_EN	1	GPIO TX source control	0b0: This GPIO source disabled for GMSL2 transmission 0b1: This GPIO source enabled for GMSL2 transmission
GPIO_OUT_DIS	0	Disables GPIO output driver	0b0: Output driver enabled 0b1: Output driver disabled

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GPIO_B (0x20D)

BIT	7	6	5	4	3	2	1	0
Field	PULL_UPDN_SEL[1:0]		OUT_TYPE	GPIO_TX_ID[4:0]				
Reset	0x2		0b1	0x04				
Access Type	Write, Read		Write, Read	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
PULL_UPDN_SEL	7:6	Buffer pullup/pulldown configuration	0b00: None 0b01: Pullup 0b10: Pulldown 0b11: Reserved
OUT_TYPE	5	Driver type selection	0b0: Open-drain 0b1: Push-pull
GPIO_TX_ID	4:0	GPIO ID for pin while transmitting	0bXXXXX: GPIO transmit ID

GPIO_C (0x20E)

BIT	7	6	5	4	3	2	1	0
Field	OVR_RES_CFG	–	–	GPIO_RX_ID[4:0]				
Reset	0x0	–	–	0x04				
Access Type	Write, Read	–	–	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
OVR_RES_CFG	7	Overrides non-GPIO port function IO setting. When set, RES_CFGr and PULL_UPDN_SEL are effective when pin is configured as non-GPIO. When cleared, non-GPIO pin function determines IO type.	0b0: Non-GPIO function determines IO type when alternate function is selected 0b1: RES_CFG and PULL_UPDN_SEL determine IO type for non-GPIO configuration
GPIO_RX_ID	4:0	GPIO ID for pin while receiving	0bXXXXX: GPIO receive ID

GPIO_A (0x20F)

BIT	7	6	5	4	3	2	1	0
Field	RES_CFG	TX_PRIO	TX_COMP_EN	GPIO_OUT	GPIO_IN	GPIO_RX_EN	GPIO_TX_EN	GPIO_OUT_DIS
Reset	0b1	0x0	0x0	0b0	0b0	0b0	0b0	0b1
Access Type	Write, Read	Write, Read	Write, Read	Write, Read	Read Only	Write, Read	Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
RES_CFG	7	Resistor pullup/pulldown strength	0b0: 40kΩ 0b1: 1MΩ
TX_PRIO	6	Priority for GPIO scheduling	0b0: Low priority 0b1: High priority
TX_COMP_EN	5	Jitter minimization compensation enable	0b0: Jitter compensation disabled 0b1: Jitter compensation enabled
GPIO_OUT	4	GPIO pin output drive value when GPIO_RX_EN = 0	0b0: This GPIO pin output is driven to 0 0b1: This GPIO pin output is driven to 1

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BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_IN	3	GPIO pin input level	0b0: This GPIO pin value is 0 0b1: This GPIO pin value is 1
GPIO_RX_EN	2	GPIO Out source control	0b0: This GPIO source disabled for GMSL2 reception 0b1: This GPIO source enabled for GMSL2 reception
GPIO_TX_EN	1	GPIO TX source control	0b0: This GPIO source disabled for GMSL2 transmission 0b1: This GPIO source enabled for GMSL2 transmission
GPIO_OUT_DIS	0	Disables GPIO output driver	0b0: Output driver enabled 0b1: Output driver disabled

GPIO_B (0x210)

BIT	7	6	5	4	3	2	1	0
Field	PULL_UPDN_SEL[1:0]		OUT_TYPE	GPIO_TX_ID[4:0]				
Reset	0x2		0b1	0x05				
Access Type	Write, Read		Write, Read	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
PULL_UPDN_SEL	7:6	Buffer pullup/pulldown configuration	0b00: None 0b01: Pullup 0b10: Pulldown 0b11: Reserved
OUT_TYPE	5	Driver type selection	0b0: Open-drain 0b1: Push-pull
GPIO_TX_ID	4:0	GPIO ID for pin while transmitting	0bXXXXX: GPIO transmit ID

GPIO_C (0x211)

BIT	7	6	5	4	3	2	1	0
Field	OVR_RES_CFG	–	–	GPIO_RX_ID[4:0]				
Reset	0x0	–	–	0x05				
Access Type	Write, Read	–	–	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
OVR_RES_CFG	7	Overrides non-GPIO port function IO setting. When set, RES_CFGr and PULL_UPDN_SEL are effective when pin is configured as non-GPIO. When cleared, non-GPIO pin function determines IO type.	0b0: Non-GPIO function determines IO type when alternate function is selected 0b1: RES_CFG and PULL_UPDN_SEL determine IO type for non-GPIO configuration
GPIO_RX_ID	4:0	GPIO ID for pin while receiving	0bXXXXX: GPIO receive ID

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GPIO_A (0x212)

BIT	7	6	5	4	3	2	1	0
Field	RES_CFG	TX_PRIO	TX_COMP_EN	GPIO_OUT	GPIO_IN	GPIO_RX_EN	GPIO_TX_EN	GPIO_OUT_DIS
Reset	0b1	0x0	0x0	0b0	0b0	0b0	0b0	0b1
Access Type	Write, Read	Write, Read	Write, Read	Write, Read	Read Only	Write, Read	Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
RES_CFG	7	Resistor pullup/pulldown strength	0b0: 40kΩ 0b1: 1MΩ
TX_PRIO	6	Priority for GPIO scheduling	0b0: Low priority 0b1: High priority
TX_COMP_EN	5	Jitter minimization compensation enable	0b0: Jitter compensation disabled 0b1: Jitter compensation enabled
GPIO_OUT	4	GPIO pin output drive value when GPIO_RX_EN = 0	0b0: This GPIO pin output is driven to 0 0b1: This GPIO pin output is driven to 1
GPIO_IN	3	GPIO pin input level	0b0: This GPIO pin value is 0 0b1: This GPIO pin value is 1
GPIO_RX_EN	2	GPIO Out source control	0b0: This GPIO source disabled for GMSL2 reception 0b1: This GPIO source enabled for GMSL2 reception
GPIO_TX_EN	1	GPIO TX source control	0b0: This GPIO source disabled for GMSL2 transmission 0b1: This GPIO source enabled for GMSL2 transmission
GPIO_OUT_DIS	0	Disables GPIO output driver	0b0: Output driver enabled 0b1: Output driver disabled

GPIO_B (0x213)

BIT	7	6	5	4	3	2	1	0
Field	PULL_UPDN_SEL[1:0]		OUT_TYPE	GPIO_TX_ID[4:0]				
Reset	0x2		0b1	0x06				
Access Type	Write, Read		Write, Read	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
PULL_UPDN_SEL	7:6	Buffer pullup/pulldown configuration	0b00: None 0b01: Pullup 0b10: Pulldown 0b11: Reserved
OUT_TYPE	5	Driver type selection	0b0: Open-drain 0b1: Push-pull
GPIO_TX_ID	4:0	GPIO ID for pin while transmitting	0bXXXXX: GPIO transmit ID

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GPIO_C (0x214)

BIT	7	6	5	4	3	2	1	0
Field	OVR_RES_CFG	–	–	GPIO_RX_ID[4:0]				
Reset	0x0	–	–	0x06				
Access Type	Write, Read	–	–	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
OVR_RES_CFG	7	Overrides non-GPIO port function IO setting. When set, RES_CFGr and PULL_UPDN_SEL are effective when pin is configured as non-GPIO. When cleared, non-GPIO pin function determines IO type.	0b0: Non-GPIO function determines IO type when alternate function is selected 0b1: RES_CFG and PULL_UPDN_SEL determine IO type for non-GPIO configuration
GPIO_RX_ID	4:0	GPIO ID for pin while receiving	0bXXXXX: GPIO receive ID

GPIO_A (0x215)

BIT	7	6	5	4	3	2	1	0
Field	RES_CFG	TX_PRIO	TX_COMP_EN	GPIO_OUT	GPIO_IN	GPIO_RX_EN	GPIO_TX_EN	GPIO_OUT_DIS
Reset	0b1	0x0	0x0	0b0	0b0	0b0	0b0	0b1
Access Type	Write, Read	Write, Read	Write, Read	Write, Read	Read Only	Write, Read	Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
RES_CFG	7	Resistor pullup/pulldown strength	0b0: 40kΩ 0b1: 1MΩ
TX_PRIO	6	Priority for GPIO scheduling	0b0: Low priority 0b1: High priority
TX_COMP_EN	5	Jitter minimization compensation enable	0b0: Jitter compensation disabled 0b1: Jitter compensation enabled
GPIO_OUT	4	GPIO pin output drive value when GPIO_RX_EN = 0	0b0: This GPIO pin output is driven to 0 0b1: This GPIO pin output is driven to 1
GPIO_IN	3	GPIO pin input level	0b0: This GPIO pin value is 0 0b1: This GPIO pin value is 1
GPIO_RX_EN	2	GPIO Out source control	0b0: This GPIO source disabled for GMSL2 reception 0b1: This GPIO source enabled for GMSL2 reception
GPIO_TX_EN	1	GPIO TX source control	0b0: This GPIO source disabled for GMSL2 transmission 0b1: This GPIO source enabled for GMSL2 transmission
GPIO_OUT_DIS	0	Disables GPIO output driver	0b0: Output driver enabled 0b1: Output driver disabled

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GPIO_B (0x216)

BIT	7	6	5	4	3	2	1	0
Field	PULL_UPDN_SEL[1:0]		OUT_TYPE	GPIO_TX_ID[4:0]				
Reset	0x2		0b1	0x07				
Access Type	Write, Read		Write, Read	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
PULL_UPDN_SEL	7:6	Buffer pullup/pulldown configuration	0b00: None 0b01: Pullup 0b10: Pulldown 0b11: Reserved
OUT_TYPE	5	Driver type selection	0b0: Open-drain 0b1: Push-pull
GPIO_TX_ID	4:0	GPIO ID for pin while transmitting	0bXXXXX: GPIO transmit ID

GPIO_C (0x217)

BIT	7	6	5	4	3	2	1	0
Field	OVR_RES_CFG	–	–	GPIO_RX_ID[4:0]				
Reset	0x0	–	–	0x07				
Access Type	Write, Read	–	–	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
OVR_RES_CFG	7	Overrides non-GPIO port function IO setting. When set, RES_CFGr and PULL_UPDN_SEL are effective when pin is configured as non-GPIO. When cleared, non-GPIO pin function determines IO type.	0b0: Non-GPIO function determines IO type when alternate function is selected 0b1: RES_CFG and PULL_UPDN_SEL determine IO type for non-GPIO configuration
GPIO_RX_ID	4:0	GPIO ID for pin while receiving	0bXXXXX: GPIO receive ID

GPIO_A (0x218)

BIT	7	6	5	4	3	2	1	0
Field	RES_CFG	TX_PRIO	TX_COMP_EN	GPIO_OUT	GPIO_IN	GPIO_RX_EN	GPIO_TX_EN	GPIO_OUT_DIS
Reset	0b1	0x0	0x0	0b0	0b0	0b0	0b0	0b1
Access Type	Write, Read	Write, Read	Write, Read	Write, Read	Read Only	Write, Read	Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
RES_CFG	7	Resistor pullup/pulldown strength	0b0: 40kΩ 0b1: 1MΩ
TX_PRIO	6	Priority for GPIO scheduling	0b0: Low priority 0b1: High priority
TX_COMP_EN	5	Jitter minimization compensation enable	0b0: Jitter compensation disabled 0b1: Jitter compensation enabled
GPIO_OUT	4	GPIO pin output drive value when GPIO_RX_EN = 0	0b0: This GPIO pin output is driven to 0 0b1: This GPIO pin output is driven to 1

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BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_IN	3	GPIO pin input level	0b0: This GPIO pin value is 0 0b1: This GPIO pin value is 1
GPIO_RX_EN	2	GPIO Out source control	0b0: This GPIO source disabled for GMSL2 reception 0b1: This GPIO source enabled for GMSL2 reception
GPIO_TX_EN	1	GPIO TX source control	0b0: This GPIO source disabled for GMSL2 transmission 0b1: This GPIO source enabled for GMSL2 transmission
GPIO_OUT_DIS	0	Disables GPIO output driver	0b0: Output driver enabled 0b1: Output driver disabled

GPIO_B (0x219)

BIT	7	6	5	4	3	2	1	0
Field	PULL_UPDN_SEL[1:0]		OUT_TYPE	GPIO_TX_ID[4:0]				
Reset	0x2		0b1	0x08				
Access Type	Write, Read		Write, Read	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
PULL_UPDN_SEL	7:6	Buffer pull up/down configuration	0b00: None 0b01: Pullup 0b10: Pulldown 0b11: Reserved
OUT_TYPE	5	Driver type selection	0b0: Open-drain 0b1: Push-pull
GPIO_TX_ID	4:0	GPIO ID for pin while transmitting	0bXXXXX: GPIO transmit ID

GPIO_C (0x21A)

BIT	7	6	5	4	3	2	1	0
Field	OVR_RES_CFG	–	–	GPIO_RX_ID[4:0]				
Reset	0x0	–	–	0x08				
Access Type	Write, Read	–	–	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
OVR_RES_CFG	7	Overrides non-GPIO port function IO setting. When set, RES_CFG and PULL_UPDN_SEL are effective when pin is configured as non-GPIO. When cleared, non-GPIO pin function determines IO type.	0b0: Non-GPIO function determines IO type when alternate function is selected 0b1: RES_CFG and PULL_UPDN_SEL determine IO type for non-GPIO configuration
GPIO_RX_ID	4:0	GPIO ID for pin while receiving	0bXXXXX: GPIO receive ID

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GPIO_A (0x21B)

BIT	7	6	5	4	3	2	1	0
Field	RES_CFG	TX_PRIO	TX_COMP_EN	GPIO_OUT	GPIO_IN	GPIO_RX_EN	GPIO_TX_EN	GPIO_OUT_DIS
Reset	0b1	0x0	0x0	0b0	0b0	0b0	0b0	0b0
Access Type	Write, Read	Write, Read	Write, Read	Write, Read	Read Only	Write, Read	Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
RES_CFG	7	Resistor pullup/pulldown strength	0b0: 40kΩ 0b1: 1MΩ
TX_PRIO	6	Priority for GPIO scheduling	0b0: Low priority 0b1: High priority
TX_COMP_EN	5	Jitter minimization compensation enable	0b0: Jitter compensation disabled 0b1: Jitter compensation enabled
GPIO_OUT	4	GPIO pin output drive value when GPIO_RX_EN = 0	0b0: This GPIO pin output is driven to 0 0b1: This GPIO pin output is driven to 1
GPIO_IN	3	GPIO pin input level	0b0: This GPIO pin value is 0 0b1: This GPIO pin value is 1
GPIO_RX_EN	2	GPIO Out source control	0b0: This GPIO source disabled for GMSL2 reception 0b1: This GPIO source enabled for GMSL2 reception
GPIO_TX_EN	1	GPIO TX source control	0b0: This GPIO source disabled for GMSL2 transmission 0b1: This GPIO source enabled for GMSL2 transmission
GPIO_OUT_DIS	0	Disables GPIO output driver	0b0: Output driver enabled 0b1: Output driver disabled

GPIO_B (0x21C)

BIT	7	6	5	4	3	2	1	0
Field	PULL_UPDN_SEL[1:0]		OUT_TYPE	GPIO_TX_ID[4:0]				
Reset	0x2		0b1	0x09				
Access Type	Write, Read		Write, Read	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
PULL_UPDN_SEL	7:6	Buffer pull up/down configuration	0b00: None 0b01: Pullup 0b10: Pulldown 0b11: Reserved
OUT_TYPE	5	Driver type selection	0b0: Open-drain 0b1: Push-pull
GPIO_TX_ID	4:0	GPIO ID for pin while transmitting	0bXXXXX: GPIO transmit ID

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GPIO_C (0x21D)

BIT	7	6	5	4	3	2	1	0
Field	OVR_RES_CFG	–	–	GPIO_RX_ID[4:0]				
Reset	0x0	–	–	0x09				
Access Type	Write, Read	–	–	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
OVR_RES_CFG	7	Overrides non-GPIO port function IO setting. When set, RES_CFG and PULL_UPDN_SEL are effective when pin is configured as non-GPIO. When cleared, non-GPIO pin function determines IO type.	0b0: Non-GPIO function determines IO type when alternate function is selected 0b1: RES_CFG and PULL_UPDN_SEL determine IO type for non-GPIO configuration
GPIO_RX_ID	4:0	GPIO ID for pin while receiving	0bXXXXX: GPIO receive ID

GPIO_A (0x21E)

BIT	7	6	5	4	3	2	1	0
Field	RES_CFG	TX_PRIO	TX_COMP_EN	GPIO_OUT	GPIO_IN	GPIO_RX_EN	GPIO_TX_EN	GPIO_OUT_DIS
Reset	0b1	0x0	0x0	0b0	0b0	0b0	0b0	0b0
Access Type	Write, Read	Write, Read	Write, Read	Write, Read	Read Only	Write, Read	Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
RES_CFG	7	Resistor pullup/pulldown strength	0b0: 40kΩ 0b1: 1MΩ
TX_PRIO	6	Priority for GPIO scheduling	0b0: Low priority 0b1: High priority
TX_COMP_EN	5	Jitter minimization compensation enable	0b0: Jitter compensation disabled 0b1: Jitter compensation enabled
GPIO_OUT	4	GPIO pin output drive value when GPIO_RX_EN = 0	0b0: This GPIO pin output is driven to 0 0b1: This GPIO pin output is driven to 1
GPIO_IN	3	GPIO pin input level	0b0: This GPIO pin value is 0 0b1: This GPIO pin value is 1
GPIO_RX_EN	2	GPIO Out source control	0b0: This GPIO source disabled for GMSL2 reception 0b1: This GPIO source enabled for GMSL2 reception
GPIO_TX_EN	1	GPIO TX source control	0b0: This GPIO source disabled for GMSL2 transmission 0b1: This GPIO source enabled for GMSL2 transmission
GPIO_OUT_DIS	0	Disables GPIO output driver	0b0: Output driver enabled 0b1: Output driver disabled

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GPIO_B (0x21F)

BIT	7	6	5	4	3	2	1	0
Field	PULL_UPDN_SEL[1:0]		OUT_TYPE	GPIO_TX_ID[4:0]				
Reset	0x2		0b1	0x0A				
Access Type	Write, Read		Write, Read	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
PULL_UPDN_SEL	7:6	Buffer pullup/pulldown configuration	0b00: None 0b01: Pullup 0b10: Pulldown 0b11: Reserved
OUT_TYPE	5	Driver type selection	0b0: Open-drain 0b1: Push-pull
GPIO_TX_ID	4:0	GPIO ID for pin while transmitting	0bXXXXX: GPIO transmit ID

GPIO_C (0x220)

BIT	7	6	5	4	3	2	1	0
Field	OVR_RES_CFG	–	–	GPIO_RX_ID[4:0]				
Reset	0x0	–	–	0x0A				
Access Type	Write, Read	–	–	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
OVR_RES_CFG	7	Overrides non-GPIO port function IO setting. When set, RES_CFG and PULL_UPDN_SEL are effective when pin is configured as non-GPIO. When cleared, non-GPIO pin function determines IO type.	0b0: Non-GPIO function determines IO type when alternate function is selected 0b1: RES_CFG and PULL_UPDN_SEL determine IO type for non-GPIO configuration
GPIO_RX_ID	4:0	GPIO ID for pin while receiving	0bXXXXX: GPIO receive ID

GPIO_A (0x221)

BIT	7	6	5	4	3	2	1	0
Field	RES_CFG	TX_PRIO	TX_COMP_EN	GPIO_OUT	GPIO_IN	GPIO_RX_EN	GPIO_TX_EN	GPIO_OUT_DIS
Reset	0b1	0x0	0x0	0b0	0b0	0b0	0b0	0b0
Access Type	Write, Read	Write, Read	Write, Read	Write, Read	Read Only	Write, Read	Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
RES_CFG	7	Resistor pullup/pulldown strength	0b0: 40kΩ 0b1: 1MΩ
TX_PRIO	6	Priority for GPIO scheduling	0b0: Low priority 0b1: High priority
TX_COMP_EN	5	Jitter minimization compensation enable	0b0: Jitter compensation disabled 0b1: Jitter compensation enabled
GPIO_OUT	4	GPIO pin output drive value when GPIO_RX_EN = 0	0b0: This GPIO pin output is driven to 0 0b1: This GPIO pin output is driven to 1

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BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_IN	3	GPIO pin input level	0b0: This GPIO pin value is 0 0b1: This GPIO pin value is 1
GPIO_RX_EN	2	GPIO Out source control	0b0: This GPIO source disabled for GMSL2 reception 0b1: This GPIO source enabled for GMSL2 reception
GPIO_TX_EN	1	GPIO TX source control	0b0: This GPIO source disabled for GMSL2 transmission 0b1: This GPIO source enabled for GMSL2 transmission
GPIO_OUT_DIS	0	Disables GPIO output driver	0b0: Output driver enabled 0b1: Output driver disabled

GPIO_B (0x222)

BIT	7	6	5	4	3	2	1	0
Field	PULL_UPDN_SEL[1:0]		OUT_TYPE	GPIO_TX_ID[4:0]				
Reset	0x0		0b1	0x0B				
Access Type	Write, Read		Write, Read	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
PULL_UPDN_SEL	7:6	Buffer pullup/pulldown configuration	0b00: None 0b01: PullupPulldown -down 0b11: Reserved
OUT_TYPE	5	Driver type selection	0b0: Open-drain 0b1: Push-pull
GPIO_TX_ID	4:0	GPIO ID for pin while transmitting	0bXXXXX: GPIO transmit ID

GPIO_C (0x223)

BIT	7	6	5	4	3	2	1	0
Field	OVR_RES_CFG	–	–	GPIO_RX_ID[4:0]				
Reset	0x0	–	–	0x0B				
Access Type	Write, Read	–	–	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
OVR_RES_CFG	7	Overrides non-GPIO port function IO setting. When set, RES_CFG and PULL_UPDN_SEL are effective when pin is configured as non-GPIO. When cleared, non-GPIO pin function determines IO type.	0b0: Non-GPIO function determines IO type when alternate function is selected 0b1: RES_CFG and PULL_UPDN_SEL determine IO type for non-GPIO configuration
GPIO_RX_ID	4:0	GPIO ID for pin while receiving	0bXXXXX: GPIO receive ID

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GPIO_A (0x224)

BIT	7	6	5	4	3	2	1	0
Field	RES_CFG	TX_PRIO	TX_COMP_EN	GPIO_OUT	GPIO_IN	GPIO_RX_EN	GPIO_TX_EN	GPIO_OUT_DIS
Reset	0b1	0x0	0x0	0b0	0b0	0b0	0b0	0b0
Access Type	Write, Read	Write, Read	Write, Read	Write, Read	Read Only	Write, Read	Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
RES_CFG	7	Resistor pullup/pulldown strength	0b0: 40kΩ 0b1: 1MΩ
TX_PRIO	6	Priority for GPIO scheduling	0b0: Low priority 0b1: High priority
TX_COMP_EN	5	Jitter minimization compensation enable	0b0: Jitter compensation disabled 0b1: Jitter compensation enabled
GPIO_OUT	4	GPIO pin output drive value when GPIO_RX_EN = 0	0b0: This GPIO pin output is driven to 0 0b1: This GPIO pin output is driven to 1
GPIO_IN	3	GPIO pin input level	0b0: This GPIO pin value is 0 0b1: This GPIO pin value is 1
GPIO_RX_EN	2	GPIO Out source control	0b0: This GPIO source disabled for GMSL2 reception 0b1: This GPIO source enabled for GMSL2 reception
GPIO_TX_EN	1	GPIO TX source control	0b0: This GPIO source disabled for GMSL2 transmission 0b1: This GPIO source enabled for GMSL2 transmission
GPIO_OUT_DIS	0	Disables GPIO output driver	0b0: Output driver enabled 0b1: Output driver disabled

GPIO_B (0x225)

BIT	7	6	5	4	3	2	1	0
Field	PULL_UPDN_SEL[1:0]		OUT_TYPE	GPIO_TX_ID[4:0]				
Reset	0x0		0b1	0x0C				
Access Type	Write, Read		Write, Read	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
PULL_UPDN_SEL	7:6	Buffer pullup/pulldown configuration	0b00: None 0b01: Pullup 0b10: Pulldown 0b11: Reserved
OUT_TYPE	5	Driver type selection	0b0: Open-drain 0b1: Push-pull
GPIO_TX_ID	4:0	GPIO ID for pin while transmitting	0bXXXXX: GPIO transmit ID

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GPIO_C (0x226)

BIT	7	6	5	4	3	2	1	0
Field	OVR_RES_CFG	–	–	GPIO_RX_ID[4:0]				
Reset	0x0	–	–	0x0C				
Access Type	Write, Read	–	–	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
OVR_RES_CFG	7	Overrides non-GPIO port function IO setting. When set, RES_CFG and PULL_UPDN_SEL are effective when pin is configured as non-GPIO. When cleared, non-GPIO pin function determines IO type.	0b0: Non-GPIO function determines IO type when alternate function is selected 0b1: RES_CFG and PULL_UPDN_SEL determine IO type for non-GPIO configuration
GPIO_RX_ID	4:0	GPIO ID for pin while receiving	0bXXXXX: GPIO receive ID

GPIO_A (0x227)

BIT	7	6	5	4	3	2	1	0
Field	RES_CFG	TX_PRIO	TX_COMP_EN	GPIO_OUT	GPIO_IN	GPIO_RX_EN	GPIO_TX_EN	GPIO_OUT_DIS
Reset	0b1	0x0	0x0	0b0	0b0	0b0	0b0	0b0
Access Type	Write, Read	Write, Read	Write, Read	Write, Read	Read Only	Write, Read	Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
RES_CFG	7	Resistor pullup/pulldown strength	0b0: 40kΩ 0b1: 1MΩ
TX_PRIO	6	Priority for GPIO scheduling	0b0: Low priority 0b1: High priority
TX_COMP_EN	5	Jitter minimization compensation enable	0b0: Jitter compensation disabled 0b1: Jitter compensation enabled
GPIO_OUT	4	GPIO pin output drive value when GPIO_RX_EN = 0	0b0: This GPIO pin output is driven to 0 0b1: This GPIO pin output is driven to 1
GPIO_IN	3	GPIO pin input level	0b0: This GPIO pin value is 0 0b1: This GPIO pin value is 1
GPIO_RX_EN	2	GPIO Out source control	0b0: This GPIO source disabled for GMSL2 reception 0b1: This GPIO source enabled for GMSL2 reception
GPIO_TX_EN	1	GPIO TX source control	0b0: This GPIO source disabled for GMSL2 transmission 0b1: This GPIO source enabled for GMSL2 transmission
GPIO_OUT_DIS	0	Disables GPIO output driver	0b0: Output driver enabled 0b1: Output driver disabled

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GPIO_B (0x228)

BIT	7	6	5	4	3	2	1	0
Field	PULL_UPDN_SEL[1:0]		OUT_TYPE	GPIO_TX_ID[4:0]				
Reset	0x0		0b1	0x0D				
Access Type	Write, Read		Write, Read	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
PULL_UPDN_SEL	7:6	Buffer pullup/pulldown configuration	0b00: None 0b01: Pullup 0b10: Pulldown 0b11: Reserved
OUT_TYPE	5	Driver type selection	0b0: Open-drain 0b1: Push-pull
GPIO_TX_ID	4:0	GPIO ID for pin while transmitting	0bXXXXX: GPIO transmit ID

GPIO_C (0x229)

BIT	7	6	5	4	3	2	1	0
Field	OVR_RES_CFG	–	–	GPIO_RX_ID[4:0]				
Reset	0x0	–	–	0x0D				
Access Type	Write, Read	–	–	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
OVR_RES_CFG	7	Overrides non-GPIO port function IO setting. When set, RES_CFG and PULL_UPDN_SEL are effective when pin is configured as non-GPIO. When cleared, non-GPIO pin function determines IO type.	0b0: Non-GPIO function determines IO type when alternate function is selected 0b1: RES_CFG and PULL_UPDN_SEL determine IO type for non-GPIO configuration
GPIO_RX_ID	4:0	GPIO ID for pin while receiving	0bXXXXX: GPIO receive ID

GPIO_A (0x22A)

BIT	7	6	5	4	3	2	1	0
Field	RES_CFG	TX_PRIO	TX_COMP_EN	GPIO_OUT	GPIO_IN	GPIO_RX_EN	GPIO_TX_EN	GPIO_OUT_DIS
Reset	0b0	0x0	0x0	0b1	0b1	0b0	0b0	0b0
Access Type	Write, Read	Write, Read	Write, Read	Write, Read	Read Only	Write, Read	Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
RES_CFG	7	Resistor pullup/pulldown strength	0b0: 40kΩ 0b1: 1MΩ
TX_PRIO	6	Priority for GPIO scheduling	0b0: Low priority 0b1: High priority
TX_COMP_EN	5	Jitter minimization compensation enable	0b0: Jitter compensation disabled 0b1: Jitter compensation enabled
GPIO_OUT	4	GPIO pin output drive value when GPIO_RX_EN = 0	0b0: This GPIO pin output is driven to 0 0b1: This GPIO pin output is driven to 1

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BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_IN	3	GPIO pin input level	0b0: This GPIO pin value is 0 0b1: This GPIO pin value is 1
GPIO_RX_EN	2	GPIO Out source control	0b0: This GPIO source disabled for GMSL2 reception 0b1: This GPIO source enabled for GMSL2 reception
GPIO_TX_EN	1	GPIO TX source control	0b0: This GPIO source disabled for GMSL2 transmission 0b1: This GPIO source enabled for GMSL2 transmission
GPIO_OUT_DIS	0	Disables GPIO output driver	0b0: Output driver enabled 0b1: Output driver disabled

GPIO_B (0x22B)

BIT	7	6	5	4	3	2	1	0
Field	PULL_UPDN_SEL[1:0]		OUT_TYPE	GPIO_TX_ID[4:0]				
Reset	0x1		0b0	0x0E				
Access Type	Write, Read		Write, Read	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
PULL_UPDN_SEL	7:6	Buffer pullup/pulldown configuration	0b00: None 0b01: Pullup 0b10: Pulldown 0b11: Reserved
OUT_TYPE	5	I^2C IO is only open-drain	0bX: NA
GPIO_TX_ID	4:0	GPIO ID for pin while transmitting	0bXXXXX: GPIO transmit ID

GPIO_C (0x22C)

BIT	7	6	5	4	3	2	1	0
Field	OVR_RES_CFG	–	–	GPIO_RX_ID[4:0]				
Reset	0x0	–	–	0x0E				
Access Type	Write, Read	–	–	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
OVR_RES_CFG	7	Overrides non-GPIO port function IO setting. When set, RES_CFG and PULL_UPDN_SEL are effective when pin is configured as non-GPIO. When cleared, non-GPIO pin function determines IO type.	0b0: Non-GPIO function determines IO type when alternate function is selected 0b1: RES_CFG and PULL_UPDN_SEL determine IO type for non-GPIO configuration
GPIO_RX_ID	4:0	GPIO ID for pin while receiving	0bXXXXX: GPIO receive ID

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GPIO_A (0x22D)

BIT	7	6	5	4	3	2	1	0
Field	RES_CFG	TX_PRIO	TX_COMP_EN	GPIO_OUT	GPIO_IN	GPIO_RX_EN	GPIO_TX_EN	GPIO_OUT_DIS
Reset	0b0	0x0	0x0	0b1	0b1	0b0	0b0	0b0
Access Type	Write, Read	Write, Read	Write, Read	Write, Read	Read Only	Write, Read	Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
RES_CFG	7	Resistor pullup/pulldown strength	0b0: 40kΩ 0b1: 1MΩ
TX_PRIO	6	Priority for GPIO scheduling	0b0: Low priority 0b1: High priority
TX_COMP_EN	5	Jitter minimization compensation enable	0b0: Jitter compensation disabled 0b1: Jitter compensation enabled
GPIO_OUT	4	GPIO pin output drive value when GPIO_RX_EN = 0	0b0: This GPIO pin output is driven to 0 0b1: This GPIO pin output is driven to 1
GPIO_IN	3	GPIO pin input level	0b0: This GPIO pin value is 0 0b1: This GPIO pin value is 1
GPIO_RX_EN	2	GPIO Out source control	0b0: This GPIO source disabled for GMSL2 reception 0b1: This GPIO source enabled for GMSL2 reception
GPIO_TX_EN	1	GPIO TX source control	0b0: This GPIO source disabled for GMSL2 transmission 0b1: This GPIO source enabled for GMSL2 transmission
GPIO_OUT_DIS	0	Disables GPIO output driver	0b0: Output driver enabled 0b1: Output driver disabled

GPIO_B (0x22E)

BIT	7	6	5	4	3	2	1	0
Field	PULL_UPDN_SEL[1:0]		OUT_TYPE	GPIO_TX_ID[4:0]				
Reset	0x1		0b0	0x0F				
Access Type	Write, Read		Write, Read	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
PULL_UPDN_SEL	7:6	Buffer pullup/pulldown configuration	0b00: None 0b01: Pullup 0b10: Pulldown 0b11: Reserved
OUT_TYPE	5	I^2C IO is only open-drain	0bX: NA
GPIO_TX_ID	4:0	GPIO ID for pin while transmitting	0bXXXXX: GPIO transmit ID

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GPIO_C (0x22F)

BIT	7	6	5	4	3	2	1	0
Field	OVR_RES_CFG	–	–	GPIO_RX_ID[4:0]				
Reset	0x0	–	–	0x0F				
Access Type	Write, Read	–	–	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
OVR_RES_CFG	7	Overrides non-GPIO port function IO setting. When set, RES_CFG and PULL_UPDN_SEL are effective when pin is configured as non-GPIO. When cleared, non-GPIO pin function determines IO type.	0b0: Non-GPIO function determines IO type when alternate function is selected 0b1: RES_CFG and PULL_UPDN_SEL determine IO type for non-GPIO configuration
GPIO_RX_ID	4:0	GPIO ID for pin while receiving	0bXXXXX: GPIO receive ID

CMU4 (0x244)

BIT	7	6	5	4	3	2	1	0
Field	–	–	–	–	GPIO_SPEED_B[1:0]		GPIO_SPEED_A[1:0]	
Reset	–	–	–	–	0b10		0b11	
Access Type	–	–	–	–	Write, Read		Write, Read	

BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_SPEED_B	3:2	Controls GPIO Speed Group B (GPIO2 to GPIO13) transition time. First value is for VDDIO = 1.8V, second value is for VDDIO = 3.3V	0: 2ns, 1ns 1: 4ns, 2ns 2: 8ns, 4ns 3: 16ns, 8ns
GPIO_SPEED_A	1:0	Controls GPIO Speed Group A (GPIO0 and GPIO1) transition time. First value is for VDDIO = 1.8V, second value is for VDDIO = 3.3V	0: 2ns, 1ns 1: 4ns, 2ns 2: 8ns, 4ns 3: 16ns, 8ns

UART_PT_0 (0x24A)

BIT	7	6	5	4	3	2	1	0
Field	BITLEN_PT_1_L[7:0]							
Reset	0xDC							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
BITLEN_PT_1_L	7:0	Custom UART bit length for pass-through UART Channel 1. Set BITLEN_MAN_CFG_1 to 1 to use this value. Set this register to the UART bit length divided by 6.666ns (LSB).	0xXX: Lower byte of custom UART bit length for pass-through UART Channel 1

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UART_PT_1 (0x24B)

BIT	7	6	5	4	3	2	1	0
Field	–	–	BITLEN_PT_1_H[5:0]					
Reset	–	–	0x05					
Access Type	–	–	Write, Read					

BITFIELD	BITS	DESCRIPTION	DECODE
BITLEN_PT_1_H	5:0	Custom UART bit length for pass-through UART Channel 1. Set BITLEN_MAN_CFG_1 to 1 to use this value. Set this register to the UART bit length divided by 6.666ns (6 MSbs).	0xXX: Upper bits of custom UART bit length for pass-through UART Channel 1

UART_PT_2 (0x24C)

BIT	7	6	5	4	3	2	1	0
Field	BITLEN_PT_2_L[7:0]							
Reset	0xDC							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
BITLEN_PT_2_L	7:0	Custom UART bit length for pass-through UART Channel 2. Set BITLEN_MAN_CFG_2 to 1 to use this value. Set this register to the UART bit length divided by 6.666ns (LSB).	0xXX: Lower byte of custom UART bit length for pass-through UART Channel 2

UART_PT_3 (0x24D)

BIT	7	6	5	4	3	2	1	0
Field	–	–	BITLEN_PT_2_H[5:0]					
Reset	–	–	0x05					
Access Type	–	–	Write, Read					

BITFIELD	BITS	DESCRIPTION	DECODE
BITLEN_PT_2_H	5:0	Custom UART bit length for pass-through UART Channel 2. Set BITLEN_MAN_CFG_2 to 1 to use this value. Set this register to the UART bit length divided by 6.666ns (6 MSbs).	0xXX: Upper bits of custom UART bit length for pass-through UART Channel 2

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I2C_PT_4 (0x252)

BIT	7	6	5	4	3	2	1	0
Field	SRC_A_1[6:0]							–
Reset	0x0							–
Access Type	Write, Read							–

BITFIELD	BITS	DESCRIPTION	DECODE
SRC_A_1	7:1	I²C address translator source A. Works in conjunction with DST_A_1. When I²C device address matches SRC_A_1, internal I²C master (on remote side) replaces the device address by DST_A_1.	0bXXXXXXX: Address of I²C device that is to be replaced by the address in DST_A_1

I2C_PT_5 (0x253)

BIT	7	6	5	4	3	2	1	0
Field	DST_A_1[6:0]							–
Reset	0x0							–
Access Type	Write, Read							–

BITFIELD	BITS	DESCRIPTION	DECODE
DST_A_1	7:1	I²C address translator destination A. Works in conjunction with SRC_A_1. See the description of SRC_A_1.	0bXXXXXXX: Address that is to replace the address of the device pointed to in SRC_A_1

I2C_PT_6 (0x254)

BIT	7	6	5	4	3	2	1	0
Field	SRC_B_1[6:0]							–
Reset	0x0							–
Access Type	Write, Read							–

BITFIELD	BITS	DESCRIPTION	DECODE
SRC_B_1	7:1	I²C address translator source B. Works in conjunction with DST_B_1. When I²C device address matches SRC_B_1, internal I²C master (on remote side) replaces the device address by DST_B_1.	0bXXXXXXX: Address of I²C device that is to be replaced by the address in DST_B_1

I2C_PT_7 (0x255)

BIT	7	6	5	4	3	2	1	0
Field	DST_B_1[6:0]							–
Reset	0x0							–
Access Type	Write, Read							–

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BITFIELD	BITS	DESCRIPTION	DECODE
DST_B_1	7:1	I²C address translator destination B. Works in conjunction with SRC_B_1. See the description of SRC_B_1.	0bXXXXXXX: Address that is to replace the address of the device pointed to in SRC_B_1

I2C_PT_8 (0x256)

BIT	7	6	5	4	3	2	1	0
Field	SRC_A_2[6:0]							–
Reset	0x0							–
Access Type	Write, Read							–

BITFIELD	BITS	DESCRIPTION	DECODE
SRC_A_2	7:1	I²C address translator source A. Works in conjunction with DST_A_2. When I²C device address matches SRC_A_2, internal I²C master (on remote side) replaces the device address by DST_A_2.	0bXXXXXXX: Address of I²C device that is to be replaced by the address in DST_A_2

I2C_PT_9 (0x257)

BIT	7	6	5	4	3	2	1	0
Field	DST_A_2[6:0]							–
Reset	0x0							–
Access Type	Write, Read							–

BITFIELD	BITS	DESCRIPTION	DECODE
DST_A_2	7:1	I²C address translator destination A. Works in conjunction with SRC_A_2. See the description of SRC_A_2.	0bXXXXXXX: Address that is to replace the address of the device pointed to in SRC_A_2

I2C_PT_10 (0x258)

BIT	7	6	5	4	3	2	1	0
Field	SRC_B_2[6:0]							–
Reset	0x0							–
Access Type	Write, Read							–

BITFIELD	BITS	DESCRIPTION	DECODE
SRC_B_2	7:1	I²C address translator source B. Works in conjunction with DST_B_2. When I²C device address matches SRC_B_2, internal I²C master (on remote side) replaces the device address by DST_B_2.	0bXXXXXXX: Address of I²C device that is to be replaced by the address in DST_B_2

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I2C_PT_11 (0x259)

BIT	7	6	5	4	3	2	1	0
Field	DST_B_2[6:0]							–
Reset	0x0							–
Access Type	Write, Read							–

BITFIELD	BITS	DESCRIPTION	DECODE
DST_B_2	7:1	I²C address translator destination A. Works in conjunction with SRC_B_2. See the description of SRC_B_2.	0bXXXXXXX: Address that is to replace the address of the device pointed to in SRC_B_2

RLMS3 (0x403, 0x503)

BIT	7	6	5	4	3	2	1	0
Field	AdaptEn	–	–	–	–	–	–	–
Reset	0x0	–	–	–	–	–	–	–
Access Type	Write, Read	–	–	–	–	–	–	–

BITFIELD	BITS	DESCRIPTION	DECODE
AdaptEn	7	Adapt process enable	0b0: Manual adaptation process disabled 0b1: Manual adaptation process enabled

RLMS4 (0x404, 0x504)

BIT	7	6	5	4	3	2	1	0
Field	EOM_CHK_AMOUNT[3:0]				EOM_CHK_THR[1:0]		EOM_PER_MODE	EOM_EN
Reset	0x3				0x2		0x0	0x1
Access Type	Write, Read				Write, Read		Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
EOM_CHK_AMOUNT	7:4	This factor (N) is used to select the order of number of observations in each eye-monitor window, as follows: Observations = 6.29 x $10^{(N + 2)}$.	0xX: N factor
EOM_CHK_THR	3:2	Eye-opening monitor number of error bits to allow in a measurement window	0b00: Allow no errors 0b01: Allow 1 error 0b10: Allow 2 errors 0b11: Allow 3 errors
EOM_PER_MODE	1	Eye-opening monitor periodic mode enable	0b0: Eye-opening monitor periodic mode disabled 0b1: Eye-opening monitor periodic mode enabled
EOM_EN	0	Eye-opening monitor enable	0b0: Eye-opening monitor disabled 0b1: Eye-opening monitor enabled

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RLMS5 (0x405, 0x505)

BIT	7	6	5	4	3	2	1	0
Field	EOM_MAN_TRG_REQ	EOM_MIN_THR[6:0]						
Reset	0x0	0x10						
Access Type	Write Only	Write, Read						

BITFIELD	BITS	DESCRIPTION	DECODE
EOM_MAN_TRG_REQ	7	Eye-opening monitor manual trigger. For use when periodic mode is disabled.	0b0: No action 0b1: EOM manual trigger request
EOM_MIN_THR	6:0	The eye-opening monitor minimum threshold as defined by the following equation: Eye-opening percentage = EOM_MIN_THR/64. If the value is zero, the EOM is disabled.	0bXXXXXXX: EOM minimum threshold factor

RLMS6 (0x406, 0x506)

BIT	7	6	5	4	3	2	1	0
Field	EOM_PV_MODE	EOM_RST_THR[6:0]						
Reset	0x1	0x00						
Access Type	Write, Read	Write, Read						

BITFIELD	BITS	DESCRIPTION	DECODE
EOM_PV_MODE	7	Eye-opening is measured vertically or horizontally	0b0: Vertical opening mode 0b1: Horizonal opening mode
EOM_RST_THR	6:0	The eye-opening monitor refresh threshold as defined by the following equation: Percent eye-opening = EOM_MIN_THR/64. If the value is zero, the EOM is disabled.	0bXXXXXXX: EOM refresh threshold factor

RLMS7 (0x407, 0x507)

BIT	7	6	5	4	3	2	1	0
Field	EOM_DONE	EOM[6:0]						
Reset	0x0	0x00						
Access Type	Read Only	Read Only						

BITFIELD	BITS	DESCRIPTION	DECODE
EOM_DONE	7	Eye-opening monitor measurement done	0b0: EOM not complete 0b1: EOM complete
EOM	6:0	Last completed eye-opening monitor observation	0bXXXXXXX: EOM measurement result

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RLMS34 (0x434, 0x534)

BIT	7	6	5	4	3	2	1	0
Field	EyeMonPerCntL[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
EyeMonPerCntL	7:0	Eye monitor period count (RxClk20) LSB	0xXX: Eye monitor period count (LSB)

RLMS35 (0x435, 0x535)

BIT	7	6	5	4	3	2	1	0
Field	EyeMonPerCntH[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
EyeMonPerCntH	7:0	Eye monitor period count (RxClk20) MSB	0xXX: Eye monitor period count (MSB)

RLMS37 (0x437, 0x537)

BIT	7	6	5	4	3	2	1	0
Field	–	–	–	EyeMonDone	EyeMonCntClr	EyeMonStart	EyeMonPh	EyeMonDPol
Reset	–	–	–	0x0	0x0	0x0	0x0	0x0
Access Type	–	–	–	Read Only	Write Only	Write Only	Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
EyeMonDone	4	Eye monitor period complete (read-only, reset on start)	0b0: Eye monitor data collection not complete 0b1: Eye monitordata collection complete
EyeMonCntClr	3	Eye monitor error/valid count clear (one-shot). Read-back is EyeMonClrPL from long pulse generation.	0b0: NA 0b1: Clear eye monitor data collection counters
EyeMonStart	2	Eye monitor start (one-shot). Read-back is EyeMonStClrPL from long pulse generation for both start and clear.	0b0: NA 0b1: Start eye monitor data collection
EyeMonPh	1	Eye monitor phase	0b0: Eye monitor search early phase 0b1: Eye monitor search late phase
EyeMonDPol	0	Eye monitor data polarity	0b0: Eye monitor search for 1s 0b1: Eye monitor search for 0s

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RLMS38 (0x438, 0x538)

BIT	7	6	5	4	3	2	1	0
Field	EyeMonErrCntL[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION	DECODE
EyeMonErrCntL	7:0	Eye monitor error count (read-only)	0xXX: Eye monitor error count (LSB)

RLMS39 (0x439, 0x539)

BIT	7	6	5	4	3	2	1	0
Field	EyeMonErrCntH[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION	DECODE
EyeMonErrCntH	7:0	Eye monitor error count (read-only)	0xXX: Eye monitor error count (MSB)

RLMS3A (0x43A, 0x53A)

BIT	7	6	5	4	3	2	1	0
Field	EyeMonValCntL[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION	DECODE
EyeMonValCntL	7:0	Eye monitor valid (hit) count (read-only)	0xXX: Eye monitor valid count (LSB)

RLMS3B (0x43B, 0x53B)

BIT	7	6	5	4	3	2	1	0
Field	EyeMonValCntH[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION	DECODE
EyeMonValCntH	7:0	Eye monitor valid (hit) count (read-only)	0xXX: Eye monitor valid count (MSB)

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RLMS3D (0x43D, 0x53D)

BIT	7	6	5	4	3	2	1	0
Field	ErrChPh[6:0]							ErrChPhTogEn
Reset	0x00							0x1
Access Type	Read Only							Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
ErrChPh	7:1	Active error channel phase command (read-only)	0bXXXXXXX: 5.6 degrees per step
ErrChPhTogEn	0	Error channel phase toggle enable	0: Use primary phase only 1: Auto-toggle phase between primary and secondary

RLMS3E (0x43E, 0x53E)

BIT	7	6	5	4	3	2	1	0
Field	ErrChPhSecTA	ErrChPhSec[6:0]						
Reset	0x1	0x33						
Access Type	Write, Read	Write, Read						

BITFIELD	BITS	DESCRIPTION	DECODE
ErrChPhSecTA	7	Error channel phase secondary timing adjust	0b0: Timing adjust disabled 0b1: Timing adjust enabled
ErrChPhSec	6:0	A 7-bit phase command used for eye monitoring and used during adaptation by the error channel. Value: 7'h3A	0bxxxxxxx: Error channel phase secondary (odd)

RLMS3F (0x43F, 0x53F)

BIT	7	6	5	4	3	2	1	0
Field	ErrChPhPriTA	ErrChPhPri[6:0]						
Reset	0x0	0x72						
Access Type	Write, Read	Write, Read						

BITFIELD	BITS	DESCRIPTION	DECODE
ErrChPhPriTA	7	Error channel phase primary timing adjust	0b0: Timing adjust disabled 0b1: Timing adjust enabled
ErrChPhPri	6:0	A 7-bit phase command used for eye monitoring and used during adaptation by the error channel (fast receive, secondary phase, 3.5Gbps).	0bxxxxxxx: 7'h79

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RLMS49 (0x449, 0x549)

BIT	7	6	5	4	3	2	1	0
Field	–	–	–	–	–	ErrChPwrUp	–	–
Reset	–	–	–	–	–	0x0	–	–
Access Type	–	–	–	–	–	Write, Read	–	–

BITFIELD	BITS	DESCRIPTION	DECODE
ErrChPwrUp	2	Error channel power-down	0b0: Error channel power disabled 0b1: Error channel power enabled

RLMS58 (0x458, 0x558)

BIT	7	6	5	4	3	2	1	0
Field	–	ErrChVTh1[6:0]						
Reset	–	0x18						
Access Type	–	Write, Read						

BITFIELD	BITS	DESCRIPTION	DECODE
ErrChVTh1	6:0	Error channel target amplitude for 1s	0bxxxxxxx: Format: {sign, Magnitude[5:0]} where sign: 1 = negative, 0 = positive Magnitude:: binary amplitude 4.7mV per count

RLMS59 (0x459, 0x559)

BIT	7	6	5	4	3	2	1	0
Field	–	ErrChVTh0[6:0]						
Reset	–	0x58						
Access Type	–	Write, Read						

BITFIELD	BITS	DESCRIPTION	DECODE
ErrChVTh0	6:0	Error channel target amplitude for 0s	0bxxxxxx: Format: {sign, Magnitude[5:0]} where sign: 1 = negative, 0 = positive Magnitude:: binary amplitude 4.7mV per count

RLMS64 (0x464, 0x564)

BIT	7	6	5	4	3	2	1	0
Field	–	–	–	–	–	–	TxSSCMode[1:0]	
Reset	–	–	–	–	–	–	0x0	
Access Type	–	–	–	–	–	–	Write, Read	

BITFIELD	BITS	DESCRIPTION	DECODE
TxSSCMode	1:0	Tx spread spectrum mode	0b00: No spread spectrum (manual phase) 0b01: Center spread

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BITFIELD	BITS	DESCRIPTION	DECODE
			0b10: Down spread 0b11: Reserved

RLMS70 (0x470, 0x570)

BIT	7	6	5	4	3	2	1	0
Field	–	TxSSCFrqCtrl[6:0]						
Reset	–	0x01						
Access Type	–	Write, Read						

BITFIELD	BITS	DESCRIPTION	DECODE
TxSSCFrqCtrl	6:0	Tx spread spectrum frequency control	0bXXXXXXX: Tx spread spectrum center frequency control

RLMS71 (0x471, 0x571)

BIT	7	6	5	4	3	2	1	0
Field	–	TxSSCCenSprSt[5:0]						RSVD
Reset	–	0x01						0x0
Access Type	–	Write, Read						Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
TxSSCCenSprSt	6:1	Tx spread spectrum center spread startup control	0bXXXXXX: Tx spread spectrum center spread startup control
RSVD	0	Tx spread spectrum enable	0b0: Tx spread spectrum disabled 0b1: Tx spread spectrum enabled

RLMS72 (0x472, 0x572)

BIT	7	6	5	4	3	2	1	0
Field	TxSSCPreSclL[7:0]							
Reset	0xCF							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
TxSSCPreSclL	7:0	Tx spread spectrum frequency pre-scaler low byte	0xXX: Tx spread spectrum frequency pre-scaler low byte

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RLMS73 (0x473, 0x573)

BIT	7	6	5	4	3	2	1	0
Field	–	–	–	–	–	TxSSCPreSclH[2:0]		
Reset	–	–	–	–	–	0x0		
Access Type	–	–	–	–	–	Write, Read		

BITFIELD	BITS	DESCRIPTION	DECODE
TxSSCPreSclH	2:0	Tx spread spectrum frequency pre-scaler high bits	0bXXX: Tx spread spectrum frequency pre-scaler high bits

RLMS74 (0x474, 0x574)

BIT	7	6	5	4	3	2	1	0
Field	TxSSCPhL[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
TxSSCPhL	7:0	Tx spread spectrum interpolator phase low byte	0xXX: Tx spread spectrum frequency interpolator phase low byte

RLMS75 (0x475, 0x575)

BIT	7	6	5	4	3	2	1	0
Field	–	TxSSCPhH[6:0]						
Reset	–	0x0						
Access Type	–	Write, Read						

BITFIELD	BITS	DESCRIPTION	DECODE
TxSSCPhH	6:0	Tx spread spectrum interpolator phase	0bXXXXXXX: Tx spread spectrum frequency interpolator phase high bits

RLMS76 (0x476, 0x576)

BIT	7	6	5	4	3	2	1	0
Field	–	–	–	–	–	–	TxSSCPhQuad[1:0]	
Reset	–	–	–	–	–	–	0x0	
Access Type	–	–	–	–	–	–	Write, Read	

BITFIELD	BITS	DESCRIPTION	DECODE
TxSSCPhQuad	1:0	Tx spread spectrum interpolator phase quadrant	0bXX: Tx spread spectrum interpolator phase quadrant

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RLMS95 (0x495, 0x595)

BIT	7	6	5	4	3	2	1	0
Field	TxAmplManEn	–	TxAmplMan[5:0]					
Reset	0x0	–	0x29					
Access Type	Write, Read	–	Write, Read					

BITFIELD	BITS	DESCRIPTION	DECODE
TxAmplManEn	7	Tx amplitude manual override	0b0: Do not manually override Tx amplitude 0b1: Manually override Tx amplitude
TxAmplMan	5:0	Tx amplitude	0bXXXXXX: Binary amplitude 10mV per count

RLMSA4 (0x4A4, 0x5A4)

BIT	7	6	5	4	3	2	1	0
Field	AEQ_PER_MULT[1:0]		AEQ_PER[5:0]					
Reset	0x3		0x8					
Access Type	Write, Read		Write, Read					

BITFIELD	BITS	DESCRIPTION	DECODE
AEQ_PER_MULT	7:6	Adaptive EQ period multiplier	0b00: 1ms 0b01: 4ms 0b10: 16ms 0b11: 64ms
AEQ_PER	5:0	Adaptive EQ period Periodic adaptation is disabled when value is 0 Adaptive EQ period is (AEQ_PER value times AEQ_PER_MULT)	0bXXXXXX: Only zeros have a special meaning. See Description

RLMSAC (0x4AC, 0x5AC)

BIT	7	6	5	4	3	2	1	0
Field	ErrChPhSecTAFR3G	ErrChPhSecFR3G[6:0]						
Reset	0x1	0x4D						
Access Type	Write, Read	Write, Read						

BITFIELD	BITS	DESCRIPTION	DECODE
ErrChPhSecTAFR3G	7	Error channel phase secondary timing adjust	0b0: Timing adjust disabled 0b1: Timing adjust enabled
ErrChPhSecFR3G	6:0	A 7-bit phase command used for eye monitoring and used during adaptation by the error channel (fast receive, secondary phase, 3Gbps)	0bXXXXXXX: 7'h4D

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RLMSAD (0x4AD, 0x5AD)

BIT	7	6	5	4	3	2	1	0
Field	ErrChPhPriTAFR3G	ErrChPhPriFR3G[6:0]						
Reset	0x0	0x0D						
Access Type	Write, Read	Write, Read						

BITFIELD	BITS	DESCRIPTION	DECODE
ErrChPhPriTAFR3G	7	Error channel phase primary timing adjust	0b0: Timing adjust disabled 0b1: Timing adjust enabled
ErrChPhPriFR3G	6:0	A 7-bit phase command used for eye monitoring and used during adaptation by the error channel (fast receive, primary phase, 3Gbps)	0bXXXXXXX: 7'h0D

RLMSAE (0x4AE, 0x5AE)

BIT	7	6	5	4	3	2	1	0
Field	ErrChPhSecTAFR1G5	ErrChPhSecFR1G5[6:0]						
Reset	0x1	0x4E						
Access Type	Write, Read	Write, Read						

BITFIELD	BITS	DESCRIPTION	DECODE
ErrChPhSecTAFR1G5	7	Error channel phase secondary timing adjust	0b0: Timing adjust disabled 0b1: Timing adjust enabled
ErrChPhSecFR1G5	6:0	A 7-bit phase command used for eye monitoring and used during adaptation by the error channel (fast receive, secondary phase, 3.5Gbps)	0bXXXXXXX: 7'h4E

RLMSAF (0x4AF, 0x5AF)

BIT	7	6	5	4	3	2	1	0
Field	ErrChPhPriTAFR1G5	ErrChPhPriFR1G5[6:0]						
Reset	0x0	0x0F						
Access Type	Write, Read	Write, Read						

BITFIELD	BITS	DESCRIPTION	DECODE
ErrChPhPriTAFR1G5	7	Error channel phase primary timing adjust	0b0: Timing adjust disabled 0b1: Timing adjust enabled
ErrChPhPriFR1G5	6:0	A 7-bit phase command used for eye monitoring and used during adaptation by the error channel (fast receive, primary phase, 3.5Gbps)	0bXXXXXXX: 7'h0F

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RLMSB0 (0x4B0, 0x5B0)

BIT	7	6	5	4	3	2	1	0
Field	ErrChPhSecTASR1G5	ErrChPhSecSR1G5[6:0]						
Reset	0x1	0x34						
Access Type	Write, Read	Write, Read						

BITFIELD	BITS	DESCRIPTION	DECODE
ErrChPhSecTASR1G5	7	Error channel phase secondary timing adjust	0b0: Timing adjust disabled 0b1: Timing adjust enabled
ErrChPhSecSR1G5	6:0	A 7-bit phase command used for eye monitoring and used during adaptation by the error channel (slow receive, secondary phase, 3.5Gbps)	0bxXXXXXX: 7'h34

RLMSB1 (0x4B1, 0x5B1)

BIT	7	6	5	4	3	2	1	0
Field	ErrChPhPriTASR1G5	ErrChPhPriSR1G5[6:0]						
Reset	0x0	0x73						
Access Type	Write, Read	Write, Read						

BITFIELD	BITS	DESCRIPTION	DECODE
ErrChPhPriTASR1G5	7	Error channel phase primary timing adjust	0b0: Timing adjust disabled 0b1: Timing adjust enabled
ErrChPhPriSR1G5	6:0	A 7-bit phase command used for eye monitoring and used during adaptation by the error channel (slow receive, primary phase, 3.5Gbps)	0bXXXXXXX: 7'h73

RLMSB2 (0x4B2, 0x5B2)

BIT	7	6	5	4	3	2	1	0
Field	ErrChPhSecTASRG75	ErrChPhSecSRG75[6:0]						
Reset	0x1	0x3E						
Access Type	Write, Read	Write, Read						

BITFIELD	BITS	DESCRIPTION	DECODE
ErrChPhSecTASRG75	7	Error channel phase secondary timing adjust	0b0: Timing adjust disabled 0b1: Timing adjust enabled
ErrChPhSecSRG75	6:0	A 7-bit phase command used for eye monitoring and used during adaptation by the error channel (slow receive, secondary phase, 750Mbps)	0bXXXXXXX: 7'h3E

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RLMSB3 (0x4B3, 0x5B3)

BIT	7	6	5	4	3	2	1	0
Field	ErrChPhPriTASRG75	ErrChPhPriSRG75[6:0]						
Reset	0x0	0x7D						
Access Type	Write, Read	Write, Read						

BITFIELD	BITS	DESCRIPTION	DECODE
ErrChPhPriTASRG75	7	Error channel phase primary timing adjust	0b0: Timing adjust disabled 0b1: Timing adjust enabled
ErrChPhPriSRG75	6:0	A 7-bit phase command used for eye monitoring and used during adaptation by the error channel (slow receive, primary phase, 750Mbps)	0bXXXXXXX: 7'h7D

RLMSB4 (0x4B4, 0x5B4)

BIT	7	6	5	4	3	2	1	0
Field	ErrChPhSecTASRG375	ErrChPhSecSRG375[6:0]						
Reset	0x1	0x3C						
Access Type	Write, Read	Write, Read						

BITFIELD	BITS	DESCRIPTION	DECODE
ErrChPhSecTASRG375	7	Error channel phase secondary timing adjust	0b0: Timing adjust disabled 0b1: Timing adjust enabled
ErrChPhSecSRG375	6:0	A 7-bit phase command used for eye monitoring and used during adaptation by the error channel (slow receive, secondary phase, 375Mbps)	0bXXXXXXX: 7'h3C

RLMSB5 (0x4B5, 0x5B5)

BIT	7	6	5	4	3	2	1	0
Field	ErrChPhPriTASRG375	ErrChPhPriSRG375[6:0]						
Reset	0x0	0x7B						
Access Type	Write, Read	Write, Read						

BITFIELD	BITS	DESCRIPTION	DECODE
ErrChPhPriTASRG375	7	Error channel phase primary timing adjust	0b0: Timing adjust disabled 0b1: Timing adjust enabled
ErrChPhPriSRG375	6:0	A 7-bit phase command used for eye monitoring and used during adaptation by the error channel (slow receive, primary phase, 375Mbps)	0bXXXXXXX: 7'h7B

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RLMSB6 (0x4B6, 0x5B6)

BIT	7	6	5	4	3	2	1	0
Field	ErrChPhSecTASRG1875	ErrChPhSecSRG1875[6:0]						
Reset	0x1	0x3B						
Access Type	Write, Read	Write, Read						

BITFIELD	BITS	DESCRIPTION	DECODE
ErrChPhSecTASRG1875	7	Error channel phase secondary timing adjust	0b0: Timing adjust disabled 0b1: Timing adjust enabled
ErrChPhSecSRG1875	6:0	A 7-bit phase command used for eye monitoring and used during adaptation by the error channel (Slowreceive, secondary phase, 187.5Mbps)	0bXXXXXXX: 7'h3B

RLMSB7 (0x4B7, 0x5B7)

BIT	7	6	5	4	3	2	1	0
Field	ErrChPhPriTASRG1875	ErrChPhPriSRG1875[6:0]						
Reset	0x0	0x7A						
Access Type	Write, Read	Write, Read						

BITFIELD	BITS	DESCRIPTION	DECODE
ErrChPhPriTASRG1875	7	Error channel phase primary timing adjust	0b0: Timing adjust disabled 0b1: Timing adjust enabled
ErrChPhPriSRG1875	6:0	A 7-bit phase command used for eye monitoring and used during adaptation by the error channel (slow receive, primary phase, 187.5Mbps)	0bXXXXXXX: 7'h7A

HDCP_0 (0x680)

BIT	7	6	5	4	3	2	1	0
Field	GMSL_BKSV_0[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION	DECODE
GMSL_BKSV_0	7:0	HDCP Receiver KSV	0xXX: Byte 0 (LSB) of the HDCP receiver KSV

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HDCP_1 (0x681)

BIT	7	6	5	4	3	2	1	0
Field	GMSL_BKSV_1[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION	DECODE
GMSL_BKSV_1	7:0	HDCP Receiver KSV	0xXX: Byte 1 of the HDCP receiver KSV

HDCP_2 (0x682)

BIT	7	6	5	4	3	2	1	0
Field	GMSL_BKSV_2[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION	DECODE
GMSL_BKSV_2	7:0	HDCP Receiver KSV	0xXX: Byte 2 of the HDCP receiver KSV

HDCP_3 (0x683)

BIT	7	6	5	4	3	2	1	0
Field	GMSL_BKSV_3[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION	DECODE
GMSL_BKSV_3	7:0	HDCP Receiver KSV	0xXX: Byte 3 of the HDCP receiver KSV

HDCP_4 (0x684)

BIT	7	6	5	4	3	2	1	0
Field	GMSL_BKSV_4[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION	DECODE
GMSL_BKSV_4	7:0	HDCP Receiver KSV	0xXX: Byte 4 (MSB) of the HDCP receiver KSV

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HDCP_5 (0x685)

BIT	7	6	5	4	3	2	1	0
Field	RIP_0[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION	DECODE
RIP_0	7:0	Link verification response (LSB)	0xXX: Byte 0 (LSB) of Ri or Ri'

HDCP_6 (0x686)

BIT	7	6	5	4	3	2	1	0
Field	RIP_1[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION	DECODE
RIP_1	7:0	Link verification response (MSB)	0xXX: Byte 1 (MSB) of Ri or Ri'

HDCP_7 (0x687)

BIT	7	6	5	4	3	2	1	0
Field	PJP[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION	DECODE
PJP	7:0	Enhanced link verification response	0xXX: Value of P_j or $P_{j'}$

HDCP_8 (0x688)

BIT	7	6	5	4	3	2	1	0
Field	AN_0[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
AN_0	7:0	Session random number	0xXX: Byte 0 (LSB) of the session random number

HDCP_9 (0x689)

BIT	7	6	5	4	3	2	1	0
Field	AN_1[7:0]							
Reset	0x00							
Access Type	Write, Read							

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BITFIELD	BITS	DESCRIPTION	DECODE
AN_1	7:0	Session random number	0xXX: Byte 1 of the session random number

HDCP_A (0x68A)

BIT	7	6	5	4	3	2	1	0
Field	AN_2[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
AN_2	7:0	Session random number	0xXX: Byte 2 of the session random number

HDCP_B (0x68B)

BIT	7	6	5	4	3	2	1	0
Field	AN_3[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
AN_3	7:0	Session random number	0xXX: Byte 3 of the session random number

HDCP_C (0x68C)

BIT	7	6	5	4	3	2	1	0
Field	AN_4[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
AN_4	7:0	Session random number	0xXX: Byte 4 of the session random number

HDCP_D (0x68D)

BIT	7	6	5	4	3	2	1	0
Field	AN_5[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
AN_5	7:0	Session random number	0xXX: Byte 5 of the session random number

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HDCP_E (0x68E)

BIT	7	6	5	4	3	2	1	0
Field	AN_6[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
AN_6	7:0	Session random number	0xXX: Byte 6 of the session random number

HDCP_F (0x68F)

BIT	7	6	5	4	3	2	1	0
Field	AN_7[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
AN_7	7:0	Session random number	0xXX: Byte 7 (MSB) of the session random number

HDCP_10 (0x690)

BIT	7	6	5	4	3	2	1	0
Field	AKSV_0[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
AKSV_0	7:0	HDCP Transmitter KSV	0xXX: Byte 0 (LSB) of the HDCP transmitter KSV

HDCP_11 (0x691)

BIT	7	6	5	4	3	2	1	0
Field	AKSV_1[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
AKSV_1	7:0	HDCP Transmitter KSV	0xXX: Byte 1 of the HDCP transmitter KSV

HDCP_12 (0x692)

BIT	7	6	5	4	3	2	1	0
Field	AKSV_2[7:0]							
Reset	0x00							
Access Type	Write, Read							

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BITFIELD	BITS	DESCRIPTION	DECODE
AKSV_2	7:0	HDCP Transmitter KSV	0xXX: Byte 2 of the HDCP transmitter KSV

HDCP_13 (0x693)

BIT	7	6	5	4	3	2	1	0
Field	AKSV_3[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
AKSV_3	7:0	HDCP Transmitter KSV	0xXX: Byte 3 of the HDCP transmitter KSV

HDCP_14 (0x694)

BIT	7	6	5	4	3	2	1	0
Field	AKSV_4[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
AKSV_4	7:0	HDCP Transmitter KSV	0xXX: Byte 4 (MSB) of the HDCP transmitter KSV

HDCP_15 (0x695)

BIT	7	6	5	4	3	2	1	0
Field	–	–	–	–	–	–	AUTH_START	ENC_EN
Reset	–	–	–	–	–	–	0b0	0b0
Access Type	–	–	–	–	–	–	Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
AUTH_START	1	HDCP Authentication started	0b0: HDCP encryption not started 0b1: HDCP encryption started
ENC_EN	0	Enable HDCP decryption	0b0: HDCP encryption disabled 0b1: HDCP encryption enabled

HDCP_16 (0x696)

BIT	7	6	5	4	3	2	1	0
Field	–	–	–	–	–	–	HDCP_NEW_DEV_CONN	HDCP_READY
Reset	–	–	–	–	–	–	0b0	0b0
Access Type	–	–	–	–	–	–	Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
HDCP_NEW	1	HDCP new device connected bit	0b0: HDCP new device not connected

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BITFIELD	BITS	DESCRIPTION	DECODE
_DEV_CON N			0b1: HDCP new device connected
HDCP_REA DY	0	HDCP repeater ready bit	0b0: HDCP repeater not ready 0b1: HDCP repeater ready

HDCP_17 (0x697)

BIT	7	6	5	4	3	2	1	0
Field	HDCP_BCAPS[6:0]							HDCP_REP EATER
Reset	0x00							0b0
Access Type	Write, Read							Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
HDCP_BCA PS	7:1	HDCP BCAPS register	0b0000000: Device is not a repeater 0b0000001: Device is a repeater 0b0000010 - 0b1111111: Reserved
HDCP_REP EATER	0	HDCP repeater enable	0b0: HDCP repeater not enabled 0b1: HDCP repeater enabled

HDCP_19 (0x699)

BIT	7	6	5	4	3	2	1	0
Field	–	–	–	–	–	–	–	DIS_AUDIO _ENC
Reset	–	–	–	–	–	–	–	0b0
Access Type	–	–	–	–	–	–	–	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
DIS_AUDIO_ ENC	0	Disable Audio Encryption	0b0: Audio encryption disabled 0b1: Audio encryption enabled

HDCP_1A (0x69A)

BIT	7	6	5	4	3	2	1	0
Field	PJFP_0[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION	DECODE
PJFP_0	7:0	Fast enhanced link verification response (lower byte)	0xXX: Byte 0 (LSB) of P_{jf} or $P_{jf'}$

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HDCP_1B (0x69B)

BIT	7	6	5	4	3	2	1	0
Field	PJFP_1[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION	DECODE
PJFP_1	7:0	Fast enhanced link verification response (middle byte)	0xXX: Byte 1 of P_{jf} or $P_{jf'}$

HDCP_1C (0x69C)

BIT	7	6	5	4	3	2	1	0
Field	PJFP_2[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION	DECODE
PJFP_2	7:0	Fast enhanced link verification response (upper byte)	0xXX: Byte 2 (MSB) of P_{jf} or $P_{jf'}$

HDCP_20 (0x6A0)

BIT	7	6	5	4	3	2	1	0
Field	VH0_0[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION	DECODE
VH0_0	7:0	H0 part of SHA-1 hash value. EN_INT_COMP = 0 → V of Tx (read). EN_INT_COMP = 1 → V' of Rx (write and read).	0xXX: Byte 0 (LSB) of V.H0 or V'.H0

HDCP_21 (0x6A1)

BIT	7	6	5	4	3	2	1	0
Field	VH0_1[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION	DECODE
VH0_1	7:0	H0 part of SHA-1 hash value. EN_INT_COMP = 0 → V of Tx (read). EN_INT_COMP = 1 → V' of Rx (write and read).	0xXX: Byte 1 of V.H0 or V'.H0

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HDCP_22 (0x6A2)

BIT	7	6	5	4	3	2	1	0
Field	VH0_2[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION	DECODE
VH0_2	7:0	H0 part of SHA-1 hash value. EN_INT_COMP = 0 → V of Tx (read). EN_INT_COMP = 1 → V' of Rx (write and read).	0xXX: Byte 2 of V.H0 or V'.H0

HDCP_23 (0x6A3)

BIT	7	6	5	4	3	2	1	0
Field	VH0_3[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION	DECODE
VH0_3	7:0	H0 part of SHA-1 hash value. EN_INT_COMP = 0 → V of Tx (read). EN_INT_COMP = 1 → V' of Rx (write and read).	0xXX: Byte 3 (MSB) of V.H0 or V'.H0

HDCP_24 (0x6A4)

BIT	7	6	5	4	3	2	1	0
Field	VH1_0[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION	DECODE
VH1_0	7:0	H1 part of SHA-1 hash value. EN_INT_COMP = 0 → V of Tx (read). EN_INT_COMP = 1 → V' of Rx (write and read).	0xXX: Byte 0 (LSB) of V.H1 or V'.H1

HDCP_25 (0x6A5)

BIT	7	6	5	4	3	2	1	0
Field	VH1_1[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION	DECODE
VH1_1	7:0	H1 part of SHA-1 hash value. EN_INT_COMP = 0 → V of Tx (read).	0xXX: Byte 1 of V.H1 or V'.H1

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BITFIELD	BITS	DESCRIPTION	DECODE
		EN_INT_COMP = 1 → V' of Rx (write and read).	

HDCP_26 (0x6A6)

BIT	7	6	5	4	3	2	1	0
Field	VH1_2[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION	DECODE
VH1_2	7:0	H1 part of SHA-1 hash value. EN_INT_COMP = 0 → V of Tx (read). EN_INT_COMP = 1 → V' of Rx (write and read).	0xXX: Byte 2 of V.H1 or V'.H1

HDCP_27 (0x6A7)

BIT	7	6	5	4	3	2	1	0
Field	VH1_3[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION	DECODE
VH1_3	7:0	H1 part of SHA-1 hash value. EN_INT_COMP = 0 → V of Tx (read). EN_INT_COMP = 1 → V' of Rx (write and read).	0xXX: Byte 3 (MSB) of V.H1 or V'.H1

HDCP_28 (0x6A8)

BIT	7	6	5	4	3	2	1	0
Field	VH2_0[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION	DECODE
VH2_0	7:0	H2 part of SHA-1 hash value. EN_INT_COMP = 0 → V of Tx (read). EN_INT_COMP = 1 → V' of Rx (write and read).	0xXX: Byte 0 (LSB) of V.H2 or V'.H2

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HDCP_29 (0x6A9)

BIT	7	6	5	4	3	2	1	0
Field	VH2_1[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION	DECODE
VH2_1	7:0	H2 part of SHA-1 hash value. EN_INT_COMP = 0 → V of Tx (read). EN_INT_COMP = 1 → V' of Rx (write and read).	0xXX: Byte 1 of V.H2 or V'.H2

HDCP_2A (0x6AA)

BIT	7	6	5	4	3	2	1	0
Field	VH2_2[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION	DECODE
VH2_2	7:0	H2 part of SHA-1 hash value. EN_INT_COMP = 0 → V of Tx (read). EN_INT_COMP = 1 → V' of Rx (write and read).	0xXX: Byte 2 of V.H2 or V'.H2

HDCP_2B (0x6AB)

BIT	7	6	5	4	3	2	1	0
Field	VH2_3[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION	DECODE
VH2_3	7:0	H2 part of SHA-1 hash value. EN_INT_COMP = 0 → V of Tx (read). EN_INT_COMP = 1 → V' of Rx (write and read).	0xXX: Byte 3 (MSB) of V.H2 or V'.H2

HDCP_2C (0x6AC)

BIT	7	6	5	4	3	2	1	0
Field	VH3_0[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION	DECODE
VH3_0	7:0	H3 part of SHA-1 hash value. EN_INT_COMP = 0 → V of Tx (read).	0xXX: Byte 0 (LSB) of V.H3 or V'.H3

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BITFIELD	BITS	DESCRIPTION	DECODE
		EN_INT_COMP = 1 → V' of Rx (write and read).	

HDCP_2D (0x6AD)

BIT	7	6	5	4	3	2	1	0
Field	VH3_1[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION	DECODE
VH3_1	7:0	H3 part of SHA-1 hash value. EN_INT_COMP = 0 → V of Tx (read). EN_INT_COMP = 1 → V' of Rx (write and read).	0xXX: Byte 1 of V.H3 or V'.H3

HDCP_2E (0x6AE)

BIT	7	6	5	4	3	2	1	0
Field	VH3_2[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION	DECODE
VH3_2	7:0	H3 part of SHA-1 hash value. EN_INT_COMP = 0 → V of Tx (read). EN_INT_COMP = 1 → V' of Rx (write and read).	0xXX: Byte 2 of V.H3 or V'.H3

HDCP_2F (0x6AF)

BIT	7	6	5	4	3	2	1	0
Field	VH3_3[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION	DECODE
VH3_3	7:0	H3 part of SHA-1 hash value. EN_INT_COMP = 0 → V of Tx (read). EN_INT_COMP = 1 → V' of Rx (write and read).	0xXX: Byte 3 (MSB) of V.H3 or V'.H3

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HDCP_30 (0x6B0)

BIT	7	6	5	4	3	2	1	0
Field	VH4_0[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION	DECODE
VH4_0	7:0	H4 part of SHA-1 hash value. EN_INT_COMP = 0 → V of Tx (read). EN_INT_COMP = 1 → V' of Rx (write and read).	0xXX: Byte 0 (LSB) of V.H4 or V'.H4

HDCP_31 (0x6B1)

BIT	7	6	5	4	3	2	1	0
Field	VH4_1[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION	DECODE
VH4_1	7:0	H4 part of SHA-1 hash value. EN_INT_COMP = 0 → V of Tx (read). EN_INT_COMP = 1 → V' of Rx (write and read).	0xXX: Byte 1 of V.H4 or V'.H4

HDCP_32 (0x6B2)

BIT	7	6	5	4	3	2	1	0
Field	VH4_2[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION	DECODE
VH4_2	7:0	H4 part of SHA-1 hash value. EN_INT_COMP = 0 → V of Tx (read). EN_INT_COMP = 1 → V' of Rx (write and read).	0xXX: Byte 2 of V.H4 or V'.H4

HDCP_33 (0x6B3)

BIT	7	6	5	4	3	2	1	0
Field	VH4_3[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION	DECODE
VH4_3	7:0	H4 part of SHA-1 hash value. EN_INT_COMP = 0 → V of Tx (read).	0xXX: Byte 3 (MSB) of V.H4 or V'.H4

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BITFIELD	BITS	DESCRIPTION	DECODE
		EN_INT_COMP = 1 → V' of Rx (write and read).	

HDCP_34 (0x6B4)

BIT	7	6	5	4	3	2	1	0
Field	HDCP_MAX_DEV_EXCEEDED	HDCP_DEV_CNT[6:0]						
Reset	0b0	0x00						
Access Type	Write, Read	Write, Read						

BITFIELD	BITS	DESCRIPTION	DECODE
HDCP_MAX_DEV_EXCEEDED	7	Maximum device count exceeded in repeater mode	0b0: Maximum number of devices not exceeded (14 or less) 0b1: Maximum number of devices exceeded (more than 14)
HDCP_DEV_CNT	6:0	Repeater device count	0bXXXXXXX: Number of devices attached

HDCP_35 (0x6B5)

BIT	7	6	5	4	3	2	1	0
Field	–	–	–	–	HDCP_MAX_CASCADE_EXCEEDED	HDCP_DEPTH[2:0]		
Reset	–	–	–	–	0b0	0x0		
Access Type	–	–	–	–	Write, Read	Write, Read		

BITFIELD	BITS	DESCRIPTION	DECODE
HDCP_MAX_CASCADE_EXCEEDED	3	HDCP repeater mode exceeded maximum depth	0b0: Maximum number of cascaded devices not exceeded (7 or less) 0b1: Maximum number of cascaded devices exceeded (more than 7)
HDCP_DEPTH	2:0	HDCP repeater depth	0bXXX: Depth of cascaded devices

HDCP_36 (0x6B6)

BIT	7	6	5	4	3	2	1	0
Field	HDCP_GPMEM[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
HDCP_GPMEM	7:0	General purpose memory byte. This has no effect on operation of the device, and can be used for handshaking between	0xXX: General purpose memory byte

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BITFIELD	BITS	DESCRIPTION	DECODE
		more than one microcontroller in the system.	

HDCP_37 (0x6B7)

BIT	7	6	5	4	3	2	1	0
Field	HDCP_PD	AH_MODE	–	–	–	–	–	–
Reset	0b1	0b1	–	–	–	–	–	–
Access Type	Write, Read	Write, Read	–	–	–	–	–	–

BITFIELD	BITS	DESCRIPTION	DECODE
HDCP_PD	7	Power down HDCP blocks	0b0: HDCP blocks powered up 0b1: Power down HDCP blocks
AH_MODE	6	Enable automatic HDCP operation	0b0: Manual HDCP or non-HDCP mode 0b1: Automatic HDCP mode

HDCP_38 (0x6B8)

BIT	7	6	5	4	3	2	1	0
Field	HDCP_INT	–	–	–	–	–	MASK_AUTH_ST	MASK_ENC_EN
Reset	0x0	–	–	–	–	–	0b0	0b0
Access Type	Read Clears All	–	–	–	–	–	Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
HDCP_INT	7	HDCP interrupt is asserted (unless masked) in the following cases: – HDCP encryption enable/disable – HDCP authentication started Read to clear the interrupt.	0b0: HDCP interrupt not asserted 0b1: HDCP interrupt asserted
MASK_AUTH_ST	1	Mask authentication start interrupt	0b0: Authentication start interrupt not masked 0b1: Authentication start interrupt masked
MASK_ENC_EN	0	Mask encryption enable/disable interrupt	0b0: HDCP encryption not masked, HDCP interrupt available 0b1: Mask HDCP encryption, disable HDCP interrupt

HDCP_39 (0x6B9)

BIT	7	6	5	4	3	2	1	0
Field	LINK_LOCK	AH_VID_LOCK	AH_RESERVED[5:0]					
Reset	0b0	0b0	0x35					
Access Type	Read Only	Read Only	Read Only					

BITFIELD	BITS	DESCRIPTION	DECODE
LINK_LOCK	7	GMSL-2 link locked (control channel is operational)	0b0: GMSL2 link is not locked 0b1: GMSL2 link is locked. control channel is operational.

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BITFIELD	BITS	DESCRIPTION	DECODE
AH_VID_LOCK	6	Video receiver has started outputting video	0b0: Video receiver not locked 0b1: Video receiver has started outputting video
AH_RESERVED	5:0	Reserved bits	0bXXXXXX: Reserved

HDCP_3A (0x6BA)

BIT	7	6	5	4	3	2	1	0
Field	KSV_LIST_0[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
KSV_LIST_0	7:0	List of key selection vectors of all downstream repeaters and receivers. A maximum of 14 downstream repeaters and receivers are supported (14 x 5 = 70).	0xXX: KSV list Byte 0

HDCP_3B (0x6BB)

BIT	7	6	5	4	3	2	1	0
Field	KSV_LIST_1[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
KSV_LIST_1	7:0	List of key selection vectors of all downstream repeaters and receivers. A maximum of 14 downstream repeaters and receivers are supported (14 x 5 = 70).	0xXX: KSV list Byte 1

HDCP_3C (0x6BC)

BIT	7	6	5	4	3	2	1	0
Field	KSV_LIST_2[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
KSV_LIST_2	7:0	List of key selection vectors of all downstream repeaters and receivers. A maximum of 14 downstream repeaters and receivers are supported (14 x 5 = 70).	0xXX: KSV list Byte 2

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HDCP_3D (0x6BD)

BIT	7	6	5	4	3	2	1	0
Field	KSV_LIST_3[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
KSV_LIST_3	7:0	List of key selection vectors of all downstream repeaters and receivers. A maximum of 14 downstream repeaters and receivers are supported (14 x 5 = 70).	0xXX: KSV list Byte 3

HDCP_3E (0x6BE)

BIT	7	6	5	4	3	2	1	0
Field	KSV_LIST_4[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
KSV_LIST_4	7:0	List of key selection vectors of all downstream repeaters and receivers. A maximum of 14 downstream repeaters and receivers are supported (14 x 5 = 70).	0xXX: KSV list Byte 4

HDCP_3F (0x6BF)

BIT	7	6	5	4	3	2	1	0
Field	KSV_LIST_5[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
KSV_LIST_5	7:0	List of key selection vectors of all downstream repeaters and receivers. A maximum of 14 downstream repeaters and receivers are supported (14 x 5 = 70).	0xXX: KSV list Byte 5

HDCP_40 (0x6C0)

BIT	7	6	5	4	3	2	1	0
Field	KSV_LIST_6[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
KSV_LIST_6	7:0	List of key selection vectors of all downstream repeaters and receivers.	0xXX: KSV list Byte 6

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BITFIELD	BITS	DESCRIPTION	DECODE
		A maximum of 14 downstream repeaters and receivers are supported (14 x 5 = 70).	

HDCP_41 (0x6C1)

BIT	7	6	5	4	3	2	1	0
Field	KSV_LIST_7[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
KSV_LIST_7	7:0	List of key selection vectors of all downstream repeaters and receivers. A maximum of 14 downstream repeaters and receivers are supported (14 x 5 = 70).	0xXX: KSV list Byte 7

HDCP_42 (0x6C2)

BIT	7	6	5	4	3	2	1	0
Field	KSV_LIST_8[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
KSV_LIST_8	7:0	List of key selection vectors of all downstream repeaters and receivers. A maximum of 14 downstream repeaters and receivers are supported (14 x 5 = 70).	0xXX: KSV list Byte 8

HDCP_43 (0x6C3)

BIT	7	6	5	4	3	2	1	0
Field	KSV_LIST_9[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
KSV_LIST_9	7:0	List of key selection vectors of all downstream repeaters and receivers. A maximum of 14 downstream repeaters and receivers are supported (14 x 5 = 70).	0xXX: KSV list Byte 9

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HDCP_44 (0x6C4)

BIT	7	6	5	4	3	2	1	0
Field	KSV_LIST_10[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
KSV_LIST_10	7:0	List of key selection vectors of all downstream repeaters and receivers. A maximum of 14 downstream repeaters and receivers are supported (14 x 5 = 70).	0xXX: KSV list Byte 10

HDCP_45 (0x6C5)

BIT	7	6	5	4	3	2	1	0
Field	KSV_LIST_11[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
KSV_LIST_11	7:0	List of key selection vectors of all downstream repeaters and receivers. A maximum of 14 downstream repeaters and receivers are supported (14 x 5 = 70).	0xXX: KSV list Byte 11

HDCP_46 (0x6C6)

BIT	7	6	5	4	3	2	1	0
Field	KSV_LIST_12[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
KSV_LIST_12	7:0	List of key selection vectors of all downstream repeaters and receivers. A maximum of 14 downstream repeaters and receivers are supported (14 x 5 = 70).	0xXX: KSV list Byte 12

HDCP_47 (0x6C7)

BIT	7	6	5	4	3	2	1	0
Field	KSV_LIST_13[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
KSV_LIST_13	7:0	List of key selection vectors of all downstream repeaters and receivers.	0xXX: KSV list Byte 13

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BITFIELD	BITS	DESCRIPTION	DECODE
		A maximum of 14 downstream repeaters and receivers are supported (14 x 5 = 70).	

HDCP_48 (0x6C8)

BIT	7	6	5	4	3	2	1	0
Field	KSV_LIST_14[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
KSV_LIST_14	7:0	List of key selection vectors of all downstream repeaters and receivers. A maximum of 14 downstream repeaters and receivers are supported (14 x 5 = 70).	0xXX: KSV list Byte 14

HDCP_49 (0x6C9)

BIT	7	6	5	4	3	2	1	0
Field	KSV_LIST_15[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
KSV_LIST_15	7:0	List of key selection vectors of all downstream repeaters and receivers. A maximum of 14 downstream repeaters and receivers are supported (14 x 5 = 70).	0xXX: KSV list Byte 15

HDCP_4A (0x6CA)

BIT	7	6	5	4	3	2	1	0
Field	KSV_LIST_16[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
KSV_LIST_16	7:0	List of key selection vectors of all downstream repeaters and receivers. A maximum of 14 downstream repeaters and receivers are supported (14 x 5 = 70).	0xXX: KSV list Byte 16

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HDCP_4B (0x6CB)

BIT	7	6	5	4	3	2	1	0
Field	KSV_LIST_17[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
KSV_LIST_17	7:0	List of key selection vectors of all downstream repeaters and receivers. A maximum of 14 downstream repeaters and receivers are supported (14 x 5 = 70).	0xXX: KSV list Byte 17

HDCP_4C (0x6CC)

BIT	7	6	5	4	3	2	1	0
Field	KSV_LIST_18[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
KSV_LIST_18	7:0	List of key selection vectors of all downstream repeaters and receivers. A maximum of 14 downstream repeaters and receivers are supported (14 x 5 = 70).	0xXX: KSV list Byte 18

HDCP_4D (0x6CD)

BIT	7	6	5	4	3	2	1	0
Field	KSV_LIST_19[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
KSV_LIST_19	7:0	List of key selection vectors of all downstream repeaters and receivers. A maximum of 14 downstream repeaters and receivers are supported (14 x 5 = 70).	0xXX: KSV list Byte 19

HDCP_4E (0x6CE)

BIT	7	6	5	4	3	2	1	0
Field	KSV_LIST_20[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
KSV_LIST_20	7:0	List of key selection vectors of all downstream repeaters and receivers.	0xXX: KSV list Byte 20

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BITFIELD	BITS	DESCRIPTION	DECODE
		A maximum of 14 downstream repeaters and receivers are supported (14 x 5 = 70).	

HDCP_4F (0x6CF)

BIT	7	6	5	4	3	2	1	0
Field	KSV_LIST_21[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
KSV_LIST_21	7:0	List of key selection vectors of all downstream repeaters and receivers. A maximum of 14 downstream repeaters and receivers are supported (14 x 5 = 70).	0xXX: KSV list Byte 21

HDCP_50 (0x6D0)

BIT	7	6	5	4	3	2	1	0
Field	KSV_LIST_22[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
KSV_LIST_22	7:0	List of key selection vectors of all downstream repeaters and receivers. A maximum of 14 downstream repeaters and receivers are supported (14 x 5 = 70).	0xXX: KSV list Byte 22

HDCP_51 (0x6D1)

BIT	7	6	5	4	3	2	1	0
Field	KSV_LIST_23[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
KSV_LIST_23	7:0	List of key selection vectors of all downstream repeaters and receivers. A maximum of 14 downstream repeaters and receivers are supported (14 x 5 = 70).	0xXX: KSV list Byte 23

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HDCP_52 (0x6D2)

BIT	7	6	5	4	3	2	1	0
Field	KSV_LIST_24[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
KSV_LIST_24	7:0	List of key selection vectors of all downstream repeaters and receivers. A maximum of 14 downstream repeaters and receivers are supported (14 x 5 = 70).	0xXX: KSV list Byte 24

HDCP_53 (0x6D3)

BIT	7	6	5	4	3	2	1	0
Field	KSV_LIST_25[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
KSV_LIST_25	7:0	List of key selection vectors of all downstream repeaters and receivers. A maximum of 14 downstream repeaters and receivers are supported (14 x 5 = 70).	0xXX: KSV list Byte 25

HDCP_54 (0x6D4)

BIT	7	6	5	4	3	2	1	0
Field	KSV_LIST_26[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
KSV_LIST_26	7:0	List of key selection vectors of all downstream repeaters and receivers. A maximum of 14 downstream repeaters and receivers are supported (14 x 5 = 70).	0xXX: KSV list Byte 26

HDCP_55 (0x6D5)

BIT	7	6	5	4	3	2	1	0
Field	KSV_LIST_27[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
KSV_LIST_27	7:0	List of key selection vectors of all downstream repeaters and receivers.	0xXX: KSV list Byte 27

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BITFIELD	BITS	DESCRIPTION	DECODE
		A maximum of 14 downstream repeaters and receivers are supported (14 x 5 = 70).	

HDCP_56 (0x6D6)

BIT	7	6	5	4	3	2	1	0
Field	KSV_LIST_28[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
KSV_LIST_28	7:0	List of key selection vectors of all downstream repeaters and receivers. A maximum of 14 downstream repeaters and receivers are supported (14 x 5 = 70).	0xXX: KSV list Byte 28

HDCP_57 (0x6D7)

BIT	7	6	5	4	3	2	1	0
Field	KSV_LIST_29[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
KSV_LIST_29	7:0	List of key selection vectors of all downstream repeaters and receivers. A maximum of 14 downstream repeaters and receivers are supported (14 x 5 = 70).	0xXX: KSV list Byte 29

HDCP_58 (0x6D8)

BIT	7	6	5	4	3	2	1	0
Field	KSV_LIST_30[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
KSV_LIST_30	7:0	List of key selection vectors of all downstream repeaters and receivers. A maximum of 14 downstream repeaters and receivers are supported (14 x 5 = 70).	0xXX: KSV list Byte 30

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HDCP_59 (0x6D9)

BIT	7	6	5	4	3	2	1	0
Field	KSV_LIST_31[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
KSV_LIST_31	7:0	List of key selection vectors of all downstream repeaters and receivers. A maximum of 14 downstream repeaters and receivers are supported (14 x 5 = 70).	0xXX: KSV list Byte 31

HDCP_5A (0x6DA)

BIT	7	6	5	4	3	2	1	0
Field	KSV_LIST_32[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
KSV_LIST_32	7:0	List of key selection vectors of all downstream repeaters and receivers. A maximum of 14 downstream repeaters and receivers are supported (14 x 5 = 70).	0xXX: KSV list Byte 32

HDCP_5B (0x6DB)

BIT	7	6	5	4	3	2	1	0
Field	KSV_LIST_33[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
KSV_LIST_33	7:0	List of key selection vectors of all downstream repeaters and receivers. A maximum of 14 downstream repeaters and receivers are supported (14 x 5 = 70).	0xXX: KSV list Byte 33

HDCP_5C (0x6DC)

BIT	7	6	5	4	3	2	1	0
Field	KSV_LIST_34[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
KSV_LIST_34	7:0	List of key selection vectors of all downstream repeaters and receivers.	0xXX: KSV list Byte 34

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BITFIELD	BITS	DESCRIPTION	DECODE
		A maximum of 14 downstream repeaters and receivers are supported (14 x 5 = 70).	

HDCP_5D (0x6DD)

BIT	7	6	5	4	3	2	1	0
Field	KSV_LIST_35[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
KSV_LIST_35	7:0	List of key selection vectors of all downstream repeaters and receivers. A maximum of 14 downstream repeaters and receivers are supported (14 x 5 = 70).	0xXX: KSV list Byte 35

HDCP_5E (0x6DE)

BIT	7	6	5	4	3	2	1	0
Field	KSV_LIST_36[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
KSV_LIST_36	7:0	List of key selection vectors of all downstream repeaters and receivers. A maximum of 14 downstream repeaters and receivers are supported (14 x 5 = 70).	0xXX: KSV list Byte 36

HDCP_5F (0x6DF)

BIT	7	6	5	4	3	2	1	0
Field	KSV_LIST_37[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
KSV_LIST_37	7:0	List of key selection vectors of all downstream repeaters and receivers. A maximum of 14 downstream repeaters and receivers are supported (14 x 5 = 70).	0xXX: KSV list Byte 37

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HDCP_60 (0x6E0)

BIT	7	6	5	4	3	2	1	0
Field	KSV_LIST_38[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
KSV_LIST_38	7:0	List of key selection vectors of all downstream repeaters and receivers. A maximum of 14 downstream repeaters and receivers are supported (14 x 5 = 70).	0xXX: KSV list Byte 38

HDCP_61 (0x6E1)

BIT	7	6	5	4	3	2	1	0
Field	KSV_LIST_39[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
KSV_LIST_39	7:0	List of key selection vectors of all downstream repeaters and receivers. A maximum of 14 downstream repeaters and receivers are supported (14 x 5 = 70).	0xXX: KSV list Byte 39

HDCP_62 (0x6E2)

BIT	7	6	5	4	3	2	1	0
Field	KSV_LIST_40[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
KSV_LIST_40	7:0	List of key selection vectors of all downstream repeaters and receivers. A maximum of 14 downstream repeaters and receivers are supported (14 x 5 = 70).	0xXX: KSV list Byte 40

HDCP_63 (0x6E3)

BIT	7	6	5	4	3	2	1	0
Field	KSV_LIST_41[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
KSV_LIST_41	7:0	List of key selection vectors of all downstream repeaters and receivers.	0xXX: KSV list Byte 41

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BITFIELD	BITS	DESCRIPTION	DECODE
		A maximum of 14 downstream repeaters and receivers are supported (14 x 5 = 70).	

HDCP_64 (0x6E4)

BIT	7	6	5	4	3	2	1	0
Field	KSV_LIST_42[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
KSV_LIST_42	7:0	List of key selection vectors of all downstream repeaters and receivers. A maximum of 14 downstream repeaters and receivers are supported (14 x 5 = 70).	0xXX: KSV list Byte 42

HDCP_65 (0x6E5)

BIT	7	6	5	4	3	2	1	0
Field	KSV_LIST_43[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
KSV_LIST_43	7:0	List of key selection vectors of all downstream repeaters and receivers. A maximum of 14 downstream repeaters and receivers are supported (14 x 5 = 70).	0xXX: KSV list Byte 43

HDCP_66 (0x6E6)

BIT	7	6	5	4	3	2	1	0
Field	KSV_LIST_44[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
KSV_LIST_44	7:0	List of key selection vectors of all downstream repeaters and receivers. A maximum of 14 downstream repeaters and receivers are supported (14 x 5 = 70).	0xXX: KSV list Byte 44

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HDCP_67 (0x6E7)

BIT	7	6	5	4	3	2	1	0
Field	KSV_LIST_45[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
KSV_LIST_45	7:0	List of key selection vectors of all downstream repeaters and receivers. A maximum of 14 downstream repeaters and receivers are supported (14 x 5 = 70).	0xXX: KSV list Byte 45

HDCP_68 (0x6E8)

BIT	7	6	5	4	3	2	1	0
Field	KSV_LIST_46[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
KSV_LIST_46	7:0	List of key selection vectors of all downstream repeaters and receivers. A maximum of 14 downstream repeaters and receivers are supported (14 x 5 = 70).	0xXX: KSV list Byte 46

HDCP_69 (0x6E9)

BIT	7	6	5	4	3	2	1	0
Field	KSV_LIST_47[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
KSV_LIST_47	7:0	List of key selection vectors of all downstream repeaters and receivers. A maximum of 14 downstream repeaters and receivers are supported (14 x 5 = 70).	0xXX: KSV list Byte 47

HDCP_6A (0x6EA)

BIT	7	6	5	4	3	2	1	0
Field	KSV_LIST_48[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
KSV_LIST_48	7:0	List of key selection vectors of all downstream repeaters and receivers.	0xXX: KSV list Byte 48

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BITFIELD	BITS	DESCRIPTION	DECODE
		A maximum of 14 downstream repeaters and receivers are supported (14 x 5 = 70).	

HDCP_6B (0x6EB)

BIT	7	6	5	4	3	2	1	0
Field	KSV_LIST_49[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
KSV_LIST_49	7:0	List of key selection vectors of all downstream repeaters and receivers. A maximum of 14 downstream repeaters and receivers are supported (14 x 5 = 70).	0xXX: KSV list Byte 49

HDCP_6C (0x6EC)

BIT	7	6	5	4	3	2	1	0
Field	KSV_LIST_50[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
KSV_LIST_50	7:0	List of key selection vectors of all downstream repeaters and receivers. A maximum of 14 downstream repeaters and receivers are supported (14 x 5 = 70).	0xXX: KSV list Byte 50

HDCP_6D (0x6ED)

BIT	7	6	5	4	3	2	1	0
Field	KSV_LIST_51[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
KSV_LIST_51	7:0	List of key selection vectors of all downstream repeaters and receivers. A maximum of 14 downstream repeaters and receivers are supported (14 x 5 = 70).	0xXX: KSV list Byte 51

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HDCP_6E (0x6EE)

BIT	7	6	5	4	3	2	1	0
Field	KSV_LIST_52[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
KSV_LIST_52	7:0	List of key selection vectors of all downstream repeaters and receivers. A maximum of 14 downstream repeaters and receivers are supported (14 x 5 = 70).	0xXX: KSV list Byte 52

HDCP_6F (0x6EF)

BIT	7	6	5	4	3	2	1	0
Field	KSV_LIST_53[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
KSV_LIST_53	7:0	List of key selection vectors of all downstream repeaters and receivers. A maximum of 14 downstream repeaters and receivers are supported (14 x 5 = 70).	0xXX: KSV list Byte 53

HDCP_70 (0x6F0)

BIT	7	6	5	4	3	2	1	0
Field	KSV_LIST_54[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
KSV_LIST_54	7:0	List of key selection vectors of all downstream repeaters and receivers. A maximum of 14 downstream repeaters and receivers are supported (14 x 5 = 70).	0xXX: KSV list Byte 54

HDCP_71 (0x6F1)

BIT	7	6	5	4	3	2	1	0
Field	KSV_LIST_55[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
KSV_LIST_55	7:0	List of key selection vectors of all downstream repeaters and receivers.	0xXX: KSV list Byte 55

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BITFIELD	BITS	DESCRIPTION	DECODE
		A maximum of 14 downstream repeaters and receivers are supported (14 x 5 = 70).	

HDCP_72 (0x6F2)

BIT	7	6	5	4	3	2	1	0
Field	KSV_LIST_56[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
KSV_LIST_56	7:0	List of key selection vectors of all downstream repeaters and receivers. A maximum of 14 downstream repeaters and receivers are supported (14 x 5 = 70).	0xXX: KSV list Byte 56

HDCP_73 (0x6F3)

BIT	7	6	5	4	3	2	1	0
Field	KSV_LIST_57[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
KSV_LIST_57	7:0	List of key selection vectors of all downstream repeaters and receivers. A maximum of 14 downstream repeaters and receivers are supported (14 x 5 = 70).	0xXX: KSV list Byte 57

HDCP_74 (0x6F4)

BIT	7	6	5	4	3	2	1	0
Field	KSV_LIST_58[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
KSV_LIST_58	7:0	List of key selection vectors of all downstream repeaters and receivers. A maximum of 14 downstream repeaters and receivers are supported (14 x 5 = 70).	0xXX: KSV list Byte 58

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HDCP_75 (0x6F5)

BIT	7	6	5	4	3	2	1	0
Field	KSV_LIST_59[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
KSV_LIST_59	7:0	List of key selection vectors of all downstream repeaters and receivers. A maximum of 14 downstream repeaters and receivers are supported (14 x 5 = 70).	0xXX: KSV list Byte 59

HDCP_76 (0x6F6)

BIT	7	6	5	4	3	2	1	0
Field	KSV_LIST_60[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
KSV_LIST_60	7:0	List of key selection vectors of all downstream repeaters and receivers. A maximum of 14 downstream repeaters and receivers are supported (14 x 5 = 70).	0xXX: KSV list Byte 60

HDCP_77 (0x6F7)

BIT	7	6	5	4	3	2	1	0
Field	KSV_LIST_61[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
KSV_LIST_61	7:0	List of key selection vectors of all downstream repeaters and receivers. A maximum of 14 downstream repeaters and receivers are supported (14 x 5 = 70).	0xXX: KSV list Byte 61

HDCP_78 (0x6F8)

BIT	7	6	5	4	3	2	1	0
Field	KSV_LIST_62[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
KSV_LIST_62	7:0	List of key selection vectors of all downstream repeaters and receivers.	0xXX: KSV list Byte 62

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BITFIELD	BITS	DESCRIPTION	DECODE
		A maximum of 14 downstream repeaters and receivers are supported (14 x 5 = 70).	

HDCP_79 (0x6F9)

BIT	7	6	5	4	3	2	1	0
Field	KSV_LIST_63[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
KSV_LIST_63	7:0	List of key selection vectors of all downstream repeaters and receivers. A maximum of 14 downstream repeaters and receivers are supported (14 x 5 = 70).	0xXX: KSV list Byte 63

HDCP_7A (0x6FA)

BIT	7	6	5	4	3	2	1	0
Field	KSV_LIST_64[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
KSV_LIST_64	7:0	List of key selection vectors of all downstream repeaters and receivers. A maximum of 14 downstream repeaters and receivers are supported (14 x 5 = 70).	0xXX: KSV list Byte 64

HDCP_7B (0x6FB)

BIT	7	6	5	4	3	2	1	0
Field	KSV_LIST_65[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
KSV_LIST_65	7:0	List of key selection vectors of all downstream repeaters and receivers. A maximum of 14 downstream repeaters and receivers are supported (14 x 5 = 70).	0xXX: KSV list Byte 65

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HDCP_7C (0x6FC)

BIT	7	6	5	4	3	2	1	0
Field	KSV_LIST_66[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
KSV_LIST_66	7:0	List of key selection vectors of all downstream repeaters and receivers. A maximum of 14 downstream repeaters and receivers are supported (14 x 5 = 70).	0xXX: KSV list Byte 66

HDCP_7D (0x6FD)

BIT	7	6	5	4	3	2	1	0
Field	KSV_LIST_67[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
KSV_LIST_67	7:0	List of key selection vectors of all downstream repeaters and receivers. A maximum of 14 downstream repeaters and receivers are supported (14 x 5 = 70).	0xXX: KSV list Byte 67

HDCP_7E (0x6FE)

BIT	7	6	5	4	3	2	1	0
Field	KSV_LIST_68[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
KSV_LIST_68	7:0	List of key selection vectors of all downstream repeaters and receivers. A maximum of 14 downstream repeaters and receivers are supported (14 x 5 = 70).	0xXX: KSV list Byte 68

HDCP_7F (0x6FF)

BIT	7	6	5	4	3	2	1	0
Field	KSV_LIST_69[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
KSV_LIST_69	7:0	List of key selection vectors of all downstream repeaters and receivers.	0xXX: KSV list Byte 69

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BITFIELD	BITS	DESCRIPTION	DECODE
		A maximum of 14 downstream repeaters and receivers are supported (14 x 5 = 70).	